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Time-sensitive networking profile for industrial automation



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Time-sensitive networking profile for industrial automation

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INTRODUCTION

This document defines time-sensitive networking profiles for industrial automation. The profile selects features, options, configurations, defaults, protocols, and procedures of bridges, end stations, and LANs to build industrial automation networks. This document also specifies YANG modules for the digital data sheet and for remote procedure calls.

The profile meets the industrial automation market objective of converging Operations Technology (OT) and Information Technology (IT) networks by defining a common, standardized network infrastructure. This objective is accomplished by taking advantage of the improvements that Time-Sensitive Networking provides to IEEE 802.1 and IEEE 802.3 standard Ethernet networks by providing guaranteed data transport with bounded low latency, low latency variation, zero congestion loss for critical traffic, and high availability.

The profile helps the convergence of industrial communication networks by referring only to international standards to build the lower layers of the communication stack and their management.

Ethernet extended with Time-Sensitive Networking technology provides the features required in the area of industrial communication networks, such as:

- meeting low latency and latency variation requirements concerning data transmission;
- efficient exchange of data records on a frequent time period;
- reliable communications with calculable downtime;
- high availability meeting application requirements;
- efficient mechanisms for bandwidth utilization of exchanges of data records, with zero congestion loss;
- improved clock synchronization mechanisms, including support of multiple gPTP domains.

1 Scope

This document defines time-sensitive networking profiles for industrial automation. The profiles select features, options, configurations, defaults, protocols, and procedures of bridges, end stations, and LANs to build industrial automation networks. This document also specifies YANG modules defining read-only information available online and offline as a digital data sheet. This document also specifies YANG modules for remote procedure calls and actions to address requirements arising from industrial automation networks.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO/IEC 9594-8:2020, *Information technology - Open systems interconnection - Part 8: The Directory: Public-key and attribute certificate frameworks*, available at: <https://www.iso.org/obp/ui/#iso:std:iso-iec:9594:-8:en> [viewed 2025-10-03]

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IEEE Std 802.1AB-2016, *IEEE Standard for Local and metropolitan area networks - Station and Media Access Control Connectivity Discovery*

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3 Terms, definitions, symbols, abbreviated terms and conventions

3.1 General

For the purposes of this document, the terms and definitions given in ITU-T G.8260, IEEE Std 802-2024TM, IEEE Std 802.3TM-2022, IEEE Std 802.1QTM-2022, IEEE Std 802.1ASTM-2020 and the following apply.

ISO, IEC, IEEE and ITU-T maintain terminology databases for use in standardization at the following addresses:

- IEC Electropedia: available at <https://www.electropedia.org/>
- ISO Online browsing platform: available at <https://www.iso.org/obp>
- IEEE Standards Dictionary Online: available at <https://dictionary.ieee.org>
- ITU-T Terms and Definitions database: available at https://www.itu.int/br_tsb_terms/#/

3.2 List of terms, abbreviated terms and definitions given in various standards

For the purposes of this document, the terms and definitions given in Table 1 apply.

For ease of understanding, the most important terms used within this document are listed in Table 1 but the definitions are not repeated.

Table 1 – List of terms

Term	Source
BTCA	IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr TM -2024
Bridge	IEEE Std 802.1Q-2022
Bridge Port	IEEE Std 802.1Q-2022
CFM	IEEE Std 802.1Q-2022
Clock	IEEE Std 802.1AS-2020
ClockTimeTransmitter	IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024
ClockTimeReceiver	IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024
ClockSource	IEEE Std 802.1AS-2020
ClockTarget	IEEE Std 802.1AS-2020
CNC	IEEE Std 802.1Q-2022
Configuration Domain	IEEE Std 802.1Qdj TM -2024
constant time error (cTE)	ITU-T G.8260
Customer Virtual Local Area Network (C-VLAN) component	IEEE Std 802.1Q-2022
CUC	IEEE Std 802.1Q-2022

Term	Source
device	IEEE Std 802.1AR™-2018
DLL	IEEE Std 802-2024
DTE	IEEE Std 802.3-2022
dynamic time error (dTE)	ITU-T G.8260
end entity (EE)	NIST Special Publication 800-57 Part 2, Revision 1
end station	IEEE Std 802-2024
Ethernet	IEEE Std 802.3-2022
FDB	IEEE Std 802.1Q-2022
FID	IEEE Std 802.1Q-2022
fingerprint	IETF RFC 7589
FQTSS	IEEE Std 802.1Q-2022
fractional frequency offset	IEEE Std 802.1AS-2020
frame	IEEE Std 802.1Q-2022
frame preemption	IEEE Std 802.1Q-2022
FRER	IEEE Std 802.1CB™-2017
gating cycle	IEEE Std 802.1Q-2022
gPTP communication path	IEEE Std 802.1AS-2020
gPTP domain	IEEE Std 802.1AS-2020
Grandmaster Clock	IEEE Std 802.1AS-2020
Grandmaster PTP Instance	IEEE Std 802.1AS-2020
IST	IEEE Std 802.1Q-2022
LAN	IEEE Std 802-2024
latency	IEEE Std 802.1Q-2022
Listener	IEEE Std 802.1Q-2022
LLDP	IEEE Std 802.1AB™-2016
LLDPDU	IEEE Std 802.1AB-2016
local clock	IEEE Std 802.1AS-2020
LocalClock	IEEE Std 802.1AS-2020
logical link	IEEE Std 802-2024
LPI	IEEE Std 802.3-2022
MAC	IEEE Std 802-2024
MMRP	IEEE Std 802.1Q-2022
MST	IEEE Std 802.1Q-2022
MVRP	IEEE Std 802.1Q-2022
NETCONF	IETF RFC 6241
PCP	IEEE Std 802.1Q-2022
PDU	IEEE Std 802.1Q-2022
PHY	IEEE Std 802.3-2022
PLS	IEEE Std 802.3-2022
Port	IEEE Std 802.1Q-2022
preciseOriginTimestamp	IEEE Std 802.1AS-2020
primary domain	IEEE Std 802.1ASdm™-2024
PTP End Instance	IEEE Std 802.1AS-2020

Term	Source
PTP Instance	IEEE Std 802.1AS-2020
PTP Link	IEEE Std 802.1AS-2020
PTP Port	IEEE Std 802.1AS-2020
PTP Relay Instance	IEEE Std 802.1AS-2020
PVID	IEEE Std 802.1Q-2022
redundancy	IEC 60050-192
residence time	IEEE Std 802.1AS-2020
secondary domain	IEEE Std 802.1ASdm-2024
signature suite	IEEE Std 802.1AR-2018
station	IEEE Std 802-2024
stream	IEEE Std 802.1Q-2022
synchronized time	IEEE Std 802.1AS-2020
Talker	IEEE Std 802.1Q-2022
time error	ITU-T G.8260
time-sensitive stream	IEEE Std 802.1Q-2022
traffic class	IEEE Std 802.1Q-2022
TLV	IEEE Std 802.3-2022
UNI	IEEE Std 802.1Q-2022
VID	IEEE Std 802.1Q-2022
VLAN	IEEE Std 802.1Q-2022
X.509	ISO/IEC 9594-8:2020
YANG	IETF RFC 6020

3.3 Terms and definitions in this document

3.3.1

application clock

clock used by the application to time events

Note 1 to entry: Events can be periodic or aperiodic.

3.3.2

Bridge component

Customer Virtual Local Area Network (C-VLAN) Bridge

3.3.3

control latency

time delay between the input to a sensor application and the output from an actuator application

Note 1 to entry: For the purposes of this document, control latency does not include latencies in the sensor, actuator, or the physical system in a process.

3.3.4

deadline

application defined reference point that represents a time when data is required by the application

3.3.5

digital data sheet

information about the capabilities of an IA-station

3.3.6

end station component

end station entity

3.3.7

Global Time

synchronized time, derived from a gPTP domain, that is traceable to the PTP timescale

3.3.8

IA-controller

industrial automation function, consisting of a comparing element and a controlling element, that performs a specified control function

Note 1 to entry: An IA-controller exchanges data with several IA-devices or other IA-controllers for the purpose of control of a system.

Note 2 to entry: The primary categories of IA-controllers are distributed control systems (DCS), programmable logic controllers (PLCs), and programmable automation controllers (PACs).

3.3.9

IA-device

industrial automation function, consisting of sensor or actuator elements to read or write process data

Note 1 to entry: An IA-device exchanges data with an IA-controller or other IA-devices for the purpose of control of a system.

3.3.10

IA-station

material element or assembly of one or more end station components, and zero, one or more bridge components

Note 1 to entry: IA-controllers and IA-devices are industrial automation functions of IA-stations.

3.3.11

imprinting

<security> equipping IA-stations with an LDevID credential, corresponding trust anchor and certificate-to-name mapping instructions

3.3.12

management entity

IA-station function responsible for configuration of Bridge components, end station components and ports

Note 1 to entry: The management entity interacts with remote management.

3.3.13

network diameter

number of links in the longest of all the calculated shortest paths between each pair of nodes in the network

3.3.14

network provisioning

process of defining a consistent network configuration, which is applied to all stations

3.3.15**nominal frequency**

ideal frequency with zero uncertainty

Note 1 to entry: The nominal frequency of the PTP timescale is further explained in IEEE Std 1588TM-2019, 7.2.1, 7.2.2, and Annex B.

3.3.16**non-proxied IA-station**

IA-station which contains a ccA Bridge component or a ccA end station component

3.3.17**ppm**

$\mu\text{Hz}/\text{Hz}$

Note 1 to entry: The term "ppm" refers to a pure multiplier of 0,000 001 and is used in the context of this document as an SI unit term to allow readable terms conformant to various rules related to expressions.

3.3.18**proxy**

IA-station whose Netconf server can be utilized by other IA-stations for the purpose of remote management

3.3.19**proxy target**

IA-station which contains only ccB Bridge or end station components

3.3.20**Working Clock**

synchronized time, derived from a gPTP domain, that is traceable to the PTP timescale, or to an ARB timescale that is continuous

Note 1 to entry: In general, the Working Clock is traceable to an ARB timescale; however, the Working Clock time can be correlated to a recognized timing standard.

3.4 Abbreviated terms

AEAD	Authenticated Encryption with Associated Data
AES	Advanced Encryption Standard
ARB	Arbitrary
ASCII	American Standard Code for Information Interchange
ASN	Abstract Syntax Notation
BTCA	Best timeTransmitter Clock Algorithm
CA	Certification Authority
CBC	Cipher Block Chaining
ccA	Conformance Class A
ccB	Conformance Class B
CFM	Connectivity Fault Management
CMLDS	Common Mean Link Delay Service
CMS	Cryptographic Message Syntax
CN	Common Name
CNC	Centralized Network Configuration
CRL	Certificate Revocation List
CRUDX	Create Read Update Delete eXecute
CSR	Certificate Signing Request

CUC	Centralized User Configuration
C-VLAN	Customer VLAN
DAC	Discretionary Access Control
DER	Distinguished Encoding Rules
DH	Diffie-Hellman
DHE	Diffie-Hellman Ephemeral
DLL	Data Link Layer
DMAC	Destination MAC Address
DNS	Domain Name Service
DSA	Digital Signature Algorithm
DTE	Data Terminal Equipment
EC	Elliptic Curve
ECC	Elliptic Curve Cryptography
ECDSA	Elliptic Curve Digital Signature Algorithm
EdDSA	Edwards-Curve Digital Signature Algorithm
EE	End Entity
FDB	Filtering Database
FID	Filtering Identifier
FQDN	Fully Qualified Domain Name
FQTSS	Forwarding and Queuing Enhancements for Time-Sensitive Streams
FRER	Frame Replication and Elimination for Reliability
GCM	Galois Counter Mode
gPTP	generalized Precision Time Protocol
HMAC	Keyed-Hashing for Message Authentication Code
HW	Hardware
IA	Industrial Automation
IDeVID	Initial Secure Device Identifier
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
I-LAN	Internal Local Area Network
ISO	International Organization for Standardization
ISS	Internal Sublayer Service
IST	Internal Spanning Tree
IT	Information Technology
ITU	International Telecommunication Union
LDeVID	Locally Significant Secure Device Identifier
LLDP	Link Layer Discovery Protocol
LLDPDU	Link Layer Discovery Protocol Data Unit
LPI	Low Power Idle
LRP	Link-local Registration Protocol
MAC	Media Access Control
MD	Media-Dependent
MDI	Medium Dependent Interface

MMRP	Multiple MAC Registration Protocol
MST	Multiple Spanning Tree
MVRP	Multiple VLAN Registration Protocol
N/A	Not applicable
NACM	Network configuration Access Control Model
NETCONF	Network Configuration Protocol
NMDA	Network Management Datastore Architecture
NPE	Network Provisioning Entity
NRR	Neighbor Rate Ratio
OCSP	Online Certificate Status Protocol
OID	Object Identifier
OMG®	Object Management Group ²
OT	Operations Technology
OUI	Organizationally Unique Identifier
PCP	Priority Code Point
PCS	Profile Conformance Statement
PDU	Protocol Data Unit
PE	Path Entity
PEM	Privacy Enhanced Mail
PFS	Perfect Forward Secrecy
PHY	Physical Layer Device
PII	Personally Identifiable Information
PKCS	Public Key Cryptography Standards
PKI	Public Key Infrastructure
PKIX	Public Key Infrastructure X.509
PLS	Physical Signaling Sublayer
PPS	Pulse Per Second
PTP	Precision Time Protocol
PVID	Port VLAN Identifier
QoS	Quality of Service
RBAC	Role-Based Access Control
RFC	Request for Comments
RPC	Remote Procedure Call
RSA	Rivest-Shamir-Adleman
RAE	Resource Allocation Entity
SAN	Subject Alternative Name
SHA	Secure Hash Algorithm
STE	Sync Tree Entity
TDE	Topology Discovery Entity

² OMG®, UML® and Unified Modeling Language™ are either registered trademarks or trademarks of a product supplied by the Object Management Group, Inc. in the United States and/or other countries. This information is given for the convenience of users of this document and does not constitute an endorsement by IEEE or IEC of the product named. Equivalent products may be used if they can be shown to lead to the same results.

TLS	Transport Layer Security
TLV	Type, Length, Value
TOFU	Trust On First Use
TSN	Time-Sensitive Networking
TSN-IA	Time-Sensitive Networking for Industrial Automation
TTP	Trusted Third Party
UML®	Unified Modeling Language™
UNI	User/Network Interface
URL	Uniform Resource Locator
URN	Uniform Resource Name
VID	VLAN Identifier
VLAN	Virtual Local Area Network
YANG	Yet Another Next Generation data modeling language

3.5 Conventions

3.5.1 Convention for capitalizations

Capitalized terms are either based on the rules given in the ISO/IEC Directives Part 2 or emphasize that these terms have a specific meaning throughout this document.

Throughout this document "bridge" is used instead of "Bridge", except when

- it occurs at the beginning of a sentence or
- it is being used as (or part of) a specific term such as "VLAN Bridge" rather than being used to identify bridges (potentially of any type) in general. If "VLAN Bridge" is meant where only "Bridge" is written, a change to "VLAN Bridge" would be appropriate.

3.5.2 Unit conventions

This document uses:

- Gb/s for gigabits per second,
- Mb/s for megabits per second and,
- kb/s for kilobits per second.

3.5.3 Conventions for YANG contents

YANG modules and XML instance data for YANG shown in this document use the following style:

Text style `higher-layer-if` text style

Contents of a YANG module use the following style:

```
<ieee802-dot1q-bridge xmlns:dot1q="urn:ieee:std:802.1Q:yang:ieee802-dot1q-bridge">
  <bridges>
    <bridge> <!-- list -->
      <name>functional-unit-x</name>
      ...
    </bridge>
  </bridges>
</ieee802-dot1q-bridge>
```


3.5.4 Conventions for YANG selection and Digital Data Sheet

The digital data sheet expresses device capabilities and therefore, it is not required that all nodes in a YANG module be included in the digital data sheet. YANG nodes in 6.4.9, marked with [m], are mandatory nodes in the digital data sheet, nodes marked with [c] are conditional mandatory if the IA-station supports the corresponding optional functionality. Nodes marked with [o] are optional nodes in the digital data sheet. These marking in no way affect whether the feature and associated YANG module is required for the IA-station. Please refer to Clause 5 for conformance criteria for the IA-station.

YANG node selections in 6.4.9 of parent nodes implicitly include all subsidiary child nodes.

4 Overview of TSN in industrial automation

4.1 Industrial application operation

Industrial network applications are based on three main types of building blocks, which can be combined in one IA-controller or provided as a combination of an IA-controller and IA-devices interconnected through a suitable communication network.

These basic building blocks are:

- IA-device Sensor subsystems, which provide input signals indicating the value of the parameter or state being monitored, such as temperature, pressure, or discrete input information;
- IA-controller subsystems, which operate on combinations of measurements and external demand settings to develop output requests, such as position corrections in a motion application;
- IA-device Actuator subsystems, which implement output requests that result in physical changes to the process or machine under control, such as a level in a storage tank, the speed of a printing press, or movement of a robot.

NOTE 1 In general, all subsystems have an internal state, based upon initial settings, and derived from execution; therefore, the application inputs are combined with the internal state to develop an updated internal state and associated outputs.

A control loop is formed when the process or machine responds to the IA-device Actuator output and produces a new measured value at the IA-device Sensor. An example of a complete loop and the data flow inside it is shown in Figure 1 and described below where an IA-controller and IA-devices are connected as end stations in the network.

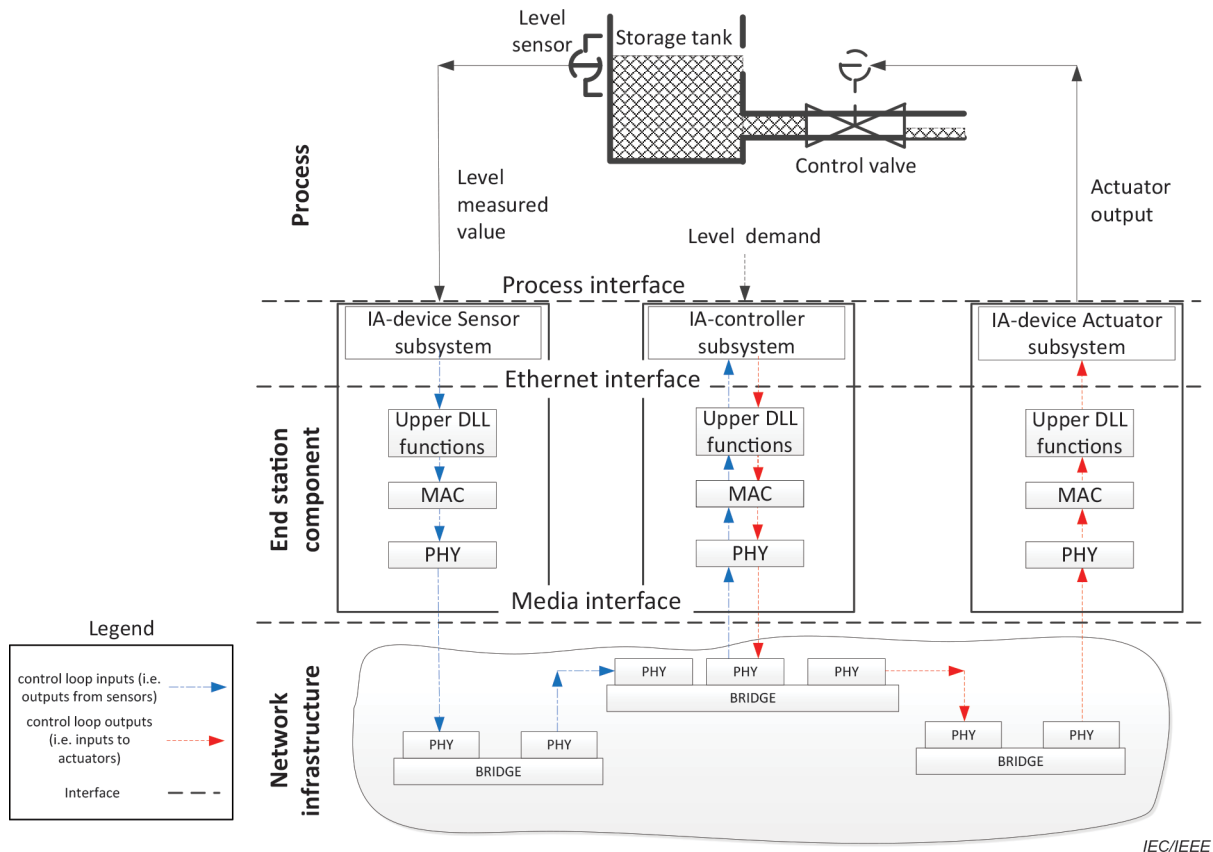


Figure 1 – Data flow in a control loop

In operation, the IA-device Sensor subsystem samples the measured value and the sampled values are transferred through the network as data packets for the IA-controller subsystem to compare with the demand value. After some computation time, the output is transferred from the IA-controller subsystem to the IA-device Actuator subsystem for implementation as a change in the external process.

This sequence repeats continuously as a regular operation using a Working Clock. The Working Clock is traceable to an ARB timescale or to the PTP timescale. Traceability to the PTP timescale is not required by all applications. For stability, the sequence repetition time (i.e., sampling time) is typically chosen to be between one fifth to one tenth of the time constant of the process response, although shorter times can also be used.

NOTE 2 The actual ratio of the process response time constant to sampling time required for stability depends on the implementation.

Control latency is a critical factor in all types of control and is bounded. Components contributing to the control latency time (as shown in Figure 1) are:

- application time for sampling, computation, and processing within each IA-controller and IA-device. These are specific to the IA-device and IA-controller and known to the IA-device or IA-controller makers,
- the time for data transfer through the upper DLL functions, MAC and PHY layers within each IA-controller and IA-device. This time depends on the implementation of these components, their situation-dependent load and performance, and configuration elements related to QoS supported by these components, and
- the End Station and Bridge schedule and transfer time through the network. These are influenced by the configuration process, which allocates available bandwidth and priorities to various types of application messages.

Offline engineering of the network is possible, including the calculation of the control latency time. During system operation, management services are provided for diagnostics and checking the performance indicators of an installed network.

4.2 Industrial applications

4.2.1 General

Industrial applications can contain multiple tasks. These tasks are executed based upon time or other events. Thus, an industrial application can have multiple tasks executing on different cycles as shown in Figure 2 and Figure 3.

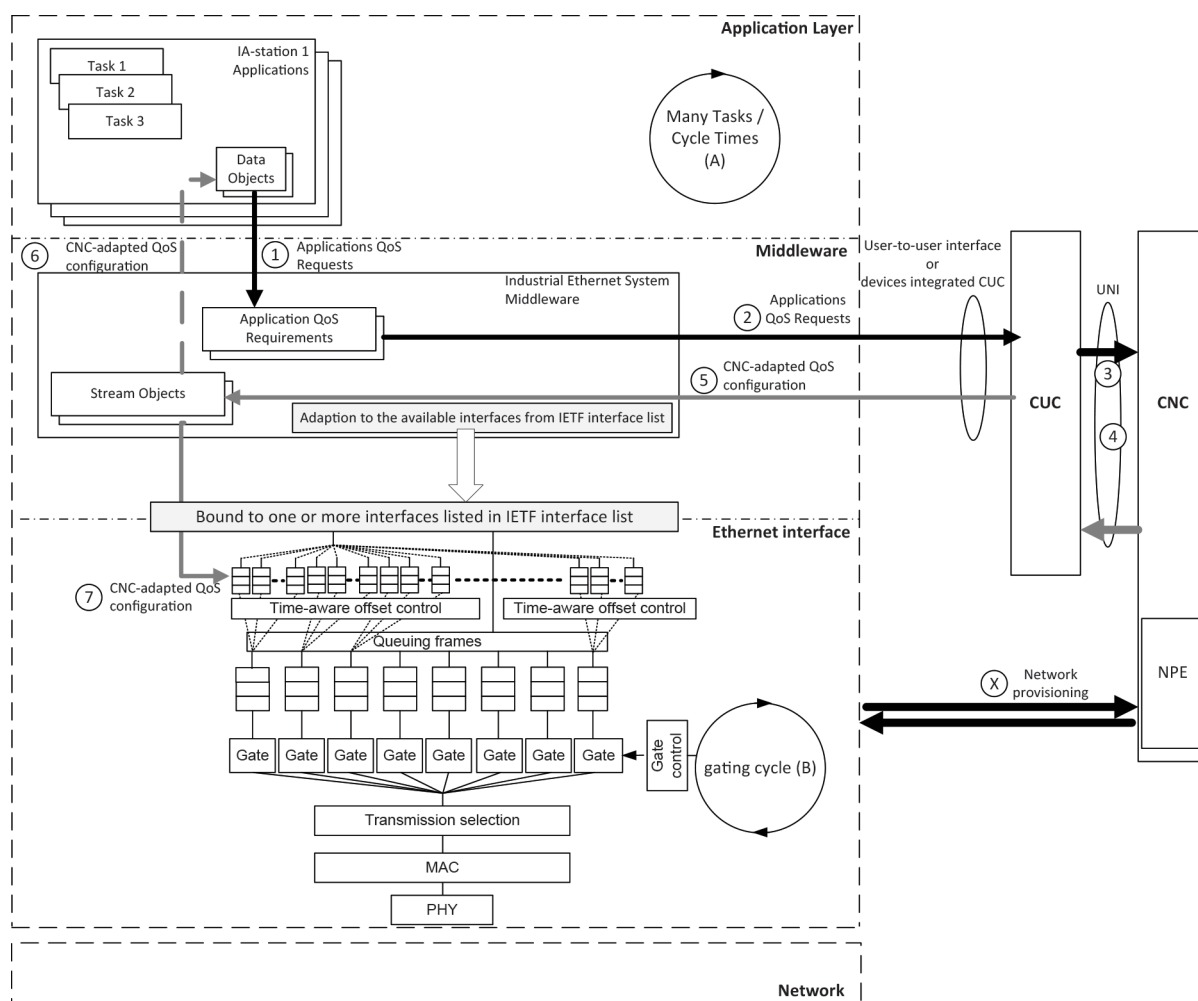
Examples of these tasks include the following.

- Background tasks, which are executed when no other task is running. There can be zero, one, or more such tasks in an industrial application.
- Main task which executes periodically. The start and execution of this task is often based upon the ARB timescale. There can be zero or one such task, in an industrial application.
- Global Time tasks. The start and execution of these tasks is often based upon Global Time (for example, at noon every day or at noon every Friday). There can be zero, one or more such tasks in an industrial application.
- Process driven tasks which are started by an event (for example, a sensor value reaches a defined point, or a process fault occurs). There can be zero, one or more such tasks in an industrial application.
- Control loop tasks which are bound to the Working Clock and started periodically. There can be zero, one or more such tasks in an industrial application.

A user defines the required automation tasks along with the data objects required as output and input for these tasks and the end station which hosts these tasks. Thus, these tasks are bound to data objects, which are exchanged between end stations per the user's definition. Many of these tasks have timing requirements, which are added as attributes to the assigned data objects. Examples of these attributes include:

- an update interval: [DataObject_Update_Interval] (time between two consecutive updates at the transmitting end station);
- a deadline: [DataObject_Deadline] (latest receive time at the end station, relative to the start of the DataObject Update Interval);
- the size of the DataObject: [DataObject_Data_Size];
- other attributes to form a stream-list request according to IEEE Std 802.1Qdj-2024, 46.1.5.

NOTE These attributes are provided for illustration purposes. The list is not representative of all industrial applications. These are not network attributes.



Legend:

(X)

IA-stations are configured by the CNC before any CUC to CNC action happens. For example, this configuration can include the gating cycle and transmission selection. Later accesses can happen concurrently with a CUC to CNC sequence.

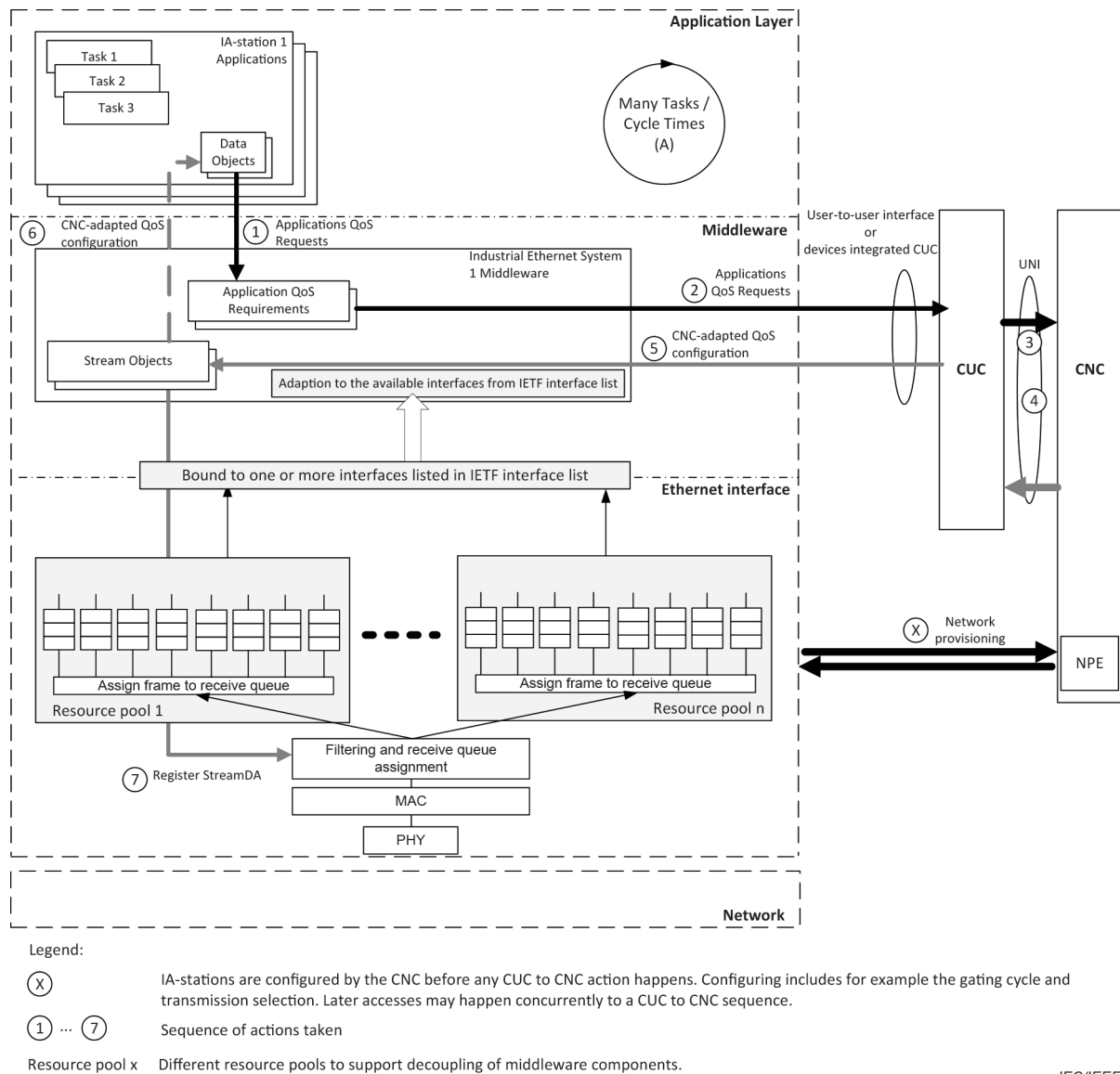
① ... ⑦

Sequence of actions taken

Time-aware offset control Queues are shared between multiple middleware components.

IEC/IEEE

Figure 2 – IA-station interaction with CNC – Transmit path



IEC/IEEE

Figure 3 – IA-station interaction with CNC – Receive path

4.2.2 Control loop tasks

Multiple control loop tasks can be implemented and run in parallel in their automation devices. For example, this behavior can be implemented by using a common Working Clock, a common starting point relative to the Working Clock and a common duration for this control loop task at the involved IA-devices and IA-controllers. The data objects associated with the control loop share common values for some attributes (for example, the same values for DataObject_Update_Interval and DataObject_Deadline). Multiple control loop tasks can be implemented and run in parallel in their automation devices.

4.2.3 Start of control loop tasks

The calculation of the starting point for a control loop task is independent from the time when the device is powered up or connected to the Configuration Domain. The start of a control loop task, which is based on the Working Clock, can be calculated in the following manner: divide the Working Clock value, expressed as an integer, by the duration of the control loop task, expressed as an integer, whenever the Working Clock value increases by one. A remainder of zero provides the basis for the start of the control loop task.

NOTE The units of the Working Clock value and the units of the duration of the control loop task are the same.

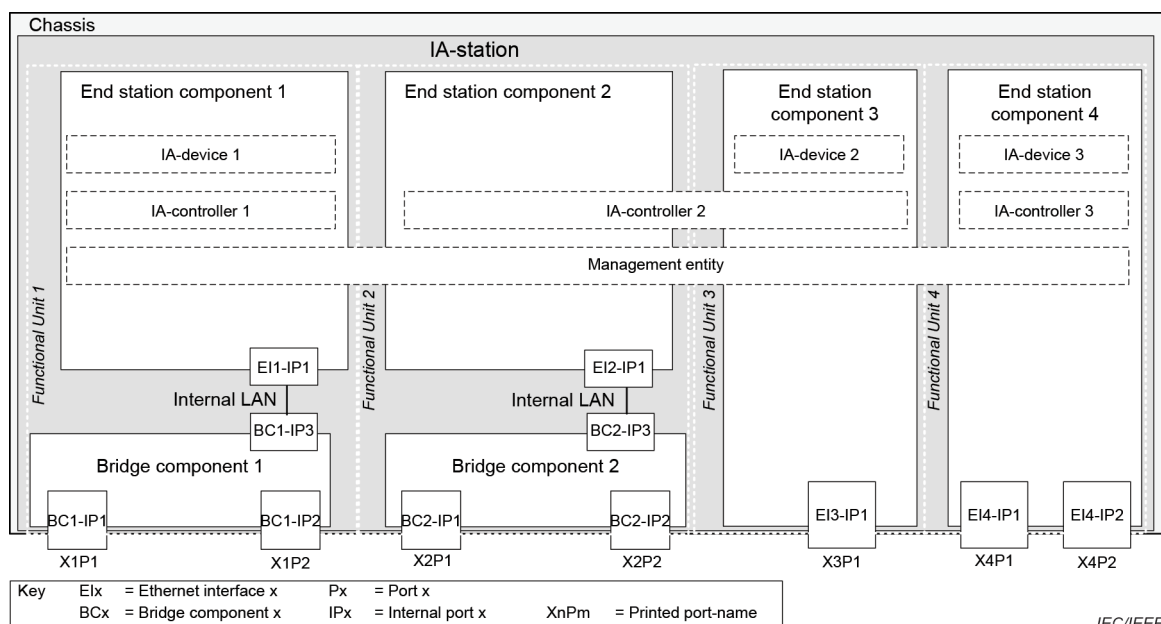
Stations in the network associated with the control loop synchronize to a Working Clock using IEEE Std 802.1AS-2020.

4.3 IA-stations

An IA-station can be a simple end station acting as source or destination for control data traffic. In addition, an IA-station can be a combined functional unit that includes an end station component together with a Bridge component in one chassis. IA-stations, incorporating multiple functional units with several end station components and Bridge components within one chassis, can also be found in industrial automation. Within this kind of combined IA-station various components can be connected by internal ports and internal LANs. All components utilize a common management entity as shown in Figure 4.

Figure 4 shows an example IA-station incorporating four functional units in one chassis. Functional unit 1 and functional unit 2 each consist of a Bridge component and an end station component. The end station components are connected by internal ports via internal LANs to the Bridge components. The Bridge components include two external ports each. Functional unit 3 includes only a single end station component with one external port. Functional unit 4 includes a single end station component with two external ports.

IA-controllers and IA-devices as well as the management entity are IA-station functions acting as source of or destination for link layer data traffic. Thus, each IA-station incorporates at least one end station component where these functions can be located. Figure 4 shows that IA-station functions can either reside in a single end station component (IA-device 1, IA-controller 1, IA-device 2, IA-device 3, IA-controller 3) or in multiple end station components (IA-controller 2, management entity).



IEC/IEEE

Figure 4 – IA-station example

4.4 Ethernet interface

One or more middleware components act as a layer between applications and the Ethernet interface. Figure 2 and Figure 3 show the relation between applications, middleware, Ethernet interface and the network. Various applications can run in parallel on an automation device. Data objects represent the information exchanged between applications running in different end stations. The application requirements contained in these data objects are translated by the middleware into stream requirements for use by the CUC. This translation can be accomplished in one or both of the following ways.

- The user defines the data objects and translates them into stream requirements and end-station communication-configurations. A user-specific mechanism is used to configure the network components, establish paths, and the time-aware offset control.
- The user defines the data objects and associates them with QoS requirements for each stream (application QoS requirements). These can be forwarded as stream requirement requests by a CUC to a CNC. The CNC responds by providing a stream configuration response. The request and response are specified in IEEE Std 802.1Qdj-2024. This information is used to configure the time-aware offset control, which utilizes per-stream queues. The CUC can be integrated into the end station or can be accessed via a user-to-user protocol. The middleware uses this information for configuring Talkers and Listeners. This information is also used to add additional timing information to the data objects for application usage.

Time-aware offset control utilizes per-stream queues (see IEEE Std 802.1Q-2022, Figure 34-1) and the traffic specification of the streams, including transmission offsets, provided by the CNC to ensure the order of stream transmission.

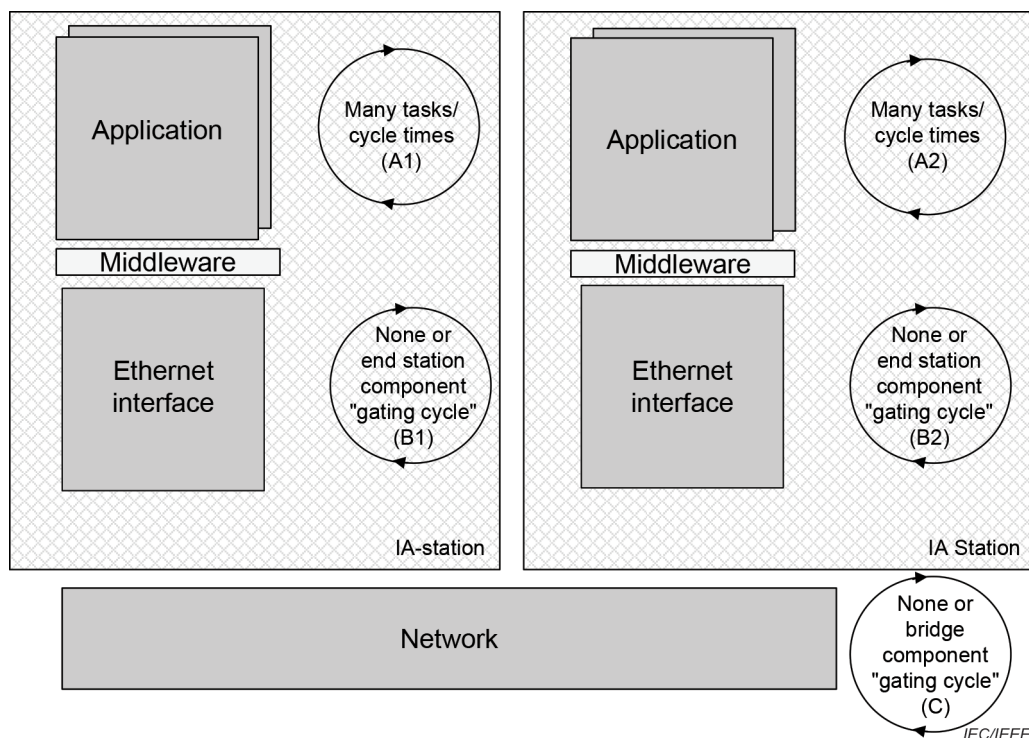


Figure 5 – Model for cycles

These automation systems, which are built from various end stations and connected via bridges, can share a common gating cycle or each station can have its own gating cycle. Alternatively, a bridge or end station can have no gating cycle (expressed as "none" in Figure 5).

4.5 Mechanisms to meet control loop latency requirements

Meeting latency requirements on a network can be accomplished using one or more combinations of the mechanisms enumerated below. The choice of a mechanism or a subset of the mechanisms listed below depends on the nature of the application(s) and the corresponding latency requirements:

- a) defining, testing, and simulating all possible application combinations and associated traffic patterns,
- b) overprovisioning the network,
- c) providing scheduled time slots for each application to transmit on the network,
- d) preempting lower priority traffic,
- e) providing scheduled time slots for certain traffic classes,
- f) time-aware offset control, and
- g) enforcing deterministic queuing delays in bridges.

NOTE This list is not comprehensive and not all mechanisms mentioned here are part of this specification. For specific mechanisms covered by this document please refer to Clause 5.

Frame preemption is specified in IEEE Std 802.1Q-2022 and IEEE Std 802.3-2022.

Reserving time on the network for certain traffic types can be done through enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 8.6.8.4. An aligned gating cycle is defined. Once a gating cycle is defined, portions of a cycle time can either be allocated to streams or classes of streams.

Multiple Talker-Listener pairs can be used for streams between end stations. Engineered time-triggered transmit can be used to coordinate transmission of all the traffic that shares a network to meet application requirements.

Creating a traffic load model in advance allows analysis of resulting traffic. It can be used to select and implement appropriate mechanisms to achieve latency requirements.

4.6 Translation between middleware and network provisioning

4.6.1 Mapping translation table

Application engineering can be done without knowledge of the network provisioning. Since the application is not aware of the network provisioning, it cannot directly map to the network configuration, for example, the use of PCP or VID as configured in the network. This problem is solved by providing a translation table, in the form of a YANG module definition, to the middleware. The IA-station's local YANG datastore contains this information.

Figure 6 shows examples of the translation models.

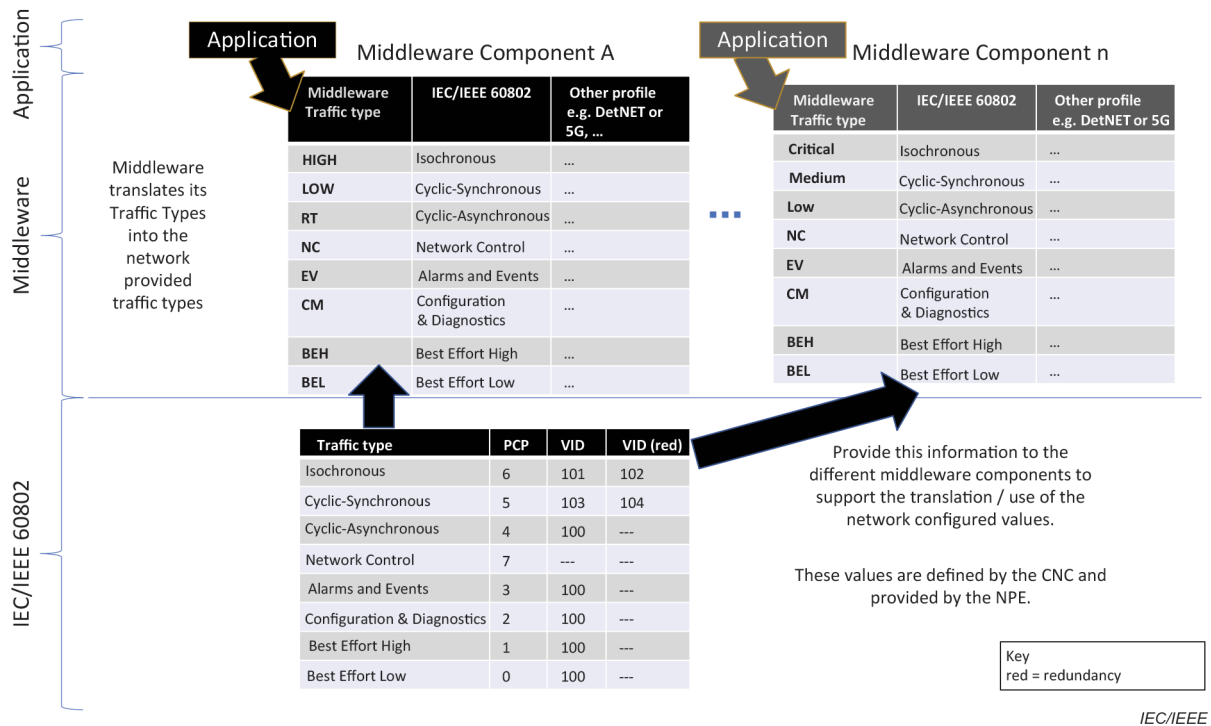


Figure 6 – Traffic type translation example

The YANG module `iecieeee60802-tt-mapping.yang` is used to provide the required mapping information to all installed middleware and applications. The information in the YANG module provides the VLAN-ID, the assigned traffic types with their PCP values and redundancy information (6.4.10.8.7).

4.6.2 PTP Instances

PTP domain numbers are also configured during network provisioning. The middleware is informed about which PTP domain is assigned to which target clock by providing `descriptionDS.userDescription` names according to IEEE Std 1588-2019, 8.2.5.5, in a translation table.

`descriptionDS.userDescription` names allow the support of multiple middleware components at one IA-station using the same PTP Instances (see 6.2.13). An IA-station's local database stores this information.

Figure 7 shows examples of the translation models.

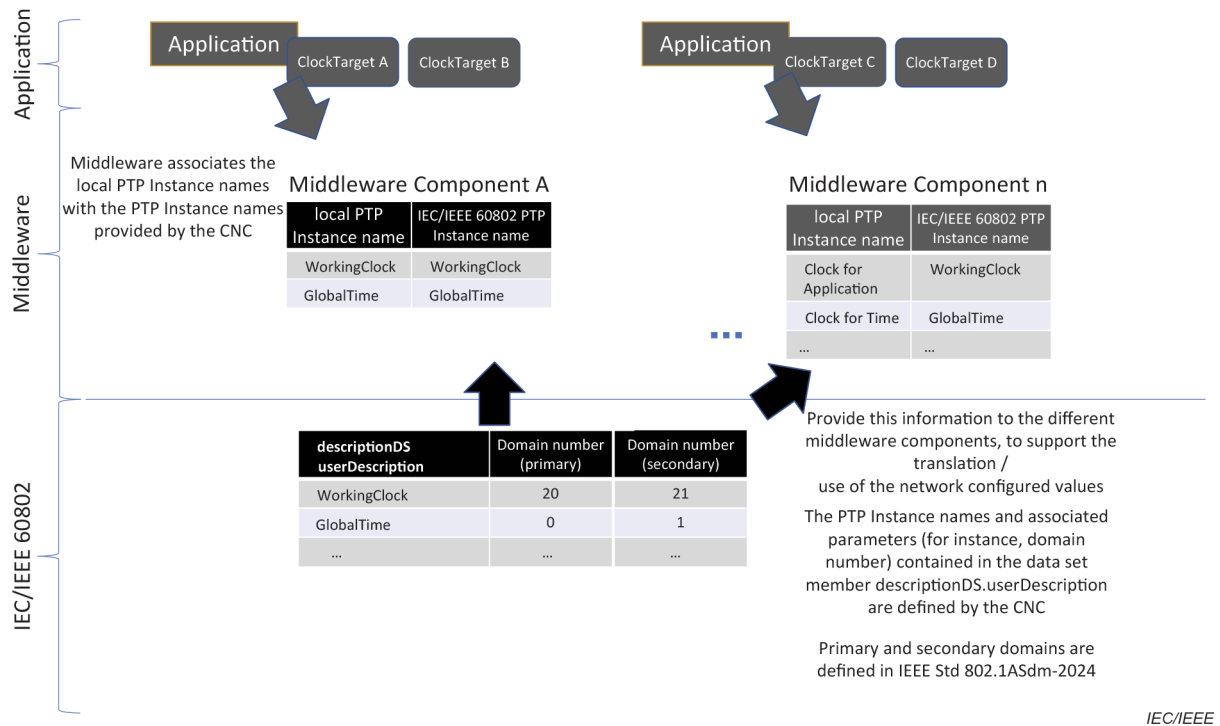


Figure 7 – PTP instance translation example

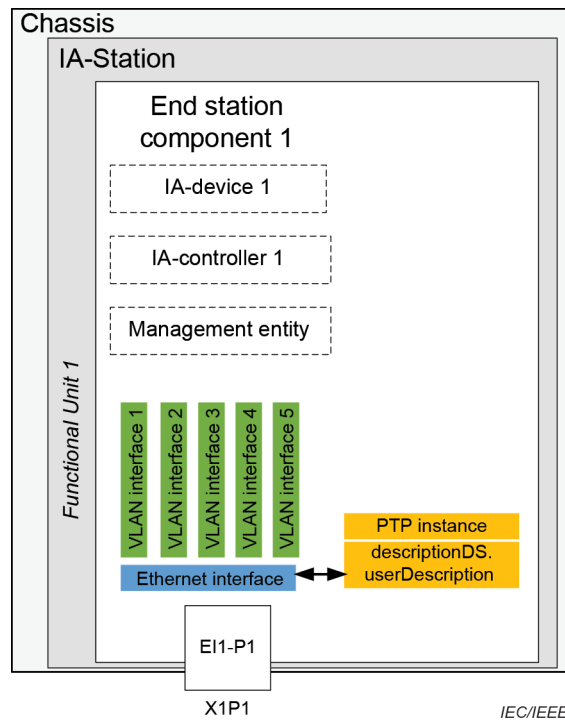


Figure 8 – descriptionDS.userDescription used for PTP instance translation

The userDescription contains the clock type (i.e., WorkingClock, GlobalTime, or both). This information is used by the middleware to align to the intended ClockTarget or ClockSource (see 6.2.13).

4.7 Industrial traffic types

4.7.1 General

Industrial automation applications make use of different traffic schemes/types for different functionalities (for example, parameterization, control, alarming). The various traffic patterns have different characteristics and thus impose different requirements on a network. To specify these traffic types, a two-step approach is used.

- a) First, define characteristics of generic traffic types (traffic type categories).
- b) Second, define instances of the generic traffic types (i.e. the traffic types).

4.7.2 Traffic type characteristics

The traffic type characteristics in Table 2 enable the identification of several distinct traffic types that are shared among sets of industrial applications.

Table 2 – Traffic type characteristics

Characteristic	Description
Cyclic	Traffic types consist of frames that can either be transmitted on a reoccurring time period (cyclic) or at no set period (acyclic). Available selections are: - required: traffic frames are transmitted cyclically, or - optional: implementation of cyclic traffic is at the discretion of the user.
Data delivery requirements	Denotes the delivery constraints for the traffic. Four options are specified. - Frame Latency: data delivery of a frame for a given Talker-Listener pair occurs within a bounded timespan. - Bandwidth: data delivery up to a certain number of frames or data size (including bursts of frames) occurs over a defined period. - Deadline: data delivery of a frame to a given Listener occurs at or before a specific point in time. - No: Denotes the case of traffic types with no special data delivery requirements
Time-triggered transmission	Talker data transmission occurs at a specific point in time based upon the Working Clock. Available selections are: - required, or - optional: implementation of time-triggered transmission is at the discretion of the user. Enhancements of scheduled traffic is only one means of achieving time-triggered transmission. Other, application-based, methods are possible
Stream-based	Indicates whether a traffic type requires registering a stream, i.e. allocating dedicated resources for the traffic before a given Talker can start transmitting data. - True: Data can only be transmitted after a Stream has been set up. - False: Data can be transmitted at any time.

4.7.3 Traffic type categories

4.7.3.1 General

The two-step approach described in 4.7.1 allows a clear differentiation between characteristics as seen from the "network" point of view and "application" point of view. Traffic type categories allow different IEEE 802 feature selections to achieve the goals of a specific network deployment. Four traffic type categories are identified in industrial automation systems:

- a) IA time-aware stream,
- b) IA stream,
- c) IA traffic engineered non-stream, and
- d) IA non-stream.

4.7.3.2 IA time-aware stream

The characteristics of this traffic type category are shown in Table 3.

Table 3 – IA time-aware stream characteristics

Characteristics	
Cyclic	Required
Data delivery requirement	Deadline or Frame Latency
Time-triggered transmission	Required
Stream-based	True

4.7.3.3 IA stream

The characteristics of this traffic type category are shown in Table 4.

Table 4 – IA stream characteristics

Characteristics	
Cyclic	Required
Data delivery requirement	Frame Latency
Time-triggered transmission	Optional
Stream-based	True

4.7.3.4 IA traffic engineered non-stream

The characteristics of this traffic type category are shown in Table 5.

Table 5 – IA traffic engineered non-stream characteristics

Characteristics	
Cyclic	Optional
Data delivery requirement	Bandwidth
Time-triggered transmission	Optional
Stream-based	False

4.7.3.5 IA non-stream

The characteristics of this traffic type category are shown in Table 6.

Table 6 – IA non-stream characteristics

Characteristics	
Cyclic	Optional
Data delivery requirement	No
Time-triggered transmission	Optional
Stream-based	False

4.7.4 Traffic types

4.7.4.1 General

Table 7 summarizes relevant industrial automation traffic types and their associated characteristics. In an industrial automation system, other applications, such as audio or video, utilizes one of these traffic types.

Table 7 – Industrial automation traffic types summary

Traffic type name	Cyclic	Data delivery requirements	Time-triggered transmission	Stream-based	Traffic type category
Isochronous	Required	Deadline	Required	True	IA time-aware stream
Cyclic-synchronous	Required	Frame Latency	Required	True	IA time-aware stream
Cyclic-asynchronous	Required	Frame Latency	Optional	True	IA stream
Alarms & events	Optional	Bandwidth	Optional	False	IA traffic engineered non-stream
Configuration & Diagnostics	Optional	Bandwidth	Optional	False	IA traffic engineered non-stream
Network Control	Optional	Bandwidth	Optional	False	IA traffic engineered non-stream
Best Effort High	Optional	No	Optional	False	IA non-stream
Best Effort Low	Optional	No	Optional	False	IA non-stream

4.7.4.2 Isochronous

Isochronous is a type of IA time-aware stream traffic. This type of traffic is transmitted cyclically using time-triggered transmission. Listeners have individual deadline requirements. Cycle times are typically in the range of microseconds to tens of milliseconds. Frame size is typically below 500 octets. Talker-Listener pairs are synchronized to the Working Clock. The network is configured by the CNC to provide zero congestion loss for this traffic type. This type of traffic is normally used in control loop tasks.

4.7.4.3 Cyclic-synchronous

Cyclic-synchronous is a type of IA time-aware stream traffic. This type of traffic is transmitted cyclically using time-triggered transmission. Talker-Listener pairs have individual latency requirements. Cycle times are typically in the range of hundreds of microseconds to hundreds of milliseconds. Frame size is unconstrained except as indicated in 5.5.1. Talker-Listener pairs are synchronized to the Working Clock. The network is configured by the CNC to provide zero congestion loss for this traffic type. This type of traffic is normally used in control loop tasks.

4.7.4.4 Cyclic-asynchronous

Cyclic-asynchronous is a type of IA stream traffic. This type of traffic is transmitted cyclically with latency requirements bounded by the interval as specified in IEEE Std 802.1Q-2022, 46.2.3.5.1. Talker-Listener pairs have individual latency requirements. Cycle times are typically in the range of milliseconds to seconds. Frame size is unconstrained except as indicated in 5.5.1. Data exchanges between Talker-Listener pairs are typically not dependent on the Working Clock. This traffic type typically tolerates limited congestion loss. The network is configured by the CNC to handle this traffic type without loss, up to a certain number of frames or data size.

4.7.4.5 Alarms and events

Alarms and events are a type of IA traffic engineered non-stream. This type of traffic is transmitted cyclically or acyclically. This traffic expects bounded latency including time for retransmission in the range of milliseconds to hundreds of milliseconds. The source of the alarm or event typically limits the bandwidth allocated to this traffic. Frame size is unconstrained except as indicated in 5.5.1. Congestion loss can occur. Retransmission to mitigate frame loss is expected. The network is configured by the CNC to handle these frames, including bursts of frames, up to a certain number of frames or data size over a defined period.

4.7.4.6 Configuration and diagnostics

Configurations and diagnostics are a type of IA traffic engineered non-stream. This type of traffic is transmitted cyclically or acyclically. This traffic expects bounded latency, up to seconds, including time for retransmission. The source of configuration or diagnostics frames typically limits the bandwidth allocated to this traffic. Frame size is unconstrained except as indicated in 5.5.1. Congestion loss can occur. Retransmission to mitigate frame loss is expected. The network is configured by the CNC to handle these frames, including bursts of frames, up to a certain number of frames or data size over a defined period.

4.7.4.7 Network control

Network control is a type of IA traffic engineered non-stream. This type of traffic can be transmitted cyclically or acyclically. This traffic expects bounded latency including time for retransmission. Frame size is unconstrained except as indicated in 5.5.1. The network is configured by the CNC to handle these frames, including bursts of frames, up to a certain number of frames or data size over a defined period. If these limits are exceeded congestion loss can occur. Network control is comprised of services required to maintain network operation. Examples include time synchronization, loop prevention, and topology detection.

4.7.4.8 Best effort

Best effort is a type of IA non-stream. The network is configured by the CNC so that these frames do not interfere with other traffic types. These frames are forwarded when resources are available. Congestion loss resulting in frame drop can occur. It is sometimes desirable to have more than one traffic class for best effort traffic (see Table 8).

4.7.4.9 Traffic class to traffic type mapping

Table 8 provides an example for the usage of traffic classes based on the traffic type:

Table 8 – Example traffic class to traffic type mapping

Traffic class	Queue Assignment (8 queues)	Queue Assignment (4 queues)	Traffic type
7	6	2	Isochronous
6	5	1	Cyclic-Synchronous
5	4	1	Cyclic-Asynchronous
4	7	3	Network Control
3	3	0	Alarms and events
2	2	0	Configuration & Diagnostics
1	1	0	Best Effort High
0	0	0	Best Effort Low
NOTE An example of mapping of PCP and traffic type to an application is provided in Figure 6.			

The traffic type categories definition allows different IEEE 802 feature selections to achieve specified goals. Moreover it helps in identification of the traffic protection mechanisms. Adherence to this example of a common mapping helps minimize potential conflicts between traffic types.

4.8 Security for TSN-IA

4.8.1 General

Subclause 4.8 describes selected aspects of TSN-IA security. Protecting the management of industrial communication is the main objective of TSN-IA security. The protection of communications that use industrial traffic types is not addressed by this document.

4.8.2 Security configuration model

Security configuration is a part of system engineering and configuration. The security configuration in this document does not encompass the supply of configuration objects for middleware and application security. Security configuration settles the prerequisites for protecting the establishment and management of communications that use industrial traffic types (see 4.7). It ensures that the security features of IA-stations (including CNCs) can be used for protecting message exchanges and authorizing the resource accesses during stream establishment and management. This security configuration supplies deployment-specific configuration objects to IA-stations. They encompass:

- instructions about cryptographic algorithms,
- credentials and trust anchors,
- instructions to interpret the outcome of peer entity authentication while enforcing resource access controls, and
- access control rules and permissions.

This security configuration uses NETCONF/YANG request/response exchanges, as follows.

- The to-be-configured IA-stations act in NETCONF server role with respect to their security configuration.
- A NETCONF client is responsible for setting-up IA-stations for security. This NETCONF client possesses information about the security relationship to be established during security configuration or about the expectations on the IA-stations in a Configuration Domain. It can be implemented as part of an interactive or automated process (for example an engineering tool, or CNC operation). As an implication, the security configuration includes options for interactive and automated setup, i.e. security configuration is done by human or non-human actors.

NOTE NETCONF notifications can also be used to recognize events such as a near-term end-of-life of certificate objects, especially EE certificate objects (see IETF RFC 4210, 3.1.1).

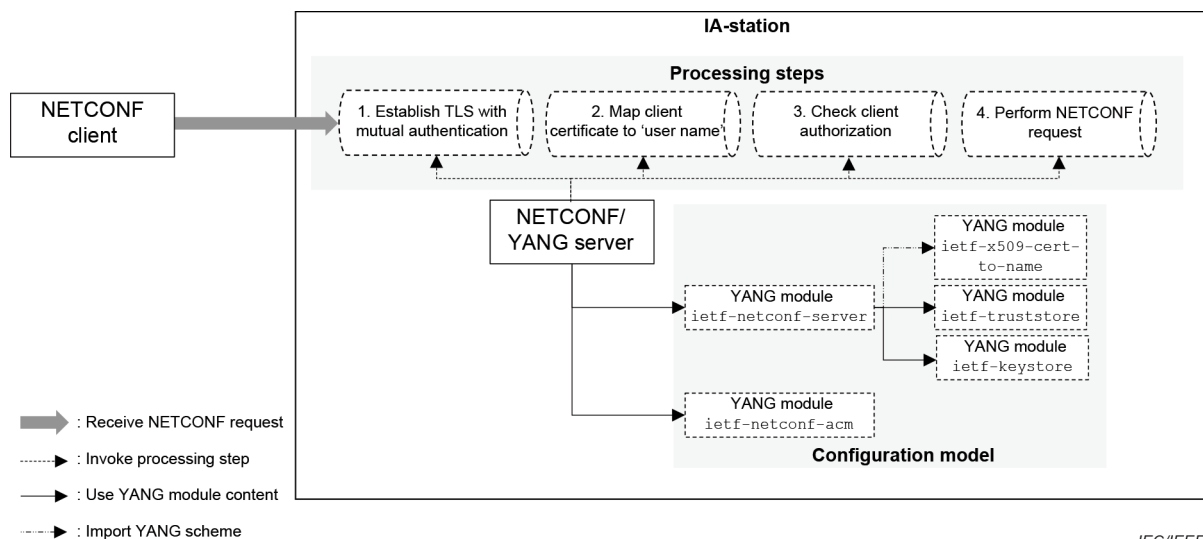
- The security configuration exchanges supply deployment-specific objects (trust anchors, credentials, etc.) to IA-stations and manages them. IA-stations that are in factory default state can only possess manufacturer-specific security objects (trust anchors, credentials, etc.) when booting initially. The protected NETCONF/YANG exchanges with IA-stations that are in factory default state are outlined in 4.8.3 to 4.8.6.

4.8.3 NETCONF/YANG processing

Securing NETCONF/YANG resources (for example, NETCONF sessions or managed objects) on NETCONF servers is specified by IETF RFC 6241 (NETCONF). Therefore, message exchange protection between NETCONF clients and servers as well as resource access authorization by NETCONF servers are utilized.

- IETF RFC 7589 and draft-ietf-netconf-over-tls13 (NETCONF-over-TLS) specify a solution to protect NETCONF message exchanges by TLS.
- IETF RFC 8341 (NACM) specifies three access control points, covering the request/response and notification model in NETCONF according to IETF RFC 8341, 2.1.

NETCONF servers enforce security as shown in Figure 9. The processing steps are executed upon the current configuration of the NETCONF server's YANG modules.



IEC/IEEE

Figure 9 – NETCONF/YANG security processing steps

The processing steps on the side of NETCONF servers are as follows.

- 1) Establish a TLS connection with mutual authentication: the NETCONF server acts as TLS server and awaits connection requests of NETCONF clients (TLS clients). At the beginning of the TLS handshake, the TLS client and server negotiate the TLS protocol version to be used. During the TLS handshake the NETCONF server authenticates itself towards the NETCONF client by a credential from its ietf-keystore YANG module. In addition, the NETCONF server challenges the NETCONF client for authentication and verifies its authentication by trust anchors in its ietf-truststore YANG module according to 6.3.4. A successful mutual authentication is a prerequisite for proceeding to the next step.
- 2) Map the client certificate to a username: the NETCONF server maps the authenticated TLS client certificate to a "NETCONF username"³ by applying an ordered list of mapping instructions. These instructions are provided in its ietf-x509-cert-to-name YANG module. The applicable list item is identified by matching its configured fingerprint (according to IETF RFC 7589, Clause 7) against the certification path that was used for TLS client authentication (an end entity certificate or a CA certificate). According to the map type of the identified list item, the NETCONF server determines the "NETCONF username". This can be done by extracting information from the end entity certificate of the NETCONF client. A successful certificate-to-"NETCONF username" mapping is a prerequisite for proceeding to the next step.
- 3) Check client authorization: the NETCONF server checks if the NETCONF client has the permission to access the requested NETCONF/YANG resource based on its "NETCONF username" and the access control rules available in its ietf-netconf-acm YANG module. See 4.8.4 for more information about NETCONF/YANG access control. A successful authorization is a prerequisite for proceeding to the next step.
- 4) Perform NETCONF request: if all preceding steps succeeded, the NETCONF server performs the NETCONF request.

4.8.4 NETCONF/YANG access control

NACM defines a YANG information model for describing permitted/denied access operations. NETCONF servers are responsible for enforcing access control to their resources according to the information in their ietf-netconf-acm YANG modules. NACM allows the description of access-controlled resources in terms of NETCONF protocol operations, nodes in YANG datastores or types of notification events. NACM uses character strings to represent the subject actors, i.e. NETCONF clients. These character strings are known as "NETCONF username". The NACM access control information of a NETCONF server is created, updated, and deleted per IA-station. The management of this information happens along the IA-station lifecycle, for example, manufacturing, bootstrapping, operation, maintaining, re-owning, destructing. Moreover, the management of the NACM access control information itself is subject to NACM access control.

This document employs multiple YANG data models for fulfilling its purposes. This extends beyond the above identified YANG modules (see 4.8.3). The NETCONF server on an IA-station enforces access control for NETCONF/YANG resources. To meet this objective, the NETCONF server on an IA-station is supplied with access control information for the used NETCONF/YANG resources. NACM is employed for this purpose and profiles default access control information for the NETCONF/YANG resources (see 6.3.2.2). This relieves other organizations or individuals for example, manufacturers, integrators, operators, owners from being responsible to create NACM access control information for the respective NETCONF/YANG resources.

³ In this document and, in almost all cases, NETCONF username values do not represent human users.

NACM relies on character strings (known as "NETCONF username") to refer to clients. NACM access control information as specified in this document, populates the "NETCONF username" character strings in NACM with role names specified in 6.3.2.1.4 c). This allows to create default NACM information without knowing actual names of individual entities. A role name can refer to 0, 1 or more individual entities. It is the responsibility of users to assign role names to individual entities. This happens by binding the assigned role names to the credentials of individual entities. The current form to express this binding is a role extension in the identity certificates of end entities defined in this document. These are NETCONF clients, i.e. these role extensions appear in the end entity certificates of LDevID credentials for NETCONF clients.

As initial step NACM maps the NETCONF username to a set of groups. The set of groups determines the set of rules to be applied for access-controlled resources.

4.8.5 Identity checking

IETF RFC 7589 (NETCONF-over-TLS) specifies that NETCONF clients check the identity of NETCONF servers (IETF RFC 7589, Clause 6) and that NETCONF servers verify the identity of NETCONF clients (IETF RFC 7589, Clause 7).

The NETCONF server identity check happens inside NETCONF clients. It matches an actual against an expectation.

- The actual server identity is established by the end entity certificate of the NETCONF server (authenticated by means of TLS).
- The expectations on server identity are established by the information that is used to connect to the NETCONF server.

IETF RFC 7589 refers to IETF RFC 6125, Clause 6, for the details of retrieving the actual and comparing it against the expected.

The NETCONF client identity check happens inside NETCONF servers. It also matches an actual against an expectation.

- The actual client identity is established by the end entity certificate of the NETCONF client (authenticated by means of TLS).
- The expectations on client identity are established by the contents of the `ietf-netconf-acm` and `ietf-x509-cert-to-name` YANG modules.

The details of this check are subject to the requested NETCONF operation. IETF RFC 7589, Clause 7, specifies the mapping of an authenticated client certificate to a "NETCONF username" whose permissions are then enforced by IETF RFC 8341 (NACM). More information is provided in 4.8.3, steps 2 and 3.

4.8.6 Secure device identity

4.8.6.1 Device identity

The term 'device' originates from IEEE Std 802.1AR-2018. It matches the term IA-station in this document.

The device identity refers to a set of information items about a device that:

- describes a device as a physical or virtual entity in a distributed system (identifier and attribute information);
- is used by a device to describe itself as such entity (identifier and attribute information);
- allows to interact with this device (addressing information, i.e. a specific identifier class).

The targeted use case, for example application data exchanges, configuration exchanges, inventory, or ordering, determines the required amount of identity information about a device.

The device identity of any single IA-station encompasses:

- MAC addresses, IP addresses, TCP ports, DNS names;
- ietf-hardware YANG module contents (IETF RFC 8348, Clause 3 and 7.1).

4.8.6.2 Verifiable device identity

Certain aspects of device identity are verified before relying on them during online interactions. These are examples.

- DNS names or IP addresses are used to call the management entity of an IA-station, i.e. its NETCONF/YANG server. Their value represents the caller's expectation on the identity of their responder in network communications. Verification of the responder's identity helps defeat DNS spoofing, component impersonation and man-in-the-middle attacks. This is specified by IETF RFC 7589 and described in IETF RFC 6125, Clause 6. Passing this check is a prerequisite before NETCONF application exchanges can happen.
- mfg-name values in instances of the ietf-hardware YANG module. These values make claims about the IA-station manufacturer. Their verification is a means to protect against counterfeiting.

The verification of IA-station identity happens according to a model that is fully specified by this document. That verification can be done in a manufacturer-agnostic manner. This verification is important before supplying locally significant credentials especially LDevID to IA-stations that are in factory-default state.

4.8.6.3 Verification support mechanisms

4.8.6.3.1 General

Subclause 4.8.6.3 considers mechanisms that support device identity verification during online interactions with IA-stations.

4.8.6.3.2 Secure transports

Sending information in plain form over a protected channel, e.g. ietf-hardware YANG module contents via NETCONF-over-TLS protects the transferred information during its transit through the network but does not vouch for the correctness of the received information, e.g. the mfg-name value.

4.8.6.3.3 Secure information

Protecting information objects by means of a cryptographic authentication code or digital signature enables verification of the authenticity and integrity of that information. These cryptographic authentication codes can use symmetric or asymmetric schemes. In case of asymmetric schemes, raw and self-signed public keys are distinguished from CA-signed public keys.

Asymmetric schemes with CA-signed public keys are preferable for the verifiable device identity use case: claimants and verifiers share a public key; the claimant possesses the corresponding private key. The establishment and storage of shared public keys uses public key certificates. For this approach self-signed CA certificates are to be established in an authentic manner. The number of self-signed CA certificates is independent from the number of verifiers (CNCs) as well as claimants (IA-stations).

4.8.6.3.4 IDevID and LDevID credentials

IDeVID and LDeVID credentials are specified by IEEE Std 802.1AR-2018. These objects are comprised of a certification path and a private key. The certification path encompasses an end entity certificate which contains verifiable device identity in a CA-signed form. The device identity verification happens after validating the certification path (IETF RFC 5280, Clause 6) and checking the proof-of-possession for the private key. The certification path validation demands trust anchors as input arguments (IETF RFC 5280, 6.1.1 input argument (d)).

Two types of credentials are distinguished by IEEE Std 802.1AR-2018:

- IDeVIDs are issued by device manufacturers. They represent an initial identity as it is known at device production-time. The initial device identity is not locally significant: it cannot contain deployment-specific information such as DNS names or IP addresses.
- LDeVIDs are issued by other actors, e.g. a device user. They represent a locally significant device identity: they can contain deployment-specific information, e.g. DNS names or IP addresses.

IEEE Std 802.1AR-2018, Clause 6, uses signature suites to describe the subject public key and the signature fields in IDeVID and LDeVID certification paths. This notion is different from TLS cipher suites.

NOTE IDeVID and LDeVID credentials also serve purposes beyond secure device identity, for instance the realization of secure transports. This facilitates the use case of NETCONF/YANG security setup from factory default state.

4.8.6.3.5 IDevID items beyond IEEE Std 802.1AR-2018

IEEE Std 802.1AR-2018 allows verification of the following identity items:

- certificate issuer (not necessarily: manufacturer) by issuer field (data type: ASN.1 Name), and
- if present: device instance by serialNumber value (data type: ASN.1 PrintableString).

NOTE 1 IEEE Std 802.1AR-2018 represents the initial device identity as an optional serialNumber attribute (OID 2.5.4.5) in the subject field of the EE certificate. This value is unique within the domain of significance of the EE certificate issuer.

NOTE 2 This verification can happen after certification path validation and the proof-of-possession checking for the private key.

The following bullet points describe options beyond IEEE Std 802.1AR-2018 for verifying the device identity of IA-stations in factory default state. It also identifies informational items for the corresponding checks:

- IA-station manufacturer check: using names that identify IA-station manufacturers, e.g. mfg-name in ietf-hardware YANG module,
- IA-station type check: using attributes that identify IA-station types, e.g. model-name, hw-revision, description in ietf-hardware YANG module, and
- IA-station instance check: using values that identify IA-station instances, e.g. serial-num in ietf-hardware YANG module.

The following model described in the bullet points applies to the verification of the initial device identity of IA-stations:

- the set of to-be-conducted checks is determined by IA-station and CNC users,
- an IA-station uses IDeVID credentials to prove its device identity. The checking happens by means of online interactions in the operational network. It happens automatically and is done by CNCs. This does not depend on Configuration Domain external repositories, and

- other stakeholders, e.g. middleware/application consortia or individual manufactures are allowed to additionally express information items in IDevID credentials to reflect their device identity model. CNCs do not assess such additional information.

4.8.6.3.6 Device identity representation in IDevID and LDevID credentials

The best practices for representing verifiable device identity information in IDevID and LDevID credentials (see 6.3.3.2.2 for more information) are as follows.

- Corresponding information (actual values or references to them) appears in EE certificates:
 - IDevID EE certificates bind initial device identity items that are known by the device manufacturer at production time, e.g. mfg-name.
 - LDevID EE certificates bind locally significant device identity items that are known by other actors such as device users, e.g. DNS names or IP addresses. They can also bind initial device identity information.
- Items that encode device naming information appear in the subjectAltName extension.

NOTE This is specified in IETF RFC 5280, 4.2.1.6. It is further explained in IETF RFC 6125, 2.3.
- A binding can take one of following forms. Multiple forms can appear in one EE certificate:
 - by-value: the verifiable device identity information is represented by its value inside the IDevID resp. LDevID EE certificate. Examples are:
 - the product serialNumber in IDevID credentials (IEEE Std 802.1AR-2018), and
 - the hostname of the NETCONF/YANG server in LDevID credentials (IETF RFC 6125, Clause 6).
 - by-ref: the verifiable device identity information is represented by a reference inside the IDevID resp. LDevID EE certificate, not by its value.
- The actual value can be provided by the device itself or by a device-external source.
- If it is provided in form of an unprotected information object, then the reference object that is embedded to EE certificates includes a digest value.

5 Conformance

5.1 General

A claim of conformance to this document is a claim that the behavior of an implementation of an IA-station (see 5.5, 5.6) with its Bridge components (see 5.7, 5.8) and end station components (see 5.9, 5.10) meets the mandatory requirements of this document and may support options identified in this document.

5.2 Requirements terminology

The verbal forms for required expressions of provisions follow the conventions below.

- Requirements terminology is provided in the ISO/IEC Directives Part 2:2021, Clause 7. This document can be found at www.iec.ch/members_experts/refdocs.
- The Profile Conformance Statement (PCS) proformas (see Annex A) reflect the occurrences of the words "shall," "may," and "should" within this document.
- This document avoids needless repetition and apparent duplication of its formal requirements by using "is", "is not", "are", and "are not" for definitions and the logical consequences of conformant behavior. Behavior that is permitted but is neither always required nor directly controlled by an implementer or administrator, or whose conformance requirement is detailed elsewhere, is described by "can". Behavior that never occurs in a conformant implementation or system of conformant implementations is described by "cannot". The word "allow" is used as a replacement for the phrase "support the ability for", and the word "capability" means "can be configured to".

5.3 Profile conformance statement (PCS)⁴

The supplier of an implementation that is claimed to conform to this document shall provide the information necessary to identify both the supplier and the implementation and shall complete a copy of the PCS proforma provided in Annex A.

5.4 Conformance classes

This document includes conformance requirements and options that are related to an entire station, as well as conformance requirements and options that are related to single Bridge or end station components within an IA-station. Figure 10 illustrates this conformance model.

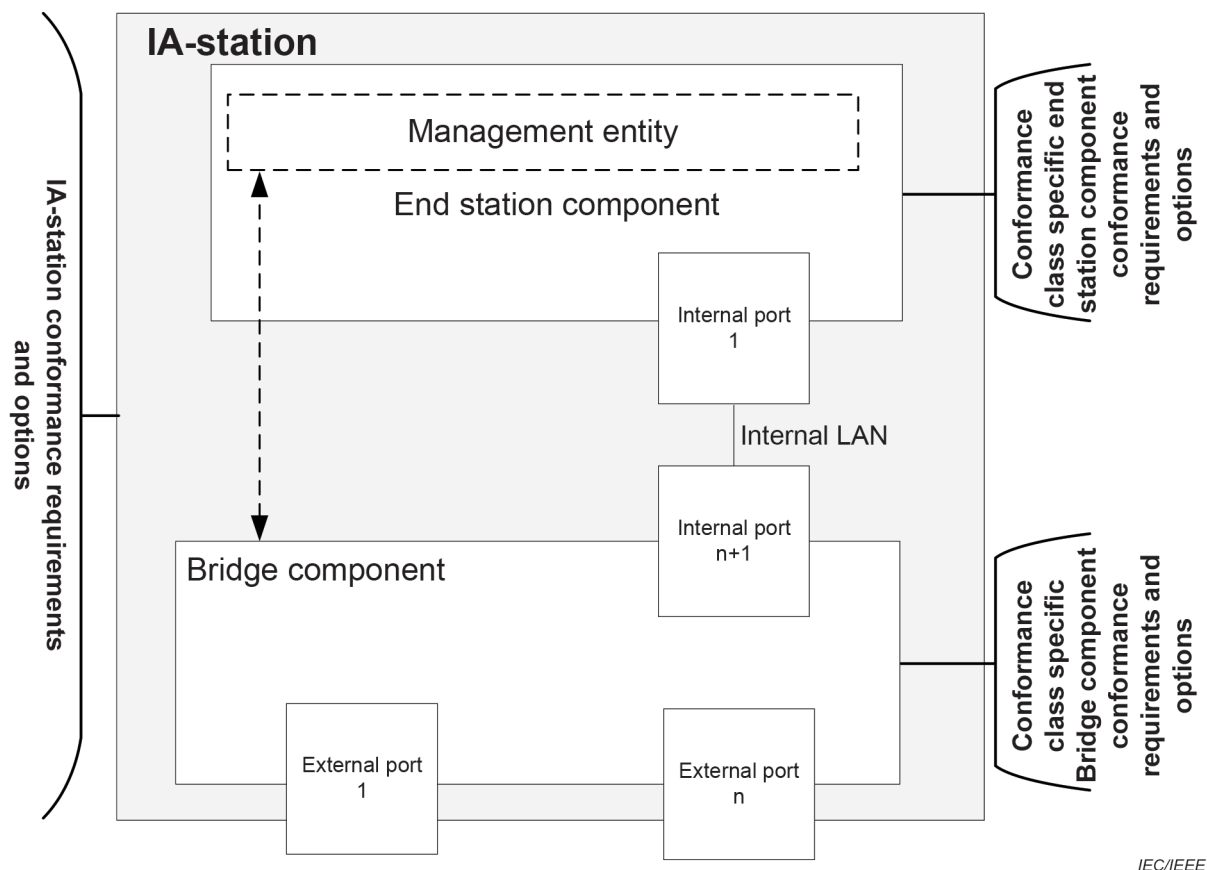


Figure 10 – IA-station conformance model

This document supports a variety of industrial use cases. In some of these use cases, support of certain TSN features can be mandatory, while in others, supporting these features can lead to non-optimal implementations. Therefore, this document defines two conformance classes that are applicable both to Bridge components and end station components. Conformance Class A (ccA) is feature-rich, i.e. tailored to use cases requiring support of many TSN-IA features. Conformance Class B (ccB) targets implementations that are more resource constrained. The details for the conformance classes are specified in 5.7 and 5.8 for Bridge components, and in 5.9 and 5.10 for end station components.

NOTE 1 It is the responsibility of the IA-station manufacturer to carefully consider the implications of mixing ccA and ccB Bridge components and end station components in a single IA-station.

⁴ An editable version of each PCS in this document can be obtained from the IEC SC 65C supporting documents website at <https://www.iec.ch/sc65c/supportingdocuments> or the IEEE 802.1 Website at <https://1.ieee802.org/conformance/>.

NOTE 2 It is the responsibility of the user to carefully consider the implications of mixing ccA and ccB Bridge components and end station components in a single Configuration Domain.

NOTE 3 Any Bridge compliant to this document is an IA-station. Any IA-station contains a management entity (i.e. an end station component).

5.5 IA-station requirements

5.5.1 IA-station PHY and MAC requirements for external ports

External ports of IA-stations that claim conformance to this document shall meet the following requirements.

- a) Support the Media Access Control (MAC) service specification according to IEEE Std 802.3-2022, Clause 2.
- b) Support the Media Access Control (MAC) frame and packet specifications according to IEEE Std 802.3-2022, Clause 3, especially the MAC Client Data field size according to IEEE Std 802.3-2022, 3.2.7, item c).
- c) Support Layer Management according to IEEE Std 802.3-2022, 5.2.4.
- d) Implement at least one IEEE Std 802.3-2022 MAC that shall operate in full-duplex mode, and associated IEEE Std 802.3-2022 PHY with a data rate of at least one of the following speeds: 10 Mb/s, 100 Mb/s, 1 000 Mb/s, 2.5 Gb/s, 5 Gb/s, or 10 Gb/s together with the corresponding managed objects:
 - 1) 10BASE-T1L MAU type according to IEEE Std 802.3-2022, Clauses 22 and 146,
 - 2) 100BASE-TX or 100BASE-FX MAU types according to IEEE Std 802.3-2022 Clauses 21, 22, 24, 25, 26 and IEEE Std 802.3-2022 23A, 28A, 28B, 28C, and 28D,
 - 3) 1000BASE-T or 1000BASE-SX MAU types according to IEEE Std 802.3-2022 Clauses 28, 34, 35, 36, 37, 38, and 40, and IEEE 802.3-2022 Annexes 28A, 28B, 28C, and 28D,
 - 4) 2.5GBASE-T or 5GBASE-T MAU types according to IEEE Std 802.3-2022, Clauses 28, 125, and 126, and IEEE Std 802.3-2022 Annexes 28A, 28B, 28C, and 28D,
 - 5) 2.5GBASE-T1 or 5GBASE-T1 MAU types according to IEEE Std 802.3-2022, Clause 149,
 - 6) 10GBASE-T or 10GBASE-SR MAU types according to IEEE Std 802.3-2022, 1462 Clauses 44, 46, 49, 51, 52, 55, and IEEE Std 802.3-2022, Annexes 48A and 55A,
 - 7) 10GBASE-T1 MAU type according to IEEE Std 802.3-2022, Clause 149,
 - 8) 100BASE-T1 MAU type according to IEEE Std 802.3-2022, Clause 96, and
 - 9) 1000BASE-T1 MAU type according to IEEE Std 802.3-2022, Clause 97.
- e) Support the YANG features and nodes of the ieee802-ethernet-interface module according to 6.4.9.2.1.
- f) Implement Ethernet support for time synchronization protocols according to IEEE Std 802.3-2022, Clause 90 as amended by IEEE Std 802.3cx™ -2023 and IEEE Std 802.3de™-2022 with the aTimeSyncSelectionDdmp set to 'symbol-after-sfd' according to IEEE Std 802.3cx-2023, 30.13.1.14.

Clauses and subclauses not mentioned can be implemented but are not obligatory.

5.5.2 IA-station topology discovery requirements

IA-stations for which a claim of conformance to this document is made shall support the following list of requirements.

- a) The required capabilities according to IEEE Std 802.1AB-2016, 5.3, and IEEE Std 802.1ABcu™-2021, 5.3.
- b) Topology discovery and verification according to 6.5.
- c) The YANG features and nodes of the ieee802-dot1ab-lldp module according to 6.4.9.2.2.

5.5.3 IA-station requirements for time synchronization

These requirements are related to the entire IA-station with all its PTP Instances and PTP Ports. IA-stations for which a claim of conformance to this document is made shall support the following list of requirements.

- a) PTP Instance requirements according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1AS-2020/Cor 1-2021, IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdnTM-2024, and IEEE Std 802.1ASdm-2024, 5.4.1 items a) through h).

NOTE A gPTP domain in a PTP End Instance can be used for Global Time, Working Clock, or both.

- b) PTP Instance defaults for BTCA according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1AS-2020/Cor 1-2021, IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdn-2024, and IEEE Std 802.1ASdm-2024, 5.4.2a.
- c) Timing and synchronization management according to IEEE Std 802.1AS-2020, 5.4.2 items j) and k).
- d) PTP Instance requirements according to 6.2.2.
- e) PTP Protocol requirements according to 6.2.3.
- f) Clock control system requirements according to 6.2.4.
- g) Error generation limits according to 6.2.5.
- h) PTP Instance synchronization status according to 6.2.6.
- i) The transmission of the Drift_Tracking TLV according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1AS-2020/Cor 1-2021, IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdn-2024, and IEEE Std 802.1ASdm-2024, 5.4.2 item n).
- j) External port configuration capability according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1AS-2020/Cor 1-2021, IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdn-2024, and IEEE Std 802.1ASdm-2024, 5.4.2 f).
- k) MAC-specific timing and synchronization methods for IEEE Std 802.3 full-duplex links according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1AS-2020/Cor 1-2021, IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdn-2024, and IEEE Std 802.1ASdm-2024, 5.5 items a) through e) and item j), and 11.2.17.
- l) The YANG features and nodes of the:
 - 1) ieee1588-ptp-tt module according to 6.4.9.2.3.1,
 - 2) ieee802-dot1as-gptp module according to 6.4.9.2.3.2, and
 - 3) ieee802-dot1as-hs module according to 6.4.9.2.3.3.
- m) The message timestamp point according to IEEE Std 802.1AS-2020, 11.3.9.
- n) The Common Mean Link Delay Service (CMLDS) according to IEEE Std 802.1AS-2020, 11.2.17.
- o) The descriptionDS according to 6.2.13 and IEEE Std 1588-2019, 8.2.5.
- p) gptpCapableStateMachinesEnabled is FALSE according to IEEE Std 802.1ASdm-2024, 5.4.2a and 14.8.54a.

NOTE Setting gptpCapableStateMachinesEnabled to FALSE allows transmission of signaling message to be disabled by design.

5.5.4 Management requirements for non-proxied IA-stations

5.5.4.1 Secure management exchanges

IA-stations which contain a ccA Bridge component or a ccA end station component and for which a claim of conformance to this document is made shall support the following list of requirements.

- a) NETCONF server functionality according to IETF RFC 6241 including:
 - 1) candidate configuration capability as described in IETF RFC 6241, 8.3,
 - 2) rollback-on-error capability as described in IETF RFC 6241, 8.5, and
 - 3) validate capability as described in IETF RFC 6241, 8.6.
- b) NETCONF-over-TLS server supporting TLS version 1.2 according to 6.3.2.1.1 and 6.3.2.1.4.
- c) Secure Device Identity according to 6.3.3 and IEEE Std 802.1AR-2018, 5.3 a), using the signature suite in IEEE Std 802.1AR-2018 9.2, 5.3 d), and 5.3 i).
- d) PKIX according to 6.3.2.1.5 and IETF RFC 5280, 4.1, 4.2.1.1-3, 4.2.1.6, 6.1 and 6.2.
- e) NACM (IETF RFC 8341) supporting four different roles according to 6.3.2.1.5 c).
- f) The YANG features and nodes of the:
 - 1) ietf-keystore module according to 6.4.9.2.4.1,
 - 2) ietf-netconf-acm module according to 6.4.9.2.4.2 and,
 - 3) ietf-truststore according to 6.4.9.2.4.3.
- g) NETCONF Event Notifications according to IETF RFC 5277 including operations according to IETF RFC 5277, Clause 2.
- h) Dynamic Subscriptions to YANG Events and Datastores over NETCONF according to 6.4.7.7.
- i) NETCONF Extensions to support the Network Management Datastore Architecture (NMDA) as described in IETF RFC 8526.
- j) DHCP client according to IETF RFC 2131, 4.1, 4.2, and 4.4.
- k) At least one of the following asymmetric key pair generation methods:
 - 1) component-internal generation according to 6.3.4.3;
 - 2) component-external generation according to 6.3.4.3;
- l) Storage of at least one IDevID credential according to 6.3.3.4.2.5 and one LDevID-NETCONF credential according to 6.3.4.1.
- m) If the IA-station supports CNC functionality, that IA-station shall also support the following list of requirements.
 - 1) TSN CNC station requirements according to IEEE Std 802.1Q-2022, 5.29.
 - 2) NETCONF-over-TLS server and related client functionality 5.5.4.2.
 - 3) The common YANG modules, features, and nodes according to 6.4.9.2.
 - 4) The optional YANG modules, features, and nodes according to 6.4.9.3.
 - 5) Support the use of at least one customer VLAN Identifier for an isolation VLAN and one VLAN identifier for a default VLAN.
 - 6) Support CUC/CNC YANG modules, features and nodes according to 6.4.9.4.
- n) If the IA-station supports CUC functionality, that IA-station shall also support the following list of requirements.
 - 1) Support NETCONF-over-TLS client functionality with client related security requirements according to 5.5.4.2.
 - 2) The TSN UNI YANG module, features, and nodes according to 6.4.9.4.1.
 - 3) The ietf-netconf-client module according to 6.4.9.4.1.

- o) If the IA-station supports proxy functionality, that IA-station shall also support the `ieciieee60802-proxy` YANG module according to 6.4.10.8.8.

IA-stations which contain a ccA Bridge component or a ccA end station component and for which a claim of conformance to this document is made should support internal key generation according to 6.3.4.3.2.

5.5.4.2 IA-station management YANG modules

IA-stations which contain a ccA Bridge component or a ccA end station component and for which a claim of conformance to this document is made shall support the YANG features and nodes for IA-station management of the:

- a) `ietf-system-capabilities` module according to 6.4.9.2.5.1,
- b) `ietf-yang-library` module according to 6.4.9.2.5.2,
- c) `ietf-yang-push` module according to 6.4.9.2.5.3,
- d) `ietf-notification-capabilities` module according to 6.4.9.2.5.4,
- e) `ietf-subscribed-notifications` module according to 6.4.9.2.5.5,
- f) diagnostics using YANG-Push subscriptions according to 6.4.7,
- g) `ietf-netconf-monitoring` module according to 6.4.9.2.5.6,
- h) `ietf-system` module according to 6.4.9.2.5.7,
- i) `ietf-hardware` module according to 6.4.9.2.5.8,
- j) `ietf-interfaces` module according to 6.4.9.2.5.9,
- k) `ieee802-dot1q-bridge` module according to 6.4.9.2.5.10,
- l) `ieciieee60802-ethernet-interface` module according to 6.4.9.2.5.11,
- m) `ietf-netconf-server` according to 6.4.9.2.5.12,
- n) `ieciieee60802-bridge` according to 6.4.9.2.5.11,
- o) `ietf-subscribed-notifications` according to 6.4.9.2.5.13,
- p) `ieciieee60802-subscribed-notifications` according to 6.4.9.2.5.13, and
- q) `ieciieee60802-ia-station` according to 6.4.9.2.5.11.
- r) `ieee802-dot1q-rstp-bridge` according to 6.4.9.2.5.14.

5.5.4.3 Digital data sheet

IA-stations for which a claim of conformance to this document is made shall provide a 60802 instance data file according to 6.4.8. The instance data file shall contain at least the YANG nodes of 6.4.9 that are marked with [m]. Nodes marked with [c] shall be included if the associated feature is supported.

NOTE It is the user's responsibility to ensure that the filename is unique by using a standardized mechanism (for example, GUID, URL, or ReverseDomainName).

5.6 IA-station options

5.6.1 IA-station PHY and MAC options for external ports

IA-stations for which a claim of conformance to this document is made may support the following list of requirements.

- a) Power over Ethernet (PoE) over 2 Pairs according to IEEE Std 802.3-2022, Clause 33.
- b) Power Interfaces according to IEEE Std 802.3-2022, Clause 104.
- c) Power over Ethernet according to IEEE Std 802.3-2022 Clause 145.

5.6.2 IA-station options for time synchronization

IA-stations for which a claim of conformance to this document is made may support the following list of requirements.

- a) The media-independent TimeTransmitter capability according to IEEE Std 802.1AS-2020, 5.4.2 item b) as amended by IEEE Std 802.1ASdr-2024.
- b) Grandmaster PTP Instance capability according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024 and IEEE Std 802.1ASdm-2024, 5.4.2 item c).
- c) More than one PTP port as a PTP Relay Instance according to IEEE Std 802.1AS-2020, 5.4.2 item d).
- d) Hot standby redundancy requirements according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdm-2024, 5.4.2, item m).

5.6.3 IA-station options for management

IA-stations for which a claim of conformance to this document is made may support the following list of requirements.

- a) Writable-Running capability according to IETF RFC 6241, 8.2.
- b) Confirmed Commit capability according to IETF RFC 6241, 8.4.
- c) Distinct Startup capability according to IETF RFC 6241, 8.7.
- d) URL capability according to IETF RFC 6241, 8.8.
- e) XPath capability according to IETF RFC 6241, 8.9.
- f) NETCONF-over-TLS server supporting TLS version 1.2, according to 6.3.2.1.2.
- g) NETCONF-over-TLS server supporting TLS version 1.3, according to 6.3.2.1.3.
- h) PKIX according to IETF RFC 5280, 4.2.1.13, Clause 5 and 6.3.

5.6.4 Proxy target options

IA-stations which contain only ccB Bridge or end station components and for which a claim of conformance to this document is made may support the following list of requirements.

- a) Secure management exchanges according to 5.5.4.1, items a) to m), including support of IA-station management YANG modules according to 5.5.4.2.
- b) Secure management exchanges using a management proxy according to 6.4.5.5 including a digital datasheet according to 5.5.4.3 including the YANG features and nodes of the iecIEEE60802-proxy-target module according to 6.4.10.8.9.

5.7 Bridge component requirements

5.7.1 Common Bridge component requirements

A Bridge component implementation of any conformance class for which a claim of conformance to this document is made shall support the following list of requirements.

- a) C-VLAN component requirements according to IEEE Std 802.1Q-2022, 5.5 and 5.4 except item o) in IEEE Std 802.1Q-2022, 5.4.
- b) The use of Customer VLAN Identifiers (C-VID).
- c) FDB to contain Static and Dynamic VLAN Registration Entries for a minimum of 10 VIDs according to IEEE Std 802.1Q-2022, 8.8.

NOTE 1 An example use case for 10 VIDs: 2 VIDs for IA time-aware stream or IA stream traffic, 2 VIDs for IA time-aware stream or IA stream redundancy, 4 VIDs for IA traffic engineered non-stream or IA non-stream traffic, 1 VID for the isolation VLAN, and 1 VID for the Default VLAN (see 6.4.5.2).

- d) Translation of VIDs through support of the VID Translation Table or through support of both the VID Translation Table and Egress VID translation table on one or more Bridge Ports according to IEEE Std 802.1Q-2022, 6.9.
 - e) The strict priority algorithm for transmission selection on each port for each traffic class according to IEEE Std 802.1Q-2022, 8.6.8.1.
 - f) The capability to disable Priority-based flow control according to IEEE Std 802.1Q-2022, Clause 36, if it is implemented.
 - g) The Priority Regeneration requirements according to IEEE Std 802.1Q-2022, 5.4.1, item o).
 - h) MST according to IEEE Std 802.1Q-2022, 5.4.1.1 a) to i) and k) to o) and 6.4.2.4.
 - i) TE-MSTID according to IEEE Std 802.1Q-2022, 5.5.2, 8.6, and 8.8.
 - j) Spanning tree, VLAN, and TE-MSTID configuration according to 6.4.2.4.
 - k) YANG features and nodes of the `ieciieee60802-tt-mapping` module according to 6.4.10.8.7.
 - l) PSFP according to IEEE Std 802.1Q-2022, 8.6.5.2.1, with support for:
 - 1) at least 3 stream filters per port according to IEEE Std 802.1Q-2022, 8.6.5.3,
 - 2) at least one stream gate according to IEEE Std 802.1Q-2022, 8.6.5.4 with at least zero gate control list entries according to IEEE Std 802.1Q-2022, 12.31.1.4, and
 - 3) at least 3 flow meters per port according to IEEE Std 802.1Q-2022, 8.6.5.5.
- NOTE 2 The flow meter parameter values Excess Information Rate (EIR) = 0, Excess burst size (EBS) = 0 and Color Mode (CM) = `color_blind`, as configurable by the CNC, are applicable to many industrial automation use cases. When CM = `color_blind`, `DropOnYellow` (IEEE Std 802.1Q-2022, 8.6.5.5, item h), `MarkAllFramesRed` (IEEE Std 802.1Q-2022, 8.6.5.1.3, item j), and `MarkAllFramesRedEnable` (IEEE Std 802.1Q-2022, 8.6.5.5, item i) are not used.
- NOTE 3 For example, an implementation can contain one flow meter for broadcast traffic, one flow meter for multicast traffic and one flow meter for unicast traffic. This example can be realized using 49 mask and match stream identification functions per Port and 49 associated stream filter instances (one-to-one relationship) that identify and map unicast (1 stream identification function for identifying all frame with the group address bit cleared in the DA), broadcast (1 stream identification function for identifying all frames with all DA bits set) and multicast (47 stream identification function with the DA group address bit set and one other DA bit cleared for all 47 combinations).
- m) The YANG features and nodes for flow meter configuration according to 6.4.9.2.5.14.
 - n) Stream identification component behaviors according to IEEE Std 802.1CB-2017, 5.3 and IEEE Std 802.1CBdbTM-2021, 5.5 d).

5.7.2 ccA Bridge component requirements

A Bridge component implementation for which a claim of conformance to ccA of this document is made shall support the following list of requirements.

- a) Common Bridge component requirements according to 5.7.1.
- b) At least 2 PTP Instances according to 5.5.3.
- c) Eight queues according to IEEE Std 802.1Q-2022, 8.6.6.
- d) Enhancements for scheduled traffic for data rates of 100 Mb/s and 1 Gb/s according to IEEE Std 802.1Q-2022, 5.4.1 items ac) and ad) including:
 - 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
 - 2) support of the YANG features and nodes of the `ieee802-dot1q-sched` module according to 6.4.9.3.2, and
 - 3) support of the YANG features and nodes of the `ieciieee60802-sched-bridge` module according to 6.4.9.3.3.

- e) Frame preemption according to IEEE Std 802.1Q-2022, 5.4.1 item ae), for data rates of 100 Mb/s and 1 Gb/s, including:
 - 1) support of Interspersing Express Traffic with preemptable traffic according to IEEE Std 802.3-2022, Clause 99 as amended by IEEE Std 802.3de-2022, including support of the Additional Ethernet Capabilities for TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7, and
 - 2) support of the YANG features and nodes of the `ieee802-dot1q-preemption` module according to 6.4.9.3.4.

5.7.3 ccB Bridge component requirements

A Bridge component implementation for which a claim of conformance to ccB of this document is made shall support the following list of requirements.

- a) Common Bridge component requirements according to 5.7.1.
- b) At least 1 PTP Instance according to 5.5.3.
- c) At least four queues according to IEEE Std 802.1Q-2022, 8.6.6.

5.8 Bridge component options

5.8.1 Common Bridge component options

A Bridge component implementation of any conformance class for which a claim of conformance to this document is made may support the operation of the credit-based shaper algorithm according to IEEE Std 802.1Q-2022, 8.6.8.2 on any Port as the transmission selection algorithm for at least 4 traffic classes including support of the YANG features and nodes of the `ieee802-dot1q-cbsa-bridge` module according to 6.4.9.3.5.

5.8.2 ccA Bridge component options

A Bridge component implementation for which a claim of conformance to ccA of this document is made may support the following list of requirements.

- a) Any or none of the common Bridge component options according to 5.8.1.
- b) More than 2 PTP Instances according to 5.5.3.
- c) Enhancements for scheduled traffic for data rates of 10 Mb/s, 2,5 Gb/s, 5 Gb/s, and 10 Gb/s according to IEEE Std 802.1Q-2022, 5.4.1 items ac) and ad) including:
 - 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
 - 2) support of the YANG features and nodes of the `ieee802-dot1q-sched` module according to 6.4.9.3.2, and
 - 3) support of the YANG features and nodes of the `ieciieee60802-sched-bridge` module according to 6.4.9.3.3.
- d) Frame preemption according to IEEE Std 802.1Q-2022, 5.4.1 item ae), for data rates of 10 Mb/s, 2,5 Gb/s, 5 Gb/s, and 10 Gb/s, including:

NOTE IEEE Std 802.3de-2022, 99.1, comprises 10 Mb/s.

- 1) support of Interspersing Express Traffic with preemptable traffic according to IEEE Std 802.3-2022, Clause 99 as amended by IEEE Std 802.3de-2022, including support of the Additional Ethernet Capabilities for TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7, and
- 2) support of the YANG features and nodes of the `ieee802-dot1q-preemption` module according to 6.4.9.3.4.

5.8.3 ccB Bridge component options

A Bridge component implementation for which a claim of conformance to ccB of this document is made may support the following list of requirements.

- a) Any or none of the common Bridge component options according to 5.8.1.
- b) Up to eight queues according to IEEE Std 802.1Q-2022, 8.6.6.
- c) More than 1 PTP Instance according to 5.5.3.
- d) Enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 5.4.1, items ac) and ad) including:
 - 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
 - 2) support of the YANG features and nodes of the `ieee802-dot1q-sched` module according to 6.4.9.3.2, and
 - 3) support of the YANG features and nodes of the `ieciieee60802-sched-bridge` module according to 6.4.9.3.3.
- e) Frame preemption according to IEEE Std 802.1Q-2022, 5.4.1, item ae), including:
 - 1) support of Interspersing Express Traffic with preemptable traffic according to IEEE Std 802.3-2022, Clause 99 as amended by IEEE Std 802.3de-2022, including support of the Additional Ethernet Capabilities for TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7, and
 - 2) support of the YANG features and nodes of the `ieee802-dot1q-preemption` module according to 6.4.9.3.4.

5.9 End station component requirements

5.9.1 Common end station Component requirements

An end station component implementation of any conformance class for which a claim of conformance to this document is made shall support the following list of requirements.

- a) The use of at least one customer VID for IA traffic engineered non-stream or IA non-stream traffic.
- b) The use of an additional customer VID for IA time-aware stream traffic if that traffic type category is supported.
- c) The use of an additional customer VID for IA stream traffic if that traffic type category is supported.
- d) The use of an additional customer VID for IA time-aware stream traffic if redundancy for that traffic type category is supported.
- e) The use of an additional customer VID for IA stream traffic if redundancy for that traffic type category is supported.
- f) Participate in only a single Configuration Domain.
- g) The use of an additional customer VID for an isolation VLAN.
- h) The use of an additional customer VID for a default VLAN

5.9.2 ccA end station component requirements

An end station component implementation for which a claim of conformance to ccA of this document is made shall support the following list of requirements.

- a) Common end station component requirements according to 5.9.1.
- b) At least 2 PTP Instances according to 5.5.3.

- c) End station requirements for enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 5.25, for data rates of 100 Mb/s and 1 Gb/s including:
 - 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
 - 2) support of the YANG features and nodes of the `ieee802-dot1q-sched` module according to 6.4.9.3.2, and
 - 3) support of the YANG features and nodes of the `ieciieee60802-sched-bridge` module according to 6.4.9.3.3.
- d) End station requirements for frame preemption according to IEEE Std 802.1Q-2022, 5.26, for data rates of 100 Mb/s, and 1 Gb/s, if the IA time-aware stream traffic or the IA stream traffic type categories are supported, including:
 - 1) support of Interspersing Express Traffic according to IEEE Std 802.3-2022, Clause 99 as amended by IEEE Std 802.3de-2022, including support of the Additional Ethernet Capabilities TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7 and Table 79-8, and
 - 2) support of the YANG features and nodes of the `ieee802-dot1q-preemption` module according to 6.4.9.3.4.

5.9.3 ccB end station component requirements

An end station component implementation for which a claim of conformance to ccB of this document is made shall support the following list of requirements:

- a) Common end station component requirements according to 5.9.1.
- b) At least 1 PTP Instance according to 5.5.3.

5.10 End station component options

5.10.1 Common end station component options

An end station component implementation of any conformance class for which a claim of conformance to this document is made may support the following list of requirements.

- a) The operation of the credit-based shaper algorithm according to IEEE Std 802.1Q-2022, 8.6.8.2 including support of the YANG features and nodes of the `ieee802-dot1q-cbs` module according to 6.4.9.3.5.
- b) Talker end system behaviors according to IEEE Std 802.1CB-2017, as amended by IEEE Std 802.1CBdb-2021 and IEEE Std 802.1CBcvTM-2021, 5.6, and 5.7 c) on one or more ports and for 1 or more Compound Streams, including support of the `ieee802-dot1cb-stream-identification` and `ieee802-dot1cb-frer` YANG modules according to 6.4.9.3.6.
- c) Listener end system behaviors according to IEEE Std 802.1CB-2017, as amended by IEEE Std 802.1CBdb-2021 and IEEE Std 802.1CBcv-2021, 5.9, on one or more ports and for 1 or more Compound Streams, including support of the `ieee802-dot1cb-stream-identification` and `ieee802-dot1cb-frer` YANG modules according to 6.4.9.3.6.

5.10.2 ccA end station component options

An end station component implementation for which a claim of conformance to ccA of this document is made may support the following list of requirements.

- a) Common end station options according to 5.10.1
- b) More than 2 PTP Instances according to 5.5.3.

- c) End station requirements for enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 5.25, for data rates of 10 Mb/s, 2,5 Gb/s, 5 Gb/s, and 10 Gb/s including:
 - 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
 - 2) support of the YANG features and nodes of the `ieee802-dot1q-sched` module according to 6.4.9.3.2, and
 - 3) support of the YANG features and nodes of the `ieciieee60802-sched-bridge` module according to 6.4.9.3.3.
- d) End station requirements for frame preemption according to IEEE Std 802.1Q-2022, 5.26, for data rates of 10 Mb/s, 2,5 Gb/s, 5 Gb/s, and 10 Gb/s including:

NOTE IEEE Std 802.3de-2022, 99.1, comprises 10 Mb/s.

- 1) support of Interspersing Express Traffic according to IEEE Std 802.3-2022, Clause 99 as amended by IEEE Std 802.3de-2022, including support of the Additional Ethernet Capabilities TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7 and Table 79-8, and
- 2) support of the YANG features and nodes of the `ieee802-dot1q-preemption` module according to 6.4.9.3.4.

5.10.3 ccB end station component options

An end station component implementation for which a claim of conformance to ccB of this document is made may support the following list of requirements.

- a) Common end station component options according to 5.10.1.
- b) One or more PTP Instances according to 5.5.3.
- c) End station requirements for enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 5.25, including:
 - 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
 - 2) support of the YANG features and nodes of the `ieee802-dot1q-sched` module according to 6.4.9.3.2, and
 - 3) support of the YANG features and nodes of the `ieciieee60802-sched-bridge` module according to 6.4.9.3.3.
- d) End station requirements for frame preemption according to IEEE Std 802.1Q-2022, 5.26, including:
 - 1) support of Interspersing Express Traffic according to IEEE Std 802.3-2022, Clause 99 as amended by IEEE Std 802.3de-2022, including support of the Additional Ethernet Capabilities TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7 and Table 79-8, and
 - 2) support of the YANG features and nodes of the `ieee802-dot1q-preemption` module according to 6.4.9.3.4.

6 Required functions for an industrial network

6.1 General

Clause 6 provides requirements specific to this document and the industrial use case.

6.2 Synchronization

6.2.1 General

An IA-station can contain more than one Grandmaster PTP Instance and PTP End Instance to support:

- a) hot-standby use cases, or
- b) Working Clock or Global Time.

For further explanation of the requirements for time synchronization, refer to Annex D.

6.2.2 PTP Instance requirements

A Grandmaster PTP Instance, a PTP Relay Instance and a PTP End Instance, and the Working Clock or Global Time clocks connected to them, shall meet the following requirements under their allowed working conditions and for their lifetime.

- a) The fractional frequency offset of the LocalClock relative to the nominal frequency shall be according to Table 9.
- b) The range of the rate of change of fractional frequency offset of the LocalClock shall be according to Table 9.
- c) During operation, the Working Clock and Global Time at Grandmaster PTP Instances and PTP End Instances shall increase monotonically, where monotonic means that for a time y that occurs after time x , the ClockTarget's timestamp of y is greater than or equal to the ClockTarget's timestamp of x .
- d) The Working Clock and Global Time at a PTP End Instance can be controlled by applying a frequency change over a period of time. The frequency applied can have a fine resolution to speed up or slow down the clock smoothly, and it has a total range of frequency adjustment.
- e) For the Global Time at a PTP End Instance, the maximum value of frequency adjustment shall be according to Table 9.
- f) For the Working Clock at a PTP End Instance, the maximum value of frequency adjustment shall be according to Table 9.

For Working Clock or Global Time, decoupled from a ClockTarget, a higher maximum value of frequency adjustment and maximum rate of change of fractional frequency offset are allowed. As soon as it is coupled (or coupled again), a) to f) apply.

Table 9 – Required values

Topic	Value
Local Clock at non-Grandmaster PTP Instance, range of fractional frequency offset relative to the nominal frequency	± 50 ppm
Local Clock at non-Grandmaster PTP Instance, range of rate of change of fractional frequency offset with respect to the nominal frequency	± 1 ppm/s
Working Clock and Global Time (acting as ClockSource) and Local Clock at Grandmaster PTP Instance, range of fractional frequency offset with respect to the nominal frequency	± 25 ppm
Working Clock and Global Time (acting as ClockSource) and Local Clock at Grandmaster PTP Instance, range of rate of change of fractional frequency offset with respect to the nominal frequency (steady state, see Annex D.2.3)	± 1 ppm/s
Working Clock and Global Time (acting as ClockSource) at Grandmaster PTP Instance, range of rate of change of fractional frequency offset (transient, see Annex D.2.3)	± 3 ppm/s
Working Clock and Global Time at PTP End Instance, maximum value of frequency adjustment	± 250 ppm over any observation interval of 1 ms
NOTE 1 If the Grandmaster PTP Instance implementation is such that its Working Clock and Local Clock are the same or otherwise locked to the same frequency, the normative requirements on the Working Clock take priority over those on the Local Clock. NOTE 2 The Maximum value of frequency adjustment represents an upper bound that limits how much a PTP End Instance can change the frequency of its Working Clock or Global Time during a given period. However, the adjustment is expected to be gradual over the defined interval rather than instantaneous. NOTE 3 The example algorithms that track clock drift use up to 4 seconds of historical data and can take that length of time to respond to changes in clock drift. The example algorithm has been used for simulating cases where no fast changes in the observed frequency drift rate were observed. In most of the real-life situations, this condition can be satisfied.	

6.2.3 PTP protocol requirements

Table 10 shows the required protocol times.

Table 10 – Protocol settings

Topic	Value
Nominal time between successive Announce messages (announce interval)	1 s
Nominal time between successive Pdelay_Req messages (Pdelay_Req message transmission interval)	125 ms
Range of allowed time between successive Pdelay_Req messages	119 ms to 131 ms
Nominal time between successive Sync messages at the Grandmaster (Sync message transmission interval)	125 ms
Range of allowed time between successive Sync messages at the Grandmaster	119 ms to 131 ms
Time between reception of a Sync message and transmission of the subsequent Sync message (i.e. residence time) at a PTP Relay instance	Maximum: 15 ms Measured Mean: ≤ 5 ms
Maximum time between transmission of a Sync message and transmission of the related Follow_Up message	2,5 ms
Time between reception of a Pdelay_Req message and transmission of the subsequent Pdelay_Resp message (i.e. Pdelay turnaround time).	Maximum: 15 ms
NOTE 1 A consequence of having a single allowed value of mean sync interval is that syncLocked mode is achieved. If the timeTransmitter port sync interval is the same as that of the timeReceiver port, syncLocked mode is achieved. NOTE 2 The values contained in this table apply to both the Working Clock and Global Time.	

6.2.4 Clock control system requirements for PTP End Instances

Table 11 shows the required clock control system characteristics at a PTP End Instance.

Table 11 – Clock control system requirements

Topic	Value
Maximum Bandwidth	1,0 Hz
Minimum Bandwidth	0,9 Hz
Maximum Gain Peaking	2,2 dB
Minimum absolute value of Roll-off	20 dB/decade
NOTE 1 For more information regarding the clock control system, see Annex C.	
NOTE 2 The values contained in this table apply to both the Working Clock and Global Time.	

6.2.5 Error generation limits

Table 12 shows the required limits on error generation at a Grandmaster PTP Instance. A limit on error generation for a Grandmaster PTP Instance is a limit on the amount of error it generates in the output Sync message compared to its Working Clock (acting as ClockSource) and Local Clock. See D.3.4.

Table 12 – Error generation limits for Grandmaster PTP Instance

Topic	Value
Accuracy of forwarded time information: (preciseOriginTimestamp + correctionField) in PTP timing message minus Working Clock at Grandmaster when Sync message is transmitted	Allowable range of the measured mean: - 10 ns to + 10 ns Range around the measured mean within which 90 % of measurements fall: ± 7 ns Range around the measured mean within which 100 % of measurements fall: ± 10 ns
Accuracy of forwarded Rate Ratio information: rateRatio field in Follow_Up information TLV minus true Rate Ratio between Working Clock at Grandmaster and Local Clock when Sync message is transmitted	Mean 0 ppm $\pm 0,1$ ppm Standard deviation $\leq 0,1$ ppm
Accuracy of syncEgressTimestamp: syncEgressTimestamp in Drift_Tracking TLV minus Local Clock when Sync message is transmitted	Range around the measured mean within which 90 % of measurements fall: ± 7 ns Range around the measured mean within which 100 % of measurements fall: ± 10 ns
NOTE 1 “Allowable range of the measured mean” specifies limits on constant error. “Range around the measured mean” and “Standard deviation” specify limits on dynamic error. A limit on the constant error of syncEgressTimestamp is not specified because constant error in this characteristic is not a source of time synchronization error.	
NOTE 2 This document specifies statistical normative requirements without a test duration. It is the responsibility of a test plan to determine the test duration and pass/fail criteria, taking sampling error into consideration, and striking an acceptable balance between test duration and accuracy.	

Table 13 shows the required limits on error generation at a PTP Relay Instance. A limit on error generation for a PTP Relay Instance is a limit on the amount of error it adds to the output Sync message compared to the input Sync message. These requirements are written for the case when errors due to change of fractional frequency offset of its Local Clock with respect to the nominal frequency and errors in the input Sync message are negligible with respect to the specified error generation limits. See D.3.5.

Table 13 – Error generation limits for PTP Relay Instance

Topic	Value
Accuracy of residenceTime measurement (no clock drift): (preciseOriginTimestamp + correctionField) in the PTP timing message transmitted by PTP Relay Instance minus Working Clock at Grandmaster when the Sync message is transmitted while - Working Clock (acting as ClockSource) at Grandmaster is stable. - Local Clock at upstream PTP Instance is stable - meanLinkDelay between upstream PTP Instance and PTP Relay Instance is negligible	Allowable range of the measured mean: - 2 ns to + 2 ns Range around the measured mean within which 90 % of measurements fall: ± 10 ns Range around the measured mean within which 100 % of measurements fall: ± 20 ns
Accuracy of Neighbor Rate Ratio measurement and correct modification of Rate Ratio (no clock drift): rateRatio field in the Follow_Up information TLV transmitted by PTP Relay Instance minus Rate Ratio from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while - Working Clock (acting as ClockSource) at Grandmaster is stable. - Local Clock at upstream PTP Instance is stable	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,02 ppm
Accuracy of Neighbor Rate Ratio measurement and correct modification of Rate Ratio (Rate Ratio drift): rateRatio field in the Follow_Up information TLV transmitted by PTP Relay Instance minus Rate Ratio from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while - WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s - Local Clock at upstream PTP Instance is stable.	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,08 ppm

Topic	Value
Accuracy of Neighbor Rate Ratio measurement and correct modification of Rate Ratio (Rate Ratio drift and Neighbor Rate Ratio drift): rateRatio field in the Follow_Up information TLV transmitted by PTP Relay Instance minus Rate Ratio from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while - WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s - Local Clock at upstream PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,08 ppm
Accuracy of Neighbor Rate Ratio Drift measurement and correct modification of Rate Ratio Drift (no clock drift): rateRatioDrift field in the Drift_Tracking TLV transmitted by PTP Relay Instance minus Rate Ratio Drift from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while - WorkingClock (acting as ClockSource) atGrandmaster is stable. - Local Clock at upstream PTP Instance is stable.	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,02 ppm
Accuracy of Neighbor Rate Ratio Drift measurement and correct modification of Rate Ratio Drift (Rate Ratio drift): rateRatioDrift field in the Drift_Tracking TLV transmitted by PTP Relay Instance minus Rate Ratio Drift from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while - WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s - Local Clock at upstream PTP Instance is stable.	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,08 ppm
Accuracy of Neighbor Rate Ratio Drift measurement and correct modification of Rate Ratio Drift (Rate Ratio drift and Neighbor Rate Ratio drift): rateRatioDrift field in the Drift_Tracking TLV transmitted by PTP Relay Instance minus Rate Ratio Drift from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while - WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s - Local Clock at upstream PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,08 ppm

Topic	Value
Accuracy of syncEgressTimestamp: syncEgressTimestamp in Drift_Tracking TLV minus Local Clock when Sync message is transmitted	Range around the measured mean within which 90 % of measurements fall: ± 7 ns Range around the measured mean within which 100 % of measurements fall: ± 10 ns
Accuracy of meanLinkDelay: meanLinkDelay measured by the PTP Relay Instance minus actual path delay	± 3 ns
Accuracy of Pdelay turnaround time measurement: (responseOriginTimestamp – requestReceiptTimestamp) for transmitted Pdelay_Resp and associated Pdelay_Resp_Follow_Up messages minus the actual interval between reception of the associated Pdelay_Req and transmission of the Pdelay_Resp as measured with respect to the Local Clock	Allowable range of the measured mean: - 2 ns to + 2 ns Range around the measured mean within which 90 % of measurements fall: ± 10 ns Range around the measured mean within which 100 % of measurements fall: ± 20 ns
NOTE 1 “Allowable range of the measured mean” specifies limits on constant error. “Range around the measure mean” and “Standard deviation” specify limits on dynamic error. A limit on the constant error of syncEgressTimestamp is not specified because constant error in this characteristic is not a source of time synchronization error. NOTE 2 This document specifies statistical normative requirements without a test duration. It is the responsibility of a test plan to determine the test duration and pass/fail criteria, taking sampling error into consideration, and striking an acceptable balance between test duration and accuracy. NOTE 3 These requirements are for errors generated by the PTP Relay Instance. They assume input Sync messages with negligible errors. However, it is important that timestamps of Pdelay_Req reception and Pdelay_Resp transmission by the upstream PTP Instance include expected timestamp errors when meanLinkDelay accuracy is assessed.	

Table 14 shows the required limits on error generation at a PTP End Instance. A limit on error generation for a PTP End Instance is a limit on the amount of error it adds to its Working Clock (acting as ClockTarget) compared to the input Sync message. These requirements are written for the case when errors due to change of fractional frequency offset of its Local Clock with respect to the nominal frequency and errors in the input Sync message are negligible with respect to the specified error generation limits. See D.3.6.

Table 14 – Error generation limits for PTP End Instance

Topic	Value
Accuracy of recovered clock (no clock drift): Working Clock (acting as ClockTarget) at PTP End Instance minus Working Clock (acting as Clock Source) at Grandmaster while - WorkingClock (acting as ClockSource) at Grandmaster is stable. - Local Clock at upstream PTP Instance is stable. - meanLinkDelay between upstream PTP Instance and PTP End Instance is negligible	Allowable range of cTE: - 10 ns to + 10 ns Allowable range of dTE: - 15 ns to + 15 ns
Accuracy of recovered clock (Rate Ratio drift): Working Clock (acting as ClockTarget) at PTP End Instance minus Working Clock (acting as Clock Source) at Grandmaster while - WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s - Local Clock at upstream PTP Instance is stable. - meanLinkDelay between upstream PTP Instance and PTP Relay Instance is negligible	Allowable range of cTE: - 10 ns to + 10 ns Allowable range of dTE: - 145 ns to + 15 ns
Accuracy of recovered clock (Rate Ratio drift and Neighbor Rate Ratio drift): Working Clock (acting as ClockTarget) at PTP End Instance minus Working Clock (acting as Clock Source) at Grandmaster while - WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s - Local Clock at upstream PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s - meanLinkDelay between upstream PTP Instance and PTP Relay Instance is negligible	Allowable range of cTE: - 10 ns to + 10 ns Allowable range of dTE: - 145 ns to + 15 ns
Accuracy of meanLinkDelay: meanLinkDelay measured by the PTP End Instance minus actual path delay	± 3 ns
NOTE 1 This document specifies statistical normative requirements without a test duration. It is the responsibility of a test plan to determine the test duration and pass/fail criteria, taking sampling error into consideration, and striking an acceptable balance between test duration and accuracy. NOTE 2 These requirements are for errors generated by the PTP End Instance. They assume input Sync messages with negligible errors. However, it is important that timestamps of Pdelay_Req reception and Pdelay_Resp transmission by the upstream PTP Instance include expected timestamp errors when meanLinkDelay accuracy is assessed.	

6.2.6 PTP Instance synchronization status

Industrial automation systems monitor the synchronization status of each PTP Instance to determine the viability of operations. This status is obtained from the isSynced global variable specified in IEEE Std 802.1ASdm-2024, 18.4.1.1.

The PtpInstanceSyncStatus state machine according to IEEE Std 802.1ASdm-2024, 18.4, shall be supported by all PTP instances independent of whether hot standby is supported. Non-Grandmaster PTP Instances shall support one or more of the application interfaces of 9.3, 9.4, 9.5, and 9.6 according to IEEE Std 802.1ASdm-2024. Grandmaster PTP Instances shall support the application interface of 9.2 according to IEEE Std 802.1ASdm-2024.

6.2.7 Application framework

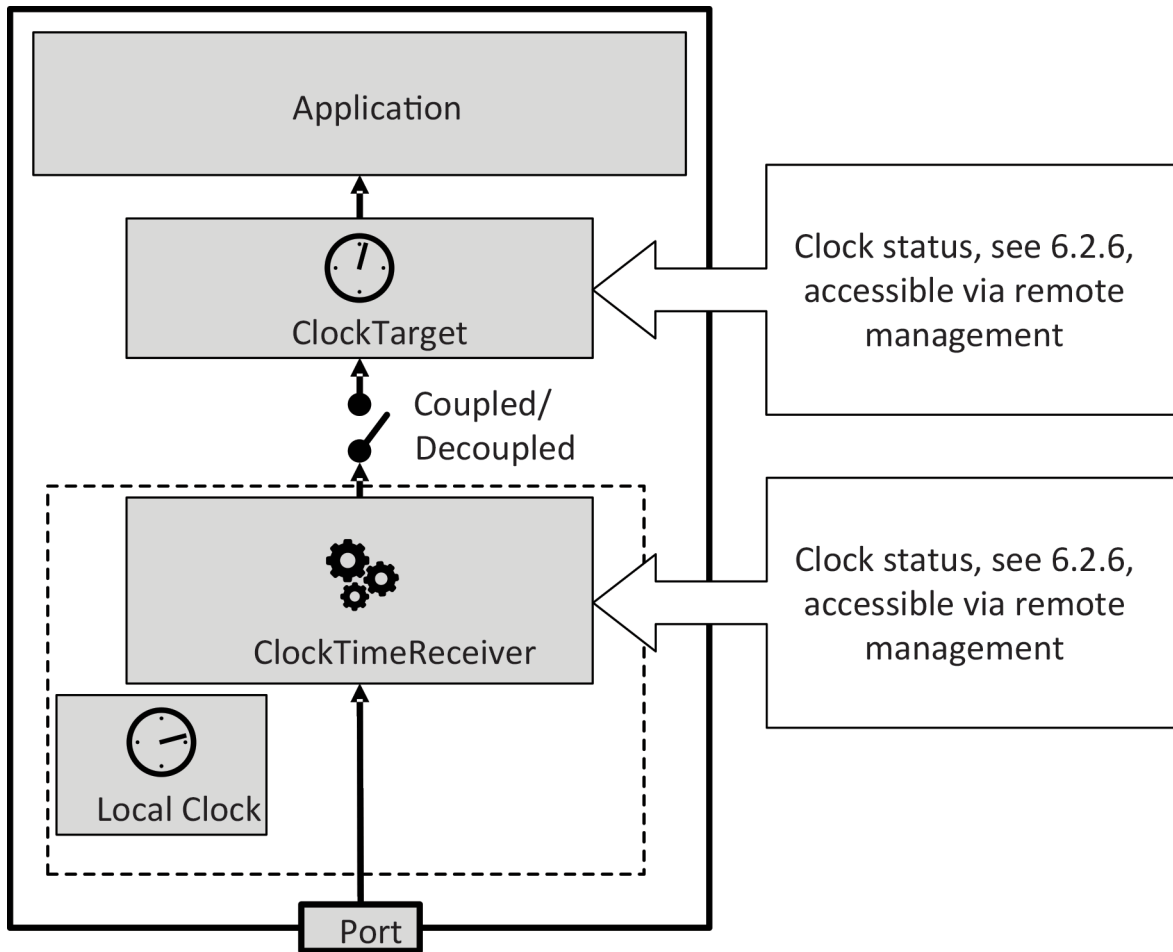
Any step change in the time of a ClockSource or ClockTarget whose absolute value exceeds a user-defined threshold (for example 1 μ s) leads to action being taken by the application or by a higher-layer entity.

If the change is in Global Time, it is desirable that all consumers of that time be made aware of this change (i.e. a jump in Global Time from the value A to the value B), so that the actual time interval between the time corresponding to A and the time corresponding to B can be evaluated.

In the case of Working Clock, a time change that exceeds the user-defined threshold (for example 1 μ s) is avoided to protect assets and prevent damage. Thus, the ClockSource or ClockTarget can be decoupled (see Figure 12 and Figure 13) from the PTP-maintained clock when such a time change occurs.

In Figure 12, two ClockTargets are traceable to a reliable source of time, which should be synchronized to Global Time or Working Clock.

The status of a ClockSource, ClockTarget, ClockTimeTransmitter or ClockTimeReceiver is given by the status of the clock (see 6.2.6) as shown in Figure 11. When timestamps are provided to the application, the current ClockSource or ClockTarget state can also be provided to the application.



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Figure 11 – Clock model**6.2.8 Working Clock domain framework**

The gPTP domainNumber of a Working Clock domain is assigned by the CNC. In industrial applications, when the number of PTP Relay Instances between the Grandmaster PTP Instance and any PTP End Instance is less than or equal to 99, $\max|TE_R|$ of the synchronized time of any ClockTarget, relative to the Grandmaster ClockSource, is less than or equal to 1 μs (see error budget A in Figure 14). Thus it is incumbent upon any PTP Instance to ensure that the requirements specified in 5.5.3, 6.2.2, 6.2.3, 6.2.4, and 6.2.5 are met.

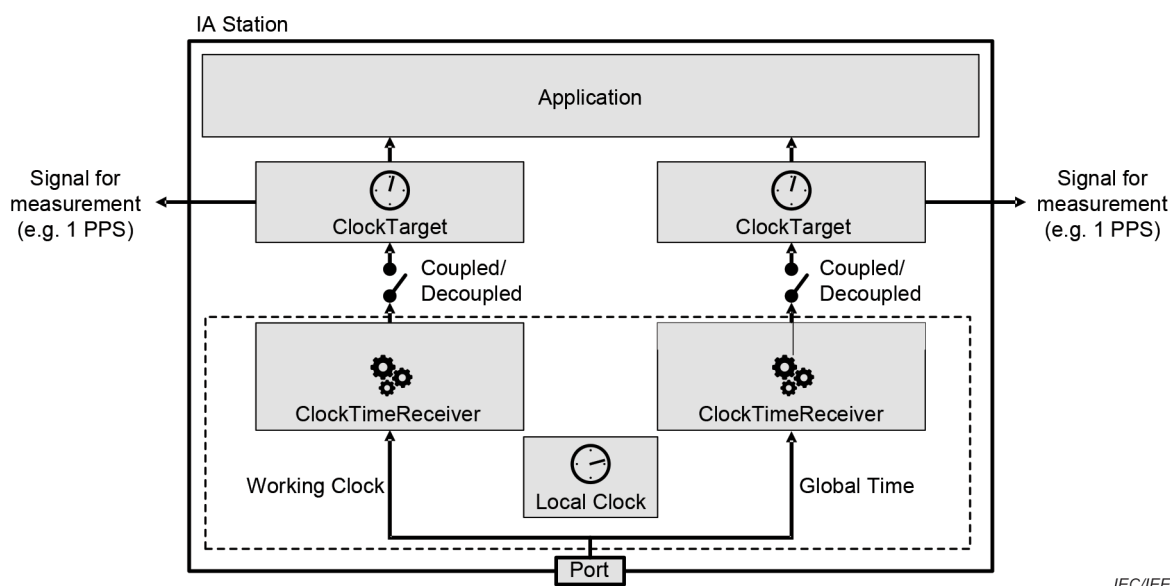
6.2.9 Global Time domain framework

The gPTP domainNumber of a Global Time domain is assigned by the CNC. In industrial applications, when the number of PTP Relay Instances between the Grandmaster PTP Instance and any PTP End Instance is less than or equal to 99, $\max|TE_R|$ of the synchronized time of any ClockTarget, relative to the Grandmaster ClockSource, is less than or equal to 100 μs (see error budget A in Figure 14). Thus it is incumbent upon any PTP Instance to ensure that the requirements specified in 5.5.3, 6.2.2, 6.2.3, 6.2.4, and 6.2.5 are met.

6.2.10 IA-station model for clocks

Industrial automation applications, as described in 4.1, require synchronized time that is traceable to a known source (i.e. Global Time) and a source of time synchronized to the Working Clock. Figure 12 and Figure 13 show examples of the IA-station internal model for clocks with the two PTP Instances. It is possible for the ClockSource or ClockTarget to start decoupled or become decoupled from the ClockTimeTransmitter or ClockTimeReceiver, respectively, of a PTP Instance; the ClockSource or ClockTarget runs independently of the availability of the network or a Grandmaster. For example, if the PTP Instance enters a state where `isSynced` is `FALSE`, the application might choose to decouple its clock from the PTP Instance and continue to run on its internal clock. If `isSynced` for the PTP Instance changes to `TRUE`, the application can choose to again synchronize to the PTP Instance.

Figure 12 shows the IA-station internal model for clocks, with the two PTP Instances used as ClockTimeReceiver/ClockTarget.



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Figure 12 – Example clock usage principles for PTP End Instances

Figure 13 shows the IA-station internal model for clocks, with the two PTP Instances used as Grandmaster.

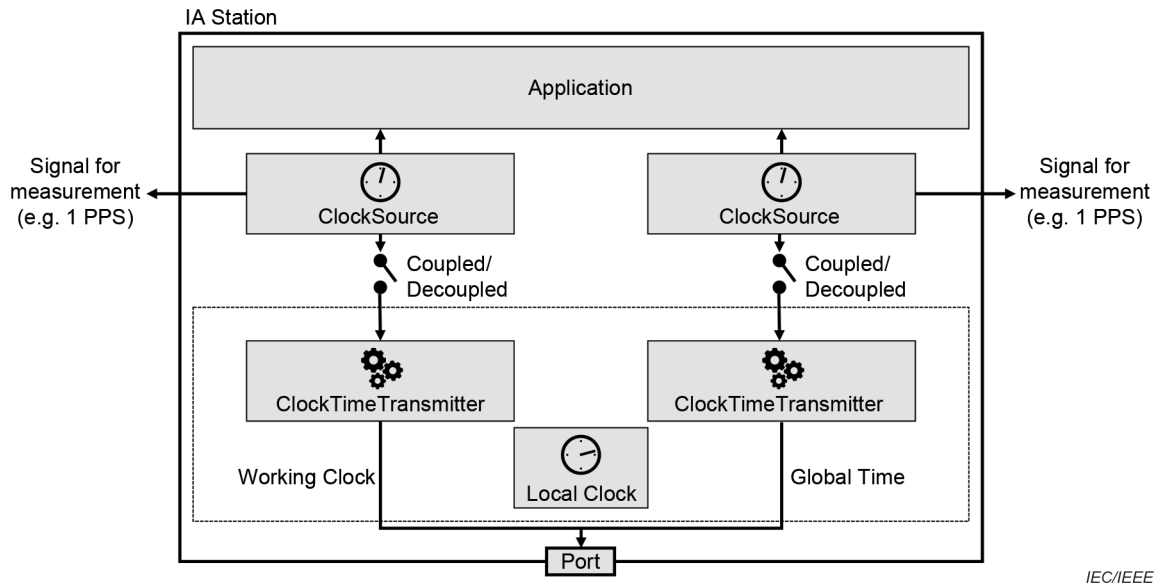


Figure 13 – Example clock usage principles for Grandmaster PTP Instances

6.2.11 Clock usage for the Ethernet interface

6.2.11.1 Time-aware offset control

Time-aware offset control (see 4.4), uses an assigned source of time and a definition when to start or to stop, which are dependent on the clock state.

The clock used is the ClockTarget or, in the case of a Grandmaster PTP Instance, the ClockSource.

IA time-aware streams are only transmitted while isSynced for the chosen ClockSource or ClockTarget is TRUE (see 6.2.6).

Thus, changes of the clock state directly influence the transmission of frames.

6.2.11.2 Gating cycle

To control the gating cycle, the gate control list uses an assigned source of time. Enabling and disabling the gate control list is dependent on the clock state.

The clock used is the ClockTarget or, in the case of a Grandmaster PTP Instance, the ClockSource.

The gating cycle is run using the chosen ClockSource or ClockTarget regardless of the value of isSynced (see 6.2.6).

6.2.12 Error model

Synchronization is transported over the entire path, from the Grandmaster PTP Instance to the PTP End Instance, through the intermediate PTP Relay Instances. All time errors, cTE and dTE, are accumulated during this process.

Time error can arise in the following processes:

- the transporting of time in PTP Instances and via PTP Links that connect PTP Instances,
- the providing of time to the Grandmaster PTP Instance, from the ClockSource entity via the ClockTimeTransmitter entity, and
- the providing of time to a ClockTarget entity (end application) via the ClockTimeReceiver entity.

NOTE Item a) includes time error introduced in a PTP End Instance between the timeReceiver port and the ClockTimeReceiver entity, and between the ClockTimeTransmitter entity and a TimeTransmitter port.

An output synchronization signal (for example, 1 pulse per second (PPS)) synchronized to the Working Clock as shown in Figure 12 and Figure 13, at any PTP Instance, is used to measure the time error between the ClockSource of the Grandmaster and the ClockTarget of a PTP Instance that is not the Grandmaster. The additional error introduced by implementation of the output synchronization signal is in the range of -10 ns to $+10$ ns. Figure 14 shows the error budget principle used. These budgets do not include any deviation from the PTP timescale. Representative budgets are provided in Annex D.

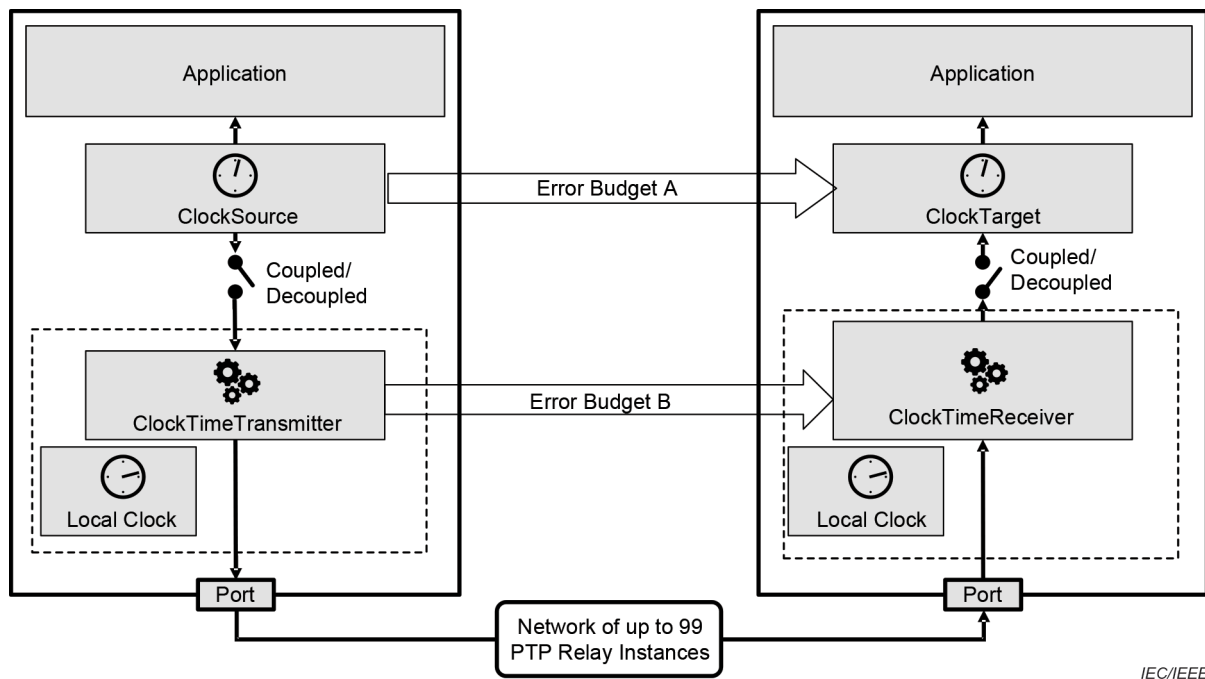


Figure 14 – Error budget scheme

Table 15 shows example values for the splitting of the available error budgets (see Figure 14).

Table 15 – Error budget

Domain	Error budget A	Error budget B
Working Clock	1 μ s	900 ns
Global Time	100 μ s	99,9 μ s

Global Time is often used for tracking events in industrial applications (i.e. sequence of events). Any usage of Global Time for time stamping of application events is allowed an error budget of 1 ms.

6.2.13 gPTP domains and PTP Instances

Any valid gPTP domain number as specified in IEEE Std 802.1AS-2020 can be used. The IEEE Std 1588-2019 attribute `descriptionDS.userDescription` shall be used according to Table 16 to support the translation of PTP Instances and middleware as described in 4.6.2. One gPTP domain can be used for both Working Clock and Global Time. If only one gPTP domain is used, then the requirements for the Working Clock apply (see 6.2.8).

Table 16 – `descriptionDS.userDescription` of gPTP Domains

gPTP Domain	<code>descriptionDS.userDescription</code>
Working Clock (no hot standby configured)	"60802-WorkingClock"
Primary Working Clock (with configured hot standby)	"60802-Primary-WorkingClock"
Secondary Working Clock (with configured hot standby)	"60802-Secondary-WorkingClock"
Global Time (no hot standby configured)	"60802-GlobalTime"
Primary Global Time (with configured hot standby)	"60802-Primary-GlobalTime"
Secondary Global Time (with configured hot standby)	"60802-Secondary-GlobalTime"
GlobalTime and WorkingClock (no hot standby configured)	"60802-GlobalTime-WorkingClock"
Primary GlobalTime and WorkingClock (with configured hot standby)	"60802-Primary-GlobalTime-WorkingClock"
Secondary GlobalTime and WorkingClock (with hot standby configured)	"60802-Secondary-GlobalTime-WorkingClock"

The `descriptionDS.userDescription` attribute is represented in the `ieee1588-ntp-tt` YANG module by the `user-description` leaf in the `description-ds` container of a PTP Instance.

The linking between a gPTP domain and the IETF interfaces is provided by the `underlying-interface`.

6.3 Security model

6.3.1 General

Subclause 6.3 specifies the security model starting with NETCONF/YANG. It describes the security functionality, the security objects in factory default state, the imprinting of Configuration Domain-specific security objects and the secure configuration based on Configuration Domain-specific security objects.

6.3.2 Security functionality

6.3.2.1 Message exchange protection

6.3.2.1.1 NETCONF-over-TLSv1.2 profile

Network configuration with NETCONF/YANG is protected by NETCONF-over-TLS according to IETF RFC 7589, where the to-be-configured IA-stations act in the NETCONF server role. TLS protocol version 1.2 according to IETF RFC 5246, 6.2.3.3, 7.4.7.2 and 8.1.2 shall be supported according to the following list of requirements.

- Mutual authentication in conjunction with the IDevID and LDevID credentials according to 6.3.4 and 6.3.5.
- The cipher suite `TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256` according to IETF RFC 5289, 3.2 and Clause 5.

NOTE IETF RFC 7589 implicitly mandates the cipher suite TLS_RSA_WITH_AES_128_CBC_SHA by referring to IETF RFC 5246. This cipher suite is not used in this document because it requires excessive asymmetric key lengths, it is not an Authenticated Encryption with Associated Data (AEAD) scheme, and it does not provide perfect forward secrecy.

- c) Signature algorithm ECDSA with SHA-256 and Curve P-256 according to NIST FIPS 186-5 Digital Signature Standard (DSS).

The TLS Certificate message from the TLS client and server contains the self-signed root certificate. This approach allows simplifying and flattening the PKI hierarchy on base of the current TLS client certificate to NETCONF username mapping algorithm in IETF RFC 7589. Implementations shall support TLS Certificate message with at least 2 certificate objects.

NETCONF-over-SSH according to IETF RFC 6242 is not used in this document.

NOTE This document selects TLS as a secure transport for NETCONF since TLS is the better match for the case of configuration clients that rely upon unattended or automated operation. This case is dominant in industrial automation.

6.3.2.1.2 NETCONF-over-TLSv1.2 options

The following list of TLS protocol version 1.2 requirements may be supported.

- a) The cipher suites TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 according to IETF RFC 5289, 3.2 and Clause 5, and TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305_SHA256 according to IETF RFC 7905, Clause 2.
- b) Signature algorithms ECDSA with SHA-512 and Curve P-521 according to NIST FIPS 186-5, Ed25519 according to IETF RFC 8032, 5.1, and Ed448 according to IETF RFC 8032, 5.2.

6.3.2.1.3 NETCONF-over-TLSv1.3 profile

Network configuration with NETCONF/YANG is protected by NETCONF-over-TLS according to IETF RFC 7589 and IETF draft-ietf-netconf-over-tls13. TLS protocol version 1.3 according to IETF RFC 8446 may be used with mutual authentication for NETCONF/YANG as follows.

- a) The cipher suites TLS_AES_128_GCM_SHA256, TLS_AES_256_GCM_SHA384 and TLS_CHACHA20_POLY1305_SHA256 according to IETF RFC 8446, 9.1.
- b) One or more of the following signature schemes:
 - 1) ecdsa_secp256r1_sha256 according to NIST FIPS 186-5 and NIST SP 800-186, 3.2.1.3, ecdsa_secp521r1_sha512 according to NIST FIPS 186-5 and NIST SP 800-186, 3.2.1.5,
 - 2) ed25519 according to IETF RFC 8032, 5.1, and
 - 3) ed448 according to IETF RFC 8032, 5.2.

The TLS Certificate message from the TLS client and server contains the self-signed root certificate.

6.3.2.1.4 Certificate-to-name mapping

The IETF RFC 7589 based certificate-to-name mapping procedure is as follows.

NOTE IETF RFC 7589, Clause 7, specifies that NETCONF servers map client certificates to "NETCONF usernames" and specifies a concrete mapping procedure for this purpose. This mapping is represented by the YANG module ietf-x509-cert-to-name.

The list of mapping entries has a single element containing:

- fingerprint: the fingerprint of the trust anchor for the Configuration Domain, and
- map_type: ext-60802-roles.

The map-type `ext-60802-roles` maps the roles provided in the `id-60802-pe-roles` extension (defined in 6.3.2.1.5) of the end entity certificate presented by the NETCONF client to a NETCONF username. The UTF-8 string representation of each role is added to the NETCONF username in chronological order of the enumeration values, whereas multiple roles are separated by ':' character.

6.3.2.1.5 Role extension

The `id-60802-pe-roles` extension in LDevID-NETCONF end entity certificates shall be constructed as follows:

a) Extension field `extnID`

The `extnID` shall provide the following OBJECT IDENTIFIER to identify the `id-60802-pe-roles` extension:

```
id-60802 OBJECT IDENTIFIER ::= { iso (1) iso-identified-organization (3) ieee (111)
standards-association-numbered-series-standards (2) lan-man-stds (802) ieee802dot1 (1)
ieee802dot1stds (2) 1 }
```

```
id-60802-pe OBJECT IDENTIFIER ::= { id-60802 1 }
```

```
id-60802-pe-roles OBJECT IDENTIFIER ::= { id-60802-pe 1 }
```

b) Extension field `critical`

The `id-60802-pe-roles` extension is marked as non-critical (`critical:= FALSE`).

c) Extension field `extnValue`

```
60802RoleNamesSyntax ::= SEQUENCE OF 60802RoleName
```

```
60802RoleName ::= ENUMERATED {
    SecurityAdminRole (0),
    ConfiguratorRole (1),
    StreamConfiguratorRole (2),
    SubscriberRole (3) }
```

6.3.2.2 Resource access authorization

Access control to NETCONF/YANG resources shall be protected by NACM according to IETF RFC 8341.

NACM specifies a YANG data model (`ietf-netconf-acm`) for expressing rules to control access to NETCONF/YANG resources. This document profiles NACM to deliver role-based access control.

NOTE 1 NACM does not natively deliver role-based access control but can be geared by profiling.

This role-based model for security resources should be applied according to the following list of requirements.

- The global switch `enable-nacm` is set to `true`.
- The set of NETCONF/YANG resources of an IA-station is partitioned according to the YANG modules specified in 6.4.9 with a permission-to-role assignment as listed below. An access operation is allowed through the keyword "permitted" and not allowed through the keyword "denied".

NOTE 2 NACM recognizes following "access-operations": `create`, `read`, `update`, `delete`, `exec` and uses the term `write` access for the access operations "create", "delete", and "update". This document uses the terms `read`, `write` and `exec` access.

- All authenticated entities (default rules): All YANG modules: read access permitted, write access denied, exec-access denied.

NOTE 3 The default rules apply for YANG modules that are listed in 6.4.9 but are not listed in the rules of the individual roles.

- Rules for StreamConfiguratorRole: YANG module `ieee802-dot1q-cnc-config`: write and execute operations permitted.
- Rules for SubscriberRole:
 - YANG module `ietf-subscribed-notifications`: write and execute operations permitted, and
 - YANG module `ietf-yang-push`: write and execute operations permitted.
- Rules for ConfiguratorRole: All YANG modules except those listed below, write and execute operations permitted:
 - YANG modules for security configuration, i.e. `ietf-truststore`, `ietf-keystore`, path to `cert-to-name` nodes of `ietf-netconf-server`, path to `tls-server-parameters` nodes of `ietf-netconf-server`,
 - YANG modules for stream configuration, i.e. `ieee802-dot1q-cnc-config`, and
 - YANG modules for subscription configuration, i.e. `ietf-subscribed-notifications`, `ietf-yang-push`.
- Rules for SecurityAdminRole:
 - YANG module `ietf-truststore`, path to certificate node of IDevID trust anchor: write and execute operations denied, and
 - YANG module `ietf-truststore` (besides path to certificate node of IDevID trust anchor): write and execute operations permitted.
 - YANG module `ietf-keystore`, path to asymmetric-key node of IDevID credential: write and execute operations denied, and
 - YANG module `ietf-keystore` (besides path to asymmetric-key node of IDevID credential): write and execute operations permitted.
 - YANG module `ietf-netconf-server` (besides path to `cert-to-name` nodes and path to `tls-server-parameters` nodes): write and execute operations denied, and
 - YANG module `ietf-netconf-server`, path to `cert-to-name` nodes: write and execute operations permitted.
 - YANG module `ietf-netconf-server`, path to `tls-server-parameters` nodes: write and execute operations permitted.

In addition, the following access control should be applied for NETCONF protocol operations:

- `<lock>`, `<unlock>`: permitted for any role specified in this document,
- `<partial-lock>`, `<partial-unlock>`: denied (not used in this document),
- `<get>` and `<get-config>`: mapped to a "read" access operation to the target datastore,
- `<edit-config>`: permitted for any role specified in this document,
- `<copy-config>`: permitted for ConfiguratorRole,
- `<delete-config>`: denied (not used in this document),
- `<commit>`: permitted for any role specified in this document,
- `<discard-changes>`: permitted for any role specified in this document,
- `<close-session>`: permitted for any role specified in this document, and
- `<kill-session>`: denied (not used in this document).

This document does not specify the assignment of role names to actual system entities. This is a duty of system owners or operators.

6.3.3 IDevID Profile

6.3.3.1 General

IA-stations shall possess IDevID credentials according to 6.3.3. CNCs shall contain trust anchors for validating IDevID credentials.

6.3.3.2 Object Contents

6.3.3.2.1 General

The IDevID credential contents shall comply to 6.3.3.2.2, 6.3.3.2.3, and IEEE Std 802.1AR-2018, Clause 6.

6.3.3.2.2 IA-station Identity

Any IDevID EE certificate of an IA-station shall take one of the following forms:

- raw form: the IDevID EE certificate complies to IEEE Std 802.1AR-2018, Clause 8, and
- extended form: the IDevID EE certificate complies to requirements provided IEEE Std 802.1AR-2018, Clause 8. The extended form of an IDevID EE certificate shall be constructed as follows:
 - the verifiable device identity shall appear as a URN in a GeneralName of type uniformResourceIdentifier in the subjectAltName extension,
 - the URN value shall be constructed according to IETF RFC 8141 and as follows:
 - namespace identifier: ieee (see IETF RFC 8069),
 - namespace-specific string: iec-ieee-60802#verifiable-device-identity, and
 - q-component (see IETF RFC 8141, 2.3.2) to parameterize the named resource: an ampersand-separated list of keyword=value tuples with following keywords and values. These tuples can appear in any order inside the q-component:
 - the keywords: hardware-rev, serial-num, mfg-name, model-name;
 - their corresponding values from the single "chassis" component list entry in the ietf-hardware YANG module (see 6.4.9.2.5.8) that represents the management entity of the IA-station respectively from its pre-material form in percent-encoding according to IETF RFC 3986, 2.1.

NOTE 1 These are the items with the YANG property config=false from the 'component' list entry that represents the management entity of the IA-station. The config=false items firmware-rev and software-rev are excluded to avoid IDevID credential updates in case of FW or SW updates.

NOTE 2 An object looks like urn:ieee:iec-ieee-60802#verifiable-device-identity?=mfg-name=<mfg-name>&model-name=<model-name>&hardware-rev=<hardware-rev>&serial-num=<serial-num>.

NOTE 3 One IDevID EE certificate can have one subjectAltName extension which can have one or more GeneralName entries. In particular, there can be one or more GeneralName entries of type uniformResourceIdentifier. This allows other organizations (e.g. middleware and application consortia or individual manufacturers) to also represent their perception of verifiable device identity in addition to the perception of this document.

6.3.3.2.3 Signature Suites

An IDevID shall utilize the signature suite: ECDSA P-256/SHA-256 according to NIST FIPS 186-5, NIST FIPS 180-4 and NIST SP 800-186 using the algorithm identifiers according to IETF RFC 5480.

An IDevID may utilize the following signature suites:

- ECDSA P-521/SHA-512 according to NIST FIPS 186-5, NIST FIPS 180-4 and NIST SP 800-186 using the algorithm identifiers according to IETF RFC 5480,
- EdDSA instance Ed25519 according to IETF RFC 8032 using Curve25519 according to IETF RFC 7748 and using the algorithm identifiers according to IETF RFC 8410, and
- EdDSA instance Ed448 according to IETF RFC 8032 using Curve448 according to IETF RFC 7748 and using the algorithm identifiers according to IETF RFC 8410.

6.3.3.3 Information Model

6.3.3.3.1 General

The information model for IDevID credentials and trust anchors shall comply to YANG and NMDA, in particular the YANG modules ietf-keystore and ietf-truststore, as well as subsequent subclauses of 6.3.3.3.

6.3.3.3.2 Entries

IDeVID credentials shall be provided in form of built-in keys of an IA-station by its manufacturer. In YANG, they are modeled as config-false nodes and are represented in the 'keystore' container that is instantiated by the YANG module ietf-keystore. The private key shall use the private-key-type choice hidden-private-key, i.e. the IDevID private key is not presented in NETCONF/YANG. The details of storing and protecting IDevID private keys as well as using them for signing purposes are implementation specific.

Trust anchors for IDevID credentials are CNC user-configured data objects; these objects shall be available as applied configuration (IETF RFC 8342) upon CNCs. In YANG, they are modeled as config-true nodes and are represented in the 'truststore' container that is instantiated by the YANG module ietf-truststore.

NOTE IA-station built-in trust anchors for use cases such as firmware/software update are not addressed in this document.

6.3.3.3.3 Entry Manifolddness

An IA-station shall possess one IDevID credential with a certification path plus trust anchor information issued under the required signature suite according to 6.3.3.2.3 as part of its factory default state.

If an IA-station supports an optional signature suite according to 6.3.3.2.3, it shall possess in addition one IDevID credential with a certification path plus trust anchor information issued under the optional signature suite as part of its factory default state.

An IA-station can have additional IDevID credential(s) with a certification path plus trust anchor information issued under a combination of any required or any supported optional DevID signature suites.

If an IA-station possesses multiple IDevID credentials, then they shall be issued by the same organization (the IA-station manufacturer). Their EE certificates shall contain the same device identity information.

A CNC shall support at least one trust anchor for IDevID credentials per supported IA-station manufacturer.

6.3.3.3.4 Entry Naming

IDeVID credentials shall be present in an 'asymmetric-key' entry that is identified as: /ietf-keystore:keystore/asymmetric-keys/asymmetric-key/name=IDeVID-<SignatureSuiteName>-<CertificateSerialNumberOfEECertificate>.

IDeVID trust anchors shall be present in 'certificate' entries that are identified as: /ietf-truststore:truststore/certificate-bags/certificate-bag/certificate/name=IDeVID-<SignatureSuiteName>-<CertificateSerialNumberOfCACertificate>.

Such entries shall be present underneath a 'certificate-bag' entry that is identified as: /ietf-truststore:truststore/certificate-bags/certificate-bag/name=IDeVID.

6.3.3.4 Processing model

6.3.3.4.1 General

The processing model for IDeVID credentials and trust anchors shall comply to IEEE Std 802.1AR-2018 and 6.3.3.4.

6.3.3.4.2 Credentials

6.3.3.4.2.1 General

IDeVID credentials are used in following use cases:

- NETCONF/YANG security setup from factory default; the number of such events scales with the number of factory resets, i.e. this use case is performed sporadically. It is conducted by CNCs and encompasses a device identity verification, and
- device identity verification happens as a subtask during NETCONF/YANG security setup from factory default. It can also happen at the discretion of the CNC user. The details of device identity verification are also subject to given policy.

In these use cases, IA-stations act in claimant role and CNCs act in verifier role:

- IA-stations shall present the certification path of and prove private key possession for an IDeVID credential, and
- CNCs shall validate the certification path, check the proof-of-possession for the private key, and verify the obtained device identity information.

6.3.3.4.2.2 Creation

IA-station manufacturers select the form factor for representing verifiable device identity in IDeVID credentials: raw or extended form. The details of the IDeVID credential issuance process are manufacturer-specific and not addressed in this document.

IA-station manufacturers are not required to offer an update feature for IDeVID credentials.

6.3.3.4.2.3 Distribution

IA-stations shall supply IDeVID credentials in form of built-in keys, see 6.3.3.3.

6.3.3.4.2.4 Use

Verifiers (CNCs) shall perform the following checks when they challenge claimants (IA-stations) to authenticate themselves by means of an IDevID credential.

- IDevID certification path validation according to IETF RFC 5280, Clause 6. Whether this validation happens with or without revocation checks is at the discretion of the CNC user.
It is the responsibility of the CNC user to supply a trust anchor configuration (set of trusted certificates or trusted public keys), a revocation check instruction (Boolean) and optionally, X.509 CRL objects according to IETF RFC 5280, Clause 5, to CNCs. The certification path validation is passed if and only if the IDevID EE certificate is the leaf of a valid certification path that ends with a CA certificate which is signed by a configured trust anchor and which is not revoked (if revocation check is enabled).
- Proof-of-possession checking for the private key. The proof-of-possession check is passed if and only if the IA-station possesses the private key which matches the public key in the IDevID EE certificate.
- It is the responsibility of the CNC user to establish and supply to CNCs: a device identity verification policy which determines the verifiable device identity subset that shall be checked by the CNC for the IA-stations in a Configuration Domain. This is a subset of {hardware-rev, serial-num, mfg-name, model-name}. The empty subset ("no-identity-check") as well as the whole set are allowed. The device identity verification for an IA-station instance shall behave according to the following list of requirements.
 - If this subset is empty, then the device identity check is passed. If the user chooses not to verify identity, information about the devices is considered unreliable. Tracking the unverified status of such devices is the responsibility of the user. It is the responsibility of the user to establish policies for the use of such devices.
 - If this subset is non-empty, then the CNC performs the following expected vs. actual check for each verifiable device identity item in this subset:
 - The check for any item in this subset is passed if the expected value (from ietf-hardware YANG module) matches the actual value (from the verifiable device identity URN value for this document in the subjectAltName extension of the IDevID EE certificate). This check fails if the IDevID has raw form.
 - The device identity check is passed if it is passed for all items in the subset.

IDevIDs in raw form (without verifiable device identity URN) can be used if the device identity verification setting option "no-identity-check" is employed. This allows to perform the NETCONF/YANG security setup from factory default for IA-stations with IDevID credentials in raw form. From CNC perspective these IA-stations remain anonymous.

NOTE This document does not specify a mechanism for device identity verification for IDevIDs in raw form. Whether and how device identity checks for such IA-stations are done in an offline mode is at the discretion of CNC users.

6.3.3.4.2.5 Storage

Credentials shall be stored persistently upon an IA-station. The details for implementing this persistent storage are IA-station manufacturer-specific and not addressed in this document.

IA-stations shall support storage of at least one IDevID credential.

6.3.3.4.2.6 Revocation

It is the responsibility of IA-station manufacturers to report revocation for the IDevID credentials issued by them in form of X.509 CRL objects according to IETF RFC 5280, Clause 5. These objects are made available in a form that allows relying parties, i.e. CNC users to retrieve them at their own discretion.

CNC users decide whether they support IDevID certification path validation with or without revocation:

- if revocation checks are disabled, then certificate path validation shall be performed according to IETF RFC 5280, 6.1, or
- if revocation checks are enabled, then certificate path validation shall be performed according to IETF RFC 5280, 6.1 and 6.3.

NOTE It is the responsibility of CNC users to obtain up-to-date X.509 CRL objects from manufactures and make them locally available for verifiers.

6.3.3.4.3 Trust anchors

6.3.3.4.3.1 General

Trust anchors are input arguments for certification path validation according to IETF RFC 5280, 6.1.1, input argument (d). Relying parties decide about these input arguments in a discretionary fashion, i.e. these objects are not created and distributed as literal trust anchor objects but in a pre-material form of, for example, self-signed certificate objects.

NOTE The digital signature in self-signed certificates do not vouch for authenticity of this object: Actor X can issue self-signed certificates featuring the name of actor A that cannot be distinguished from self-signed certificates issued by A. The mechanisms to verify the authenticity of self-signed certificates are not addressed in this document.

The trust anchors for use cases where IA-stations act in claimant role are determined by CNC users.

6.3.3.4.3.2 Creation

The details of the issuance and update processes for trust anchors for validation of IDevID credentials are not addressed by this document.

6.3.3.4.3.3 Distribution

With respect to use cases where IA-stations act in claimant role, e.g. NETCONF/YANG security setup and device identity verification, the following model applies.

- Issuers (IA-station manufacturers) create and distribute trust anchors. Issuers also provide out-of-band means that allow relying parties to check the authenticity of these objects, and
- Relying parties (CNC users) check the authenticity of trust anchors and decide about their acceptance as trust anchors for certification path validation in a discretionary manner and configure their verifiers (CNCs) accordingly.

The details of distribution and validation of trust anchors are not addressed by this document.

6.3.3.4.3.4 Use

Trust anchors for IDevID credentials are used for certification path validation according to IETF RFC 5280, 6.1.1 d). This concerns CNCs with respect to the use cases NETCONF/YANG security setup from factory default, device identity verification.

6.3.3.4.3.5 Storage

Trust anchors for IDevID credentials shall be stored persistently upon CNCs. The details for implementing this persistent storage are not addressed in this document.

6.3.3.4.3.6 Revocation

IA-station manufacturers are not required to support an authority revocation feature for IDevID credential certification authorities.

6.3.4 Security setup based on LDevID

6.3.4.1 General

IA-stations in factory default state shall conduct a security setup sequence for the Configuration Domain. This sequence consists of the following steps, each step is described in 6.3.4.

- imprintTrustAnchor: imprint of a Configuration Domain specific trust anchor to an IA-station that allows to validate LDevID-NETCONF certificates presented by communication partners.
- imprintCredential: imprint of a Configuration Domain specific credential to an IA-station, i.e. a private key and the corresponding X.509 v3 end entity certificate according to ISO/IEC 9594-8 as profiled in IETF RFC 5280, Clause 4, (plus intermediate CA certificates, if applicable) plus self-signed root CA certificate that serves as own LDevID credential.
- imprintCertToNameMapping: imprint a Configuration Domain specific certificate-to-name mapping to an IA-station.

Credentials shall be stored persistently upon an IA-station. The details for implementing this persistent storage are IA-station manufacturer-specific and not addressed in this document.

IA-stations shall support storage of at least one LDevID-NETCONF credential.

6.3.4.2 imprintTrustAnchor

IA-stations in factory default state shall support the imprinting of a single Configuration Domain specific trust anchor according to IETF RFC 6024 via NETCONF-over-TLS according to a procedure called "provisional accept of client certificate", which uses an LDevID credential on NETCONF and TLS server side (IA-station) and a LDevID credential on NETCONF and TLS client side (for example, a CNC) and operates as follows at the NETCONF and TLS server.

- a) Challenge the client for TLS client authentication according to IETF RFC 7589 by sending a CertificateRequest message with an empty certificate_authorities entry.
- b) Perform certification path validation according to IETF RFC 5280, Clause 6, for the contents of the client's Certificate message. This certification path validation fails due to a missing trust anchor for the LDevID credential.
- c) Provisionally accept the failing certification path validation when the reason is "no matching trust anchor" (and only this reason) and proceed with the TLS exchange.
- d) Expect the client to send a trust anchor for LDevID over the provisionally accepted TLS session (no other object type).
- e) If the trust anchor in the NETCONF application payload was accepted, then redo the priorly failing certification path validation using this trust anchor, see step b).
- f) If this certification path revalidation is successful, then keep the TLS session alive and send an <rpc-reply> with success. The client then is expected to perform the NETCONF exchanges for imprintCredential (described in 6.3.4.3) and for imprintCertToNameMapping (described in 6.3.4.4) via the already established TLS session.
- g) If this certification path revalidation is not successful, then terminate the TLS session. The usual NETCONF/YANG hygiene applies. This is expected to remove the entry in the ietf-truststore that was created in step d).

NOTE This "provisional accept of client certificate" is a mirrored version of the "provisional accept of server cert" in IETF RFC 8995.

The "provisional accept of client cert" in factory default state shall skip the certificate-to-name mapping and shall use the NACM recovery session, i.e. skip permission checking. In this model all authenticated clients are accepted as authorized for doing the first imprinting of the LDevID credential and the corresponding trust anchor. Only contextual checks such as "once only when being in factory default state" are feasible. This model is also known as "trust on first use" (TOFU) and, e.g. also allows to read contents of the ietf-hardware module by the client for an extended identity check.

The imprinting NETCONF client checks the actual server identity that is stated by the IA-station on TLS level.

The NETCONF client checks the LDevID end entity certificate presented by the NETCONF server on TLS level for existence of subjectAltName extension with GeneralName entries of type uniformResourceIdentifier. If an entry contains the namespace identifier and the namespace-specific string as defined in 6.3.3.2.2, the presented server certificate is in extended form, otherwise it is in raw form.

In case that the server certificate is in raw form, the following matching can be done:

- Match a list of accepted (or blocked) manufacturers against the issuer or subject field entries of the certificate.
- Match a list of accepted (or blocked) product instances against the product serial number from the subject field per accepted manufacturer.
- Match the end entity certificate object as a whole against a list of pinned certificates.

In case that the server certificate is in extended form, the following additional matching can be done: match the q-components included in the verifiable device identity according to 6.3.3.2.2 against those that can be read out from the corresponding leaves of the YANG module ietf-hardware or against reference values obtained by a method not addressed in this document.

Details of how the matching happens depend on the implementation of the client that performs this imprinting.

The LDevID-NETCONF trust anchor certificate shall be imprinted using the truststore container of the ietf-truststore module with:

- /ts:truststore/ts:certificate-bags/ts:certificate-bag/ts:name = IEC60802,
- /ts:truststore/ts:certificate-bags/ts:certificate-bag/[ts:name=IEC60802]/,
 - ts:certificate/ts:name = IEC60802-LDevID,
 - ts:certificate/ts:cert-data containing the IEC60802-LDevID trust anchor certificate data object of type trust-anchor-cert-cms according to IETF RFC 9640, i.e., enveloped in Base64-encoded CMS SignedData in degenerated form "certs-only" (no signature value), and
 - The imprintTrustAnchor step shall use the NETCONF operation <edit-config> according to IETF RFC 6241 for the truststore container. The NETCONF operation <commit> is not yet applied, but rather after successful completion of all security setup sequence steps.

6.3.4.3 imprintCredential

6.3.4.3.1 General

The LDevID-NETCONF end entity certificate shall be provided as X.509 v3 public key certificate according to ISO/IEC 9594-8 as profiled in IETF RFC 5280, Clause 4, with the following criteria:

- contains the FQDN of the NETCONF server in its subjectAltName extension according to IETF RFC 7589, Clause 6, and IETF RFC 6125, 2.2 and B.7;
- contains a public key and is signed by a signature suite according to 5.5.4.2 or 5.6.3;
- contains a digitalSignature in its keyUsage extension;
- has a finite validity period.

NOTE The actual length of the validity period is at the discretion of the user of the Configuration Domain.

Depending on the key generation capabilities, different steps are applied to this keystore container.

6.3.4.3.2 Internal key generation

For IA-station with internal key generation capabilities, two NETCONF exchanges are performed. Processing steps for the first NETCONF exchange shall be applied as follows at the NETCONF server.

- a) Receive and process the NETCONF request message with action <generate-csr> and input values as follows:
 - 1) /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/[ks:name=LDevID_NETCONF]/ks:generate-csr/ks:input/ks:csr-format containing derived identity p10-csr-format according to IETF RFC 9640, and
 - 2) /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/[ks:name=LDevID_NETCONF]/ks:generate-csr/ks:input/ks:csr-info containing a Base64-encoded PKCS#10 CertificationRequestInfo according to IETF RFC 2986, Clause 4.
- b) Base64-decode the <csr-info> value and parse it as a PKCS#10 CertificationRequestInfo object.
- c) Extract the algorithm information from the child element SubjectPublicKeyInfo of CertificationRequestInfo and randomly generate a key pair for the specified algorithm.
- d) Internally store the private key together with its metadata for example, algorithm information, <name> value in a secure manner.
- e) Put the public key into the (parsed) PKCS#10 CertificationRequestInfo.
- f) Serialize the PKCS#10 CertificationRequestInfo (including the public key).
- g) Use the private key to create signature value for the (serialized) PKCS#10 CertificationRequestInfo (including the public key).
- h) Create a NETCONF reply message with /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/[ks:name=LDevID-NETCONF]/ks:generate-csr/ks:output/ks:p10-csr containing the data object of the previous step.

In the second NETCONF exchange, the LDevID-NETCONF end entity certificate (plus intermediate CA certificates) shall be imprinted using the keystore container of the ietf-keystore module with:

- /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/ks:name = LDevID-NETCONF,
- /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/[ks:name=LDevID-NETCONF]/,
 - ks:certificates/ks:certificate/ks:name = LDevID-NETCONF, and
 - ks:certificates/ks:certificate/ks:cert-data containing the certificate chain LDevID-NETCONF end entity certificate (plus intermediate CA certificates, if applicable) plus self-signed root CA certificate as data object of type end-entity-cert-cms according to IETF RFC 9640

The imprintCredential step shall use the NETCONF operation <edit-config> according to IETF RFC 6241 for the keystore container. The NETCONF operation <commit> is not yet applied, but rather after successful completion of all security setup sequence steps.

6.3.4.3.3 External key generation

External key generation can be used for IA-stations without internal key generation capability. For external key generation, one NETCONF exchange is performed.

The LDevID-NETCONF private key and end entity certificate (plus intermediate CA certificates) shall be imprinted using the keystore container of the ietf-keystore module with:

- /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/ks:name = LDevID-NETCONF,
- /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/[ks:name=LDevID-NETCONF]/,
 - ks:certificates/ks:certificate/ks:name = LDevID-NETCONF,
 - ks:certificates/ks:certificate/ks:public-key-format describing the encoding of the public key of the selected cryptographic algorithm according to IETF RFC 9640,
 - ks:certificates/ks:certificate/ks:public-key containing the public key value in the selected public-key-format,
 - ks:certificates/ks:certificate/ks:private-key-format describing the encoding of the private key of the selected cryptographic algorithm according to IETF RFC 9640,
 - ks:certificates/ks:certificate/ks:cleartext-private-key containing the private key value in the selected private-key-format,

NOTE The private key is confidentiality protected by NETCONF-over-TLS even if the option <cleartext-private-key> is used.

- ks:certificates/ks:certificate/ks:name = LDevID-NETCONF, and
- ks:certificates/ks:certificate/ks:cert-data containing the certificate chain LDevID-NETCONF end entity certificate (plus intermediate CA certificates, if applicable) plus self-signed root CA certificate as data object of type end-entity-cert-cms according to IETF RFC 9640.

The imprintCredential step shall use the NETCONF operation <edit-config> according to IETF RFC 6241 for the keystore container. The NETCONF operation <commit> is not yet applied, but rather after successful completion of all security setup sequence steps.

External key generation can introduce security vulnerabilities during the generation and loading process. Ensuring those processes are secure is the responsibility of the user and not addressed in this document.

6.3.4.4 imprintCertToNameMapping

The Configuration Domain specific certificate-to-name mapping is imprinted in the ietf-netconf-server YANG module under the following node.

- /ncs:netconf-server/ncs:listen/ncs:endpoint/ncs:tls/ncs:netconf-server-parameters/ncs:client-identity-mappings/ncs:cert-to-name, with the following leaves:
 - id = 1,
 - fingerprint = Configuration Domain specific fingerprint of the LDevID-NETCONF trust anchor using the hash algorithm sha256 according to IETF RFC 7589, Clause 7, and
 - map-type = ext-60802-roles.

The application of this map-type is described in 6.3.5, steps e) and f).

The imprintCertToNameMapping step uses the NETCONF operation <edit-config> according to IETF RFC 6241 for the certificate-to-name mapping. Afterwards the NETCONF operation <commit> is applied to finalize the security setup sequence steps and to leave the factory default state.

6.3.5 Secure configuration based on LDevID-NETCONF

Configuration by NETCONF/YANG is protected by NETCONF-over-TLS as described in 6.3.2.1 and NACM as described in 6.3.2.2. The NETCONF/YANG servers and clients shall use LDevID credentials for authentication.

The procedure called "provisional accept of client certificate" as described in 6.3.4.2 is not applied anymore if the IA-station has left the factory default state. Instead, after successful establishment of a TLS session according to IETF RFC 7589 and IETF draft-ietf-netconf-over-tls13, the NETCONF server shall perform a certificate-to-name mapping and authorization check as follows.

- a) Compare the fingerprint of the trust anchor of the NETCONF client's certification path with the fingerprint contained in cert-to-name list entries of the x509c2n container for equal values.
- b) If no cert-name list entry match is found, then terminate the TLS session.
- c) If a cert-to-name list entry match is found, then verify if the map-type is equal to ext-60802-roles.
- d) If the map-type does not match, then terminate the TLS session.
- e) If the map-type value matches, then extract the role values from the id-60802-pe-roles certificate extension of the NETCONF client's TLS-authenticated end entity certificate. The output is a list of string values from the enumeration of specified role names according to this document.
- f) The list of role name string values is provided as input to NACM for permission checking. The access to the requested resource is checked according to the rules configured in the nacm container of the ietf-netconf-acm YANG module.

The NETCONF client checks if the expected identity to address the NETCONF server (IP address or DNS name) matches to the actual server identity that is stated by the IA-station on TLS level. This shall be done by comparing the expected identity with the subjectAltName extension of the TLS authenticated LDevID-NETCONF end entity certificate of the NETCONF server.

6.4 Management

6.4.1 General

Subclause 6.4 describes a model for configuration, deployment, and management of an industrial automation network.

6.4.2 IA-station management model

6.4.2.1 General

The management model of IA-stations covers simple end station IA-stations as well as combined IA-stations as described in 4.3. The IA-station management model is applied for topology discovery, network provisioning and stream establishment.

6.4.2.2 IEEE 802.1Q management model

In industrial automation both Bridge and end station components make use of IEEE 802.1Q specified functionality (for example, traffic classes, gate control). Thus, the IEEE 802.1Q management model is the basic management model to be applied to all IA-stations. Figure 15 shows the implementation of the IEEE Std 802.1Q Bridge model in YANG as specified in IEEE Std 802.1Q-2022, Clause 48. The IETF Interface Management YANG data model is specified in IETF RFC 8343.

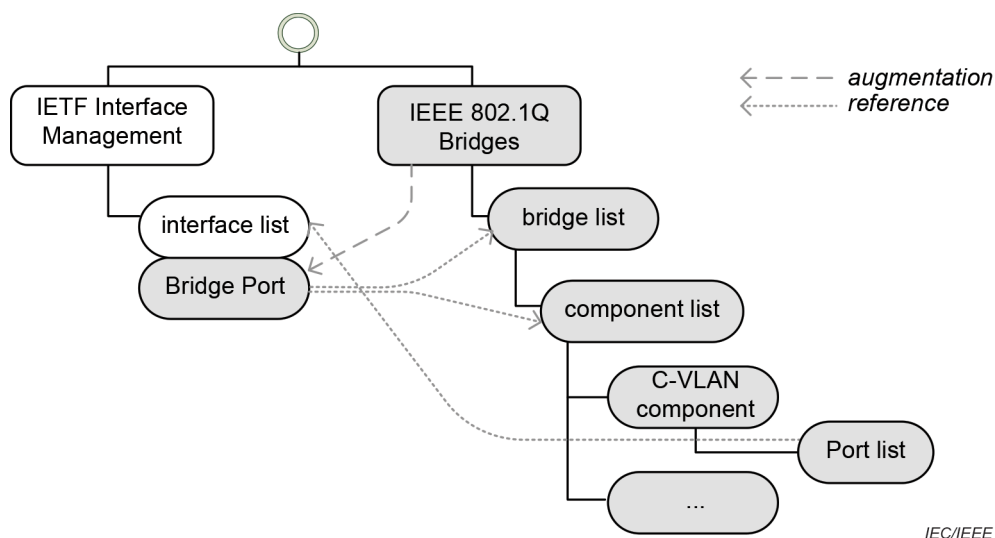


Figure 15 – Generic IEEE 802.1Q YANG Bridge management model

The IEEE 802.1Q Bridge model is organized as a bridge list where each bridge includes an underlying component list (for example, C-VLAN components). Each component has a Port list attached with references to the representation of the ports in the IETF interface list. The managed data of the ports is defined as Bridge Port augmentation to the IETF interface model. Each Bridge Port includes a reference to its bridge and component instances in the IEEE 802.1Q Bridge model.

The YANG data model for an IEEE 802.1Q Bridge is applied to IA-stations as follows.

- Each functional unit of an IA-station is modeled as bridge entry in the bridge list.
- Each Bridge and end station component of an IA-station is modeled as C-VLAN component.
- IA-station components belonging to the same functional unit are added to the component list of this functional unit's bridge entry.
- Each IA-station external or internal port is modeled as Bridge Port.

IA-station ports belonging to the same component are added to the Port list of the related component list entry.

Further YANG data models which are relevant for IA-stations are described in 6.4.9.

6.4.2.3 Internal LAN connection model

The modeling of internal connections between C-VLAN components within an IA-station is aligned to IEEE Std 802.1Q-2022, 17.3.2.2, which includes an I-LAN interface. As shown in Figure 16, the I-LAN interface is modeled as an ilan IETF interface object (see IETF RFC 7224) together with appropriate `higher-layer-if` and `lower-layer-if` reference objects to describe the internal connection.

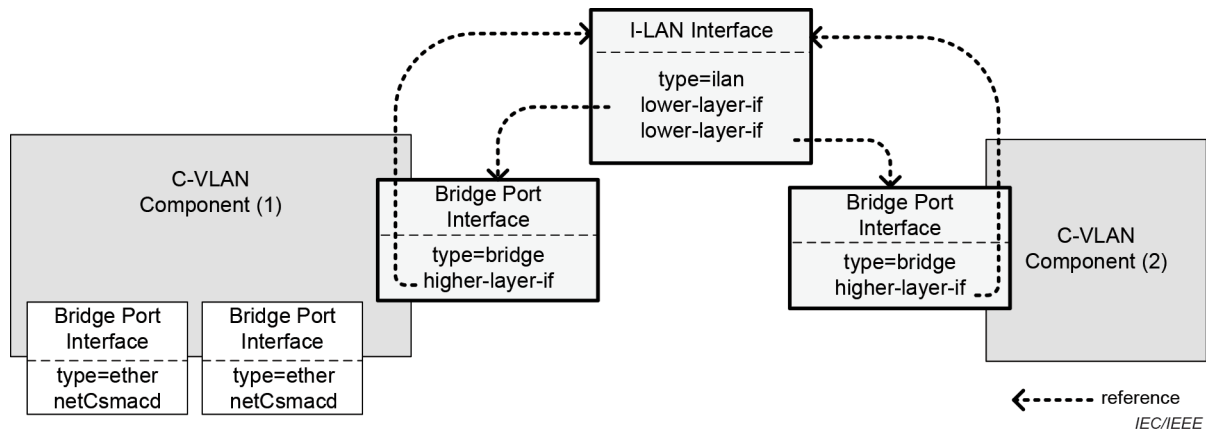


Figure 16 – Internal LAN connection management model

This internal LAN connection model comprises three configuration steps.

- The internal Ports of the C-VLAN components are modelled as IETF interfaces of type bridge with Bridge Port augmentation.
- An additional I-LAN interface of type ilan as described in IETF RFC 7224 is created.
- The I-LAN interface references the internal Bridge Port interfaces of the connected C-VLAN components as lower-layer-if, and the internal Bridge Port interfaces of the connected C-VLAN components reference the I-LAN interface as higher-layer-if.

Figure 17 shows the application of this model to the example IA-station of Figure 16.

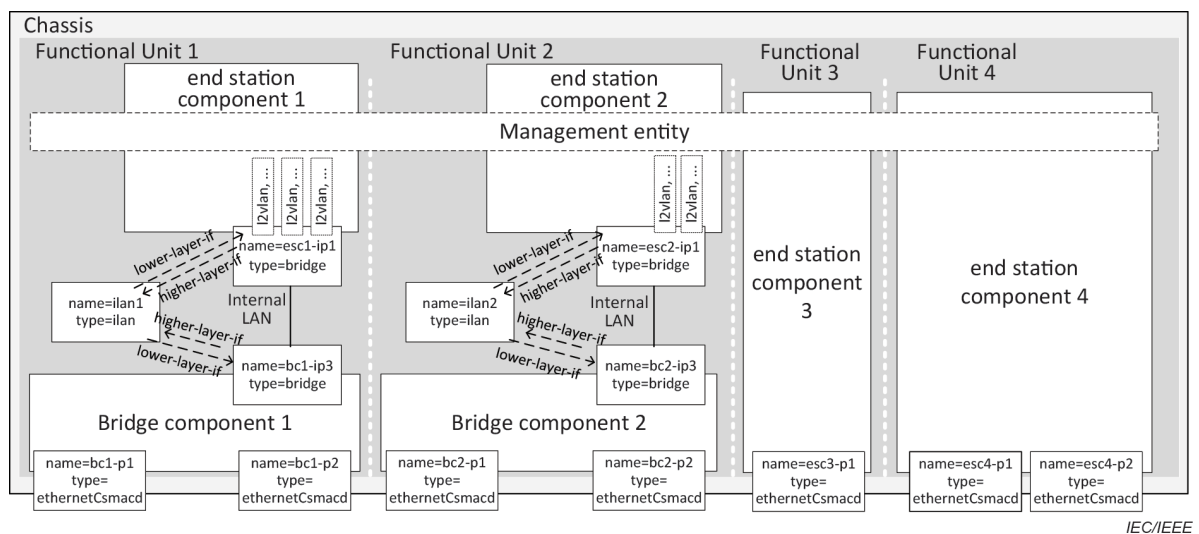


Figure 17 – IA-station example with IETF interfaces

NOTE Figure 17 represents an abstract model and is not intended to imply a particular implementation or partitioning.

Late binding of IA-station applications to the configured VLANs and priorities is performed using the `iecieeee60802-tt-mapping` YANG module in 6.4.10.8.7.

6.4.2.4 Spanning Tree, VLAN and TE-MSTID configuration

C-VLAN Bridge components of IA-stations shall support:

- the Common and Internal Spanning Tree (CIST) calculated by the Multiple Spanning Tree Algorithm and Protocol (MSTP), and
- the Traffic Engineering Multiple Spanning Tree Instance Identifier (TE-MSTID) as specified in IEEE Std 802.1Q-2022, 5.5.2.

The MSTP configuration is either default or accomplished by IA-station specific means.

CNCs configure VLANs in the vlan list in the bridge-vlan container of the ieee802-dot1q-bridge YANG module. Ports are assigned to a vlan as static-filtering-entries in a filtering-database.

NOTE vlan, in lowercase, refers to a YANG element.

VLANs are assigned to filtering databases in the vid-to-fid list of the bridge-vlan container. The filtering databases, and in consequence the VLANs, are by default assigned to the MSTP calculated Internal Spanning Tree and can be assigned to the TE-MSTID by management. IA-time-aware streams and IA-streams are assigned to the TE-MSTID.

TE-MSTID assignment is accomplished via the bridge-mst container of the ieee802-dot1q-bridge YANG module.

The following example shows the VID/FID/MSTID configuration of an IA-station's C-VLAN Bridge component, which supports three VLANs in three Forwarding Databases (VID 100 in FID 1, VID 101 in FID 2 and VID 102 in FID 3). FID 2 and FID 3 – and in consequence VID 101 and VID 102 – are assigned to the TE-MSTID. FID 1 – and in consequence VID 100 – is not assigned to a MSTID and thus, is implicitly assigned to the Internal Spanning Tree (IST).

Figure 18 shows the representation of this example configuration in the MST configuration table.

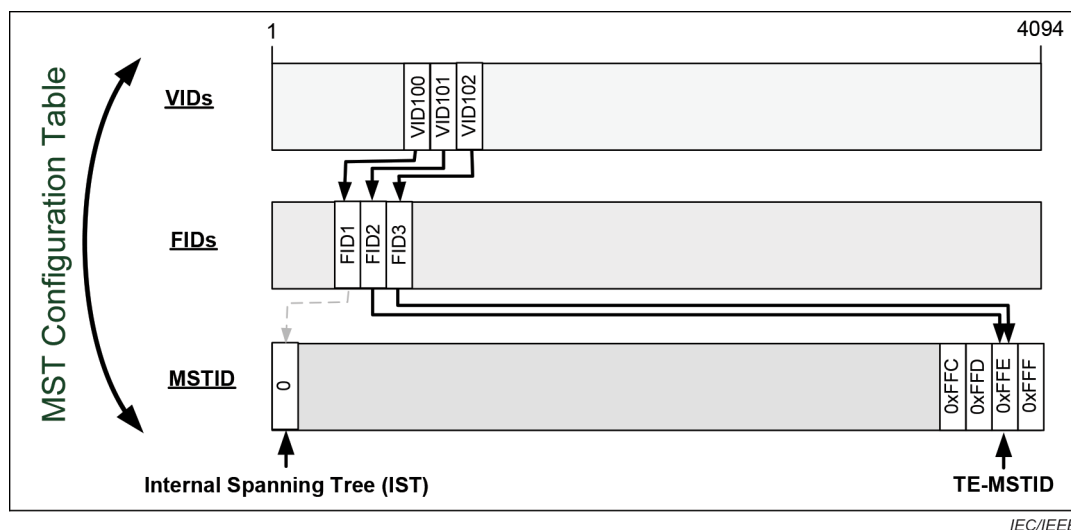


Figure 18 – VID/FID/MSTID example

The YANG-based configuration of this example is shown as YANG instance data snippet of the ieee802-dot1q-bridge YANG module. Herein the MST configuration table is included in component "bridge-component-x", which is part of bridge "functional-unit-x". It is the responsibility of the user to ensure that traffic type names are consistent with those provided in the YANG module iecIEEE60802-tt-mapping (see 6.4.10.8.7).

```

<ieee802-dot1q-bridge xmlns:dot1q="urn:ieee:std:802.1Q:yang:ieee802-dot1q-bridge">
  <bridges>
    <bridge> <!-- list -->
      <name>functional-unit-x</name>
      ...
      <component> <!-- list -->
        <name>bridge-component-x</name>
        ...
        <bridge-vlan>
          <version>2</version> <!-- MST supported -->
          ...
          <vlan>
            <vid>100</vid>
            <name>60802-best-effort-hl-100</name> <!-- best effort high and
low -->
          </vlan>
          <vlan>
            <vid>101</vid>
            <name>60802-isochronous-101</name> <!-- isochronous -->
          </vlan>
          <vlan>
            <vid>102</vid>
            <name>60802-isochronous-102R</name> <!-- isochronous -->
          </vlan>
          ...
          <vid-to-fid>
            <vid>100</vid>
            <fid>1</fid>
          </vid-to-fid>
          <vid-to-fid>
            <vid>101</vid>
            <fid>2</fid>
          </vid-to-fid>
          <vid-to-fid>
            <vid>102</vid>
            <fid>3</fid>
          </vid-to-fid>
        </bridge-vlan>
        ...
        <bridge-mst>
          ...
          <fid-to-mstid> <!-- list -->
            <!-- fid 1 is implicitly assigned to mstid 0 -->
            <fid>2</fid>
            <mstid>4094</mstid> <!-- TE-MSTID -->
          </fid-to-mstid>
          <fid-to-mstid> <!-- list -->
            <fid>3</fid>
            <mstid>4094</mstid> <!-- TE-MSTID -->
          </fid-to-mstid>
        </bridge-mst>
        ...
      </component>
    </bridge>
  </bridges>
</ieee802-dot1q-bridge>

```

6.4.3 Discovery of IA-station internal structure

LLDP provides information about the external connectivity of IA-stations. To identify the internal structure of complex IA-stations (see 4.3) the IEEE 802.1Q management model (see 6.4.2.2) and the IETF Interface management model are applied.

- The functional units of an IA-station are represented as bridge entries in the bridge-list.
- The components of a functional unit are represented as component entries in the associated bridge entry's component-list.

- Internal LAN connections between components of a functional unit are identified by I-LAN entries in the IETF interface list (6.4.2.3).

6.4.4 Network engineering model

To understand the requirements for network configuration, deployment and management, an engineering model covering industrial use cases is required. The "fully centralized model" described in IEEE Std 802.1Q-2022, 46.1.3.3 includes two functional entities: the CUC and the CNC. The relationship between user and network configuration is described in IEEE Std 802.1Q-2022, Clause 46. This document further elaborates this relationship to address use cases for industrial automation. A conceptual block diagram of a CNC is shown in Figure 19, which adds further details to the CNC specified in IEEE Std 802.1Q-2022 to serve the industrial automation use case. The following functional entities are introduced.

a) The Topology Discovery Entity (TDE)

The topology discovery entity is responsible for the topology discovery (i.e. Bridge component and end station component discovery). The TDE also performs a topology verification in cases where an expected topology is provided by the engineering tool. The resulting topology information is used by the CNC. The TDE detects added or removed IA-stations, including internal structure and connectivity. Thus, the CNC becomes aware of them. Overall, the TDE discovers and maintains an inventory of the devices, including their capabilities and the topology they form.

b) The Path Entity (PE)

The PE computes, establishes and maintains the forwarding paths for the IA time-aware stream and IA stream traffic type categories according to 4.7.3.

c) The Sync Tree Entity (STE)

The STE computes, establishes and maintains the sync trees. For example, for Working Clock and Global Time.

d) The Resource Allocation Entity (RAE)

The RAE is responsible for the allocation of the resources that are necessary for all traffic type categories, according to 4.7.3, to meet their requirements via their forwarding paths. For example, frame buffers at egress ports and FDB entries.

e) The Network Provisioning Entity (NPE)

The NPE applies a network policy provided by the Engineering Tool to the IA-stations within the Configuration Domain. It uses the information discovered by the TDE to create a network configuration based upon this policy which is then applied to all IA-stations. The CNC uses the chosen network configuration together with the discovered IA-stations and their capabilities as input for its stream calculation and deployment.

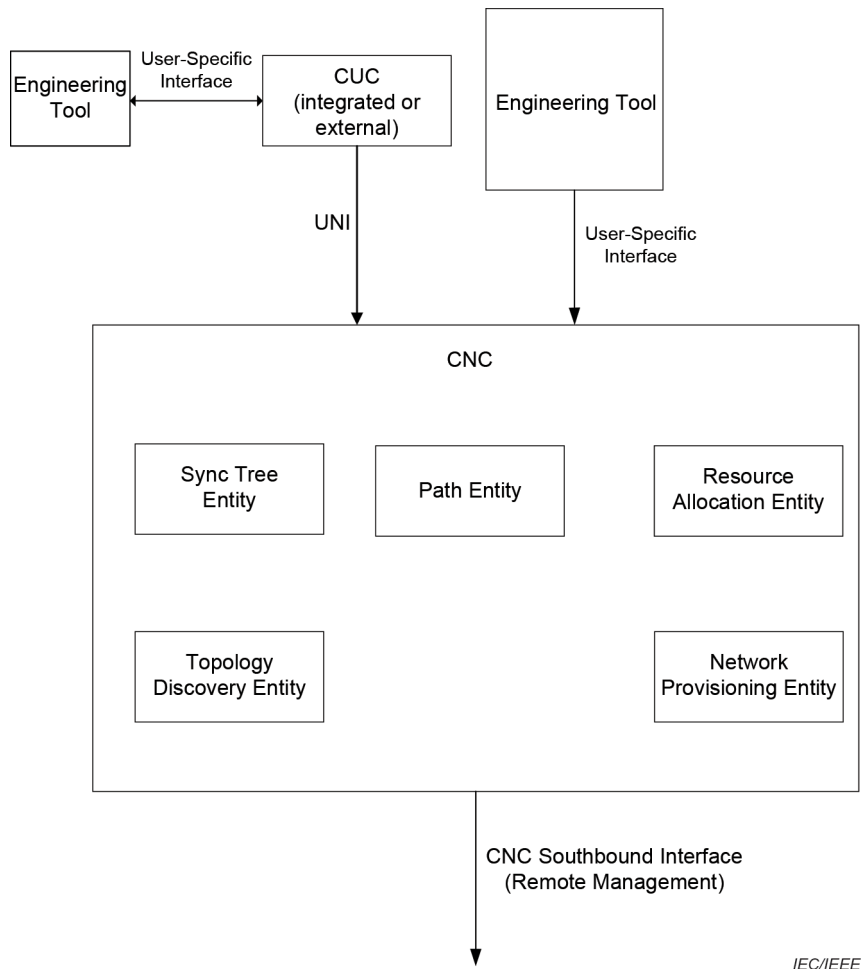
A CNC includes these functional entities. The implementation of these functional entities and the CNC can vary. The means of communication among these functional entities is implementation dependent.

If there are multiple CNCs in one Configuration Domain, then, by some means not addressed by this document, only a single CNC is in charge at any time in the given Configuration Domain.

The CNC can be in a dedicated station or integrated into any IA-controller or IA-device. Generally, its engineering tool interface is user-specific and can only work with the compatible engineering tools. The definition of this interface is not addressed in this document.

The CUC can be in a dedicated station or integrated into any IA-controller or IA-device. Generally, the CUC is user-specific. In industrial automation use cases, an IA-controller integrated CUC is very likely.

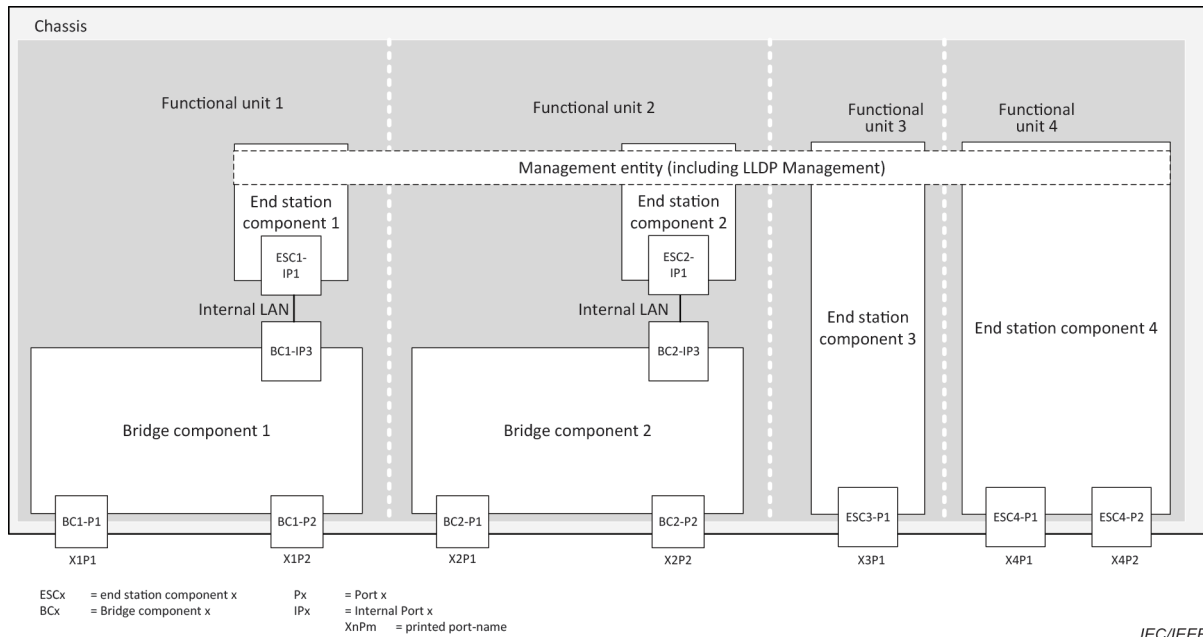
For stream establishment, the UNI of the CNC component is exposed.



IEC/IEEE

Figure 19 – Structure and interfaces of a CNC

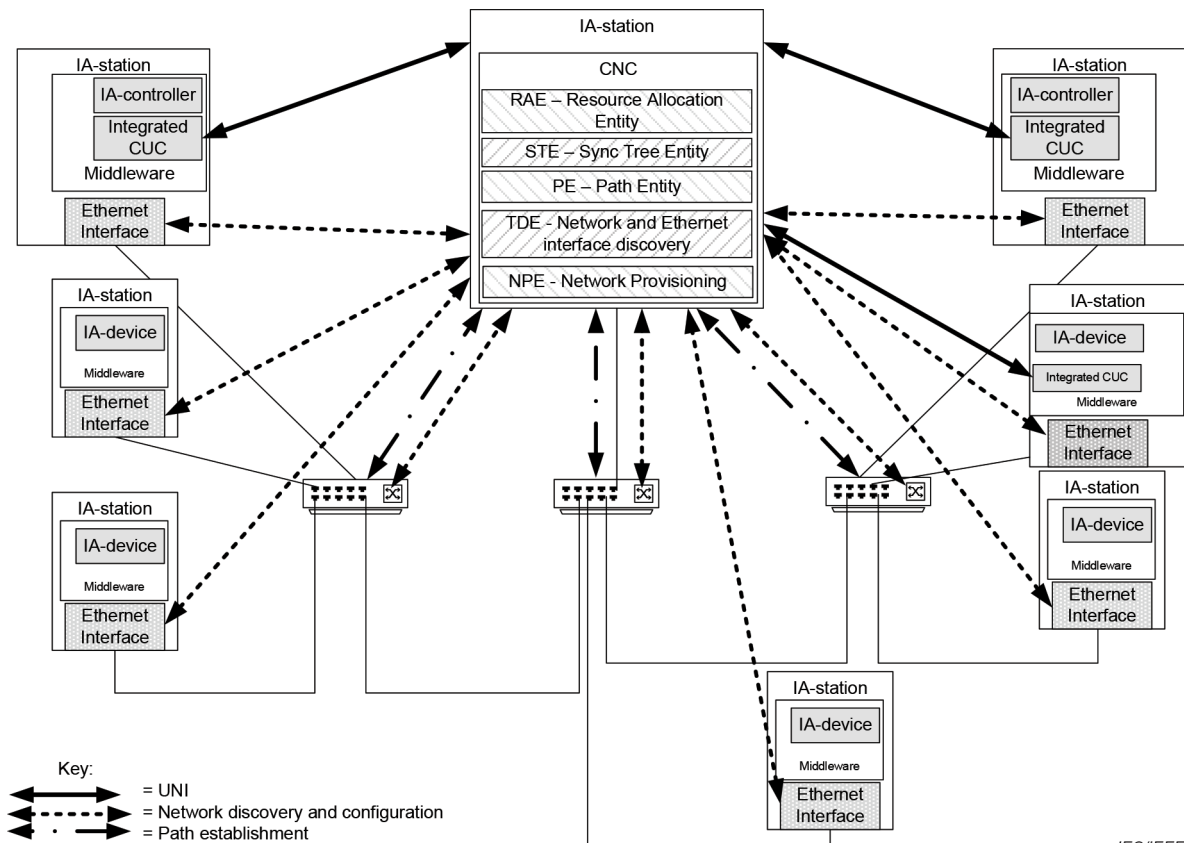
Figure 20 shows an example of the structure of an IA-station which the CNC can discover and manage.



IEC/IEEE

Figure 20 – IA-station structure example

Figure 21 shows the interaction of IA-stations with the CNC.



IEC/IEEE

Figure 21 – CNC interaction

6.4.5 Operation

6.4.5.1 General

A representative model for network configuration is shown in Figure 22. This diagram maintains the traditional role of the IA-controller and the IA-device in an industrial automation network. IA-devices and IA-controllers require configuration from engineering tools (refer to engineering tools A, B, D, and E). These tools and associated interfaces are not addressed by this document. In this example, engineering tool C communicates directly with the CNC to provide traffic requirements for the network. The protocols that the engineering tool uses for communication with end stations are specific to the user application.

The UNI is the interface to the CNC which is serviced by NETCONF over TLS. The UNI service recognizes that industrial automation communications are typically connection oriented. There is a communication initiator, typically in an IA-controller, which is responsible for establishing those connections, determining what data is of interest and providing the required update rate. So, while an application/middleware of an IA-station (for example a Drive) understands what information it can produce and the maximum rate at which that information can be provided, until an IA-controller establishes a connection with that device, it does not know where that information goes and what update rate is required to close the control loop. The IA-controller gets this information from its engineering tool. There can be multiple IA-controllers in each Configuration Domain. The CNC uses the topology, the device capabilities, the device configuration, and the traffic specifications from the user to calculate a path for each Talker-Listener pair. The UNI then provides stream identification (VLAN, DMAC, etc.) to the middleware.

The operational management model, see Figure 22, reflects the model used in industrial automation. Figure 22 shows an active CNC managing multiple IA-stations. Each station can wholly incorporate a CUC and interact with the CNC directly.

Security requirements (see 6.3) are an important consideration for these networks and are integrated into the design, configuration, and deployment of any management model.

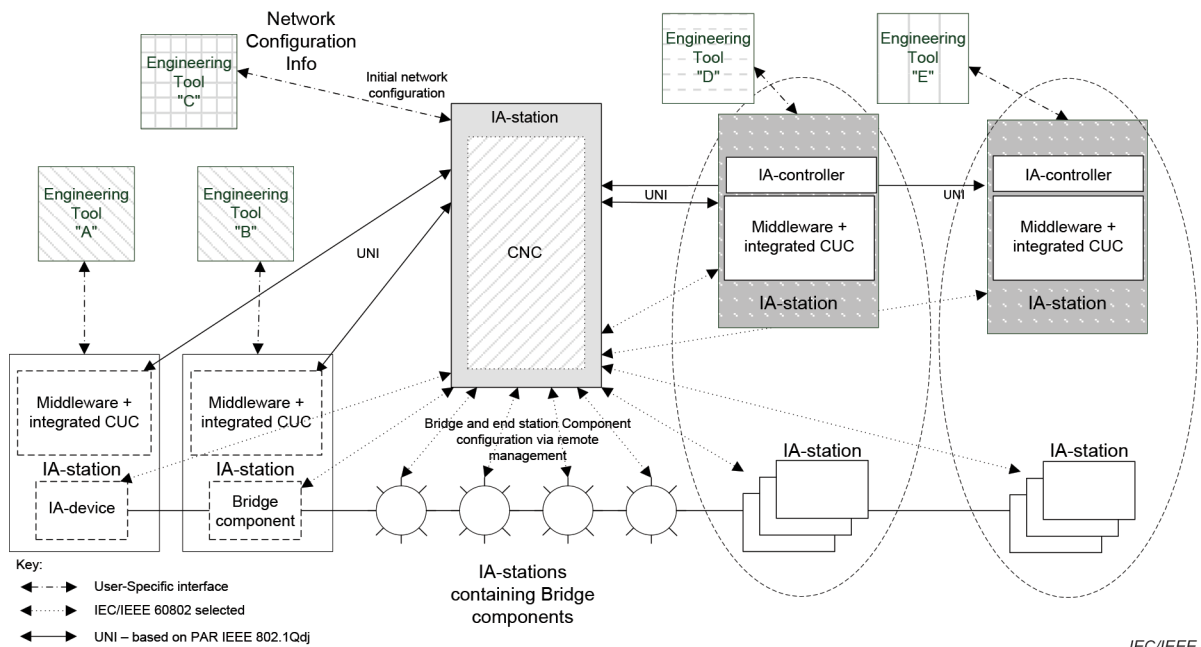


Figure 22 – Operational management model

Figure 23 shows the steps that are typically performed in the scope of the CUC-CNC interaction.

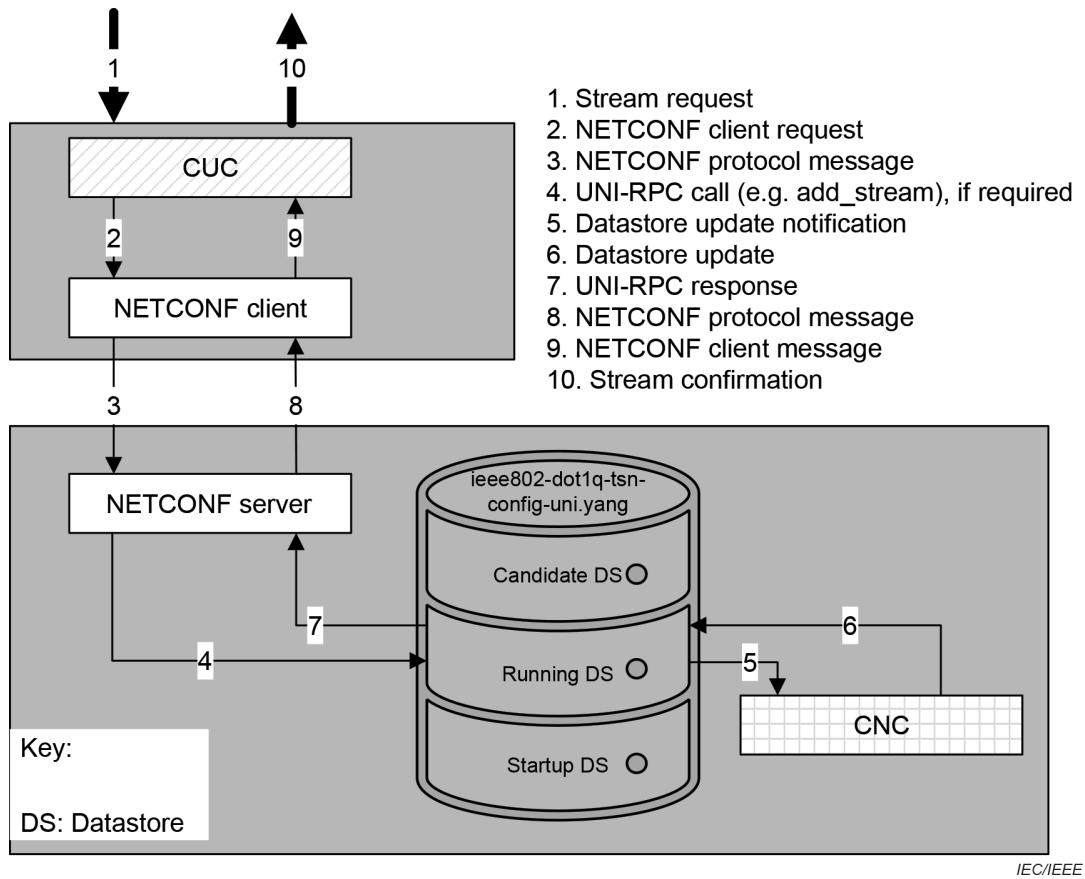


Figure 23 – UNI service model

After the computation of the paths and the scheduling or shaping configuration have been done, the CNC configures the IA-stations via NETCONF client. The typical steps that are performed in this process are shown in Figure 24 below.

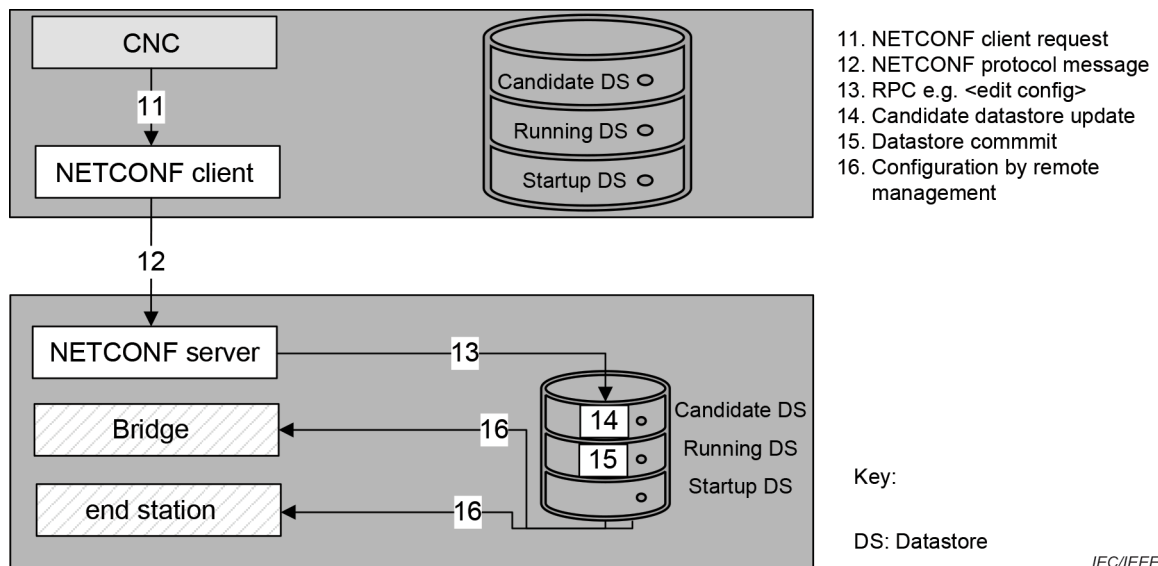
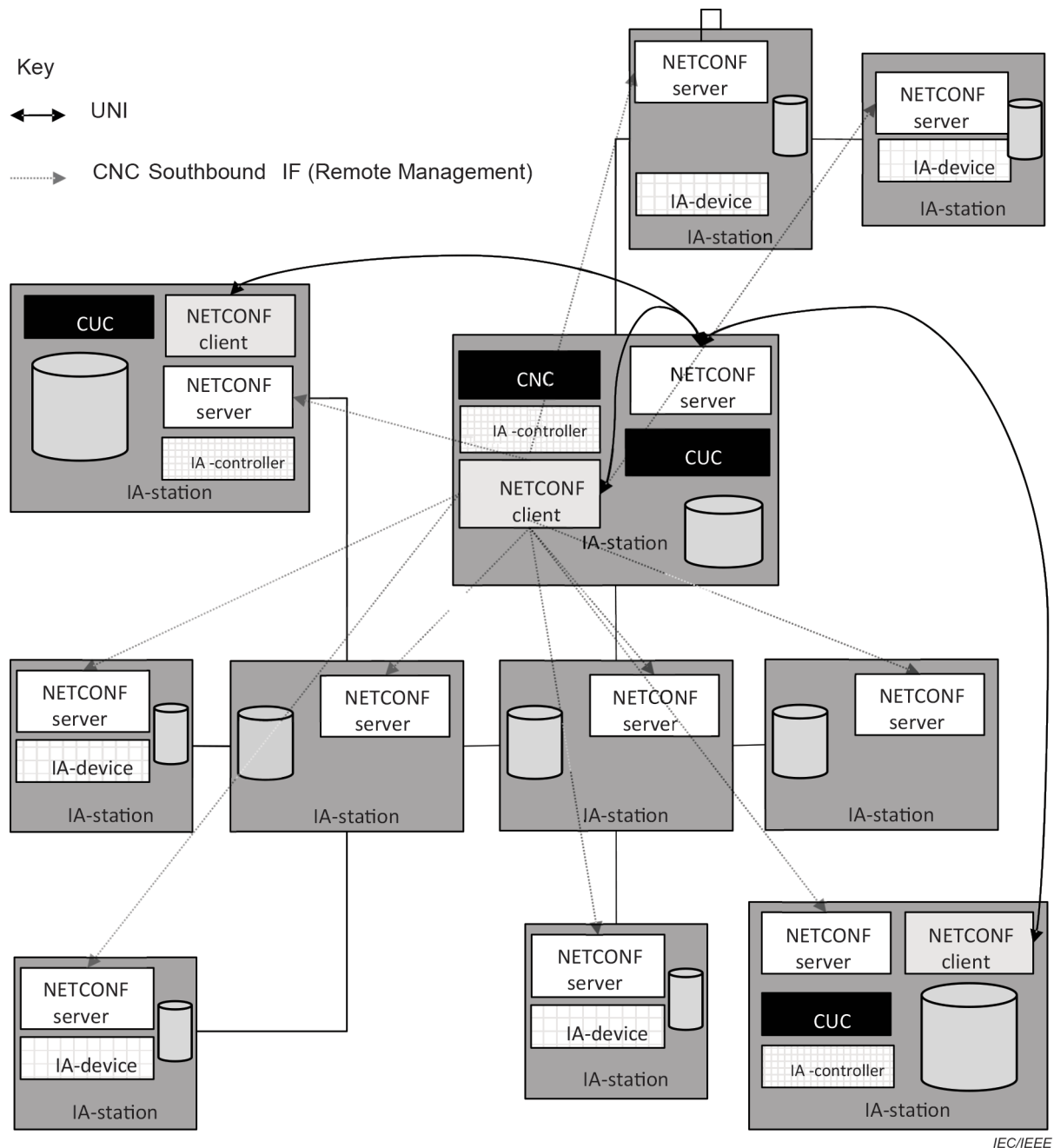


Figure 24 – CNC southbound

Instances of NETCONF servers and clients within a Configuration Domain are shown in Figure 25. IA-stations that contain a CNC or CUC entity contain both a NETCONF server and a NETCONF client. The CUC utilizes a NETCONF client for the UNI. The CNC utilizes a NETCONF server to accommodate the UNI as well as remote network management of the end stations and bridges that are contained in the same chassis as the CNC entity. The CNC utilizes a NETCONF client for the southbound interface of the CNC, i.e. for the remote management of the bridges and end stations in the scope of stream configuration. All IA-stations have a NETCONF server to make remote management possible. The NETCONF server used by the CNC serves multiple NETCONF Clients (CUCs) within a single Configuration Domain whose requests clients can occur simultaneously.



IEC/IEEE

Figure 25 – NETCONF usage in a Configuration Domain

6.4.5.2 Domain port states

A CNC manages available network resources and assigns them to the IA-stations. Management of the network resources is only possible if the CNC owns these resources. Thus, no connected station is allowed to make use of network resources that are not granted by the CNC. The security configuration of a connected station allows remote access for the CNC.

Protection of the network resources is done by managing the ports (see Figure 26) at the boundary of the Configuration Domain. The state of any newly connected station is unknown. The CNC is responsible for determining if the newly connected station is added to the Configuration Domain and configuring the IA-station appropriately.

This port state model avoids any assumptions about configuration of added stations or network portions.

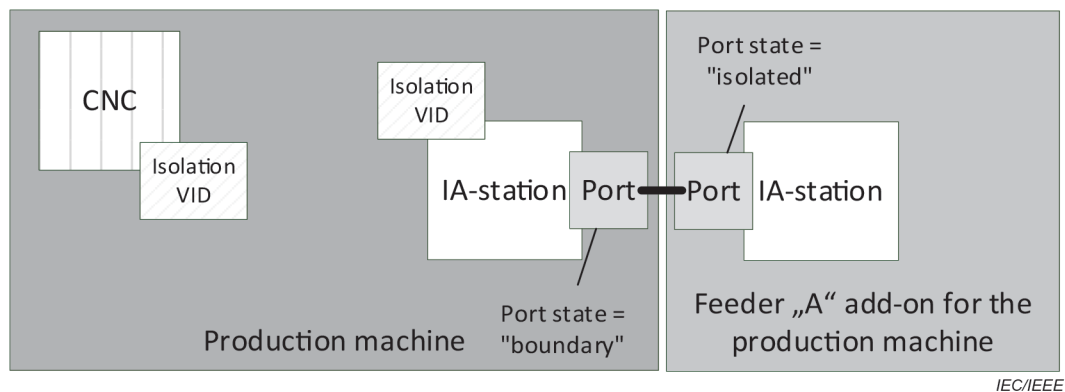


Figure 26 – Boundary port model

Ports of an IA-station that is a member of a Configuration Domain have different states:

- Isolated – a station connected via this port can only exchange information with a CNC. The CNC is responsible for establishing an isolation VID and for onboarding the station. In the isolated state:
 - the port gets to or remains in isolated state in case of a link down event, e.g. when nothing is connected, or no link is established;
 - the port gets to or remains in isolated state in case of a link up event;
 - the port stays in isolated state as long as the neighbor is unknown, not able to enter Boundary state.
- Boundary – a station connected via this port is not part of the Configuration Domain, but is allowed to access devices inside the Configuration Domain and to pass traffic through the Configuration Domain
- Inside – a station connected via this port is part of the Configuration Domain

The determination of whether a given port of an IA-station remains in the Isolated state or transitions to the Boundary or Inside state is performed by the CNC using remote management. A port acts as a domain boundary if it is in the Isolated or Boundary state.

For example, a port could be configured as follows:

- Isolated state:
 - port is IST boundary,
 - port is not part of a sync tree,
 - port uses VLAN stripping for egress,
 - port uses VLAN assignment and priority regeneration to assign all traffic to an isolated VLAN, and
 - port uses an ingress rate limiter to control the amount of traffic for the Configuration Domain.
- Boundary state:
 - port is part of IST,
 - port is part of a sync tree,
 - port uses VLAN stripping for egress,
 - port uses VLAN assignment and priority regeneration to assign all traffic to a default VLAN, and
 - port uses an ingress rate limiter to control the amount of traffic for the Configuration Domain.
- Inside state:
 - port is part of IST,
 - port is part of a sync tree, and
 - port is part of the active topology for stream and non-stream traffic.

An example workflow includes the following steps executed by the CNC:

- a) Topology discovery:
 - 1) Case A: Link down / Port not connected:
 - i) set port to isolated state;
 - ii) configure a NETCONF subscription "on data change" to the port state leaf
 - 2) Case B: Neighbor is not a Configuration Domain member:
 - i) set port to boundary state;
 - ii) configure a NETCONF subscription "on data change" to the port state leaf.
 - 3) Case C: Neighbor is not a Configuration Domain member – but part of expected topology:
 - i) set port to boundary state;
 - ii) configure the neighbor station as Configuration Domain member;
 - iii) set port to inside state.
- b) NETCONF subscription trigger:

Issued to the CNC upon change of subscribed YANG data.

6.4.5.3 Engineered network

For an offline engineered (based on the available digital data sheets of the used IA-stations) centralized approach with fixed topology, fixed stations and fixed paths, the user provides traffic requirements, path information, topology information and expected network configuration to the CNC. The CNC then uses the TDE, RAE and the NPE to perform the calculation of paths, resources, and stream schedules necessary to meet the specified traffic requirements and deploys the result of these calculations via remote management. The CNC also provides the relevant results to the CUC via the UNI. The CUC then configures the end stations using the user-to-user interface (see Figure 3).

The workflow for this example consists of the following steps:

- a) The user determines:
 - 1) the expected network topology,
 - 2) the expected stations and its capabilities, value ranges and quantities,
 - 3) the expected paths and resources,
 - 4) the required streams, and
 - 5) the requirements for IA non-stream traffic.

This step focuses on network capabilities including the Ethernet interface of the end stations. For example, if the end station is a sensor, the user considers the Ethernet interface capabilities of the sensor as they apply to the physical world.

- b) Engineering Tool provides this information to the CNC via a user-specific interface.

Although the communication between the CNC and any Engineering Tool is user-specific, the CNC obtains all necessary information via the integrated TDE and NPE.

- c) The CNC uses the TDE to discover the topology and checks it against the expected topology. The NPE is used to configure the IA-stations of the Configuration Domain.
- d) The CNC uses STE and NPE to setup, validate, and monitor synchronization configuration in the Configuration Domain.
- e) The CNC uses the information from engineering item a), steps 1 to 5, above to respond to requests from middleware (with integrated CUC) using UNI. These requests are handled using the already established communication paths received from the user.

If the CNC is not required after commissioning, then the CNC can be removed after setting up the IA-stations. That requires that all IA-stations have a persistent storage for the data provided by the CNC.

6.4.5.4 Dynamic topology

6.4.5.4.1 General

For a centralized approach with a dynamic topology and dynamic paths, the user provides the network policy to the CNC. The TDE performs topology discovery including IA-station capabilities (YANG representation of the digital data sheet) and the NPE performs network configuration for the CNC. IA-stations then provide traffic requirements via the middleware to the CNC via the UNI. The CNC then uses the TDE, RAE, and NPE to perform the calculation of paths, resources, and stream schedules necessary to meet the specified traffic requirements and deploys the result of these calculations via remote management. The CNC also provides the relevant results to the CUC via the UNI. The CUC then configures the end stations using the user-to-user interface (see Figure 3).

The workflow for this example consists of the following steps.

- a) The user determines the network policy and provides it to the CNC.
- b) The TDE continuously discovers the physical network topology and station capabilities of each station using remote management.
- c) The NPE uses the information gathered in steps a) to b) to configure the IA-stations in the Configuration Domain.
- d) The CNC uses STE and NPE to setup, validate and monitor time synchronization configuration in the Configuration Domain.

The CNC uses the information from steps a) to d) to respond to requests from middleware using UNI. The CNC establishes streams in the bridges via a remote management protocol.

6.4.5.4.2 Adding an IA-station

Each IA-station added to the Configuration Domain is discovered by the TDE and receives the initial network configuration from the NPE. After this, the middleware can request stream establishment.

When an IA-station is added to the network, it is isolated until the CNC determines that its traffic requirements can be accommodated without disrupting other traffic (see 6.4.5.2).

6.4.5.4.3 Removing an IA-station

Each IA-station removed from the Configuration Domain is discovered by the TDE. A neighboring station can receive an updated network configuration by the NPE. After this, the removed IA-station is no longer part of the Configuration Domain.

6.4.5.4.4 Replacing an IA-station

In the simplest case, replacing an IA-station is simply the sequence of removing an IA-station (6.4.5.4.3) and adding an IA-station (6.4.5.4.2). In more complex cases, the user can take other precautions or actions following deployment.

6.4.5.5 Management proxies

6.4.5.5.1 General

Implementing NETCONF server functionality can be burdensome for resource constrained IA-stations. IA-stations which contain only ccB Bridge or end station components may utilize the NETCONF server of another IA-station as a management proxy. IA-stations supporting proxy functionality (proxies), maintain a list of the types of proxy functionality supported as well as the IA-stations for which they act as proxy (proxy targets). Proxy types are described in E.1.3. The information is maintained in the YANG module `iecieeee60802-proxy` as described in 6.4.10.8.8.

6.4.5.5.2 Proxy operation

When onboarding a new proxy target, a different process is followed. Proxies communicate with proxy targets via the isolation VLAN. When a new IA station is connected, the following sequence is followed.

- a) The IA-station to which the proxy target is connected (neighbor), which is already part of the configuration domain, reports a change in link status to the CNC.
- b) The CNC reads the stored LLDP information related to the proxy target from the neighbor via NETCONF. The neighbor can be a proxied device in which case the LLDP information about the proxy target would be retrieved using the neighbor's proxy.
- c) Each proxy target identifies itself using the 60802 device info TLV described in Annex E.
- d) Based on this TLV data, the CNC determines whether the connected station:
 - 1) natively speaks NETCONF/YANG (no information regarding proxy operation is included), or
 - 2) requires a proxy, and if so, which one.
- e) The proxy onboards the new device via a user-specific protocol. This user-specific protocol is not described in this document. Ensuring the security of this user-specific mechanism is not described in this document. If security equivalent to the secure management exchanges as described in 5.5.4.1 is not implemented in the user-specific protocol, security of the network can be compromised.

- f) The proxy communicates the result of the Authentication to the CNC. For instance, the CNC can utilize dynamic subscriptions and the on-change feature of the ietf-yang-push module to automatically be notified when the new proxy target is added to the proxy's list of proxy targets. Based upon the results of the authentication, the CNC sets the boundary port state appropriately
- g) The CNC communicates with the appropriate proxy and configures the device accordingly. The proxy handles the NETCONF parameterization and translates it into the required organization-defined or vendor-specific protocol and configures the proxy target.

6.4.5.6 Engineered network extended by dynamic topology

The engineered and dynamic topology workflows can be used together. For instance, modular machines, robot tool changers or more general plug & produce can add or remove modules. The basic machine is handled as an engineered network. Additional modules or removed modules are handled dynamically.

6.4.6 Engineered time-synchronization spanning tree

6.4.6.1 General

Engineered time-synchronization spanning tree (sync tree) for a given gPTP domain refers to the usage of external port configuration instead of BTCA for the construction of a desired sync tree with the Grandmaster PTP Instance as the root (see IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024 and IEEE Std 802.1ASdm-2024, 10.3.1). The Grandmaster PTP Instance can reside in a dedicated grandmaster-capable IA-station.

One of the advantages of engineered sync trees is to enable a planned, deterministic, and stable configuration of the IEEE Std 802.1AS-2020 sync tree for a given gPTP domain. For example, this approach prevents sync tree changes in case of IA-station addition or removal from the network. Hot standby (see IEEE Std 802.1ASdm-2024) is a use case of an engineered sync tree.

6.4.6.2 Sync tree requirements

If an engineered synchronization spanning tree is used, the sync tree requirements for all participating PTP Instances in a gPTP domain are specified in 5.5.3, h).

6.4.6.3 STE phases

6.4.6.3.1 General

The STE follows the logical sequence described in 6.4.6.3 if an engineered sync tree is utilized in a gPTP domain. Each STE phase describes an externally observable behavior of the participating PTP Instances in a gPTP domain.

6.4.6.3.2 Discovery phase

In discovery phase, STE utilizes the topology discovered by the TDE to verify the capabilities and status of participating IA-stations via a diagnostics entity (see 6.4.7.1) by reading the following managed objects.

- The status of oper-status parameter is up (see IETF RFC 8343) for all participating Ethernet links.
- The status of isMeasuringDelay (see IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdm-2024, 14.16.4) is TRUE for all PTP Ports.
- The status of asCapable (see IEEE Std 802.1AS-2020, 14.8.7) is TRUE for all PTP Ports.
- The status of asCapableAcrossDomains (see IEEE Std 802.1AS-2020, 14.16.5) is TRUE for all LinkPorts.

- The status of gmCapable (see IEEE Std 802.1AS-2020, 14.2.7) is TRUE, only applicable to the Grandmaster PTP Instance.

STE should use the information collected via managed objects and the discovered topology to verify the constraints on the gPTP domain, for example:

- Verify that the number of PTP Relay Instances (hops) between the Grandmaster PTP Instance and any given timeReceiver PTP End Instance is within the limit prescribed by for example, CNC.

6.4.6.3.3 Provisioning phase

In provisioning phase, STE should apply the desired configuration to all participating PTP Instances, for example:

- externalPortConfigurationEnabled is set to true,
- the desiredState of all PTP ports of the Grandmaster PTP Instance is set to TimeTransmitterPort,
- the desiredState of exactly one PTP port of all the other PTP Instances is set to TimeReceiverPort,
- the desiredState of remaining PTP ports that are part of sync tree in non-Grandmaster PTP Relay Instances is set to TimeTransmitterPort, and
- the desiredState of all other PTP ports is set to PassivePort.

Then STE should validate, for example, the syncLocked (see IEEE Std 802.1AS-2020, 14.8.52) parameter is TRUE for all PTP ports of PTP Relay Instances that are in TimeTransmitterPort state.

6.4.6.4 Adding an IA-station

Each IA-station added to the gPTP domain is discovered by STE via TDE. It is the responsibility of the CNC to on-board this newly added station. IA-stations can receive an updated gPTP configuration via STE.

A newly installed IA-station can disrupt the operation of a gPTP domain. The extent of disruption is dependent on the location of the IA-station in the gPTP domain and the type of PTP Instance running on that IA-station. For example, if PTP Instances are arranged in a daisy-chain formation and an IA-station with a non-Grandmaster Relay Instance is installed in the middle of a daisy-chain, then this change can disrupt the operation of downstream Instances.

6.4.6.5 Removing an IA-station

The removal of a station from the gPTP domain is detected by STE via TDE. IA-stations can receive an updated gPTP configuration via STE.

6.4.6.6 Replacing an IA-station

An IA-station replacement follows the sequence of removing an IA-station according to 6.4.6.5 and adding an IA-station according to 6.4.6.4.

6.4.7 Diagnostics

6.4.7.1 General

Diagnosis for an IA-station is done by monitoring YANG representation of the IA-station's local database.

A vendor can implement an observer in a diagnostics entity, which could reside in the CNC. This diagnostics entity uses the information provided by remote management to define the monitored objects and set up fitting notifications.

6.4.7.2 Observer model

A diagnostic entity can select any objects described via YANG and observe them via NETCONF. The NETCONF binding is specified in IETF RFC 8640, and the subscription model in IETF RFC 8641. NETCONF messages can be pipelined, i.e. a client can invoke multiple RPCs without having to wait for RPC result messages first. RPC messages are specified in IETF RFC 6241, and notification messages are specified in IETF RFC 5277. To reduce the load on the diagnostic entity when many stations are providing notifications, the diagnostic objects can be monitored and notifications can be retrieved from individual IA-stations.

Figure 27 shows the model of a diagnostic observer.

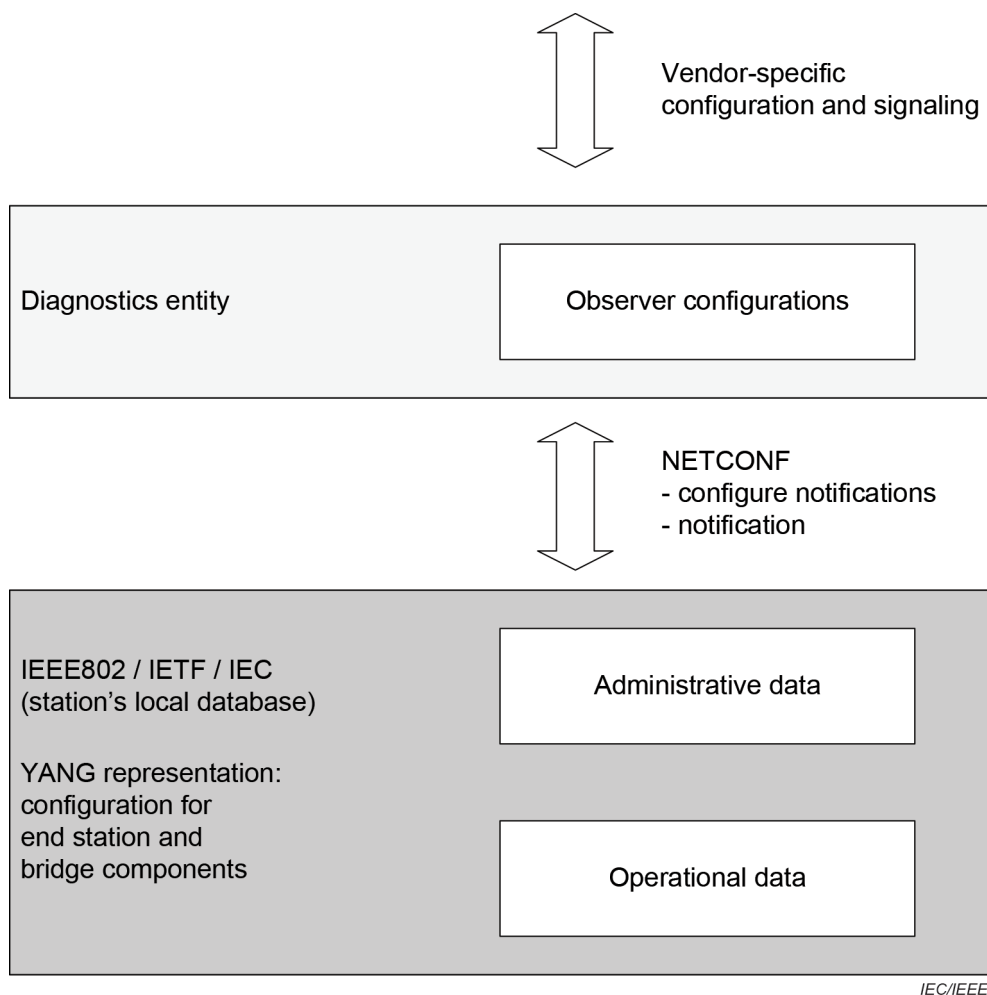


Figure 27 – Observer model

6.4.7.3 Usage of YANG Push

For diagnostics, an IA-station shall support YANG-Push subscriptions according to IETF RFC 8641 (YANG Push) and IETF RFC 8639 (Subscribed Notifications).

IA-stations shall support the "subtree" selection filter as specified in IETF RFC 8641, 3.6

6.4.7.4 Mandatory RPCs

An IA-station shall support following RPCs as specified in IETF RFC 8641:

- a) establish-subscription,
- b) modify-subscription,
- c) delete-subscription, and
- d) kill-subscription.

6.4.7.5 Mandatory notifications

An IA-station shall support following notifications as specified in IETF RFC 8641:

- a) subscription-resumed,
- b) subscription-modified,
- c) subscription-terminated,
- d) subscription-suspended,
- e) push-update, and
- f) push-change-update.

6.4.7.6 Mandatory diagnostics data nodes

An IA-station shall provide the following data nodes for NETCONF subscriptions.

- a) Change of link-status per Ethernet port:
`/ietf-interfaces/interface/oper-status`
- b) Change of MAU-type per Ethernet port:
`/ieee802-ethernet-lldp/lldp/port/ operational-mau-type`
- c) Change of sync-status:
 - 1) per PTP Instance:
 for every application-clock:
`/ieciieee60802-bridge/bridges/bridge/component/clock/is-synced`
 - 2) per hot standby Instance:
`/dot1as-hs/ptp/common-services/hss/hot-standby-system-list/hot-standby-system-ds/hot-standby-system-state`
- d) Data to be provided as periodic time-aligned subscriptions:
 - 1) Dropped frames statistic counters per Ethernet interface
`/ietf-interfaces/interface/statistics/in-octets`
`/ietf-interfaces/interface/statistics/in-discards`
`/ietf-interfaces/interface/statistics/in-errors`
`/ietf-interfaces/interface/statistics/out-octets`
`/ietf-interfaces/interface/statistics/out-discards`
`/ietf-interfaces/interface/statistics/out-errors`

2) VLAN specific counters per Ethernet Interface and VLAN ID

```

/ieee802-dot1q-bridge/interfaces/interface/bridge-
port/statistics/octets-rx
/ieee802-dot1q-bridge/interfaces/interface/bridge-
port/statistics/octets-tx
/ieee802-dot1q-bridge/interfaces/interface/bridge-
port/statistics/forward-outbound
/ieee802-dot1q-bridge/interfaces/interface/bridge-
port/statistics/discard-inbound

```

6.4.7.7 Usage of NETCONF notifications

IA-stations shall implement the binding of a stream of events according to IETF RFC 8640 (NETCONF Notifications) using the "encode-xml" feature and the "NETCONF" event stream of IETF RFC 8639.

An IA-station shall support dynamic subscriptions according to 6.4.7.6, IETF RFC 8640 Clauses 5, 6 and 7. The number of dynamic subscriptions shall be reported.

6.4.8 Data sheet**6.4.8.1 General**

Data sheets containing the capabilities, value ranges and quantities of IA-stations allow a user to select appropriate IA-stations and enable users to configure a system using online and offline engineering. See Annex B for quantities in a representative Configuration Domain.

Online data sheets are modeled using YANG. YANG modeling is used for the offline data sheet to keep the offline (6.4.5.3) and online (6.4.5.4) format the same.

6.4.8.2 Digital data sheet of an IA-station

Both engineering models, offline via an engineering tool and online with plug & produce by the CNC, require information about the capabilities of an IA-station, for example, states, configurations, or supported features. An example depicting the creation of a digital data sheet is provided in Figure 28.

This data is extracted from the implemented YANG modules, which are available in the local database of the IA-station.

The data from the implemented YANG modules is also available offline in the form of a digital data sheet of an IA-station as a digital data sheet file.

The digital data sheet of an IA-station provides a collection of all instantiated data nodes of all YANG modules that are required by this document (see 6.4.9). A manufacturer may reduce the instance data set by removing statistical config-false YANG nodes from the digital data sheet.

The digital data sheet does not contain any additional information that is not modeled by the YANG modules that exist in the local database of the IA-station.

The data sheet contains a single instance data set. It carries complete configuration and state data of each YANG module that is present in the local database of the IA-station.

The identity of the datastore with which the instance data set is associated and the format of the YANG instance data set shall be reported according to IETF RFC 9195. The file format is based on the XML encoding. It is created by applying the respective XML encoding rules for the YANG structure of all YANG modules included in the digital data sheet.

A user uses the information from the digital data sheet to understand the quantities and capabilities of an IA-station, which is required for successful offline engineering of the network.

The features of a CNC are used for offline and online engineering or diagnostics. For this purpose, YANG modules are used that allow structured access to the local database of the CNC according to 6.4.9.2.5.11.

Any IA-station can include a CNC entity in which case the collection of YANG modules of such IA-station includes all CNC specific YANG modules for example, the `ieee802-dot1q-cnc-config` YANG module. Since all IA-stations meet the requirements from 5.5.4, the CNC related YANG instance data is automatically included in the digital data sheet of the IA-station that hosts the CNC as described in 6.4.9.2.

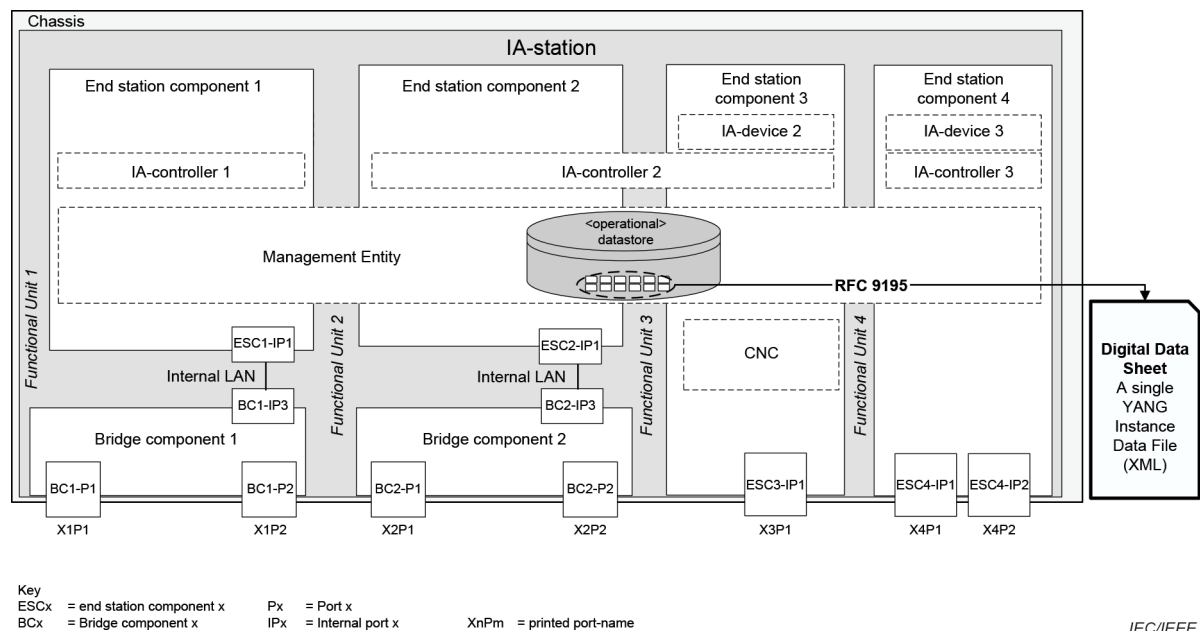


Figure 28 – Creation of the digital data sheet of an IA-station

6.4.9 YANG representation of managed objects and nodes

6.4.9.1 General

All managed objects shall be represented in the YANG 1.1 format as described in IETF RFC 7950. The markings (i.e. [m], [o], [c]) indicate whether the node is included in the digital data sheet (see 3.5.4). These markings are independent of the conformance criteria for an IA-station (see 5.2).

6.4.9.2 Common YANG modules, features, and nodes

6.4.9.2.1 IEEE standard for Ethernet

IA-stations shall support the `ieee802-ethernet-interface` YANG module according to IEEE Std 802.3.2™-2025 with the following nodes:

```
[o] /ieee802-ethernet-interface/interfaces/interface/ethernet/duplex
[o] /ieee802-ethernet-interface/interfaces/interface/ethernet/speed
[o] /ieee802-ethernet-interface/interfaces/interface/ethernet/flow-control/pause/direction (if the feature "ethernet-pause" is supported)
```

- [o] ieee802-ethernet-lldp/lldp/port/operational-mau-type
- [o] ieee802-ethernet-lldp/lldp/port/preemption-supported
- [o] ieee802-ethernet-lldp/lldp/port/preemption-enabled
- [o] ieee802-ethernet-lldp/lldp/port/preemption-active
- [o] ieee802-ethernet-lldp/lldp/port/additional-fragment-size
- [o] ieee802-ethernet-lldp/lldp/port/local-max-frame-size

6.4.9.2.2 Station and media access control connectivity discovery

IA-stations shall support the following nodes from the ieee802-dot1ab-lldp YANG module according to IEEE Std 802.1ABcu-2021 with values and value ranges according to 6.5.

- [o] /ieee802-dot1ab-lldp/lldp/message-fast-tx
- [o] /ieee802-dot1ab-lldp/lldp/message-tx-hold-multiplier
- [o] /ieee802-dot1ab-lldp/lldp/message-tx-interval
- [o] /ieee802-dot1ab-lldp/lldp/reinit-delay
- [o] /ieee802-dot1ab-lldp/lldp/tx-credit-max
- [o] /ieee802-dot1ab-lldp/lldp/tx-fast-init
- [o] /ieee802-dot1ab-lldp/lldp/notification-interval
- [o] /ieee802-dot1ab-lldp/lldp/remote-statistics
- [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/last-change-time
- [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/remote-inserts
- [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/remote-deletes
- [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/remote-drops
- [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/remote-ageouts
- [m] /ieee802-dot1ab-lldp/lldp/local-system-data
- [o] /ieee802-dot1ab-lldp/lldp/local-system-data/chassis-id-subtype
- [o] /ieee802-dot1ab-lldp/lldp/local-system-data/chassis-id
- [o] /ieee802-dot1ab-lldp/lldp/local-system-data/system-name
- [o] /ieee802-dot1ab-lldp/lldp/local-system-data/system-description
- [m] /ieee802-dot1ab-lldp/lldp/local-system-data/system-capabilities-supported
- [o] /ieee802-dot1ab-lldp/lldp/local-system-data/system-capabilities-enabled
- [o] /ieee802-dot1ab-lldp/lldp/port
- [o] /ieee802-dot1ab-lldp/lldp/port/name
- [o] /ieee802-dot1ab-lldp/lldp/port/dest-mac-address

```
[o] /ieee802-dot1ab-lldp/lldp/port/admin-status
[o] /ieee802-dot1ab-lldp/lldp/port/notification-enable
[o] /ieee802-dot1ab-lldp/lldp/port/tlvs-tx-enable
[o] /ieee802-dot1ab-lldp/lldp/port/message-fast-tx
[o] /ieee802-dot1ab-lldp/lldp/port/message-tx-hold-multiplier
[o] /ieee802-dot1ab-lldp/lldp/port/message-tx-interval
[o] /ieee802-dot1ab-lldp/lldp/port/reinit-delay
[o] /ieee802-dot1ab-lldp/lldp/port/tx-credit-max
[o] /ieee802-dot1ab-lldp/lldp/port/tx-fast-init
[o] /ieee802-dot1ab-lldp/lldp/port/notification-interval
[o] /ieee802-dot1ab-lldp/lldp/port/management-address-tx-port
[o] /ieee802-dot1ab-lldp/lldp/port/port-id-subtype
[o] /ieee802-dot1ab-lldp/lldp/port/port-id
[o] /ieee802-dot1ab-lldp/lldp/port/port-desc
[o] /ieee802-dot1ab-lldp/lldp/port/remote-systems-data
```

6.4.9.2.3 Synchronization

6.4.9.2.3.1 Timesync

IA-stations shall support the `ieee1588-ptp-tt` YANG module according to IEEE Std 1588e™-2024 with the following features:

- `cmlds` (Common Mean Link Delay Service), and
- `external-port-config`.

IA-stations shall support the `ieee1588-ptp-tt` YANG module according to IEEE Std 1588e-2024 with the following nodes:

```
[o] /ieee1588-ptp-tt/ptp/instances/instance/instance-index
[o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/clock-identity
[o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/number-ports
[o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/priority1
[o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/domain-number
[o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/time-receiver-only
[o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/sdo-id
[o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/instance-enable
[o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/external-port-config-enable
[o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/instance-type
```


[o] /ieee1588-ntp-tt/ntp/instances/instance/description-ds/user-description

[o] /ieee1588-ntp-tt/ntp/instances/ports/port/port-index

[o] /ieee1588-ntp-tt/ntp/instances/ports/port/underlying-interface

[o] /ieee1588-ntp-tt/ntp/instances/ports/port/port-ds/port-state

[o] /ieee1588-ntp-tt/ntp/instances/ports/port/port-ds/delay-mechanism

[o] /ieee1588-ntp-tt/ntp/instances/ports/port/port-ds/port-enable

[o] /ieee1588-ntp-tt/ntp/instances/ports/port/external-port-config-port-ds/desired-state

[o] /ieee1588-ntp-tt/ntp/common-services/cmls/default-ds/clock-identity

[o] /ieee1588-ntp-tt/ntp/common-services/cmls/default-ds/number-link-ports

[o] /ieee1588-ntp-tt/ntp/common-services/cmls/ports/port/port-index

[o] /ieee1588-ntp-tt/ntp/common-services/cmls/ports/port/underlying-interface

[o] /ieee1588-ntp-tt/ntp/common-services/cmls/ports/port/link-port-ds/port-identity/clock-identity

[o] /ieee1588-ntp-tt/ntp/common-services/cmls/ports/port/link-port-ds/port-identity/port-number

[o] /ieee1588-ntp-tt/ntp/common-services/cmls/ports/port/link-port-ds/domain-number

[o] /ieee1588-ntp-tt/ntp/common-services/cmls/ports/port/link-port-ds/service-measurement-valid

[o] /ieee1588-ntp-tt/ntp/common-services/cmls/ports/port/link-port-ds/mean-link-delay

[o] /ieee1588-ntp-tt/ntp/common-services/cmls/ports/port/link-port-ds/scaled-neighbor-rate-ratio

[o] /ieee1588-ntp-tt/ntp/common-services/cmls/ports/port/link-port-ds/log-min-pdelay-req-interval

[m] /ieee1588-ntp-tt/ntp/common-services/cmls/ports/port/link-port-ds/version-number

[m] /ieee1588-ntp-tt/ntp/common-services/cmls/ports/port/link-port-ds/minor-version-number

[o] /ieee1588-ntp-tt/ntp/common-services/cmls/ports/port/link-port-ds/delay-asymmetry

6.4.9.2.3.2 Timesync (ieee802-dot1as-gptp)

IA-stations shall support the ieee802-dot1as-gptp YANG module according to IEEE Std 802.1ASdn-2024™ with the following nodes:

```
[o] /ieee802-dot1as-gptp/ptp/instances/instance/default-ds/gm-capable
[o] /ieee802-dot1as-gptp/ptp/instances/instance/default-ds/current-
    utc-offset-valid
[o] /ieee802-dot1as-gptp/ptp/instances/instance/default-ds/ptp-
    timescale
[o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/sync-
    receipt-timeout
[o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/current-one-
    step-tx-oper
[o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/use-mgt-one-
    step-tx-oper
[o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/mgt-one-
    step-tx-oper
[o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/sync-locked
[o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
    port-ds/cmls-link-port-enabled
[o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
    port-ds/is-measuring-delay
[o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
    port-ds/as-capable-across-domains
[o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
    port-ds/mean-link-delay-thresh
[o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
    port-ds/current-log-pdelay-req-interval
[o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
    port-ds/use-mgt-log-pdelay-req-interval
[o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
    port-ds/mgt-log-pdelay-req-interval
[o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
    port-ds/current-compute-rate-ratio
[o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
    port-ds/use-mgt-compute-rate-ratio
[o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
    port-ds/mgt-compute-rate-ratio
[o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
    port-ds/current-compute-mean-link-delay
[o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
    port-ds/use-mgt-compute-mean-link-delay
```

```
[o] /ieee802-dot1as-gptp/ptp/common-services/cmlds/ports/port/link-
port-ds/mgt-compute-mean-link-delay
```

```
[o] /ieee802-dot1as-gptp/ptp/common-services/cmlds/ports/port/link-
port-ds/allowed-lost-responses
```

```
[o] /ieee802-dot1as-gptp/ptp/common-services/cmlds/ports/port/link-
port-ds/allowed-faults
```

6.4.9.2.3.3 Timesync (ieee802-dot1as-hs)

IA-stations shall support the ieee802-dot1as-hs YANG module according to IEEE Std 802.1ASdm-2024 with the following nodes:

```
[o] /ieee802-dot1as-hs/ptp/instances/instance/parent-ds/gmPresent
```

```
[o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/issynced
```

```
[o] /ieee802-dot1as-hs/ptp/instances/instance/ports/port/portDS/nrr-
pdelay
```

```
[o] /ieee802-dot1as-hs/ptp/instances/instance/ports/port/portDS/nrr-
sync
```

```
[o] /ieee802-dot1as-hs/ptp/instances/instance/ports/port/portDS/nrr-
comp-method
```

```
[o] /ieee802-dot1as-hs/ptp/instances/instance/ports/port/portDS/as-
capable-across-domains
```

```
[o] /ieee802-dot1as-hs/ptp/instances/instance/ports/port/portDS/gptp-
capable-state-machines-enabled
```

6.4.9.2.4 Security configuration modules

6.4.9.2.4.1 YANG module for a keystore

IA-stations shall support the ietf-keystore YANG module according to IETF RFC 9642 with the following features:

- central-truststore-supported, and
- asymmetric-keys.

IA-stations shall support the ietf-keystore YANG module according to IETF RFC 9642 with the following nodes:

```
[o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/name
```

```
[o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/public-key-
format
```

```
[o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/public-key
```

```
[o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/private-
key-format
```

```
[o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/hidden-
private-key
```

```
[o] /ietf-keystore/certificates/certificate/name
```

```
[o] /ietf-keystore/certificates/certificate/cert-data
```

```
[o] /ietf-keystore/certificates/certificate/expiration-date
[o] /ietf-keystore/certificates/certificate/csr-info
[o] /ietf-keystore/certificates/certificate/certificate-signing-
request
```

6.4.9.2.4.2 Network configuration access control

IA-stations shall support the ietf-netconf-acm YANG module according to IETF RFC 8341 with the following nodes:

```
[o] /ietf-netconf-acm/nacm/enable-nacm
[o] /ietf-netconf-acm/nacm/read-default
[o] /ietf-netconf-acm/nacm/write-default
[o] /ietf-netconf-acm/nacm/exec-default
[o] /ietf-netconf-acm/nacm/enable-external-groups
[o] /ietf-netconf-acm/nacm/groups
[o] /ietf-netconf-acm/nacm/rule-list
```

6.4.9.2.4.3 A YANG data module for a truststore

IA-stations shall support the ietf-truststore YANG module according to IETF RFC 9641 with the following features:

- central-keystore-supported, and
- certificates.

IA-stations shall support the ietf-truststore YANG module according to IETF RFC 9641 with the following nodes:

```
[o] /ietf-truststore/truststore/certificate-bags/certificate-bag/name
[o] /ietf-truststore/truststore/certificate-bags/certificate-
bag/certificate/name
[o] /ietf-truststore/truststore/certificate-bags/certificate-
bag/certificate/cert-data
[o] /ietf-truststore/truststore/certificate-bags/certificate-
bag/certificate/expiration-date
```

6.4.9.2.5 IA-station management

6.4.9.2.5.1 System capabilities

IA-stations shall support the ietf-system-capabilities YANG modules according to IETF RFC 9196 with the following nodes:

```
[m] /ietf-system-capabilities/system-capabilities/datastore-
capabilities/datastore
[m] /ietf-system-capabilities/system-capabilities/datastore-
capabilities/per-node-capabilities
[m] /ietf-system-capabilities/system-capabilities/datastore-
capabilities/on-change-supported
```

6.4.9.2.5.2 YANG library

IA-stations shall support the `ietf-yang-library` YANG module according to IETF RFC 8525 with the following nodes:

```
[m] /ietf-yang-library/yang-library/module-set
[m] /ietf-yang-library/yang-library/schema
[m] /ietf-yang-library/yang-library/datastore
[m] /ietf-yang-library/yang-library/content-id
```

6.4.9.2.5.3 YANG push

IA-stations shall support the `ietf-yang-push` YANG module according to IETF RFC 8641, 4.1, with the `on-change` feature.

IA-stations shall support the `ietf-yang-push` YANG module according to IETF RFC 8641, 4.1, with the following nodes:

```
[o] /ietf-subscribed-notifications/filters/selection-filter
[o] /ietf-subscribed-
notifications/subscriptions/subscription/target/datastore
[o] /ietf-subscribed-notifications/subscriptions/subscription/update-
trigger
```

6.4.9.2.5.4 YANG notification capabilities

IA-stations shall support the `ietf-notification-capabilities` YANG module according to IETF RFC 9196 with the following nodes:

```
[m] /ietf-notification-capabilities/system-capabilities/subscription-
capabilities
[m] /ietf-notification-capabilities/system-capabilities/datastore-
capabilities/per-node-capabilities/subscription-capabilities
```

6.4.9.2.5.5 YANG notifications

IA-stations shall support the `ietf-subscribed-notifications` YANG module according to IETF RFC 8639 with the following features:

- `Configured`,
- `encode-xml`, and
- `subtree`.

IA-stations shall support the `ietf-subscribed-notifications` YANG module according to IETF RFC 8639 with the following nodes:

```
[o] /ietf-subscribed-notifications/streams/stream/name
[o] /ietf-subscribed-notifications/streams/stream/description
[o] /ietf-subscribed-notifications/streams/stream/replay-support
[o] /ietf-subscribed-notifications/streams/stream/replay-log-creation-
time
```

- [o] /ietf-subscribed-notifications/streams/stream/replay-log-aged-time
- [o] /ietf-subscribed-notifications/filters/stream-filter/name
- [o] /ietf-subscribed-notifications/filters/stream-filter/filter-spec
- [o] /ietf-subscribed-notifications/subscriptions/subscription/id
- [o] /ietf-subscribed-notifications/subscriptions/subscription/target
- [o] /ietf-subscribed-notifications/subscriptions/subscription/stop-time
- [o] /ietf-subscribed-notifications/subscriptions/subscription/dscp
- [o] /ietf-subscribed-notifications/subscriptions/subscription/weighting
- [o] /ietf-subscribed-notifications/subscriptions/subscription/dependency
- [o] /ietf-subscribed-notifications/subscriptions/subscription/transport
- [o] /ietf-subscribed-notifications/subscriptions/subscription/encoding
- [o] /ietf-subscribed-notifications/subscriptions/subscription/purpose
- [o] /ietf-subscribed-notifications/subscriptions/subscription/notification-message-origin
- [o] /ietf-subscribed-notifications/subscriptions/subscription/configured-subscription-state
- [o] /ietf-subscribed-notifications/subscriptions/subscription/receivers

6.4.9.2.5.6 NETCONF monitoring

IA-stations shall support the ietf-netconf-monitoring YANG module according to IETF RFC 6022 with the following nodes:

- [m] /ietf-netconf-monitoring/netconf-state/capabilities
- [m] /ietf-netconf-monitoring/netconf-state/datastores
- [m] /ietf-netconf-monitoring/netconf-state/schemas

6.4.9.2.5.7 System management

IA-stations shall support the ietf-system YANG module according to IETF RFC 7317 with the following nodes:

- [o] /ietf-system/system/contact
- [o] /ietf-system/system/hostname
- [o] /ietf-system/system/location

6.4.9.2.5.8 Hardware management

IA-stations shall support the ietf-hardware YANG module according to IETF RFC 8348 with the following nodes:

```
[m] /ietf-hardware/hardware/component/name
[m] /ietf-hardware/hardware/component/class
[m] /ietf-hardware/hardware/component/description
[m] /ietf-hardware/hardware/component/hardware-rev
[m] /ietf-hardware/hardware/component/software-rev
[o] /ietf-hardware/hardware/component/serial-num
[m] /ietf-hardware/hardware/component/mfg-name
[m] /ietf-hardware/hardware/component/model-name
```

An IA-station shall provide exactly one /ietf-hardware/component with class "chassis" and may provide further components with other classes.

6.4.9.2.5.9 Interface management

IA-stations shall support the ietf-interfaces YANG module according to IETF RFC 8343 with the following nodes:

```
[m] /ietf-interfaces/interfaces/interface/name
[m] /ietf-interfaces/interfaces/interface/description
[m] /ietf-interfaces/interfaces/interface/type
[o] /ietf-interfaces/interfaces/interface/enabled
[o] /ietf-interfaces/interfaces/interface/oper-status
[o] /ietf-interfaces/interfaces/interface/phys-address
[o] /ietf-interfaces/interfaces/interface/higher-layer-if
[o] /ietf-interfaces/interfaces/interface/lower-layer-if
[o] /ietf-interfaces/interfaces/interface/speed
[o] /ietf-interfaces/interfaces/interface/statistics/discontinuity-
time
[o] /ietf-interfaces/interfaces/interface/statistics/in-octets
[o] /ietf-interfaces/interfaces/interface/statistics/in-discards
[o] /ietf-interfaces/interfaces/interface/statistics/in-errors
[o] /ietf-interfaces/interfaces/interface/statistics/out-octets
[o] /ietf-interfaces/interfaces/interface/statistics/out-discards
[o] /ietf-interfaces/interfaces/interface/statistics/out-errors
```

6.4.9.2.5.10 Bridge and end station component management**6.4.9.2.5.10.1 General**

IA-stations shall support the `ieee802-dot1q-bridge` YANG module according to IEEE Std 802.1Q-2022, Clause 48, as amended by IEEE Std 802.1Qcw™-2023 with the following feature: `ingress-filtering`.

IA-stations shall support the `ieee802-dot1q-bridge` YANG module according to IEEE Std 802.1Q-2022, Clause 48, as amended by IEEE Std 802.1Qcw-2023 with the following nodes. A distinction is made between nodes that shall be supported by bridge and end station components, or by bridge components only.

6.4.9.2.5.10.2 Bridge nodes to be supported by bridge and end station components

```
[m] /ieee802-dot1q-bridge/bridges/bridge/name
[o] /ieee802-dot1q-bridge/bridges/bridge/address
[m] /ieee802-dot1q-bridge/bridges/bridge/bridge-type
[m] /ieee802-dot1q-bridge/bridges/bridge/ports
[m] /ieee802-dot1q-bridge/bridges/bridge/components
[m] /ieee802-dot1q-bridge/bridges/bridge/component/name
[o] /ieee802-dot1q-bridge/bridges/bridge/component/id
[m] /ieee802-dot1q-bridge/bridges/bridge/component/type
[o] /ieee802-dot1q-bridge/bridges/bridge/component/traffic-class-
enabled
[m] /ieee802-dot1q-bridge/bridges/bridge/component/ports
[o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-port
[m] /ieee802-dot1q-bridge/bridges/bridge/component/capabilities
[m] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
database/size
[o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
database/static-vlan-registration-entries
[o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
database/vlan-registration-entry
```

6.4.9.2.5.10.3 Filtering-database nodes to be supported by bridge components

```
[o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
database/aging-time
[o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
database/static-entries
[o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
database/dynamic-entries
[o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
database/dynamic-vlan-registration-entries
```


[o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-database/mac-address-registration-entries

[o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-database/filtering-entry

6.4.9.2.5.10.4 Permanent-database nodes to be supported by bridge components

[m] /ieee802-dot1q-bridge/bridges/bridge/component/permanent-database/size

[o] /ieee802-dot1q-bridge/bridges/bridge/component/permanent-database/static-entries

[o] /ieee802-dot1q-bridge/bridges/bridge/component/permanent-database/static-vlan-registration-entries

[o] /ieee802-dot1q-bridge/bridges/bridge/component/permanent-database/filtering-entry

6.4.9.2.5.10.5 Bridge-vlan nodes to be supported by bridge and end station components

[m] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/version

[m] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/max-vids

[o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/override-default-pvid

[o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/vlan

6.4.9.2.5.10.6 Bridge-vlan nodes to be supported by bridge components

[m] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/max-msti

[o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/vid-to-fid-allocation

[o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/fid-to-vid-allocation

[o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/vid-to-fid

6.4.9.2.5.10.7 Bridge-mst nodes to be supported by bridge components

[o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-mst

6.4.9.2.5.10.8 Bridge-port nodes to be supported by bridge and end station components

[m] /ietf-interfaces/interfaces/interface/bridge-port/bridge-name

[m] /ietf-interfaces/interfaces/interface/bridge-port/component-name

[m] /ietf-interfaces/interfaces/interface/bridge-port/port-type

[o] /ietf-interfaces/interfaces/interface/bridge-port/pvid

[o] /ietf-interfaces/interfaces/interface/bridge-port/default-priority

[m] /ietf-interfaces/interfaces/interface/bridge-port/traffic-class

```
[o] /ietf-interfaces/interfaces/interface/bridge-port/statistics
[m] /ietf-interfaces/interfaces/interface/bridge-port/capabilities
[m] /ietf-interfaces/interfaces/interface/bridge-port/type-capabilities
[o] /ietf-interfaces/interfaces/interface/bridge-port/transmission-
selection-algorithm-table
```

6.4.9.2.5.10.9 Bridge-port nodes to be supported by bridge component ports

```
[o] /ietf-interfaces/interfaces/interface/bridge-port/priority-
regeneration
[o] /ietf-interfaces/interfaces/interface/bridge-port/acceptable-frame
[o] /ietf-interfaces/interfaces/interface/bridge-port/enable-ingress-
filtering
[o] /ietf-interfaces/interfaces/interface/bridge-port/enable-vid-
translation-table
[o] /ietf-interfaces/interfaces/interface/bridge-port/vid-translations
[o] /ietf-interfaces/interfaces/interface/bridge-port/enable-egress-
vid-translation-table
[o] /ietf-interfaces/interfaces/interface/bridge-port/egress-vid-
translations
```

6.4.9.2.5.11 IEC/IEEE 60802 YANG modules

IA-stations shall support the `ieciieee60802-ethernet-interface` YANG module according to this document with the following nodes:

```
[m] /ieciieee60802-ethernet-
interface/interfaces/interface/ethernet/supported-mau-types/mau-type
[m] /ieciieee60802-ethernet-
interface/interfaces/interface/ethernet/supported-mau-
types/preemption-supported
```

IA-stations shall support the `ieciieee60802-bridge` YANG module according to this document with the following nodes:

```
[m] /ieciieee60802-bridge/interfaces/interface/bridge-port/max-burst-
params
[m] /ieciieee60802-bridge/interfaces/interface/bridge-port/committed-
data-rates
[m] /ieciieee60802-bridge/interfaces/interface/bridge-
port/transmission-selection-algorithm
[m] /ieciieee60802-bridge/interfaces/interface/bridge-port/supported-resource-
pools
[m] /ieciieee60802-bridge/bridges/bridge/component/frer-supported
[m] /ieciieee60802-bridge/bridges/bridge/component/max-redundant-
streams
[m] /ieciieee60802-bridge/bridges/bridge/component/max-fids
```

```
[m] /ieciieee60802-bridge/bridges/bridge/component/max-fdb-entries
[m] /ieciieee60802-bridge/bridges/bridge/component/delay-variance
[m] /ieciieee60802-bridge/bridges/bridge/component/max-ptp-instances
[m] /ieciieee60802-bridge/bridges/bridge/component/max-hot-standby-
systems
[m] /ieciieee60802-bridge/bridges/bridge/component/clock
[m] /ieciieee60802/interfaces/interface/bridge-port/phy-latencies/mau-
type
[m] /ieciieee60802/interfaces/interface/bridge-port/phy-
latencies/latency
[m] /ieciieee60802/interfaces/interface/bridge-port/phy-
latencies/latency-variation
```

IA-stations shall support the ieciieee60802-ia-station YANG module according to this document with the following nodes:

```
[m] /ieciieee60802-ia-station/ia-station-capabilities/capability-lldp
[m] /ieciieee60802-ia-station/ia-station-capabilities/capability-
timesync
[m] /ieciieee60802-ia-station/ia-station-capabilities/capability-
keystore
[m] /ieciieee60802-ia-station/ia-station-capabilities/capability-
truststore
[m] /ieciieee60802-ia-station/ia-station-capabilities/capability-nacm
[m] /ieciieee60802-ia-station/ia-station-capabilities/capability-yang-
library
[m] /ieciieee60802-ia-station/ia-station-capabilities/capability-yang-
push
[m] /ieciieee60802-ia-station/ia-station-capabilities/capability-yang-
notifications
[m] /ieciieee60802-ia-station/ia-station-capabilities/capability-
netconf-monitoring
[m] /ieciieee60802-ia-station/ia-station-capabilities/capability-
netconf-client
[m] /ieciieee60802-ia-station/ia-station-capabilities/capability-cnc-
config
[m] /ieciieee60802-ia-station/ia-station-capabilities/capability-sched-
traffic
[m] /ieciieee60802-ia-station/ia-station-capabilities/capability-frame-
preemption
```

IA-stations shall support the `ieciieee60802-subscribed-notifications` YANG module according to this document with the following nodes:

```
[m] /ieciieee60802-subscribed-notifications/subscriptions/max-
subscriptions
```

```
[m] /ieciieee60802-subscribed-notifications/subscriptions/max-on-
change-subscription-leaves
```

```
[m] /ieciieee60802-subscribed-notifications/subscriptions/max-periodic-
subscription-leaves
```

```
[m] /ieciieee60802-subscribed-notifications/subscriptions/max-periodic-
subscription-interval
```

IA-stations shall support the `ieciieee60802-proxy` YANG module according to this document with the following nodes:

```
[c] /ieciieee60802-proxy/proxy/max-num-proxy-targets
```

```
[c] /ieciieee60802-proxy/proxy/proxy-types
```

```
[o] /ieciieee60802-proxy/proxy/proxy-targets
```

IA-stations shall support the `ieciieee60802-proxy-target` YANG module according to this document with the following nodes:

```
[c] /ieciieee60802-proxy-target/proxy-target/required-proxy-type
```

6.4.9.2.5.12 NETCONF server

IA-stations shall support the `ietf-netconf-server` YANG module according to draft-ietf-netconf-netconf-client-server, 3.1.1, with the following features:

- `tls-call-home`, and
- `central-netconf-server-supported`.

IA-stations shall support the `ietf-netconf-server` YANG module according to draft-ietf-netconf-netconf-client-server, 3.3, with the following nodes:

```
[o] /ietf-netconf-server/netconf-server/listen/idle-timeout
```

```
[o] /ietf-netconf-server/netconf-server/listen/endpoint/name
```

```
[o] /ietf-netconf-server/netconf-
server/listen/endpoint/transport/tls/netconf-server-parameters
```

```
[o] /ietf-netconf-server/netconf-
server/listen/endpoint/transport/tls/tls-server-parameters
```

```
[o] /ietf-netconf-server/netconf-server/call-home/netconf-client/name
```

```
[o] /ietf-netconf-server/netconf-server/call-home/netconf-
client/endpoints/endpoint/name
```

```
[o] /ietf-netconf-server/netconf-server/call-home/netconf-
client/endpoints/endpoint/transport/tls/netconf-server-parameters
```

```
[o] /ietf-netconf-server/netconf-server/call-home/netconf-
client/endpoints/endpoint/transport/tls/tls-server-parameters
```

6.4.9.2.5.13 Flow Meter Management

IA-stations which incorporate a bridge component shall support the `ieee802-dot1q-stream-filters-gates` YANG module according to IEEE Std 802.1Qcz-2023 as amended by IEEE Std 802.1Qcw-2023 with the following nodes:

```
[o] /ieee802-dot1q-bridge/bridges/bridge/component/stream-
filters/stream-filter-instance-table/stream-filter-instance-id

[o] /ieee802-dot1q-bridge/bridges/bridge/component/stream-
filters/stream-filter-instance-table/stream-handle

[o] /ieee802-dot1q-bridge/bridges/bridge/component/stream-
filters/stream-filter-instance-table/flow-meter-ref

[o] /ieee802-dot1q-bridge/bridges/bridge/component/stream-
filters/stream-filter-instance-table/flow-meter-enable

[m] /ieee802-dot1q-bridge/bridges/bridge/component/stream-filters/max-
stream-filter-instances

[o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/stream-
gates/stream-gate-instance-table

[m] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/stream-
gates/max-stream-gate-instances

[m] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/stream-
gates/supported-list-max
```

IA-stations which incorporate a bridge component shall support the `ieee802-dot1cb-stream-identification` YANG module according to IEEE Std 802.1CBcv-2021 as amended by IEEE Std 802.1CBdb-2021 with the following nodes:

```
[o] /ieee802-dot1cb-stream-identification/stream-identity/index

[o] /ieee802-dot1cb-stream-identification/stream-identity/handle

[o] /ieee802-dot1cb-stream-identification/stream-identity/out-
facing/input-port

[o] /ieee802-dot1cb-stream-identification/stream-
identity/parameters/mask-and-match-stream-identification/destination-
mac-mask

[o] /ieee802-dot1cb-stream-identification/stream-
identity/parameters/mask-and-match-stream-identification/destination-
mac-match
```

NOTE For example, an implementation can contain one flow meter for broadcast traffic, one flow meter for multicast traffic and one flow meter for unicast traffic. This example can be realized using 49 mask and match stream identification functions per Port and 49 associated stream filter instances (one-to-one relationship) that identify and map unicast (1 stream identification function for identifying all frame with the group address bit cleared in the DA), broadcast (1 stream identification function for identifying all frames with all DA bits set) and multicast (47 stream identification function with the DA group address bit set and one other DA bit cleared for all 47 combinations).

IA-stations which incorporate a bridge component shall support the `ieee802-dot1q-psfp-bridge` YANG module according to IEEE Std 802.1Qcw-2023 with the following nodes:

```
[o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
meters/flow-meter-instance-table/flow-meter-instance-id

[o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
meters/flow-meter-instance-table/committed-information-rate
```

[o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-meters/flow-meter-instance-table/committed-burst-size

[o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-meters/flow-meter-instance-table/excess-information-rate

[o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-meters/flow-meter-instance-table/excess-burst-size

[o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-meters/flow-meter-instance-table/coupling-flag

[o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-meters/flow-meter-instance-table/color-mode

[o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-meters/flow-meter-instance-table/drop-on-yellow

[m] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-meters/max-flow-meter-instances

6.4.9.2.5.14 RSTP/MSTP

IA-stations shall support the ieee802-dot1q-rstp-bridge YANG module according to IEEE Std 802.1Qdy™-2025 with the following nodes.

[o] /ieee802-dot1q-rstp-bridge/bridges/bridge/rstp

[o] /ieee802-dot1q-rstp-bridge/interfaces/interface/bridge-port/rstp

IA-stations shall support the ieee802-dot1q-mstp-bridge YANG module according to IEEE Std 802.1Qdy-2025 with the following nodes.

[o] /ieee802-dot1q-mstp-bridge/bridges/bridge/component/bridge-mst/bridge-mstp

[o] /ieee802-dot1q-mstp-bridge/interfaces/interface/bridge-port/port-mstp

6.4.9.3 Optional YANG data models, features, and nodes

6.4.9.3.1 General

The following YANG modules, features and nodes shall be supported by IA-stations if the functionality they describe is included.

6.4.9.3.2 Scheduled traffic

IA-stations supporting the enhancements for scheduled traffic shall support the ieee802-dot1q-sched-bridge YANG module according to IEEE Std 802.1Qcw-2023 with the following feature: scheduled-traffic.

IA-stations supporting the enhancements for scheduled traffic shall support the ieee802-dot1q-sched-bridge YANG module according to IEEE Std 802.1Qcw-2023 with the following nodes:

[o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-parameter-table/queue-max-sdu-table

[o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-parameter-table/gate-enabled

```

[o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/admin-gate-states

[o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/oper-gate-states

[o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/admin-control-list

[o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/oper-control-list

[o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/admin-cycle-time

[o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/oper-cycle-time

[o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/admin-cycle-time-extension

[o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/oper-cycle-time-extension

[o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/admin-base-time

[o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/oper-base-time

[o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/config-change

[o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/config-change-time

[o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/tick-granularity

[o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/current-time

[o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/config-pending

[o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/config-change-error

[c] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/supported-list-max

[c] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/supported-cycle-max

[c] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/supported-interval-max

```

6.4.9.3.3 IEC/IEEE 60802 YANG modules

IA-stations that support enhancements for scheduled traffic shall support the `iecieeee60802-sched-bridge` YANG module according to this document with the following nodes:

```

[c] /iecieeee60802-sched-bridge/interfaces/interface/bridge-port/gate-
parameter-table/min-gating-times

```

6.4.9.3.4 Frame preemption

IA-stations supporting frame preemption according to IEEE Std 802.1Q-2022, 5.4.1, ae), shall support the `ieee802-dot1q-preemption-bridge` YANG module according to IEEE Std 802.1Qcw-2023 with the following feature: `frame-preemption`.

IA-stations supporting frame preemption according to IEEE Std 802.1Q-2022, 5.4.1, ae), shall support the `ieee802-dot1q-preemption-bridge` YANG module according to IEEE Std 802.1Qcw-2023 with the following nodes:

```
[o] ieee802-dot1q-preemption-bridge/interfaces/interface/bridge-
port/frame-preemption-parameters/frame-preemption-status-table
```

```
[o] ieee802-dot1q-preemption-bridge/interfaces/interface/bridge-
port/frame-preemption-parameters/preemption-active
```

6.4.9.3.5 Credit-based shaper

IA-stations supporting the credit-based shaper according to IEEE Std 802.1Q-2022, 8.6.8.2, shall support the `ieee802-dot1q-cbsa-bridge` YANG module according to IEEE Std 802.1Qdx™-2024 with the following nodes.

```
[c] /ieee802-dot1q-cbsa-bridge/interfaces/interface/bridge-
port/cbsa/cbsa-parameter-table/traffic-class
```

```
[c] /ieee802-dot1q-cbsa-bridge/interfaces/interface/bridge-
port/cbsa/cbsa-parameter-table/admin-idle-slope
```

```
[c] /ieee802-dot1q-cbsa-bridge/interfaces/interface/bridge-
port/cbsa/cbsa-parameter-table/oper-idle-slope
```

6.4.9.3.6 FRER

IA-stations supporting FRER according to 5.10.1 item b) or item c), shall support the `ieee802-dot1cb-stream-identification` and `ieee802-dot1cb-frer` YANG modules according to IEEE Std 802.1CBcv-2021 with the following nodes:

```
[o] /ieee802-dot1cb-stream-identification/stream-identity/index
```

```
[o] /ieee802-dot1cb-stream-identification/stream-identity/handle
```

```
[o] /ieee802-dot1cb-stream-identification /stream-identity/out-
facing/input-port
```

```
[o] /ieee802-dot1cb-stream-identification /stream-identity/out-
facing/output-port
```

```
[o] /ieee802-dot1cb-stream-identification /stream-
identity/parameters/null-stream-identification
```

```
[o] /ieee802-dot1cb-frer/frer/sequence-generation/index
```

```
[o] /ieee802-dot1cb-frer/frer/sequence-generation/stream
```

```
[o] /ieee802-dot1cb-frer/frer/sequence-generation/direction-out-facing
```

```
[o] /ieee802-dot1cb-frer/frer/sequence-recovery/index
```

```
[o] /ieee802-dot1cb-frer/frer/sequence-recovery/stream
```

```
[o] /ieee802-dot1cb-frer/frer/sequence-recovery/port
```



```
[o] /ieee802-dot1cb-frer/frer/sequence-recovery/direction-out-facing
[o] /ieee802-dot1cb-frer/frer/sequence-recovery/algorithm/vector
[o] /ieee802-dot1cb-frer/frer/sequence-identification/port
[o] /ieee802-dot1cb-frer/frer/sequence-identification/direction-out-facing
[o] /ieee802-dot1cb-frer/frer/sequence-identification/stream
[o] /ieee802-dot1cb-frer/frer/sequence-identification/encapsulation/r-tag
[o] /ieee802-dot1cb-frer/frer/stream-split
```

6.4.9.4 CUC/CNC YANG

6.4.9.4.1 NETCONF Client

IA-stations with CNC or CUC functionality shall support the ietf-netconf-client YANG module according to draft-ietf-netconf-netconf-client-server, 2.1.1, with the following features:

- tls-initiate,
- tls-listen, and
- central-netconf-client-supported.

6.4.9.4.2 YANG module for CNC-Config

IA-stations with CNC or CUC functionality shall support the ieee802-dot1q-cnc-config YANG module according to IEEE Std 802.1Qdj-2024 with the node:

```
[o] /ieee802-dot1q-cnc-config/cnc-config
```

6.4.10 YANG Data Model

6.4.10.1 General

Subclause 6.4.10 specifies the YANG data model for IA-stations. YANG (IETF RFC 7950) is a data modelling language used to model configuration data and state data for remote network management protocols. The selected YANG-based remote network management protocol is NETCONF (IETF RFC 6241). A YANG module specifies the organization and rules for the management data, and a mapping from YANG to the specific encoding enables the data to be understood correctly by both client (e.g. network manager) and server (e.g. IA-stations).

6.4.10.2 YANG framework

The core of the YANG module for IEC/IEEE 60802 IA-stations consists of YANG "augment" statements, used to add members to the tree of existing YANG modules plus one new module for IEC/IEEE 60802 specific objects.

6.4.10.3 IEC/IEEE 60802 Specific managed objects

6.4.10.3.1 General

Subclause 6.4.10.3 defines the set of managed objects, and their functionality, that provides additional information about an IA-station that is required by a CNC to calculate network configurations.

IEC/IEEE 60802 specific managed objects are specified:

- per Ethernet interface, i.e. external port, in 6.4.10.3.2,
- per end station component internal or external port in 6.4.10.3.3 and 6.4.10.3.4,
- per bridge component internal or external port in 6.4.10.3.4,
- per end station component in 6.4.10.3.5 and 6.4.10.3.7,
- per bridge component in 6.4.10.3.6 and 6.4.10.3.7, and
- per IA-station in 6.4.10.3.8.

IEC/IEEE 60802 specific managed objects for CNC entities are specified in 6.4.10.3.9.

6.4.10.3.2 IEC/IEEE 60802 managed objects per Ethernet interface

6.4.10.3.2.1 supportedMauTypes

The list of supported MAU Types including the data:

a) mauType

The value is the supported MAU Type derived from the list position of the corresponding dot3MauType shall be as listed in IETF RFC 4836, Clause 5.

b) preemptionSupported

The Boolean value indicates if preemption is supported by the MAU Type.

NOTE The operational MAU Type of an Ethernet interface is provided as leaf operational-mau-type of the ieee802-ethernet-Ildp YANG module. The operational MAU Type is included in the supportedMauTypes list.

6.4.10.3.3 IEC/IEEE 60802 managed objects per end station component port

6.4.10.3.3.1 worstCasePacketGap

The value is the worst case maximum inter-packet gap between consecutive frames in a traffic burst expressed in bit-times.

NOTE Minimum interPacketGap is defined in IEEE Std 802.3-2022, 1.4.362. The worst-case-packet-gap is never less than the minimum interPacketGap.

6.4.10.3.3.2 maxBurstFrames

The value is the maximum number of frames that can be sent with minimal inter packet gap.

6.4.10.3.3.3 maxBurstBytes

The value is the maximum number of octets that can be sent with minimal inter packet gap.

6.4.10.3.3.4 committedDataRates

The list of committed data rates per traffic class and supported line speed including the data:

a) committedInformationRate

The value is the bandwidth limit in kbit/s.

b) committedBurstSize

The value is the burst size limit in octets.

6.4.10.3.4 IEC/IEEE 60802 managed objects per bridge or end station component port**6.4.10.3.4.1 transmissionSelectionAlgorithm**

The list of supported transmission selection algorithms according to IEEE Std 802.1Q-2022, 8.6.8, per traffic class.

6.4.10.3.4.2 supportedResourcePools

The list of supported buffer resource pools including the data:

- a) resourcePoolName
The value is the name of a resource pool.
- b) coveredTimeInterval
The value specifies the covered buffering time given as rational number of seconds for the highest supported link speed.
- c) resourcePoolTrafficClasses
The list of the traffic classes to be served by the resource pool.

6.4.10.3.4.3 minGatingTimes

The list of minimum gating times per supported line speed including the data:

- a) minCycleTime
The value is the minimum value supported by this port of the AdminCycleTime and OperCycleTime parameters given as rational number of seconds.
- b) minIntervalTime
The value is the minimum value supported by this port of the TimeIntervalValue parameter in nanoseconds.

6.4.10.3.5 IEC/IEEE 60802 managed objects per end station component**6.4.10.3.5.1 frerSupported**

The value indicates if FRER is supported.

6.4.10.3.5.2 maxRedundantStreams

The value is the maximum number of supported redundant streams.

6.4.10.3.6 IEC/IEEE 60802 managed objects per bridge component**6.4.10.3.6.1 delayVariance**

The value indicates variance in delay depending upon the use of a singleValue or multipleValues (see 6.4.10.3.6.2).

6.4.10.3.6.2 delayTimes

The list of minimum and maximum frame length independent and frame length dependent delay time values of frames as they pass through a bridge component. These values are given:

- per supported MAU Type pair and traffic class, if delayVariance is singleValue, or
- per port pair with supported MAU Types and traffic class, if delayVariance is multipleValues.

The list includes the data:

a) independentDelayMin

The value is the minimum delay portion that is independent of frame length according to IEEE Std 802.1Q-2022, 12.32.1.1.

b) independentDelayMax

The value is the maximum delay portion that is independent of frame length according to IEEE Std 802.1Q-2022, 12.32.1.1.

c) dependentDelayMin

The value is the minimum delay portion that is dependent on frame length according to IEEE Std 802.1Q-2022, 12.32.1.2.

d) dependentDelayMax

The value is the maximum delay portion that is dependent on frame length according to IEEE Std 802.1Q-2022, 12.32.1.2.

6.4.10.3.6.3 latency

The value is the PHY latency expressed in ns.

6.4.10.3.6.4 latencyVariation

The value is the PHY latency variation expressed in ns.

6.4.10.3.7 IEC/IEEE 60802 managed objects per bridge or end station component

6.4.10.3.7.1 maxFids

The value is the maximum number of supported FIDs.

6.4.10.3.7.2 maxFdbEntries

The list of the maximum number of static (6.4.10.3.7.3) and dynamic (6.4.10.3.7.4) FDB entries per FDB.

6.4.10.3.7.3 maxStaticFdbEntries

The value is the maximum number of static FDB entries.

6.4.10.3.7.4 maxDynamicFdbEntries

The value is the maximum number of dynamic FDB entries.

6.4.10.3.7.5 maxPtpInstances

The value is the maximum number of supported PTP Instances.

6.4.10.3.7.6 maxHotStandbySystems

The value is the maximum number of supported HotStandbySystem entities (see IEEE Std 802.1ASdm-2024).

6.4.10.3.7.7 clockList

The list of supported application clock entities including the data:

- a) clockIdentity
The clock identity of the application clock.
- b) clockTarget
The Boolean value indicates if the application clock is a clock target (TRUE) or clock source (FALSE).
- c) arbSupported
The Boolean value indicates if the application clock supports the ARB timescale.
- d) ptpSupported
The Boolean value indicates if the application clock supports the PTP timescale.
- e) hotStandbySupported
The Boolean value indicates if the application clock supports hot standby.
- f) attachedPtpInstance
The value is a reference to the PTP or hot standby Instance, that is attached to the application clock.
- g) isSynced
The Boolean value indicates if the application clock is either synchronized to the attached PTP Instance (TRUE) or to an internal/external ClockSource (FALSE).

6.4.10.3.8 IEC/IEEE 60802 managed objects per IA-station**6.4.10.3.8.1 maxSubscriptions**

The value is the maximum number of supported NETCONF Server subscriptions.

6.4.10.3.8.2 maxOnChangeSubscriptionLeaves

The value is the maximum number of supported leaves for NETCONF Server on-change subscriptions according to IETF RFC 8641.

6.4.10.3.8.3 maxPeriodicSubscriptionLeaves

The value is the maximum number of supported leaves for NETCONF Server periodic subscriptions according to IETF RFC 8641.

6.4.10.3.8.4 minPeriodicSubscriptionInterval

The value is the minimum periodic subscription interval in centiseconds (0,01 seconds) for NETCONF Server periodic subscriptions according to IETF RFC 8641.

6.4.10.3.8.5 capabilityLLDP

This Boolean value indicates if LLDP is supported.

6.4.10.3.8.6 capabilityTimesync

This Boolean value indicates if Timesync is supported.

6.4.10.3.8.7 capabilityKeystore

This Boolean value indicates if Keystore is supported.

6.4.10.3.8.8 capabilityNACM

This Boolean value indicates if NACM is supported.

6.4.10.3.8.9 capabilityTruststore

This Boolean value indicates if Truststore is supported.

6.4.10.3.8.10 capabilityYangLibrary

This Boolean value indicates if YANG library is supported.

6.4.10.3.8.11 capabilityYangPush

This Boolean value indicates if Yang Push is supported.

6.4.10.3.8.12 capabilityYangNotifications

This Boolean value indicates if YANG notifications is supported.

6.4.10.3.8.13 capabilityNetconfMonitoring

This Boolean value indicates if NETCONF Monitoring is supported.

6.4.10.3.8.14 capabilityNetconfClient

This Boolean value indicates if NETCONF client is supported.

6.4.10.3.8.15 capabilityCncConfig

This Boolean value indicates if CNC configuration is supported.

6.4.10.3.8.16 capabilitySchedTraffic

This Boolean value indicates if scheduled traffic is supported.

6.4.10.3.8.17 capabilityFramePreemption

This Boolean value indicates if frame preemption is supported.

6.4.10.3.9 IEC/IEEE 60802 managed objects per proxy station**6.4.10.3.9.1 maxNumProxyTargets**

The value indicates how many proxy targets can be handled by this proxy.

6.4.10.3.9.2 proxyType

This value indicates OID of proxy targets handled by this proxy.

6.4.10.3.9.3 device

This value indicates the identification of the device.

6.4.10.3.9.4 deviceType

This value indicates the type of the device.

6.4.10.3.9.5 proxyAddr

This value indicates the ip-address that is used by the CNC to configure the proxy target.

6.4.10.3.9.6 port

This value indicates the port-number that is used by the CNC to configure the proxy target.

6.4.10.3.9.7 deviceAddr

This value indicates the ip-address that is used by the proxy to address the proxy target.

6.4.10.3.9.8 macAddr

This value indicates the mac-address that is used by the proxy to address the proxy target.

6.4.10.3.10 IEC/IEEE 60802 managed objects per proxy target**6.4.10.3.10.1 proxyType**

This value indicates OID of proxy that this proxy target requires.

6.4.10.3.11 IEC/IEEE 60802 managed objects for CNC entities**6.4.10.3.11.1 maxConfigurationDomains**

The value is the maximum number of supported Configuration Domains.

6.4.10.3.11.2 maxCUCs

The value is the maximum number of supported CUC entities.

6.4.10.3.11.3 maxIAstations

The value is the maximum number of supported IA-stations.

6.4.10.3.11.4 maxNetworkDiameter

The value is the maximum supported network diameter.

6.4.10.3.11.5 maxStreams

The value is the maximum number of supported streams.

6.4.10.3.11.6 maxNumSeamlessTrees

The value is the maximum number of trees supported for seamless redundancy of a stream.

6.4.10.3.11.7 hotStandbySupported

The Boolean value indicates if hot standby is supported.

6.4.10.4 RPCs and actions specific to this document**6.4.10.4.1 RPC ia-factory-reset****6.4.10.4.1.1 General**

In contrast to the original factory-reset RPC in IETF RFC 8808, this RPC puts the device into a state where a subsequent configuration by a CNC component results in a functioning IA-station according to this document. Depending on the factory default configuration, after being reset, the device can become unreachable on the network.

6.4.10.4.1.2 Input

There is no input.

6.4.10.4.1.3 Output

There is no output.

6.4.10.4.2 Action add-streams**6.4.10.4.2.1 General**

This Action requests a CNC to add a list of streams.

6.4.10.4.2.2 Input

The RPC requires the following input parameters.

- a) Cuid – The ID of the CUC for which the streams are to be added.
- b) StreamId – The Stream ID is a unique identifier of a Stream request and corresponding configuration.
- c) Container Talker – The Talker container contains:
 - 1) Talker's behavior for Stream (how/when transmitted),
 - 2) Talker's requirements from the network, and
 - 3) TSN capabilities of the Talker's interface(s).
- d) List Listener – Each Listener list entry contains:
 - 1) Listener's requirements from the network, and
 - 2) TSN capabilities of the Listener's interface(s).

6.4.10.4.2.3 Output

Result – Status information indicating if Stream addition has been successful.

6.4.10.4.3 Action remove-listener**6.4.10.4.3.1 General**

This Action removes listeners from a stream.

6.4.10.4.3.2 Input

List Listener – A list of indices of listeners to be removed from a stream.

6.4.10.4.3.3 Output

Result – Status information indicating if Stream addition has been successful.

6.4.10.5 IEC/IEEE 60802 YANG data models

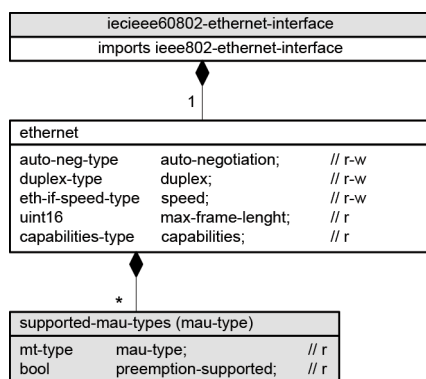
A representation similar to UML® is used to provide an overview of the hierarchy of the IEC/IEEE 60802 YANG data model.

A UML-like representation of the management model is provided in Figure 29, Figure 30, Figure 31, Figure 32, Figure 33, Figure 34, Figure 35, Figure 36, and Figure 37. The purpose of a UML-like diagram is to express the model design in a concise manner. The structure of the UML-like representation shows the name of the object followed by a list of properties for the object. The properties indicate their type and accessibility. It should be noted that UML-like representation is meant to express simplified semantics for the properties. It is not meant to provide the specific datatype used to encode the object in either MIB or YANG.

NOTE OMG® UML® 2.5 conventions together with C++ language constructs are used as a representation to convey model structure and relationships.

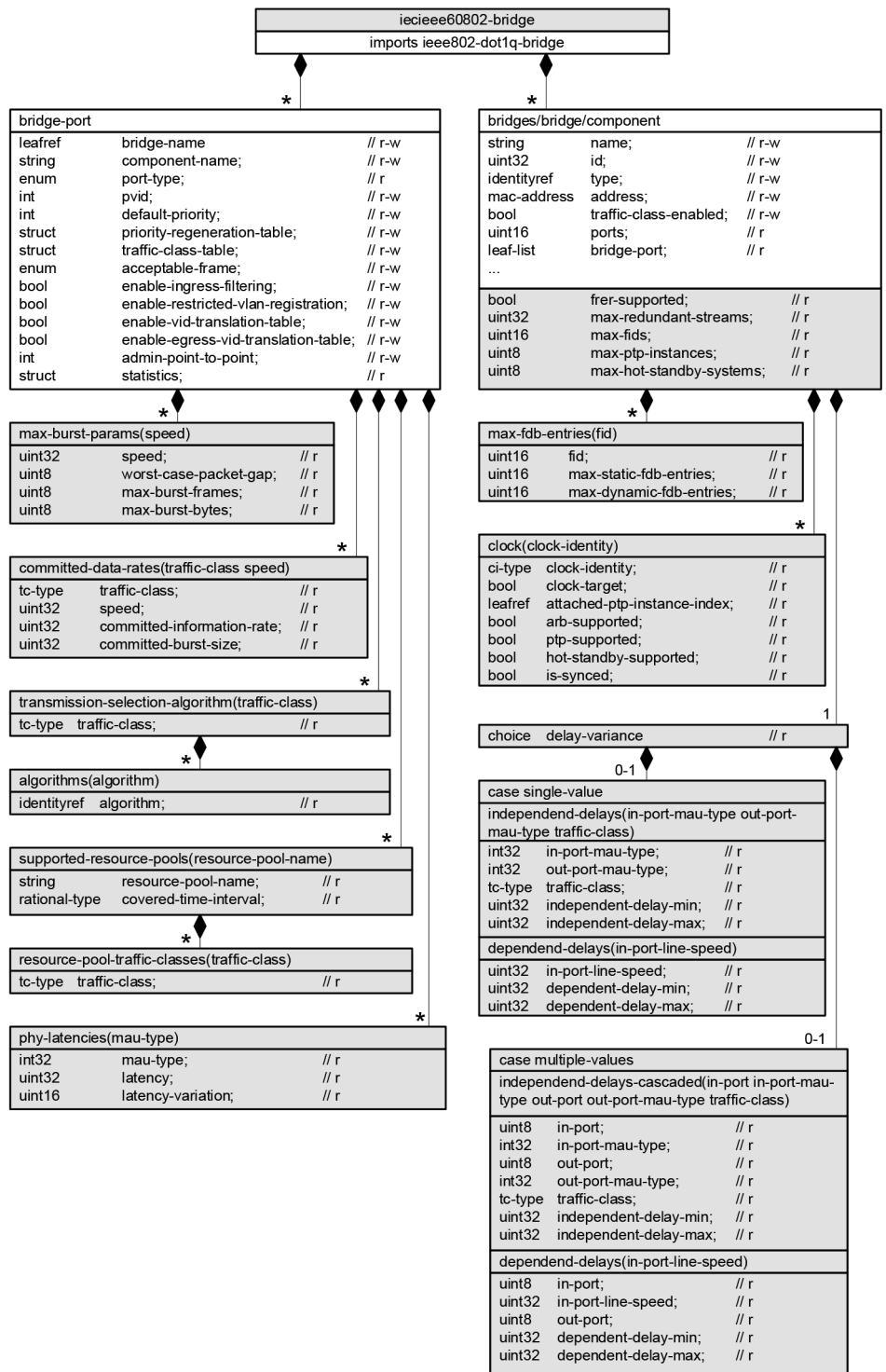
For all UML® figures, data that is imported from original modules is shown in white, and data that is augmented according to IEC/IEEE 60802 is shown in grey.

Figure 29, Figure 30, Figure 31, Figure 32, Figure 33, Figure 34, Figure 35, Figure 36, and Figure 37 provide an overview of the IEC/IEEE 60802 augmentations.



IEC/IEEE

Figure 29 – Module iecieeee60802-ethernet-interface



IEC/IEEE

Figure 30 – Module iecieee60802-bridge

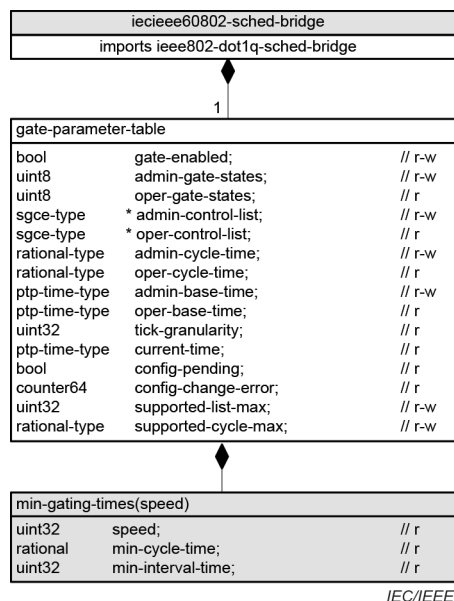


Figure 31 – Module iecieee60802-dot1-sched-bridge

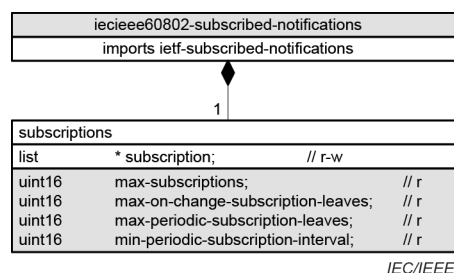


Figure 32 – Module iecieee60802-subscribed-notifications

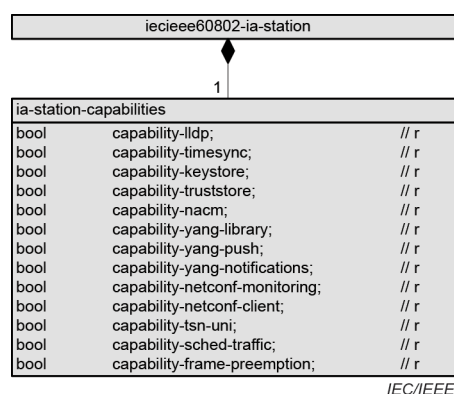


Figure 33 – Module iecieee60802-ia-station

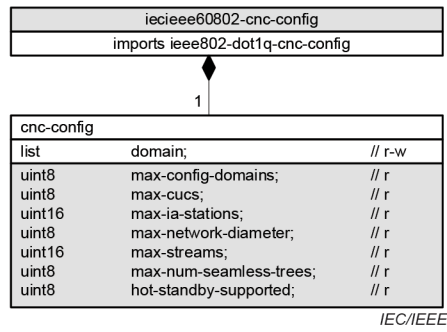


Figure 34 – Module iecieeee60802-cnc-config

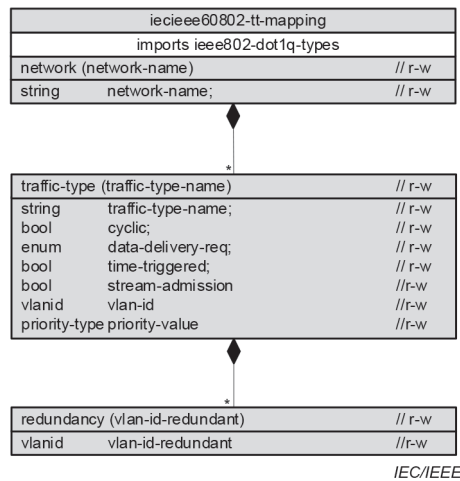


Figure 35 – Module iecieeee60802-tt-mapping

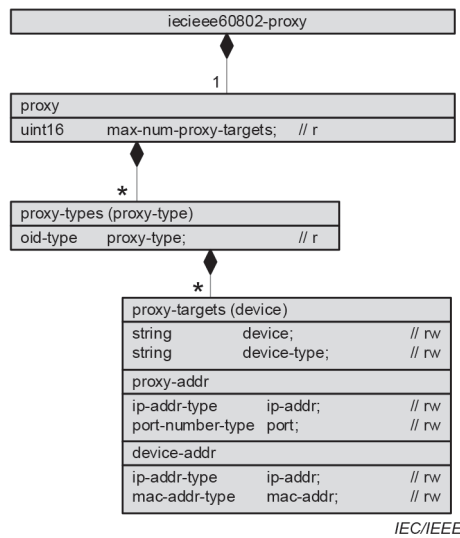
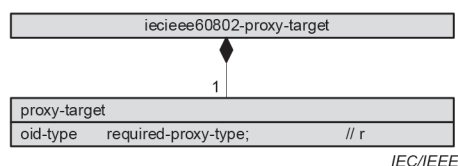


Figure 36 – Module iecieeee60802-proxy

**Figure 37 – iecieeee60802-proxy-target****6.4.10.6 Structure of IEC/IEEE 60802 YANG data models**

The YANG data models specified by this standard use the YANG modules summarized in Table 17.

In the YANG module definitions, if any discrepancy between the "description" text and the corresponding definition in any other part of this standard occurs, the definitions outside Clause 6 take precedence.

Table 17 – Summary of the YANG modules

Module	Description
ieee802-ethernet-interface	This module contains YANG definitions for configuring IEEE Std 802.3 Ethernet Interfaces.
ietf-interfaces	This module contains a collection of YANG definitions for managing network interfaces.
ieciieee60802-ethernet-interface	This module augments ieee802-ethernet-interface.
ieee802-types	This module contains a collection of generally useful derived data types for IEEE YANG data models.
ieee802-dot1q-bridge	This module describes the bridge configuration model for IEEE 802.1Q Bridges.
ieee802-dot1q-types	This module contains common types used within dot1q-bridge modules.
ieciieee60802-bridge	This module augments ieee802-dot1q-bridge.
ieee802-dot1q-sched-bridge	This module provides for management of IEEE Std 802.1Q Bridges that support Scheduled Traffic Enhancements.
ieciieee60802-dot1q-sched-bridge	This module augments ieee802-dot1q-sched-bridge.
ieee802-dot1cb-frer	This module provides management objects that control the frame replication and elimination from IEEE Std 802.1CB-2017.
ieee1588-ptp-tt	This module defines a data model for the configuration and state of IEEE Std 1588 clocks.
ietf-netconf-acm	This module provides management for the Network Configuration Access Control Model.
ieee802-dot1q-cnc-config	This module provides the Time-Sensitive Networking (TSN) User/Network Interface (UNI) for the exchange of information between CUC and CNC that are required to configure TSN Streams in a TSN network.
ieciieee60802-cnc-config	This module augments ieee802-dot1q-cnc-config.
ieciieee60802-ia-station	This module provides read-only information about the capabilities and RPCs for IEC/IEEE 60802 IA-stations.
ietf-subscribed-notifications	This module defines a YANG data model for subscribing to event records and receiving matching content in notification messages.
ieciieee60802-subscribed-notifications	This module augments ietf-subscribed-notifications.
ieciieee60802-tt-mapping	This module provides information on characteristics of traffic classes configured in a network to allow applications to map application traffic classes to them
ieciieee60802-proxy.yang	This module contains configuration and capability information for IEC/IEEE 60802 proxy stations
ieciieee60802-proxy-target.yang	This module contains capability information for IEC/IEEE 60802 proxy targets

6.4.10.7 YANG schema tree definitions

6.4.10.7.1 General

The schema tree is provided as an overview of the YANG modules. The symbols and their meaning are specified in YANG Tree Diagrams (IETF RFC 8340).

6.4.10.7.2 Module ieciieee60802-ethernet-interface

module: ieciieee60802-ethernet-interface

```

augment /if:interfaces/if:interface/eth-if:ethernet:
  +-ro supported-mau-types* [mau-type]
    +-ro mau-type                int32
    +-ro preemption-supported?   boolean

```

6.4.10.7.3 Module iecieee60802-bridge

module: iecieee60802-bridge

```

augment /if:interfaces/if:interface/bridge:bridge-port:
  +-ro max-burst-params* [speed]
  | +-ro speed uint32
  | +-ro worst-case-packet-gap? uint8
  | +-ro max-burst-frames? uint8
  | +-ro max-burst-bytes? uint8
  +-ro committed-data-rates* [traffic-class speed]
  | +-ro traffic-class dot1q-types:traffic-class-type
  | +-ro speed uint32
  | +-ro committed-information-rate? uint32
  | +-ro committed-burst-size? uint32
  +-ro transmission-selection-algorithm* [traffic-class]
  | +-ro traffic-class dot1q-types:traffic-class-type
  | +-ro algorithms* [algorithm]
  | | +-ro algorithm identityref
  +-ro supported-resource-pools* [resource-pool-name]
  | +-ro resource-pool-name string
  | +-ro covered-time-interval
  | | +-ro numerator? uint32
  | | +-ro denominator? uint32
  | +-ro resource-pool-traffic-classes* [traffic-class]
  | | +-ro traffic-class dot1q-types:traffic-class-type
  +-ro phy-latencies* [mau-type]
  | +-ro mau-type int32
  | +-ro latency? uint32
  | +-ro latency-variation? uint16
augment /bridge:bridges/bridge:bridge/bridge:component:
  +-ro frer-supported? boolean
  +-ro max-redundant-streams? uint32
  +-ro max-fids? uint16
  +-ro max-fdb-entries* [fid]
  | +-ro fid uint16
  | +-ro max-static-fdb-entries? uint16
  | +-ro max-dynamic-fdb-entries? uint16
  +-ro (delay-variance)?
  | +--:(single-value)
  | | +-ro independent-delays* [in-port-mau-type out-port-mau-type
traffic-class]
  | | | +-ro in-port-mau-type int32
  | | | +-ro out-port-mau-type int32
  | | | +-ro traffic-class dot1q-types:traffic-class-type
  | | | +-ro independent-delay-min? uint32
  | | | +-ro independent-delay-max? uint32
  | | +-ro dependent-delays* [in-port-line-speed]
  | | | +-ro in-port-line-speed uint32
  | | | +-ro dependent-delay-min? uint32
  | | | +-ro dependent-delay-max? uint32
  | | +--:(multiple-values)
  | | +-ro independent-delays-cascaded* [in-port in-port-mau-type out-
port out-port-mau-type traffic-class]
  | | | +-ro in-port uint8
  | | | +-ro in-port-mau-type int32
  | | | +-ro out-port uint8
  | | | +-ro out-port-mau-type int32
  | | | +-ro traffic-class dot1q-types:traffic-class-type
  | | | +-ro independent-delay-min? uint32
  | | | +-ro independent-delay-max? uint32

```

```

|      +-ro dependent-delays-cascaded* [in-port in-port-line-speed out-
port]
|      +-ro in-port                uint8
|      +-ro in-port-line-speed      uint32
|      +-ro out-port                uint8
|      +-ro dependent-delay-min?    uint32
|      +-ro dependent-delay-max?    uint32
+-ro max-ptp-instances?            uint8
+-ro max-hot-standby-systems?      uint8
+-ro clock* [clock-identity]
|      +-ro clock-identity          ptp-tt:clock-identity
|      +-ro clock-target?          boolean
|      +-ro attached-ptp-instance-index? -> /ptp-
tt:ptp/instances/instance/instance-index
|      +-ro arb-supported?          boolean
|      +-ro ptp-supported?          boolean
|      +-ro hot-standby-supported?  boolean
|      +-ro is-synced?             boolean

```

6.4.10.7.4 Module iecieee60802-sched-bridge

```
module: iecieee60802-sched-bridge
```

```
augment /if:interfaces/if:interface/bridge:bridge-port/sched-bridge:gate-
parameter-table:
```

```

+-ro min-gating-times* [speed]
|      +-ro speed                uint32
|      +-ro min-cycle-time
|      |      +-u ieee802:rational-grouping
|      +-ro min-interval-time?  uint32

```

6.4.10.7.5 Module iecieee60802-cnc-config

```
module: iecieee60802-cnc-config
```

```
augment /dot1q-cnc-config:cnc-config:
```

```

+-ro max-config-domains?      uint8
+-ro max-cucs?                uint8
+-ro max-ia-stations?         uint16
+-ro max-network-diameter?    uint8
+-ro max-streams?             uint16
+-ro max-num-seamless-trees?  uint8
+-ro hot-standby-supported?    uint8
+-x add_streams
|      +-w input
|      |      +-w cuc-id?      string
|      |      +-w stream-list* [stream-id]
|      |      |      +-w stream-id    tsn-types:stream-id-type
|      |      |      +-w talker
|      |      |      |      +-w stream-rank
|      |      |      |      |      +-w rank?    uint8
|      |      |      |      |      +-w end-station-interfaces* [mac-address interface-name]
|      |      |      |      |      |      +-w mac-address    string
|      |      |      |      |      |      +-w interface-name  string
|      |      |      |      |      +-w data-frame-specification* [index]
|      |      |      |      |      |      +-w index          uint8
|      |      |      |      |      |      +-w (field)?
|      |      |      |      |      |      |      +-:(ieee802-mac-addresses)
|      |      |      |      |      |      |      |      +-w ieee802-mac-addresses
|      |      |      |      |      |      |      |      +-w destination-mac-address?  string
|      |      |      |      |      |      |      |      +-w source-mac-address?      string
|      |      |      |      |      +-:(ieee802-vlan-tag)
|      |      |      |      |      |      +-w ieee802-vlan-tag

```



```

|         |         |         |         +-w priority-code-point?    uint8
|         |         |         |         +-w vlan-id?                uint16
|         |         |         +-: (ipv4-tuple)
|         |         |         +-w ipv4-tuple
|         |         |         +-w source-ip-address?      inet:ipv4-
address
|         |         |         +-w destination-ip-address?  inet:ipv4-
address
|         |         |         +-w dscp?                  uint8
|         |         |         +-w protocol?              uint16
|         |         |         +-w source-port?           uint16
|         |         |         +-w destination-port?       uint16
|         |         +-: (ipv6-tuple)
|         |         +-w ipv6-tuple
|         |         +-w source-ip-address?                inet:ipv6-
address
|         |         +-w destination-ip-address?          inet:ipv6-
address
|         |         +-w dscp?                            uint8
|         |         +-w protocol?                        uint16
|         |         +-w source-port?                    uint16
|         |         +-w destination-port?               uint16
|         +-w traffic-specification
|         | +-w interval
|         | | +-w numerator?      uint32
|         | | +-w denominator?    uint32
|         | +-w max-frames-per-interval?  uint16
|         | +-w max-frame-size?          uint16
|         | +-w transmission-selection?   uint8
|         | +-w time-aware!
|         | +-w earliest-transmit-offset?  uint32
|         | +-w latest-transmit-offset?    uint32
|         | +-w jitter?                   uint32
|         +-w user-to-network-requirements
|         | +-w num-seamless-trees?      uint8
|         | +-w max-latency?             uint32
|         +-w interface-capabilities
|         | +-w vlan-tag-capable?        boolean
|         | +-w cb-stream-iden-type-list* uint32
|         | +-w cb-sequence-type-list*   uint32
|         +-w listener* [index]
|         | +-w index                    uint32
|         +-w end-station-interfaces* [mac-address interface-name]
|         | +-w mac-address              string
|         | +-w interface-name           string
|         +-w user-to-network-requirements
|         | +-w num-seamless-trees?      uint8
|         | +-w max-latency?             uint32
|         +-w interface-capabilities
|         | +-w vlan-tag-capable?        boolean
|         | +-w cb-stream-iden-type-list* uint32
|         | +-w cb-sequence-type-list*   uint32
+-rw output
  +-rw result?    boolean
augment /dot1q-cnc-config:cnc-config/dot1q-cnc-config:domain/dot1q-cnc-
config:cuc/dot1q-cnc-config:stream:
  +-x remove_listener
  +-w input
  | +-w listener* [index]
  | +-w index      uint32
  +-rw output
    +-rw result?    boolean

```

6.4.10.7.6 Module iecieee60802-ia-station

```

module: iecieee60802-ia-station
  +-ro ia-station-capabilities
    +-ro capability-lldp?                boolean
    +-ro capability-timesync?            boolean
    +-ro capability-keystore?            boolean
    +-ro capability-truststore?          boolean
    +-ro capability-nacm?                boolean
    +-ro capability-yang-library?        boolean
    +-ro capability-yang-push?          boolean
    +-ro capability-yang-notifications?  boolean
    +-ro capability-netconf-monitoring?  boolean
    +-ro capability-netconf-client?      boolean
    +-ro capability-cnc-config?          boolean
    +-ro capability-sched-traffic?       boolean
    +-ro capability-frame-preemption?    boolean

  rpcs:
    +--x ia-factory-reset

```

6.4.10.7.7 Module iecieee60802-subscribed-notifications

```

module: iecieee60802-subscribed-notifications

augment /sn:subscriptions:
  +-ro max-subscriptions?                uint16
  +-ro max-on-change-subscription-leaves? uint16
  +-ro max-periodic-subscription-leaves? uint16
  +-ro min-periodic-subscription-interval? uint16

```

6.4.10.7.8 Module iecieee60802-tt-mapping

```

module: iecieee60802-tt-mapping
  +-rw tt-mapping
    +-rw network* [network-name]
      +-rw network-name      string
    +-rw traffic-type* [traffic-type-name]
      +-rw traffic-type-name  string
      +-rw cyclic?            boolean
      +-rw data-delivery-req? enumeration
      +-rw time-triggered?    boolean
      +-rw stream-based?      boolean
      +-rw vlan-id?           dot1q-types:vlanid
      +-rw redundancy* [vlan-id-redundant]
        | +-rw vlan-id-redundant  dot1q-types:vlanid
      +-rw priority-value?      dot1q-types:priority-type

```

6.4.10.7.9 Module iecieee60802-proxy

```

module: iecieee60802-proxy
  +--rw proxy!
    +--ro max-num-proxy-targets?  uint16
    +--ro proxy-types* [proxy-type]
      +--ro proxy-type            string
    +--ro proxy-targets* [device]
      +--ro device                string
      +--ro device-type?          string
      +--ro proxy-addr
        | +--ro ip-addr?          inet:ip-address
        | +--ro port?             inet:port-number
      +--ro device-addr
        +--ro ip-addr?            inet:ip-address
        +--ro mac-addr?           ieee:mac-address

```

6.4.10.7.10 Module iecieee60802-proxy-target

```

module: iecieee60802-proxy-target
  +--rw proxy-target!
    +--ro required-proxy-type?   string

```

6.4.10.8 YANG modules⁵**6.4.10.8.1 Module iecieee60802-ethernet-interface**

```

module iecieee60802-ethernet-interface {
  yang-version 1.1;
  namespace
    "urn:ieee:std:60802:yang:iecieee60802-ethernet-interface";
  prefix ia-eth-if;

  import ieee802-ethernet-interface {
    prefix eth-if;
  }
  import ietf-interfaces {
    prefix if;
  }

  organization
    "IEEE 802.1 Working Group and IEC subcommittee 65C:
    Industrial networks, of IEC technical committee 65:
    Industrial-process measurement, control and automation";
  contact
    "WG-URL: http://ieee802.org/1/
    WG-EMail: stds-802-1-1@ieee.org

    Contact: IEEE 802.1 Working Group Chair
    Postal: C/O IEEE 802.1 Working Group
    IEEE Standards Association
    445 Hoes Lane
    Piscataway, NJ 08854
    USA

    E-mail: stds-802-1-chairs@ieee.org";
  description
    "Management objects that provide information about IEC/IEEE 60802
    IA-Stations as specified in IEC/IEEE 60802.

    Copyright (C) IEC/IEEE (2025).
    This version of this YANG module is part of IEC/IEEE 60802;
    see the standard itself for full legal notices.";

  revision 2025-03-12 {
    description "Published as part of IEC/IEEE 60802-2025.
    The following reference statement identifies each referenced
    IEEE Standard as updated by applicable amendments.";
    reference
      "IEC/IEEE 60802 TSN profile for industrial automation:
      IEC/IEEE 60802-2025.
      IEEE Std 802.1Q Bridges and Bridged Networks:
      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
      IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
      IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
  }
}

```

⁵ An ASCII version of each YANG module in this document can be obtained from the IEC SC 65C supporting documents website at <https://www.iec.ch/sc65c/supportingdocuments> or the IEEE 802.1 website at <https://1.ieee802.org/yang-modules/>.

```

}

augment "/if:interfaces/if:interface/eth-if:ethernet" {
  description
    "Augment IEEE Std 802.3 ethernet.";
  list supported-mau-types {
    key "mau-type";
    config false;
    description
      "Contains a list of supported MAU parameters.";
    leaf mau-type {
      type int32;
      config false;
      description
        "The value is the supported MAU Type derived from the list
        position of the corresponding dot3MauType as listed in
        Clause 5 of IETF RFC 4836.";
      reference
        "Item a) in 6.4.10.3.2.1 of IEC/IEEE 60802";
    }
    leaf preemption-supported {
      type boolean;
      config false;
      description
        "The Boolean value indicates if preemption is supported by
        the MAU Type.";
      reference
        "Item b) in 6.4.10.3.2.1 of IEC/IEEE 60802";
    }
  }
}
}

```

6.4.10.8.2 Module iecieee60802-bridge

```

module iecieee60802-bridge {
  yang-version 1.1;
  namespace "urn:ieee:std:60802:yang:ieciieee60802-bridge";
  prefix ia-bridge;

  import ieee802-types {
    prefix ieee802;
  }
  import ieee802-dot1q-bridge {
    prefix bridge;
  }
  import ietf-interfaces {
    prefix if;
  }
  import ieee802-dot1q-types {
    prefix dot1q-types;
  }
  import ieee1588-ptp-tt {
    prefix ptp-tt;
  }

  organization
    "IEEE 802.1 Working Group and IEC subcommittee 65C:
    Industrial networks, of IEC technical committee 65:
    Industrial-process measurement, control and automation";
  contact
    "WG-URL: http://ieee802.org/1/

```

WG-EMail: stds-802-1-1@ieee.org

Contact: IEEE 802.1 Working Group Chair
 Postal: C/O IEEE 802.1 Working Group
 IEEE Standards Association
 445 Hoes Lane
 Piscataway, NJ 08854
 USA

E-mail: stds-802-1-chairs@ieee.org";
 description

"Management objects that provide information about
 IEC/IEEE 60802 IA-Stations as specified in IEC/IEEE 60802.

Copyright (C) IEC/IEEE (2025).
 This version of this YANG module is part of IEC/IEEE 60802;
 see the standard itself for full legal notices.";

revision 2025-03-12 {

description

"Published as part of IEC/IEEE 60802-2025.
 The following reference statement identifies each referenced
 IEEE Standard as updated by applicable amendments.";

reference

"IEC/IEEE 60802 TSN profile for industrial automation:
 IEC/IEEE 60802-2025.
 IEEE Std 802.1Q Bridges and Bridged Networks:
 IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
 IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
 IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";

}

augment "/if:interfaces/if:interface/bridge:bridge-port" {

description

"Augment IEEE Std 802.1 bridge.";

list max-burst-params {

key "speed";

config false;

description

"The list of maximum burst parameters per supported line
 speed.";

leaf speed {

type uint32;

description

"This value is the line speed in Mbps.";

}

leaf worst-case-packet-gap {

type uint8;

config false;

description

"The value is the worst case maximum inter-packet gap
 between consecutive frames in a traffic burst expressed
 in bit-times.";

reference

"Item a) in 6.4.10.3.3.1 of IEC/IEEE 60802";

}

leaf max-burst-frames {

type uint8;

config false;

description

"The value is the maximum number of frames that can be sent
 with minimal inter packet gap.";

```

        reference
            "Item b) in 6.4.10.3.3.1 of IEC/IEEE 60802";
    }
    leaf max-burst-bytes {
        type uint8;
        config false;
        description
            "The value is the maximum number of octets that can be sent
            with minimal inter packet gap.";
        reference
            "Item c) in 6.4.10.3.3.1 of IEC/IEEE 60802";
    }
}
list committed-data-rates {
    key "traffic-class speed";
    config false;
    description
        "The list of committed data rates per traffic class and
        supported line speed.";
    leaf traffic-class {
        type dot1q-types:traffic-class-type;
        description
            "The traffic class of the entry (0..7).";
        reference
            "8.6.6 of IEEE Std 802.1Q";
    }
    leaf speed {
        type uint32;
        description
            "This value is the line speed in Mbps.";
    }
    leaf committed-information-rate {
        type uint32;
        config false;
        description
            "The value is the bandwidth limit in kbit/s.";
        reference
            "Item a) in 6.4.10.3.3.2 of IEC/IEEE 60802";
    }
    leaf committed-burst-size {
        type uint32;
        config false;
        description
            "The value is the burst size limit in bytes.";
        reference
            "Item b) in 6.4.10.3.3.2 of IEC/IEEE 60802";
    }
}
list transmission-selection-algorithm {
    key "traffic-class";
    config false;
    description
        "The list of supported transmission selection algorithms
        according to 8.6.8 of IEEE Std 802.1Q per traffic class.";
    leaf traffic-class {
        type dot1q-types:traffic-class-type;
        config false;
        description
            "Traffic class. (0..7)";
        reference
            "8.6.6 of IEEE Std 802.1Q";
    }
}

```

```

list algorithms {
  key "algorithm";
  config false;
  description
    "The list of supported transmission selection algorithms
    according to 8.6.8 of IEEE Std 802.1Q for this traffic
    class.";
  leaf algorithm {
    type identityref {
      base dot1q-types:transmission-selection-algorithm;
    }
    config false;
    description
      "Transmission selection algorithm";
    reference
      "8.6.8 of IEEE Std 802.1Q";
  }
}

list supported-resource-pools {
  key "resource-pool-name";
  config false;
  description
    "The list of supported buffer resource pools.";
  leaf resource-pool-name {
    type string;
    config false;
    description
      "The value is the name of a resource pool.";
    reference
      "Item a) in 6.4.10.3.4.2 of IEC/IEEE 60802";
  }
}

container covered-time-interval {
  config false;
  uses ieee802:rational-grouping;
  description
    "The value is the covered buffering time given as rational
    number of seconds for the highest supported link speed.";
  reference
    "Item b) in 6.4.10.3.4.2 of IEC/IEEE 60802";
}

list resource-pool-traffic-classes {
  key "traffic-class";
  config false;
  description
    "The list of the traffic classes to be served by the
    resource pool.";
  reference
    "Item c) in 6.4.10.3.4.2 of IEC/IEEE 60802";
  leaf traffic-class {
    type dot1q-types:traffic-class-type;
    description
      "The traffic class of the entry.";
    reference
      "8.6.6 of IEEE Std 802.1Q";
  }
}

list phy-latencies {
  key "mau-type";
  config false;
  description

```

```

    "Latency information for each MAU type";
  leaf mau-type {
    type int32;
    config false;
    description
      "The MAU type";
  }
  leaf latency {
    type uint32;
    config false;
    description
      "The PHY latency in ns";
    reference
      "6.4.10.3.6.3 of IEC/IEEE 60802";
  }
  leaf latency-variation {
    type uint16;
    config false;
    description
      "The PHY latency variation in ns";
    reference
      "6.4.10.3.6.4 of IEC/IEEE 60802";
  }
}
}

augment "/bridge:bridges/bridge:bridge/bridge:component" {
  description
    "Augment IEEE Std 802.1 bridge component.";
  leaf frer-supported {
    type boolean;
    config false;
    description
      "The Boolean value indicates if FRER is supported.";
    reference
      "6.4.10.3.5.1 of IEC/IEEE 60802";
  }
  leaf max-redundant-streams {
    type uint32;
    config false;
    description
      "The value is the maximum number of supported redundant
        streams.";
    reference
      "6.4.10.3.5.2 of IEC/IEEE 60802";
  }
  leaf max-fids {
    type uint16;
    config false;
    description
      "The value is the maximum number of supported FIDs.";
    reference
      "6.4.10.3.7.1 of IEC/IEEE 60802";
  }
  list max-fdb-entries {
    key "fid";
    config false;
    description
      "The list of the maximum number of static and dynamic
        FDB entries per FID.";
    reference
      "6.4.10.3.7.2 of IEC/IEEE 60802";
  }
}

```



```

leaf fid {
    type uint16;
    config false;
    description
        "The FID number";
}
leaf max-static-fdb-entries {
    type uint16;
    config false;
    description
        "The value is the maximum number of static FDB
        entries.";
    reference
        "6.4.10.3.7.3 of IEC/IEEE 60802";
}
leaf max-dynamic-fdb-entries {
    type uint16;
    config false;
    description
        "The value is the maximum number of dynamic FDB entries.";
    reference
        "6.4.10.3.7.4 of IEC/IEEE 60802";
}
}
choice delay-variance {
    config false;
    description
        "The value indicates variance in delay depending upon the use
        of a singleValue or multipleValues.";
    reference
        "6.4.10.3.6.1 of IEC/IEEE 60802";
    case single-value {
        list independent-delays {
            key "in-port-mau-type out-port-mau-type traffic-class";
            config false;
            description
                "The list of minimum and maximum frame length
                independent delay time values of frames as they pass
                through a bridge component.";
            reference
                "6.4.10.3.6.2 of IEC/IEEE 60802";
            leaf in-port-mau-type {
                type int32;
                config false;
                description
                    "The MAU type of the input port";
            }
            leaf out-port-mau-type {
                type int32;
                config false;
                description
                    "The MAU type of the input port";
            }
        }
        leaf traffic-class {
            type dot1q-types:traffic-class-type;
            config false;
            description
                "The traffic class of the entry.";
            reference
                "8.6.6 of IEEE Std 802.1Q";
        }
        leaf independent-delay-min {

```

```

    type uint32;
    config false;
    description
        "The value is the minimum delay portion that is
        independent of frame length according to 12.32.1.1.
        of IEEE 802.1Q";
    reference
        "Item a) in 6.4.10.3.6.2 of IEC/IEEE 60802";
}
leaf independent-delay-max {
    type uint32;
    config false;
    description
        "The value is the maximum delay portion that is
        independent of frame length according to 12.32.1.1.
        of IEEE 802.1Q";
    reference
        "Item b) in 6.4.10.3.6.2 of IEC/IEEE 60802";
}
}
list dependent-delays {
    key "in-port-line-speed";
    config false;
    description
        "The list of minimum and maximum frame length dependent
        delay time values of frames as they pass through a
        bridge component";
    reference
        "6.4.10.3.6.2 of IEC/IEEE 60802";
    leaf in-port-line-speed {
        type uint32;
        config false;
        description
            "This value is the line speed in Mbps.";
    }
    leaf dependent-delay-min {
        type uint32;
        config false;
        description
            "The value is the minimum delay portion that is
            dependent on frame length according to 12.32.1.2.
            of IEEE 802.1Q";
        reference
            "Item c) in 6.4.10.3.6.2 of IEC/IEEE 60802";
    }
    leaf dependent-delay-max {
        type uint32;
        config false;
        description
            "The value is the maximum delay portion that is
            dependent on frame length according to 12.32.1.2.
            of IEEE 802.1Q";
        reference
            "Item d) in 6.4.10.3.6.2 of IEC/IEEE 60802";
    }
}
}
}
case multiple-values {
    list independent-delays-cascaded {
        key "in-port in-port-mau-type out-port out-port-mau-type
        traffic-class";
        config false;
    }
}

```

```

description
    "The list of minimum and maximum frame length
    independent delay time values of frames as they pass
    through a bridge component.";
reference
    "6.4.10.3.6.2 of IEC/IEEE 60802";
leaf in-port {
    type uint8;
    config false;
    description
        "The port number of the input port";
}
leaf in-port-mau-type {
    type int32;
    config false;
    description
        "The MAU type of the input port";
}
leaf out-port {
    type uint8;
    config false;
    description
        "The port number of the output port";
}
leaf out-port-mau-type {
    type int32;
    config false;
    description
        "The MAU type of the input port";
}
leaf traffic-class {
    type dot1q-types:traffic-class-type;
    config false;
    description
        "The traffic class of the entry.";
    reference
        "8.6.6 of IEEE Std 802.1Q";
}
leaf independent-delay-min {
    type uint32;
    config false;
    description
        "The value is the minimum delay portion that is
        independent of frame length according to 12.32.1.1.
        of IEEE 802.1Q";
    reference
        "Item a) in 6.4.10.3.6.2 of IEC/IEEE 60802";
}
leaf independent-delay-max {
    type uint32;
    config false;
    description
        "The value is the maximum delay portion that is
        independent of frame length according to 12.32.1.1.
        of IEEE 802.1Q";
    reference
        "Item b) in 6.4.10.3.6.2 of IEC/IEEE 60802";
}
}
list dependent-delays-cascaded {
    key "in-port in-port-line-speed out-port";
    config false;

```

```

description
    "The list of minimum and maximum frame length dependent
    delay time values of frames as they pass through a
    bridge component";
reference
    "6.4.10.3.6.2 of IEC/IEEE 60802";
leaf in-port {
    type uint8;
    config false;
    description
        "The port number of the input port";
}
leaf in-port-line-speed {
    type uint32;
    config false;
    description
        "This value is the line speed in Mbps.";
}
leaf out-port {
    type uint8;
    config false;
    description
        "The port number of the output port";
}
leaf dependent-delay-min {
    type uint32;
    config false;
    description
        "The value is the minimum delay portion that is
        dependent on frame length according to 12.32.1.2.
        of IEEE 802.1Q";
    reference
        "Item c) in 6.4.10.3.6.2 of IEC/IEEE 60802";
}
leaf dependent-delay-max {
    type uint32;
    config false;
    description
        "The value is the maximum delay portion that is
        dependent on frame length according to 12.32.1.2.
        of IEEE 802.1Q";
    reference
        "Item d) in 6.4.10.3.6.2 of IEC/IEEE 60802";
}
}
}
leaf max-ptp-instances {
    type uint8;
    config false;
    description
        "The value is the maximum number of supported PTP
        Instances.";
    reference
        "6.4.10.3.7.5 of IEC/IEEE 60802";
}
leaf max-hot-standby-systems {
    type uint8;
    config false;
    description
        "The value is the maximum number of supported
        HotStandbySystem entities";
}

```

```

reference
  "6.4.10.3.7.6 of IEC/IEEE 60802";
}
list clock {
  key "clock-identity";
  config false;
  description
    "The list of supported application clock entities.";
  reference
    "6.4.10.3.7.7 of IEC/IEEE 60802";
  leaf clock-identity {
    type ptp-tt:clock-identity;
    config false;
    description
      "The clock identity of the application clock.";
    reference
      "Item a) in 6.4.10.3.7.7 of IEC/IEEE 60802";
  }
  leaf clock-target {
    type boolean;
    config false;
    description
      "The Boolean value indicates if the application clock is a
        clock target (TRUE) or clock source (FALSE).";
    reference
      "Item b) in 6.4.10.3.7.7 of IEC/IEEE 60802";
  }
  leaf attached-ptp-instance-index {
    type leafref {
      path "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-
tt:instance-index";
    }
    config false;
    description
      "The value is a reference to the index of the PTP or hot
        standby Instance, that is attached to the application
        clock.";
    reference
      "Item f) in 6.4.10.3.7.7 of IEC/IEEE 60802";
  }
  leaf arb-supported {
    type boolean;
    config false;
    description
      "The Boolean value indicates if the application clock
        supports the ARB timescale.";
    reference
      "Item c) in 6.4.10.3.7.7 of IEC/IEEE 60802";
  }
  leaf ptp-supported {
    type boolean;
    config false;
    description
      "The Boolean value indicates if the application clock
        supports the PTP timescale.";
    reference
      "Item d) in 6.4.10.3.7.7 of IEC/IEEE 60802";
  }
  leaf hot-standby-supported {
    type boolean;
    config false;
    description

```

```

        "The Boolean value indicates if the application clock
        supports the hot standby.";
    reference
        "Item e) in 6.4.10.3.7.7 of IEC/IEEE 60802";
}
leaf is-synced {
    type boolean;
    config false;
    description
        "The Boolean value indicates if the application clock is
        either synchronized to the attached PTP Instance (TRUE)
        or to an internal/external ClockSource (FALSE).";
    reference
        "Item g) in 6.4.10.3.7.7 of IEC/IEEE 60802";
}
}
}
}

```

6.4.10.8.3 Module **ieciieee60802-sched-bridge**

```

module iecieee60802-sched-bridge {
    yang-version 1.1;
    namespace "urn:ieee:std:60802:yang:ieciieee60802-sched-bridge";
    prefix ia-sched-bridge;

    import ieee802-types {
        prefix ieee802;
    }
    import ieee802-dot1q-bridge {
        prefix bridge;
    }
    import ieee802-dot1q-sched-bridge {
        prefix sched-bridge;
    }
    import ietf-interfaces {
        prefix if;
    }

    organization
        "IEEE 802.1 Working Group and IEC subcommittee 65C:
        Industrial networks, of IEC technical committee 65:
        Industrial-process measurement, control and automation";
    contact
        "WG-URL: http://ieee802.org/1/
        WG-EMail: stds-802-1-1@ieee.org

        Contact: IEEE 802.1 Working Group Chair
        Postal: C/O IEEE 802.1 Working Group
        IEEE Standards Association
        445 Hoes Lane
        Piscataway, NJ 08854
        USA

        E-mail: stds-802-1-chairs@ieee.org";
    description
        "Management objects that provide information about IEC/IEEE 60802
        IA-Stations as specified in IEC/IEEE 60802.

        Copyright (C) IEC/IEEE (2025).
        This version of this YANG module is part of IEC/IEEE 60802;
        see the standard itself for full legal notices.";
}

```

```

revision 2025-03-12 {
  description "Published as part of IEC/IEEE 60802-2025.
    The following reference statement identifies each referenced
    IEEE Standard as updated by applicable amendments.";
  reference
    "IEC/IEEE 60802 TSN profile for industrial automation:
    IEC/IEEE 60802-2025.
    IEEE Std 802.1Q Bridges and Bridged Networks:
    IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
    IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
    IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
}

augment "/if:interfaces/if:interface/bridge:bridge-port/sched-
bridge:gate-parameter-table" {
  description
    "Augment IEEE Std 802.1 bridge/gate-parameter-table.";
  list min-gating-times {
    key "speed";
    config false;
    description
      "The list of minimum gating times per supported line speed.";
    reference
      "6.4.10.3.4.3 of IEC/IEEE 60802";
    leaf speed {
      type uint32;
      config false;
      description
        "This value is the line speed in Mbps.";
    }
    container min-cycle-time {
      uses ieee802:rational-grouping;
      description
        "The value is the minimum value supported by this port of
        the AdminCycleTime and OperCycleTime parameters given as
        rational number of seconds.";
      reference
        "Item a) in 6.4.10.3.4.3 of IEC/IEEE 60802";
    }
    leaf min-interval-time {
      type uint32;
      description
        "The value is the minimum value supported by this port of
        the TimeIntervalValue parameter in nanoseconds.";
      reference
        "Item b) in 6.4.10.3.4.3 of IEC/IEEE 60802";
    }
  }
}

```

6.4.10.8.4 Module iecieee60802-cnc-config

```

module iecieee60802-cnc-config {
  yang-version 1.1;
  namespace "urn:ieee:std:60802:yang:iecieee60802-cnc-config";
  prefix ia-cnc-config;

  import ieee802-dot1q-cnc-config {
    prefix dot1q-cnc-config;
  }
  import ieee802-dot1q-tsn-types {
    prefix tsn-types;
  }
}

```

organization

"IEEE 802.1 Working Group and IEC subcommittee 65C:
Industrial networks, of IEC technical committee 65:
Industrial-process measurement, control and automation";

contact

"WG-URL: <http://ieee802.org/1/>
WG-EMail: stds-802-1-1@ieee.org

Contact: IEEE 802.1 Working Group Chair
Postal: C/O IEEE 802.1 Working Group
IEEE Standards Association
445 Hoes Lane
Piscataway, NJ 08854
USA

E-mail: stds-802-1-chairs@ieee.org";

description

"Management objects that provide information about IEC/IEEE 60802
IA-Stations as specified in IEC/IEEE 60802.

Copyright (C) IEC/IEEE (2025).
This version of this YANG module is part of IEC/IEEE 60802;
see the standard itself for full legal notices.";

revision 2025-01-10 {

description "Published as part of IEC/IEEE 60802-2025.
The following reference statement identifies each referenced
IEEE Standard as updated by applicable amendments.";

reference

"IEC/IEEE 60802 TSN profile for industrial automation:
IEC/IEEE 60802-2025.
IEEE Std 802.1Q Bridges and Bridged Networks:
IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";

}

augment "/dot1q-cnc-config:cnc-config" {

description

"Augment main container in cnc-config.";

leaf max-config-domains {

type uint8;

config false;

description

"The value is the maximum number of supported configuration
domains.";

reference

"6.4.10.3.9.1 of IEC/IEEE 60802";

}

leaf max-cucs {

type uint8;

config false;

description

"The value is the maximum number of supported CUC entities.";

reference

"6.4.10.3.9.2 of IEC/IEEE 60802";

}

leaf max-ia-stations {

type uint16;

config false;

description


```

    "The value is the maximum number of supported IA-stations.";
  reference
    "6.4.10.3.9.3 of IEC/IEEE 60802";
}
leaf max-network-diameter {
  type uint8;
  config false;
  description
    "The value is the maximum supported network diameter.";
  reference
    "6.4.10.3.9.4 of IEC/IEEE 60802";
}
leaf max-streams {
  type uint16;
  config false;
  description
    "The value is the maximum number of supported streams.";
  reference
    "6.4.10.3.9.5 of IEC/IEEE 60802";
}
leaf max-num-seamless-trees {
  type uint8;
  config false;
  description
    "The value is the maximum number of trees supported for
    seamless redundancy of a stream.";
  reference
    "6.4.10.3.9.6 of IEC/IEEE 60802";
}
leaf hot-standby-supported {
  type uint8;
  config false;
  description
    "The Boolean value indicates if PTP hot standby is
    supported.";
  reference
    "6.4.10.3.9.7 of IEC/IEEE 60802";
}
action add_streams {
  description
    "This Action requests a CNC to add a list of streams.";
  input {
    leaf cuc-id {
      type string;
      description
        "The CUC ID where the streams are to be added";
    }
    list stream-list {
      key "stream-id";
      description
        "List of Streams that should be added.";
      leaf stream-id {
        type tsn-types:stream-id-type;
        description
          "The Stream ID is a unique identifier of a Stream
          request and corresponding configuration. It is used to
          associate a CUC's Stream request with a CNC's
          corresponding response.";
      }
    }
    container talker {
      description
        "The Talker container contains: - Talker's behavior for

```

```

        Stream (how/when transmitted) - Talker's requirements
        from the network - TSN capabilities of the Talker's
        interface(s).";
    uses tsn-types:group-talker;
}
list listener {
    key "index";
    description
        "Each Listener list entry contains: - Listener's
        requirements from the network - TSN capabilities of
        the Listener's interface(s).";
    leaf index {
        type uint32;
        description
            "This index is provided in order to provide a unique
            key per list entry.";
    }
    uses tsn-types:group-listener;
}
}
}
output {
    leaf result {
        type boolean;
        description
            "Returns status information indicating if Stream addition
            has been successful.";
    }
}
}
}

augment "/dot1q-cnc-config:cnc-config/dot1q-cnc-config:domain/dot1q-cnc-
config:cuc/dot1q-cnc-config:stream" {
    description
        "Augment stream list in tsc-config-uni.";
    action remove_listener {
        description
            "This Action removes listeners from a stream.";
        input {
            list listener {
                key "index";
                description
                    "Each Listener list entry contains: - Listener's
                    requirements from the network - TSN capabilities of the
                    Listener's interface(s).";
                leaf index {
                    type uint32;
                    description
                        "This index is provided in order to provide a unique
                        key per list entry.";
                }
            }
        }
        output {
            leaf result {
                type boolean;
                description
                    "Returns status information indicating if listene removal
                    has been successful.";
            }
        }
    }
}

```

```

    }
  }
}

```

6.4.10.8.5 Module `iecieeee60802-ia-station`

```

module iecieeee60802-ia-station {
  yang-version 1.1;
  namespace "urn:ieee:std:60802:yang:iecieeee60802-ia-station";
  prefix ias;

  import ietf-datastores {
    prefix ds;
    reference
      "IETF RFC 8342: Network Management Datastore Architecture
       (NMDA)";
  }
  import ietf-netconf-acm {
    prefix nacm;
    reference
      "IETF RFC 8341: Network Configuration Access Control Model";
  }

  organization
    "IEEE 802.1 Working Group and IEC subcommittee 65C:
     Industrial networks, of IEC technical committee 65:
     Industrial-process measurement, control and automation";
  contact
    "WG-URL: http://ieee802.org/1/
     WG-EMail: stds-802-1-1@ieee.org

     Contact: IEEE 802.1 Working Group Chair
     Postal: C/O IEEE 802.1 Working Group
     IEEE Standards Association
     445 Hoes Lane
     Piscataway, NJ 08854
     USA

     E-mail: stds-802-1-chairs@ieee.org";
  description
    "Capability information and reset to factory defaults
     functionality for IEC/IEEE 60802 IA-Stations as specified in
     IEC/IEEE 60802.

     Copyright (C) IEC/IEEE (2025).
     This version of this YANG module is part of IEC/IEEE 60802;
     see the standard itself for full legal notices.";

  revision 2025-03-12 {
    description "Published as part of IEC/IEEE 60802-2025.
     The following reference statement identifies each referenced
     IEEE Standard as updated by applicable amendments.";
    reference
      "IEC/IEEE 60802 TSN profile for industrial automation:
       IEC/IEEE 60802-2025.
       IEEE Std 802.1Q Bridges and Bridged Networks:
       IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
       IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
       IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
  }

  feature ia-factory-default-datastore {
    description

```

```

    "Indicates that the factory default configuration is
    available as a datastore.";
}

identity ia-factory-default {
    if-feature "ia-factory-default-datastore";
    base ds:datastore;
    description
        "This read-only datastore contains the factory default
        configuration for the device that will be used to replace
        the contents of the read-write conventional configuration
        datastores during a 'ia-factory-reset' RPC operation.";
}

container ia-station-capabilities {
    config false;
    description
        "This container provides read only information about an
        ia-station's capabilities.";
    reference
        "IEC/IEEE 60802 - YANG Data Model";
    leaf capability-lldp {
        type boolean;
        config false;
        description
            "The value is true if the device supports LLDP.";
        reference
            "6.4.10.3.8.5 of IEC/IEEE 60802";
    }
    leaf capability-timesync {
        type boolean;
        config false;
        description
            "The value is true if the device supports Timesync.";
        reference
            "6.4.10.3.8.6 of IEC/IEEE 60802";
    }
    leaf capability-keystore {
        type boolean;
        config false;
        description
            "The value is true if the device supports Keystore.";
        reference
            "6.4.10.3.8.7 of IEC/IEEE 60802";
    }
    leaf capability-truststore {
        type boolean;
        config false;
        description
            "The value is true if the device supports Truststore.";
        reference
            "6.4.10.3.8.9 of IEC/IEEE 60802";
    }
    leaf capability-nacm {
        type boolean;
        config false;
        description
            "The value is true if the device supports NACM.";
        reference
            "6.4.10.3.8.8 of IEC/IEEE 60802";
    }
    leaf capability-yang-library {

```

```

    type boolean;
    config false;
    description
        "The value is true if the device supports YANG library.";
    reference
        "6.4.10.3.8.10 of IEC/IEEE 60802";
}
leaf capability-yang-push {
    type boolean;
    config false;
    description
        "The value is true if the device supports YANG push.";
    reference
        "6.4.10.3.8.11 of IEC/IEEE 60802";
}
leaf capability-yang-notifications {
    type boolean;
    config false;
    description
        "The value is true if the device supports YANG
        notifications.";
    reference
        "6.4.10.3.8.12 of IEC/IEEE 60802";
}
leaf capability-netconf-monitoring {
    type boolean;
    config false;
    description
        "The value is true if the device supports NETCONF
        monitoring.";
    reference
        "6.4.10.3.8.13 of IEC/IEEE 60802";
}
leaf capability-netconf-client {
    type boolean;
    config false;
    description
        "The value is true if the device supports NETCONF client.";
    reference
        "6.4.10.3.8.14 of IEC/IEEE 60802";
}
leaf capability-cnc-config {
    type boolean;
    config false;
    description
        "The value is true if the device supports the CNC configuration.";
    reference
        "6.4.10.3.8.15 of IEC/IEEE 60802";
}
leaf capability-sched-traffic {
    type boolean;
    config false;
    description
        "The value is true if the device supports scheduled
        traffic.";
    reference
        "6.4.10.3.8.16 of IEC/IEEE 60802";
}
leaf capability-frame-preemption {
    type boolean;
    config false;
    description

```

```

        "The value is true if the device supports frame preemption.";
    reference
        "6.4.10.3.8.17 of IEC/IEEE 60802";
}
}

rpc ia-factory-reset {
    nacm:default-deny-all;
    description
        "The server resets all datastores to their factory
        default contents and any nonvolatile storage back to
        factory condition, deleting all dynamically
        generated files, including those containing keys,
        certificates, logs, and other temporary files.

        Depending on the factory default configuration, after
        being reset, the device may become unreachable on the
        network.

        In contrast to the original factory-reset RPC in IETF RFC
        8808, this RPC puts the device into a state where a
        subsequent configuration by a CNC component results in a
        functioning 60802 IA-station";
}
}

```

6.4.10.8.6 Module **iecieeee60802-subscribed-notifications**

```

module iecieeee60802-subscribed-notifications {
    yang-version 1.1;
    namespace
        "urn:ieee:std:60802:yang:iecieeee60802-subscribed-notifications";
    prefix ia-sn;

    import ietf-subscribed-notifications {
        prefix sn;
    }

    organization
        "IEEE 802.1 Working Group and IEC subcommittee 65C:
        Industrial networks, of IEC technical committee 65:
        Industrial-process measurement, control and automation";
    contact
        "WG-URL: http://ieee802.org/1/
        WG-EMail: stds-802-1-l@ieee.org

        Contact: IEEE 802.1 Working Group Chair
        Postal: C/O IEEE 802.1 Working Group
        IEEE Standards Association
        445 Hoes Lane
        Piscataway, NJ 08854
        USA

        E-mail: stds-802-1-chairs@ieee.org";
    description
        "Management objects that provide information about IEC/IEEE 60802
        IA-Stations as specified in IEC/IEEE 60802.

        Copyright (C) IEC/IEEE (2025).
        This version of this YANG module is part of IEC/IEEE 60802;
        see the standard itself for full legal notices.";

    revision 2024-02-19 {
        description "Published as part of IEC/IEEE 60802-2025.
        The following reference statement identifies each referenced
        IEEE Standard as updated by applicable amendments.";
        reference
            "IEC/IEEE 60802 TSN profile for industrial automation:
            IEC/IEEE 60802-2025.
            IEEE Std 802.1Q Bridges and Bridged Networks:
            IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,

```

```

IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
}

augment "/sn:subscriptions" {
  description
    "Augment subscriptions in ietf-subscribed-notifications.";
  leaf max-subscriptions {
    type uint16;
    config false;
    description
      "The value is the maximum number of supported NETCONF Server
      subscriptions.";
    reference
      "6.4.10.3.8.1 of IEC/IEEE 60802";
  }
  leaf max-on-change-subscription-leaves {
    type uint16;
    config false;
    description
      "The value is the maximum number of supported leaves for
      NETCONF Server on-change subscriptions according to IETF
      RFC 8641.";
    reference
      "6.4.10.3.8.2 of IEC/IEEE 60802";
  }
  leaf max-periodic-subscription-leaves {
    type uint16;
    config false;
    description
      "The value is the maximum number of supported leaves for
      NETCONF Server periodic subscriptions according to IETF
      RFC 8641.";
    reference
      "6.4.10.3.8.3 of IEC/IEEE 60802";
  }
  leaf min-periodic-subscription-interval {
    type uint16;
    config false;
    description
      "The value is the minimum periodic subscription interval in
      centiseconds (0.01 seconds) for NETCONF Server periodic
      subscriptions according to IETF RFC 8641.";
    reference
      "6.4.10.3.8.4 of IEC/IEEE 60802";
  }
}
}

```

6.4.10.8.7 Module **ieciieee60802-tt-mapping**

```

module iecieee60802-tt-mapping {
  yang-version "1.1";
  namespace "urn:ieee:std:60802:yang:ieciieee60802-tt-mapping";
  prefix tt-mapping;

  import ieee802-dot1q-types {
    prefix dot1q-types;
  }

  organization
    "IEEE 802.1 Working Group and IEC subcommittee 65C:
    Industrial networks, of IEC technical committee 65:
    Industrial-process measurement, control and automation";
  contact
    "WG-URL: http://ieee802.org/1/
    WG-EMail: stds-802-1-1@ieee.org

    Contact: IEEE 802.1 Working Group Chair
    Postal: C/O IEEE 802.1 Working Group
           IEEE Standards Association
           445 Hoes Lane

```

Piscataway, NJ 08854
USA

```

E-mail: stds-802-1-chairs@ieee.org";
description
  "This module provides information on characteristics of traffic classes
  configured in a network to allow applications to map application
traffic
  classes to them.

Copyright (C) IEC/IEEE (2025).

This version of this YANG module is part of IEC/IEEE 60802;
see the standard itself for full legal notices.";
revision 2024-12-13 {
  description
    "Published as part of IEC/IEEE 60802-2025.

    The following reference statement identifies each referenced
    IEEE Standard as updated by applicable amendments.";
  reference
    "IEC/IEEE 60802 TSN profile for industrial automation:
    IEC/IEEE 60802-2025.
    IEEE Std 802.1Q Bridges and Bridged Networks:
    IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
    IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
    IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
}

container tt-mapping {
  description
    "Top-level container for the traffic class module.";
  list network {
    key "network-name";
    description
      "A list of networks, each containing one or more sets of traffic
class
      characteristics.";
    leaf network-name {
      type string;
      description
        "The network-name uniquely identifies a network.";
    }
    list traffic-type {
      key "traffic-type-name";
      description
        "A list of traffic types in the network, each with its associated
        traffic type characteristics.";
      leaf traffic-type-name {
        type string;
        description
          "The traffic-type-name uniquely identifies the traffic type.";
      }
      leaf cyclic {
        type boolean;
        description
          "Indicates if cyclic transmission is mandatory (TRUE) or
optional (FALSE)
          for this traffic type.";
        reference
          "4.7.2 of IEC/IEEE 60802-2025";
      }
    }
  }
}

```



```

leaf data-delivery-req {
  type enumeration {
    enum frame-latency {
      value 0;
      description
        "Data delivery of a frame for a given Talker-Listener pair
        occurs within a bounded timespan.";
    }
    enum flow-latency {
      value 1;
      description
        "Data delivery up to a certain number of frames or data
        (including bursts of frames) occurring over a defined
        period.";
    }
    enum deadline {
      value 2;
      description
        "Data delivery of a frame to a given Listener occurs at or
        before a specific point in time.";
    }
    enum no {
      value 3;
      description
        "No special data delivery requirements.";
    }
  }
  description
    "The data-delivery-req provides information on the data
    delivery requirements for this traffic type.";
  reference
    "4.7.2 of IEC/IEEE 60802-2025";
}
leaf time-triggered {
  type boolean;
  description
    "Indicates if it is required (TRUE) or optional (FALSE) that
    transmission occurs at a specific point in time based upon
    the Working Clock.";
  reference
    "4.7.2 of IEC/IEEE 60802-2025";
}
leaf stream-based {
  type boolean;
  default true;
  description
    "Indicates if traffic of this traffic type requires
    registering a stream, i.e., allocating dedicated
    resources, for the traffic, before a given Talker
    can start transmitting data (TRUE), or if registering
    a stream is not required (FALSE).";
  reference
    "4.7.2 of IEC/IEEE 60802-2025";
}
leaf vlan-id {
  type dot1q-types:vlanid;
  description
    "The VID this traffic type is mapped to.";
}
list redundancy {
  key "vlan-id-redundant";

```

```

    description
      "A list of VIDs this traffic type is mapped to for redundant
      transmission. There is one entry in the list per redundant
      transmission ";
    leaf vlan-id-redundant {
      type dot1q-types:vlanid;
      description
        "The VID this traffic type is mapped for a redundant
transmission.";
    }
  }
  leaf priority-value {
    type dot1q-types:priority-type;
    description
      "The PCP this traffic type is mapped to.";
  }
}
}
}
}
}

```

6.4.10.8.8 Module iecieee60802-proxy

```

module iecieee60802-proxy {
  yang-version 1.1;
  namespace "urn:ieee:std:60802:yang:ieciieee60802-proxy";
  prefix pr;

  import ieee802-types {
    prefix ieee;
  }
  import ietf-inet-types {
    prefix inet;
  }

  organization
    "IEEE 802.1 Working Group and IEC subcommittee 65C:
    Industrial networks, of IEC technical committee 65:
    Industrial-process measurement, control and automation";
  contact
    "WG-URL: http://ieee802.org/1/
    WG-EMail: stds-802-1-1@ieee.org

    Contact: IEEE 802.1 Working Group Chair
    Postal: C/O IEEE 802.1 Working Group
    IEEE Standards Association
    445 Hoes Lane
    Piscataway, NJ 08854
    USA

    E-mail: stds-802-1-chairs@ieee.org";
  description
    "Configuration and capability information for IEC/IEEE 60802 proxy
    targets as specified in IEC/IEEE 60802.

    Copyright (C) IEC/IEEE (2025).
    This version of this YANG module is part of IEC/IEEE 60802;
    see the standard itself for full legal notices.";

  revision 2025-03-12 {
    description
      "Published as part of IEC/IEEE 60802-2025.
      The following reference statement identifies each referenced

```

```

    IEEE Standard as updated by applicable amendments.";
reference
    "IEC/IEEE 60802 TSN profile for industrial automation:
    IEC/IEEE 60802-2025.
    IEEE Std 802.1Q Bridges and Bridged Networks:
    IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
    IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
    IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
}

container proxy {
    presence "The presence of this container indicates that this IA station
    provides proxy functionality for proxy targets";
    description
        "This container provides information about a
        60802 IA station with proxy functionality.";
    reference
        "IEC/IEEE 60802 - YANG Data Model";
    leaf max-num-proxy-targets {
        type uint16;
        config false;
        description
            "The value indicates how many proxy targets can be handled by this
            proxy component.";
        reference
            "6.4.10.3.9.1 of IEC/IEEE 60802";
    }
    list proxy-types {
        key "proxy-type";
        config false;
        description
            "Supported proxy types";
        leaf proxy-type {
            type string {
                pattern '[0-9A-F]{2}(-[0-9A-F]{2}){2}';
            }
            config false;
            description
                "OID of proxy targets handled by this proxy.";
            reference
                "6.4.10.3.9.2 of IEC/IEEE 60802";
        }
    }
    list proxy-targets {
        key "device";
        description
            "List of proxy targets that are connected to this proxy";
        leaf device {
            type string;
            description
                "The identification of the device.";
            reference
                "6.4.10.3.9.3 of IEC/IEEE 60802";
        }
        leaf device-type {
            type string;
            description
                "The type of the device.";
            reference
                "6.4.10.3.9.4 of IEC/IEEE 60802";
        }
    }
    container proxy-addr {
        description

```

```
module iecieeee60802-proxy-target {
  yang-version 1.1;
  namespace "urn:ieee:std:60802:yang:iecieeee60802-proxy-target";
  prefix cd;
```

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 445 Hoes Lane
 Piscataway, NJ 08854
 USA

```
E-mail: stds-802-1-chairs@ieee.org";
description
  "Capability information for IEC/IEEE 60802 proxy targets as
  specified in IEC/IEEE 60802.

  Copyright (C) IEC/IEEE (2025).
  This version of this YANG module is part of IEC/IEEE 60802;
  see the standard itself for full legal notices.";

revision 2025-01-08 {
  description
    "Published as part of IEC/IEEE 60802-2025.
    The following reference statement identifies each referenced
    IEEE Standard as updated by applicable amendments.";
  reference
    "IEC/IEEE 60802 TSN profile for industrial automation:
    IEC/IEEE 60802-2025.
    IEEE Std 802.1Q Bridges and Bridged Networks:
    IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
    IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
    IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
}

container proxy-target {
  presence "The presence of this container indicates that an IA station
  is a proxy target that requires a proxy device";
  description
    "This container provides information about a
    60802 proxy target device.";
  reference
    "IEC/IEEE 60802 - YANG Data Model";
  leaf required-proxy-type {
    type string {
      pattern '[0-9A-F]{2}(-[0-9A-F]{2}){2}';
    }
    config false;
    description
      "OID of proxy that this proxy target requires.";
    reference
      "6.4.10.3.10.1 of IEC/IEEE 60802";
  }
}
}
```

6.5 Topology discovery and verification

6.5.1 Topology discovery and verification requirements

Electrical engineering of machines with multiple IA-stations includes the definition of the machine internal network topology (i.e. the engineered topology).

The machine internal network topology includes type specific data of IA-stations (for example model name or manufacturer name) as well as instance specific data (for example IP addresses or DNS names).

The electrical engineering data of the network topology is used:

- during commissioning so that machine planning and installation are identical;
- by the TDE during operation to verify that the actual topology of the Configuration Domain matches the engineered topology;
- by maintenance staff during repair to easily identify failed IA-stations, ports, or links to be replaced.

Repair and replacement of an IA-station do not require verification of the updated engineered topology so that the TDE does not produce a verification error.

IA-stations are necessarily pre-configured when they are repaired or replaced. IA-stations report type and instance data as described in 6.5.3.

6.5.2 Topology discovery overview

6.5.2.1 General

LLDP enables the discovery of IA-stations, their external ports, and their external connectivity. A Topology Discovery Entity can query LLDP data by remote management to derive the physical network topology.

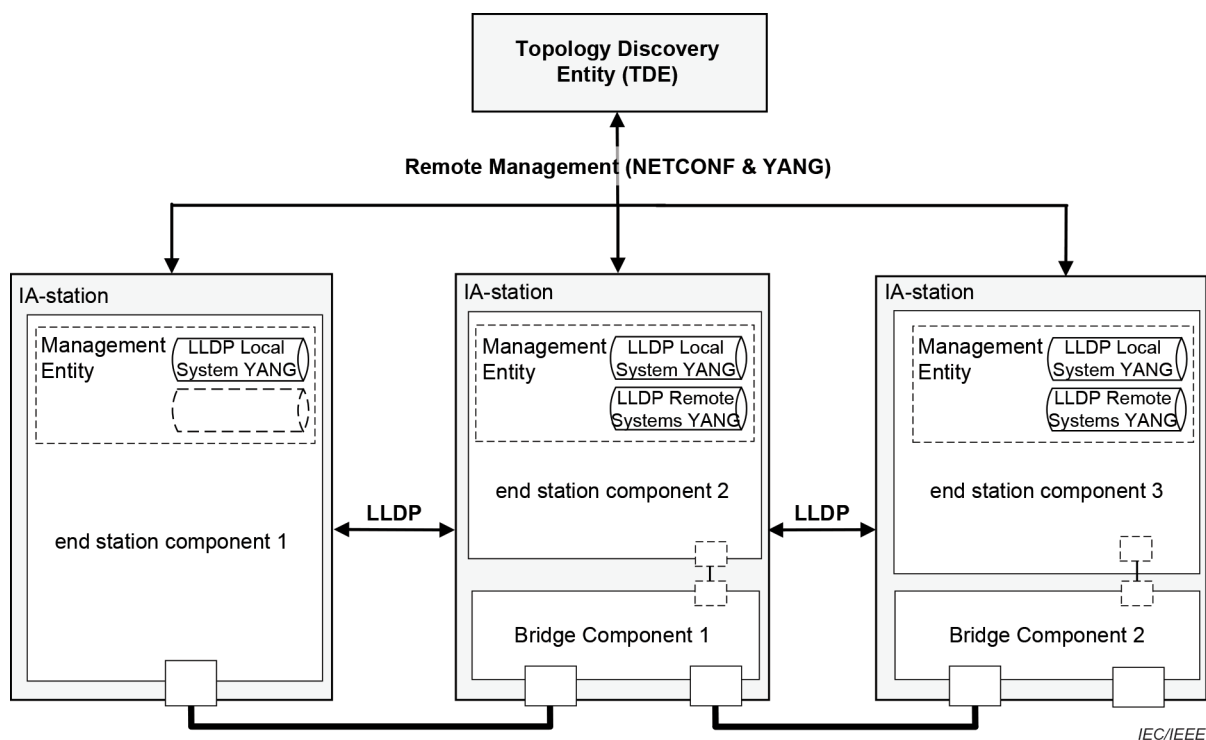


Figure 38 – Usage example of LLDP

Figure 38 illustrates a network showing the LLDP agent implementations in an IA-station consisting of a single end station component and two IA-stations with end station and Bridge components (see 4.3). The LLDP protocol is used to convey neighborhood information among peers, and NETCONF is used between the TDE and the IA-stations to query this neighborhood information from the IA-stations. This information allows the TDE to discover IA-stations and the physical network topology.

NOTE A Topology Discovery Entity (TDE) can be run from anywhere in the network with reachability to the to-be-discovered devices.

IA-stations announce themselves via LLDP to support discovery by the TDE. Announcements contain the management address (see 6.5.2.4.6) and system capabilities (see 6.5.2.4.5) for the discovery operation. The announced system capabilities information enables the TDE to identify IA-stations with multiple end station and Bridge components. The TDE can use the definitions in 6.4.3 for the discovery of the internal structure of such IA-stations.

To allow for operational behavior and exchanged information, IA-stations support the local system YANG (see 6.4.9.2.2). IA-stations that include a Bridge component additionally support the processing of received LLDP messages and support the remote systems YANG (see 6.4.9.2.2).

6.5.2.2 LLDP operational control parameters

LLDP defines several operational parameters that control the protocol behavior (see IEEE Std 802.1AB-2016, 10.5.1). These parameter definitions apply to all external ports of an IA-station.

NOTE According to IEEE Std 802.1AB-2016, 9.1.1 c), changes to the local system that impact information exchanged via LLDP immediately trigger the transmission of an LLDPDU to communicate the local changes as quickly as possible to any neighboring systems.

An IA-station shall support LLDP transmit mode (adminStatus enabledTxOnly) on an external end station component port and may support transmit and receive mode (adminStatus enabledRxTx) on that port (see IEEE Std 802.1AB-2016, 10.5.1).

An IA-station shall support LLDP transmit and receive mode (adminStatus enabledRxTx) on an external Bridge component port (see IEEE Std 802.1AB-2016, 10.5.1).

6.5.2.3 LLDPDU transmission, reception, and addressing

The destination address to be used for LLDPDU transmission (dest-mac-address) shall be the nearest bridge group MAC address, i.e., 01-80-C2-00-00-0E, on all ports to limit the scope of LLDPDU propagation to a single physical link (see IEEE Std 802.1AB-2016, 7.1 a).

NOTE IEEE Std 802.1AB-2016 defines LLDPDUs to be transmitted untagged, i.e. frames do not carry priority information for traffic class selection. At the same time, IEEE Std 802.1AB-2016 neither specifies a well-defined device-internal priority nor management capabilities for the configuration of the traffic class to be used for the transmission of LLDPDUs. It is the user's responsibility to prevent LLDPDUs from interfering with the transmission of time-critical control data.

6.5.2.4 LLDP TLV selection

6.5.2.4.1 General

An IA-station transmitting LLDPDUs shall include the LLDP TLVs selected in 6.5.2.4 and may include additional TLVs (tlvs-tx-enable). An IA-station receiving LLDPDUs shall process LLDPDUs.

Each LLDPDU shall contain the following LLDP TLVs specified in IEEE Std 802.1AB-2016, 8.5:

- exactly one Chassis ID TLV according to 6.5.2.4.2,
- exactly one Port ID TLV according to 6.5.2.4.3,
- exactly one Time To Live TLV according to 6.5.2.4.4,
- exactly one System Capabilities TLV according to 6.5.2.4.5, and
- one or more Management Address TLVs according to 6.5.2.4.6.

NOTE The concatenation of the Chassis ID and Port ID fields enables the recipient of an LLDPDU to identify the sending LLDP agent/port.

In addition, each LLDPDU shall contain the 60802 device info TLV described in Annex E.

6.5.2.4.2 Chassis ID TLV

The Chassis ID field shall contain the same value for all transmitted LLDPDUs independent from the transmitting port of the IA-station, i.e. be a non-volatile identifier which is unique within the context of the administrative domain.

The Chassis ID subtype field (chassis-id-subtype) should contain subtype 4, indicating that the Chassis ID field (chassis-id) contains a MAC address to achieve the Chassis ID's desired uniqueness. For IA-stations with multiple unique MAC addresses, any one of the IA-station's MAC addresses may be used and shall be the same for all external ports of that IA-station.

6.5.2.4.3 Port ID TLV

The Port ID field shall contain the same value for all transmitted LLDPDUs for a given external port, i.e. be a non-volatile, IA-station-unique identifier of the LLDPDU-transmitting port.

The Port ID subtype field (port-id-subtype) should contain subtype 5, indicating that the Port ID field contains the port interface name (name) according to IETF RFC 8343.

IA-stations should restrict the system-defined port ID to read-only access and a maximum name length of 255 characters. The names should match the port names printed on the chassis.

6.5.2.4.4 Time To Live TLV

The Time To Live value shall be set according to IEEE Std 802.1AB-2016, 8.5.4.

6.5.2.4.5 System capabilities TLV

An IA-station consisting of a single end station component shall set the system capabilities and enabled capabilities fields (system-capabilities-supported, system-capabilities-enabled) to Station Only (i.e. bit 8 set to 1) for all transmitted LLDPDUs.

An IA-station consisting of at least one end station component and at least one Bridge component shall set the system capabilities and enabled capabilities fields to Station Only (i.e. bit 8 set to 1) and C-VLAN component (i.e. bit 9 set to 1) for all transmitted LLDPDUs.

NOTE The combination of the Station Only and C-VLAN component flags is used as a marker indicating to the TDE that the internal structure of the IA-station consists of multiple components. This is a deliberate deviation from IEEE Std 802.1AB-2016, Table 8-4, which states in a footnote: "The Station Only capability is intended for devices that implement only an end station capability, and for which none of the other capabilities in the table apply. Bit 8 therefore is not set in conjunction with any other bits."

6.5.2.4.6 Management address TLV

An IA-station shall announce at least one IPv4 address by which its Management entity (see 4.3) can be reached (management-address-tx-port).

6.5.2.5 LLDP remote systems data

An IA-station supporting the remote systems YANG shall be able to store information from at least one neighbor per external port.

Receiving LLDPDUs from more neighbors than supported on a given port shall result in the last one received being saved to the remote systems YANG as described in IEEE Std 802.1AB-2016, 9.2.7.7.5.

6.5.3 Topology verification overview

Topology verification checks discovered topologies against engineered topologies. Topology verification data includes for every IA-station:

- model name,
- manufacturer name, and
- management address.

Topology verification data includes for every external port of an IA-station:

- port name, and
- remote connection (i.e. management address and port name of connected IA-station).

To support topology verification IA-stations shall support LLDP YANG data as specified in 6.4.9.2.2 and Hardware Management YANG data as specified in 6.4.9.2.5.8.

IA-station hardware instance specific data like MAC addresses or serial numbers are not considered for topology verification. This kind of data changes after a repair and replacement operation and thus, induces a topology verification error.

6.6 CNC

6.6.1 General

Subclause 6.6 describes stream destination MAC address handling at the CNC.

6.6.2 Stream destination MAC address range

A CNC manages the destination MAC address for requested streams. This destination MAC address together with the VID identifies the path used for these streams. Thus, a stream destination MAC address is unique together with the VID in a Configuration Domain.

Figure 39 shows the possible selections of a CNC for a contiguous address range. The CNC selects an OUI and an offset of the address range for the stream destination MAC addresses.

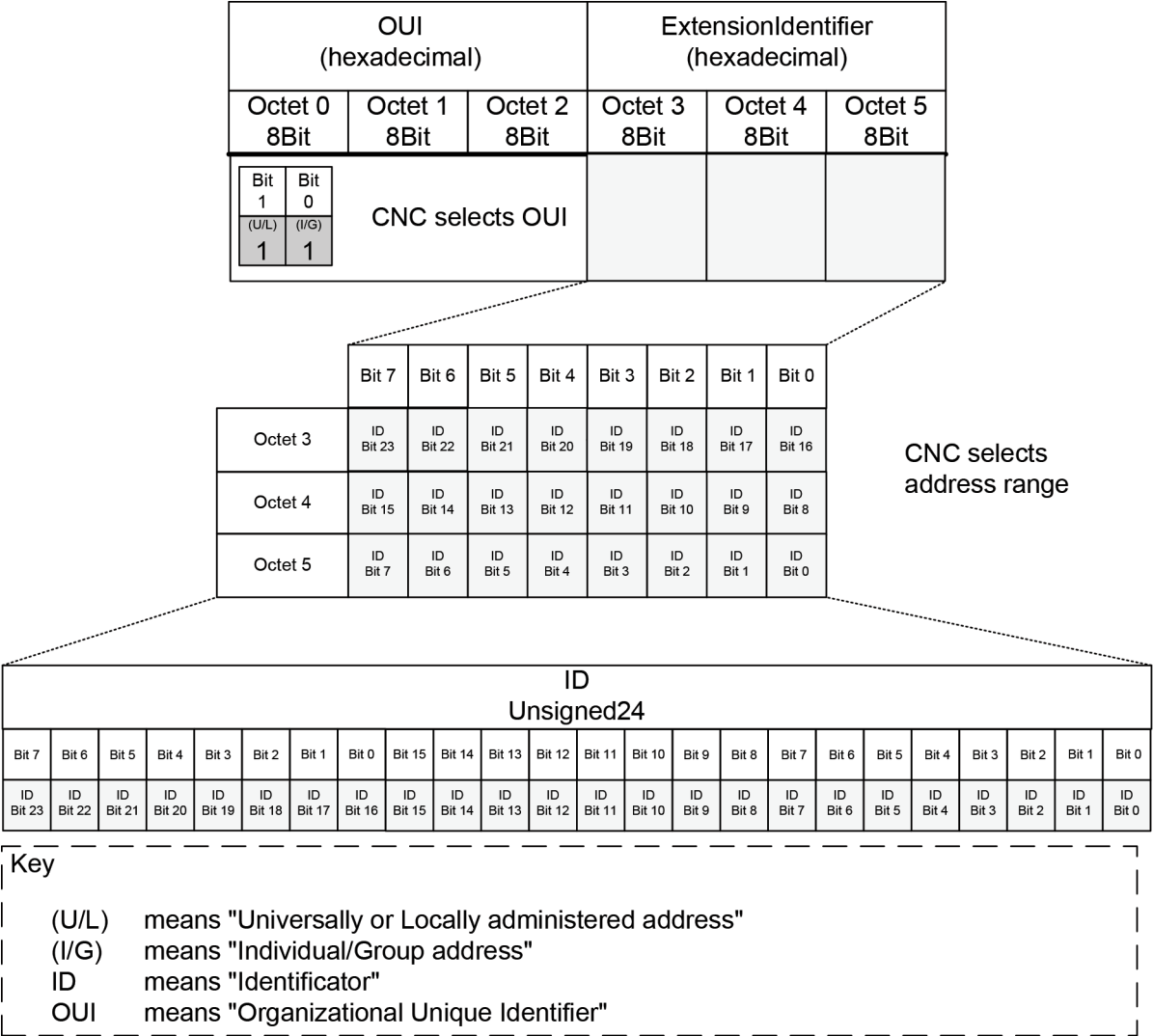
An address range of 2 048 stream destination MAC addresses allows together with a VID the usage of 2 048 streams. Each additional VID used for streams allows an additional 2 048 streams.

EXAMPLE

CNC selected OUI:= 00-80-C2

CNC selected address range:= 0.2047

CNC selected VID:= 101



IEC/IEEE

Figure 39 – Stream destination MAC Address

Annex A

(normative)

PCS proforma – Time-sensitive networking profile for industrial automation

A.1 General⁶

The supplier of an implementation that is claimed to conform to the profile specified in this document shall complete the corresponding Profile Conformance Statement (PCS) proforma, which is presented in a tabular format based on the format used for Protocol Implementation Conformance Statement (PICS) proformas.

The tables do not contain an exhaustive list of all requirements that are stated in the referenced standards; for example, if a row in a table asks whether the implementation is conformant to Standard X, and the answer "Yes" is chosen, then it is assumed that it is possible, for that implementation, to fill out the PCS proforma specified in Standard X to show that the implementation is conformant; however, the tables in this document only further refine those elements of conformance to Standard X where particular answers are required for the profiles specified here.

A completed PCS proforma is the PCS for the implementation in question. The PCS is a statement of which capabilities and options of the protocol have been implemented. The PCS can have several uses, including use by the following.

- a) Protocol implementer, as a checklist to reduce the risk of failure to conform to the document through oversight.
- b) Supplier and acquirer, or potential acquirer, of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard PCS proforma.
- c) User, or potential user, of the implementation, as a basis for initially checking the possibility of interworking with another implementation.

NOTE While interworking can never be guaranteed, failure to interwork can often be predicted from incompatible PCS.

- d) Protocol tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.
- e) The user, to verify whether the IA-station, as described by the PCS, fulfills use-case requirements.

⁶ An editable version of each PCS in this document can be obtained from the IEC SC 65C supporting documents website at <https://www.iec.ch/sc65c/supportingdocuments> or the IEEE 802.1 Website at <https://1.ieee802.org/conformance/>.

A.2 Abbreviated terms and special symbols

A.2.1 Status symbols

M	mandatory
O	optional
O.n	optional, but support of at least one of the group of options labeled by the same numeral n is required
X	prohibited
pred	conditional-item symbol, including predicate identification (see A.3.4)
¬	logical negation, applied to a conditional item's predicate

A.2.2 General abbreviated terms

N/A	not applicable
PCS	Profile Conformance Statement

A.3 Instructions for completing the PCS proforma

A.3.1 General structure of the PCS proforma

The first part of the PCS proforma, implementation identification and protocol summary, is to be completed as indicated with the information necessary to identify fully both the supplier and the implementation.

The main part of the PCS proforma is a fixed-format questionnaire, divided into several subclauses, each containing a number of individual items. Answers to the questionnaire items are to be provided in the rightmost column, either by simply marking an answer to indicate a restricted choice (usually Yes or No) or by entering a value or a set or range of values. There are some items where two or more choices from a set of possible answers can apply; all relevant choices are to be marked. Each item is identified by an item reference in the first column. The second column contains the question to be answered; the third column records the status of the item – whether support is mandatory, optional, or conditional (see also A.3.4). The fourth column contains the reference or references to the material that specifies the item in the main body of this document, and the fifth column provides the space for the answers.

The PCS indicates support of one of the conformance classes, ccA or ccB, per bridge and end-station component, specified in this profile.

A single IA-station can incorporate the functionality of one or more of the functions listed in this PCS. For example, an IA-station could have both an end station component and a Bridge component.

A supplier can also provide (or be required to provide) further information, categorized as either additional information (see A.3.2) or exception information (see A.3.3). When present, each kind of further information is to be provided in a further subclause of items labeled Ai or Xi, respectively, for cross-referencing purposes, where (i) is any unambiguous identification for the item (for example, simply a numeral). There are no other restrictions on its format and presentation.

A completed PCS proforma, including any Additional Information and Exception Information, is the Protocol Implementation Conformation Statement for the implementation in question.

NOTE Where an implementation is capable of being configured in more than one way, a single PCS can be used to describe all such configurations. However, the supplier has the choice of providing more than one PCS, each covering some subset of the implementation's configuration capabilities, in case that makes for easier and clearer presentation of the information.

A.3.2 Additional information

Items of Additional Information allow a supplier to provide further information intended to assist the interpretation of the PCS. It is not intended or expected that a large quantity be supplied, and a PCS can be considered complete without any such information. Examples can be an outline of the ways in which a (single) implementation can be set up to operate in a variety of environments and configurations, or information about aspects of the implementation that are outside the scope of this document but that have a bearing on the answers to some items.

References to items of Additional Information can be entered next to any answer in the questionnaire and can be included in items of Exception Information.

A.3.3 Exception information

It can occasionally happen that a supplier wishes to answer an item with mandatory status (after any conditions have been applied) in a way that conflicts with the indicated requirement. No preprinted answer will be found in the Support column for this item. Instead, the supplier shall write the missing answer into the Support column, together with an Xi reference to an item of Exception Information and shall provide the appropriate rationale in the Exception item itself.

An implementation for which an Exception item is required in this way does not conform to this document.

NOTE A possible reason for the situation described previously is that a defect in this document has been reported, a correction for which is expected to change the requirement not met by the implementation.

A.3.4 Conditional status

A.3.4.1 Conditional items

The PCS proforma contains a number of conditional items. These are items for which both the applicability of the item itself, and its status if it does apply (mandatory or optional) are dependent on whether certain other items are supported.

Where a group of items is subject to the same condition for applicability, a separate preliminary question about the condition appears at the head of the group, with an instruction to skip to a later point in the questionnaire if the "Not Applicable" (N/A) answer is selected. Otherwise, individual conditional items are indicated by a conditional symbol in the Status column.

A conditional symbol is of the form "pred: S" where pred is a predicate as described in A.3.4.2, and S is a status symbol, M or O.

If the value of the predicate is true (see A.3.4.2), the conditional item is applicable, and its status is indicated by the status symbol following the predicate: the answer column is to be marked in the usual way. If the value of the predicate is false, the "Not Applicable" (N/A) answer is to be marked.

A.3.4.2 Predicates

A predicate is one of the following:

- a) an item-reference for an item in the PCS proforma: the value of the predicate is true if the item is marked as supported and is false otherwise,
- b) a predicate-name, for a predicate defined as a Boolean expression constructed by combining item-references using the Boolean operator OR: the value of the predicate is true if one or more of the items is marked as supported, or
- c) the logical negation symbol "¬" prefixed to an item-reference or predicate-name: the value of the predicate is true if the value of the predicate formed by omitting the "¬" symbol is false, and vice versa.

Each item whose reference is used in a predicate or predicate definition, or in a preliminary question for grouped conditional items, is indicated by an asterisk in the Item column.

A.4 Common requirements

A.4.1 Instructions

One instance of Clause A.4 shall be filled out per IA-station.

A.4.2 Implementation identification

The entire PCS proforma is a form that shall be filled out by a supplier according to Table A.1.

Table A.1 – Implementation identification template

Supplier	
Contact point for queries about the PCS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification, for example, name(s) and version(s) of machines and operating system names	

Only the first three items are required for all implementations; other information can be completed as appropriate in meeting the requirement for full identification. The terms "Name" and "Version" should be interpreted appropriately to correspond with a supplier's terminology (for example, Type, Series, Model).

A.4.3 Profile summary

Table A.2 shows the profile summary template.

Table A.2 – Profile summary template

Identification of profile specification	IEC/IEEE 60802 – Time-Sensitive Networking profile for industrial automation			
Identification of amendments (Amd) and corrigenda (Corr) to the PCS proforma that have been completed as part of the PCS	Amd	:	Corr	:
	Amd	:	Corr	:
Have any Exception items been required? (See A.3.3: the answer "Yes" means that the implementation does not conform to IEC/IEEE 60802)	No	[]	Yes	[]
Date of Statement				

A.4.4 Implementation summary

The form in Table A.3 is used to indicate the type of system that the PCS describes.

Table A.3 – Implementation type

Item	Feature	Status	References	Support
BC-CCA-N	State the number of Conformance Class A bridge components implemented by the IA-station.	O	5.7.2, 5.8.2	Number _____
BC-CCB-N	State the number of Conformance Class B bridge components implemented by the IA-station.	O	5.7.3, 5.8.3	Number _____
ESC-CCA-N	State the number of Conformance Class A end station components implemented by the IA-station.	O.1	5.9.2, 5.10.2	Number _____
ESC-CCB-N	State the number of Conformance Class B end station components implemented by the IA-station.	O.1	5.9.3, 5.10.3	Number _____
CNC	Does the IA-station include a CNC?	O	5.5.4.1 m)	Yes [] No []
CUC	Does the IA-station include a CUC?	O	5.5.4.1 n)	Yes [] No []
PRX	Does the IA-station support proxy functionality?	O	5.5.4.1 o)	Yes [] No []

A.5 IA-station requirements and options

A.5.1 Instructions

One instance of Clause A.5 shall be filled out for an IA-station.

A.5.2 IA-station requirements

The form in Table A.4 is used to indicate the IA-station requirements.

Table A.4 – IA-station requirements

Item	Feature	Status	References	Support
IATA-1	Does the IA-station support PHY and MAC requirements for external ports?	M	5.5.1	Yes []
IATA-2	Does the IA-station support topology discovery requirements?	M	5.5.2	Yes []
IATA-3	Does the IA-station support requirements for time synchronization?	M	5.5.3	Yes []
IATA-4	Does the IA-station provide a digital data sheet?	M	5.5.4.3	Yes []

A.5.3 Management Requirements for non-proxied IA-stations

The form in Table A.5 is used to indicate management requirements for IA-stations containing conformance class A components

Table A.5 – Additional requirements for non-proxied IA-stations

Item	Feature	Status	References	Support
IATA-5	Does the IA-station support requirements for Secure management exchanges?	M	5.5.4.1 a) to l)	Yes []
IATA-6	Number of of Dynamic Subscriptions to YANG Events and Datastores over NETCONF	M	5.5.4.1 h)	Number _____
IATA-7	Does the IA-station support management YANG modules?	M	5.5.4.2	Yes []

A.5.4 IA-station PHY and MAC options for external ports

The form in Table A.6 is used to indicate PHY and MAC options for external ports.

Table A.6 – IA-station PHY and MAC options

Item	Feature	Status	References	Support
DOT3-1	Does the IA-station support PoE over 2 pairs?	O	5.6.1 a)	Yes [] No [] N/A []
DOT3-2	Does the IA-station support Power Interfaces?	O	5.6.1 b)	Yes [] No [] N/A []
DOT3-3	Does the IA-station support PoE?	O	5.6.1 c)	Yes [] No [] N/A []

A.5.5 IA-station options for time synchronization

The form in Table A.7 is used to indicate options for time synchronization.

Table A.7 – IA-station time synchronization options

Item	Feature	Status	References	Support
PTP-1	Does the IA-station support media-independent TimeTransmitter capability according to IEEE Std 802.1AS-2020, 5.4.2 item b) as amended by IEEE Std 802.1ASdr-2024?	O	5.6.2 a)	Yes [] No []
PTP-2	Does the IA-station support Grandmaster PTP Instance capability Grandmaster PTP Instance capability according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024 and IEEE Std 802.1ASdm-2024, 5.4.2 item c)?	O	5.6.2 b)	Yes [] No []
PTP-3	Does the IA-station support more than one PTP port as a PTP Relay Instance according to IEEE Std 802.1AS-2020, 5.4.2 item d)?	O	5.6.2 c)	Yes [] No []
PTP-4	Does the IA-station support hot standby redundancy requirements according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdm-2024, 5.4.2, item m)?	O	5.6.2 d)	Yes [] No []

A.5.6 IA-station secure management exchange options

The form in Table A.8 is used to indicate options for secure management exchange.

Table A.8 – IA-station secure management exchange options

Item	Feature	Status	References	Support
SECMGMT-5	Does the IA-station support Writable-Running capability?	O	5.6.3 a)	Yes [] No []
SECMGMT-6	Does the IA-station support Confirmed Commit capability?	O	5.6.3 b)	Yes [] No []
SECMGMT-7	Does the IA-station support Distinct Startup capability?	O	5.6.3 c)	Yes [] No []
SECMGMT-8	Does the IA-station support URL capability?	O	5.6.3 d)	Yes [] No []
SECMGMT-9	Does the IA-station support XPath capability?	O	5.6.3 e)	Yes [] No []
SECMGMT-10	NETCONF-over-TLS server supporting TLS version 1.2?	O	5.6.3 f)	Yes [] No []
SECMGMT-11	Does the IA-station support NETCONF-over-TLS server supporting TLS version 1.3?	O	5.6.3 g)	Yes [] No []
SECMGMT-12	Does the IA-station support PKIX?	O	5.6.3 h)	Yes [] No []

A.5.7 Proxy target requirements

The form in Table A.9 is used to indicate requirements for proxy targets.

Table A.9 – Proxy target requirements

Item	Feature	Status	References	Support
PT-1	Does the IA-station support Secure management exchanges?	O	5.6.4 a)	Yes [] No []
PT-1	Does the IA-station support Secure management exchanges using a management proxy?	O	5.6.4 b)	Yes [] No []

A.6 Bridge component

A.6.1 Bridge component requirements

The form in Table A.10 is used to indicate bridge component requirements.

Table A.10 – Bridge component requirements

Item	Feature	Status	References	Support
BC-1	Does the bridge component support the common bridge component requirements?	M	5.7.1	Yes []
BC-2	Does the bridge component support ccA bridge component requirements?	O.2	5.7.2	Yes [] No []
BC-3	Does the bridge component support ccB bridge component requirements?	O.2	5.7.3	Yes [] No []

A.6.2 Instructions

One instance of Clause A.6 shall be filled out per bridge component implemented by an IA-station.

A.6.3 Bridge Common bridge component options

The form in Table A.11 is used to indicate bridge component options.

Table A.11 – Common bridge component options

Item	Feature	Status	References	Support
BC-4	Does the bridge component support the operation of the credit-based shaper algorithm?	O	5.8.1	Yes [] No []

A.6.4 Conformance class A bridge component options

The form in Table A.12 is used to indicate options for bridge components conforming to conformance class A.

Table A.12 – Conformance class A bridge component options

Item	Feature	Status	References	Support
CCA-BC-1	Does the bridge component support any of the common bridge component options?	O	5.8.2 a)	Yes [] No [] N/A []
CCA-BC-2	Does the bridge component support more than 2 PTP instances?	O	5.8.2 b)	Yes [] No [] N/A []
CCA-BC-3	State the number of PTP instances supported by the bridge component.	CCA-BC-2:M	5.8.2 b)	Number ____
CCA-BC-4	Does the bridge component support enhancements for scheduled traffic for the 10 Mb/s, 2,5 Gb/s, 5 Gb/s, or 10 Gb/s data rates?	O	5.8.2 c)	Yes [] No [] N/A []
CCA-BC-5	Does the bridge component support frame preemption for the 10 Mb/s, 2,5 Gb/s, 5 Gb/s, or 10 Gb/s data rates?	O	5.8.2 d)	Yes [] No [] N/A []

A.6.5 Conformance class B bridge component options

The form in Table A.13 is used to indicate options for bridge components conforming to conformance class B.

Table A.13 – Conformance class B bridge component options

Item	Feature	Status	References	Support
CCB-BC-1	Does the bridge component support any of the common bridge component options?	O	5.8.3 a)	Yes [] No [] N/A []
CCB-BC-2	Does the bridge component support more than 4 but not more than 8 egress queues?	O	5.8.3 b)	Yes [] No [] N/A []
CCB-BC-3	State the number of egress queues supported by the bridge component.	CCB-BC-2:M	5.8.3 b)	Number ____
CCB-BC-4	Does the bridge component support more than 1 PTP Instance?	O	5.8.3 c)	Yes [] No [] N/A []
CCB-BC-5	State the number of PTP Instances supported by the bridge component.	CCB-BC-4:M	5.8.3 c)	Number ____
CCB-BC-6	Does the bridge component support enhancements for scheduled traffic?	O	5.8.3 d)	Yes [] No [] N/A []
CCB-BC-7	Does the bridge component support frame preemption?	O	5.8.3 e)	Yes [] No [] N/A []

A.7 End station component

A.7.1 Instructions

One instance of Clause A.7 shall be filled out per end station component implemented by an IA-station.

A.7.2 Common end station component requirements

The form in Table A.14 is used to indicate common requirements for end station components.

Table A.14 – Common end station component requirements

Item	Feature	Status	References	Support
ESC-1	Does the end station component support the common end station component requirements?	M	5.9.1	Yes []
ESC-2	Does the end station component support the ccA end station component requirements?	O.3	5.9.2	Yes [] No []
ESC-3	Does the end station component support the ccB end station component requirements?	O.3	5.9.3	Yes [] No []

A.7.3 Common end station component options

The form in Table A.15 is used to indicate options for end station components.

Table A.15 – Common end station component options

Item	Feature	Status	References	Support
ESC-4	Does the end station component support the operation of the credit-based shaper?	O	5.10.1 a)	Yes [] No []
ESC-5	Does the end station component support talker end system behaviors?	O	5.10.1 b)	Yes [] No []
ESC-6	Does the end station component support listener end system behaviors?	O	5.10.1 c)	Yes [] No []

A.7.4 Conformance class A end station component options

The form in Table A.16 is used to indicate options for end station components conforming to conformance class A.

Table A.16 – Conformance class A end station component options

Item	Feature	Status	References	Support
CCA-ESC-1	Does the end station component support any of the common end station component options?	O	5.10.2 a)	Yes [] No [] N/A []
CCA-ESC-2	Does the end station component support more than 2 PTP Instances?	O	5.10.2 b)	Yes [] No [] N/A []
CCA-ESC-3	State the number of PTP Instances supported by the end-station component.	CCA-ESC-2:M	5.10.2 b)	Number ____
CCA-ESC-4	Does the end station component support enhancements for scheduled traffic for data rates 10 Mb/s, 2,5 Gb/s, 5 Gb/s, or 10 Gb/s?	O	5.10.2 c)	Yes [] No [] N/A []
CCA-ESC-5	Does the end station component support requirements for frame preemption for data rates 10 Mb/s, 2,5 Gb/s, 5 Gb/s, or 10 Gb/s?	O	5.10.2 d)	Yes [] No [] N/A []

A.7.5 Conformance class B end station component options

The form in Table A.17 is used to indicate options for end station components conforming to conformance class B.

Table A.17 – Conformance class B end station component options

Item	Feature	Status	References	Support
CCB-ESC-1	Does the end station component support any of the common end station component options?	O	5.10.3 a)	Yes [] No [] N/A []
CCB-ESC-2	Does the end station component support one or more PTP Instances?	O	5.10.3 b)	Yes [] No [] N/A []
CCB-ESC-3	State the number of PTP Instances supported by the end-station component.	CCB-ESC-2:M	5.10.3 b)	Number ____
CCB-ESC-4	Does the end station component support enhancements for scheduled traffic?	O	5.10.3 c)	Yes [] No [] N/A []
CCB-ESC-5	Does the end station component support requirements for frame preemption?	O	5.10.3 d)	Yes [] No [] N/A []

Annex B (informative)

Representative Configuration Domain

The following quantities are representative of what can be supported in a single Configuration Domain.

- IA-stations: 1 024.
- Network diameter: 64.
- Streams per IA-controller for IA-controller to IA-device (C2D) communication:
 - 512 Talker and ≥ 512 Listener streams, and
 - 1 024 Talker and $\geq 1\,024$ Listener streams in case of seamless redundancy.
- Streams per IA-controller for IA-controller to IA-controller (C2C) communication:
 - 64 Talker and ≥ 64 Listener streams, and
 - 128 Talker and ≥ 128 Listener streams in case of seamless redundancy.
- Streams per IA-device for IA-device-to-IA-device (D2D) communication:
 - 2 Talker and 2 Listener streams, and
 - 4 Talker and 4 Listener streams in case of seamless redundancy.
- Example calculation of data flow quantities for eight PLCs – without seamless redundancy:
 - $8 \times 512 \times 2 = 8\,192$ streams for C2D communication,
 - $8 \times 64 \times 2 = 1\,024$ streams for C2C communication, and
 - $(8\,192 + 1\,024) \times 2\,000 = 18\,432\,000$ octets data for all streams.

Annex C (informative)

Description of clock control system

C.1 Clock control system introduction

Annex C provides an introductory discussion of a basic clock control system. For more detailed information, see the Bibliography References ITU-T Recommendation G.8260 and ITU-T Series G Supplement 65.

Figure C.1 shows a basic control system model that uses a proportional plus integral (PI) controller. This is meant to be reference model, i.e. it is not meant to specify an implementation. Requirements for the clock control system can be expressed using parameters (e.g. 3 dB bandwidth, gain peaking, frequency response) that are based on this reference model. Any implementation whose parameters are within the requirements is considered to be acceptable. For example, the model of Figure C.1 is expressed in the analog domain (i.e. s-domain), and is shown to be second order. An actual implementation can be digital, and can be higher order, as long as it meets the respective requirements.

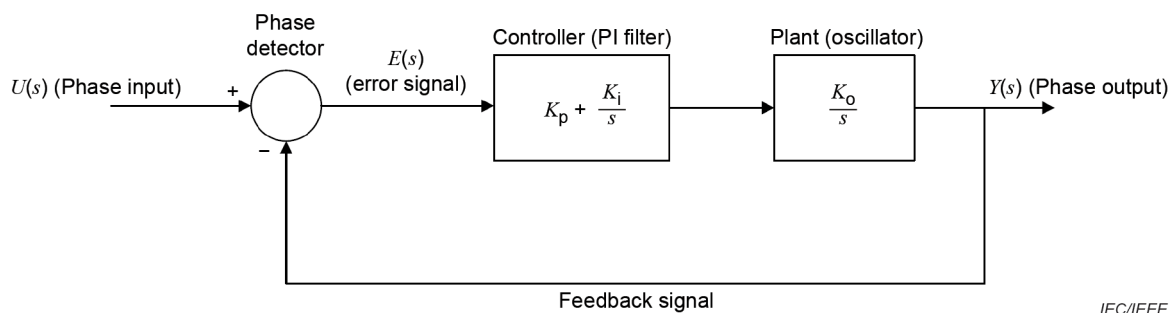


Figure C.1 – Reference model for clock control system

In Figure C.1, the plant, i.e. the entity being controlled, represents the clock oscillator. It is desired that the phase output, $y(t)$, of the oscillator follows the phase input, $u(t)$, as closely as possible (the signals are shown in the frequency domain (i.e. as Laplace Transforms) in Figure C.1; however, they can equivalently be expressed in the time domain, with t representing time). The parameter K_o is the oscillator gain; the oscillator frequency is equal to the oscillator input multiplied by K_o . In some implementations the input signal to the oscillator is a voltage, and the oscillator is referred to as a voltage-controlled oscillator (VCO). However, other implementations are possible, e.g. digital implementations, where the oscillator is a digital controlled oscillator (DCO). Since the input to the oscillator depends on the implementation, it is not labeled in Figure C.1.

The control system of Figure C.1 uses negative feedback to enable the phase output to follow the phase input. The phase detector computes the difference between the input and output signals to produce the error signal $e(t)$. The error signal is then filtered by the PI filter to produce the input to the oscillator. The filter is referred to as a PI filter because its output is the sum of the proportional gain, K_p , multiplied by the error signal and the integral gain, K_i , multiplied by the integral of the error signal. The gains K_o , K_p , and K_i must be chosen such that the performance of the control system is acceptable, i.e. the time-domain behavior of the output with respect to the input is acceptable. However, an alternative set of parameters, which are more convenient, can be defined in terms of K_o , K_p , and K_i ; this is done in Clause C.2.

C.2 Transfer function for control system

From the block diagram of Figure C.1, the input and output are related by:

$$Y(s) = \left(K_p + \frac{K_i}{s} \right) \left(\frac{K_o}{s} \right) (U(s) - Y(s)) \quad (C.1)$$

where

$Y(s)$ is the phase output, expressed in the s-domain;

$U(s)$ is the phase input, in the s-domain;

K_p is the proportional gain;

K_i is the integral gain;

K_o is the oscillator gain.

or

$$Y(s) = \frac{\left(K_p + \frac{K_i}{s} \right) \left(\frac{K_o}{s} \right)}{1 + \left(K_p + \frac{K_i}{s} \right) \left(\frac{K_o}{s} \right)} U(s) \quad (C.2)$$

This can be simplified by multiplying the numerator and denominator by s^2 to produce:

$$Y(s) = H(s)U(s) \quad (C.3)$$

where the transfer function $H(s)$ is given by:

$$H(s) = \frac{K_p K_o s + K_i K_o}{s^2 + K_p K_o s + K_i K_o} \quad (C.4)$$

In Formula (C.4), the parameter K_o does not appear independently of K_p and K_i ; rather, only the products $K_p K_o$ and $K_i K_o$ appear. The plant and PI filter could have been combined in the model of Figure C.1; this is consistent with the fact that the exact nature of the signal between the PI filter and plant is unimportant in this reference model. The units of $K_p K_o$ are (time)⁻¹ and the units of $K_i K_o$ are (time)⁻². The frequency units are the same as the units of s , e.g. if s has units rad/s, then $K_p K_o$ has units rad/s and $K_i K_o$ has units (rad/s)². The integration operation in the plant results in the transfer function being dimensionless, which is consistent with the fact that the input and output of the control system both have units of phase.

The transfer function can be expressed in an equivalent form by defining the undamped natural frequency, ω_n , and damping ratio, ζ :

$$H(s) = \frac{2\zeta\omega_n s + \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (\text{C.5})$$

where

ζ is the damping ratio;

ω_n is the undamped natural frequency expressed in rad/s.

And, ζ and ω_n are given by:

$$\omega_n = \sqrt{K_i K_o} \quad (\text{C.6})$$

$$\zeta = \frac{K_p K_o}{2\sqrt{K_i K_o}} = \frac{K_p}{2} \sqrt{\frac{K_o}{K_i}} \quad (\text{C.7})$$

In the Formula (C.7), the first form shows explicitly that ζ depends only on the products $K_p K_o$ and $K_i K_o$.

C.3 Frequency response for control system

The frequency response is obtained by setting $s = j\omega$ in Formula (C.5) and taking the absolute value (here j rather than i is used for $\sqrt{-1}$ to avoid confusion with other uses of i), where ω is the frequency in rad/s. The result is:

$$|H(j\omega)| = \left| \frac{2\zeta\omega_n \omega j + \omega_n^2}{-\omega^2 + \omega_n^2 + 2\zeta\omega_n \omega j} \right| = \left(\frac{4\zeta^2 \omega_n^2 \omega^2 + \omega_n^4}{(\omega_n^2 - \omega^2)^2 + 4\zeta^2 \omega_n^2 \omega^2} \right)^{1/2} \quad (\text{C.8})$$

Dividing the numerator and denominator of Formula (C.8) by ω_n^4 and defining the dimensionless frequency $x = \omega/\omega_n$ produces:

$$|H(j\omega)| = \left(\frac{4\zeta^2 x^2 + 1}{(1 - x^2)^2 + 4\zeta^2 x^2} \right)^{1/2} \quad (\text{C.9})$$

where

ζ is the damping ratio;

ω is the frequency expressed in rad/s;

$$x = \omega/\omega_n.$$

Figure C.2 contains plots of frequency response (Formula (C.9)) versus dimensionless frequency x , on a log-log scale, for damping ratio ζ equal to 0,3, 0,5, 0,707, 1,0, 2,0, 3,0, 4,0, and 5,0. It is seen that the frequency response is very close to 1 for values of dimensionless frequency much less than 1 (i.e. for $\omega \ll \omega_n$). The frequency response increases as the frequency approaches the undamped natural frequency (i.e. as dimensionless frequency approaches 1) and reaches a peak for dimensionless frequency slightly less than 1. The frequency response then decreases, eventually having a slope (i.e. roll-off) of 20 dB/decade (i.e. frequency response decreases by a factor of 10 for every factor of 10 increase in x for $x \gg 1$). Figure C.3 shows the detail of frequency response for x in the range 0,1 to 10.

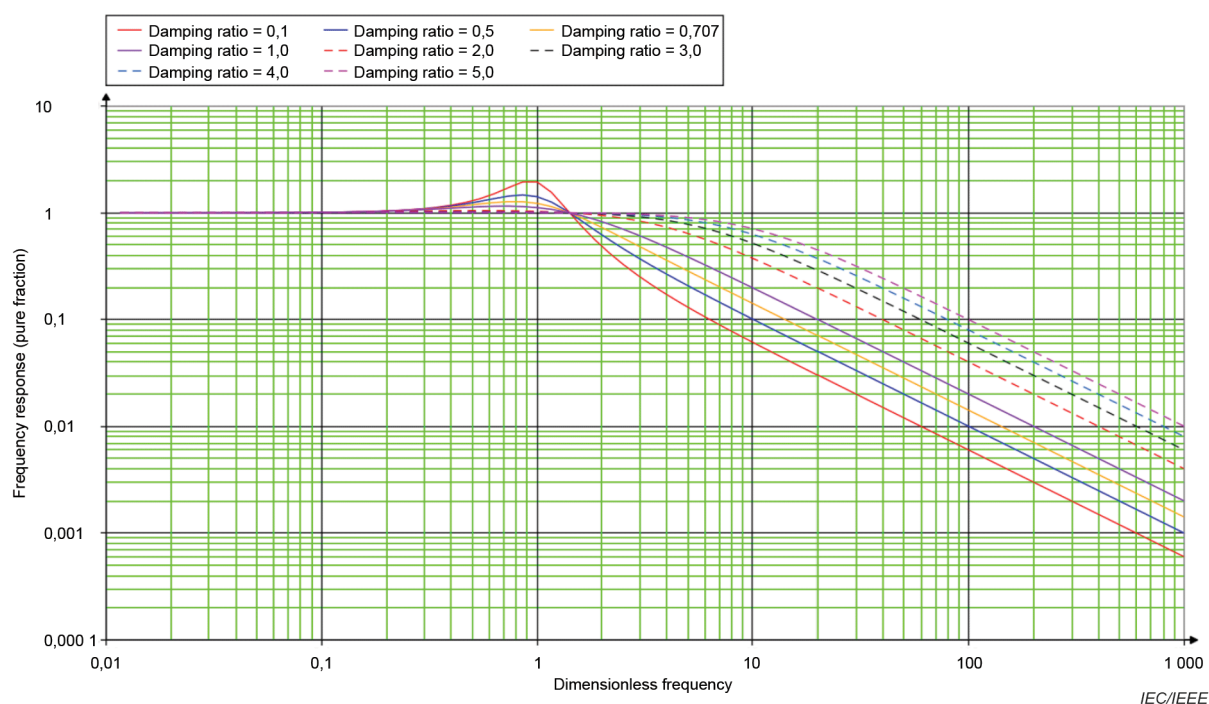


Figure C.2 – Frequency response for the control system of Figure C.1

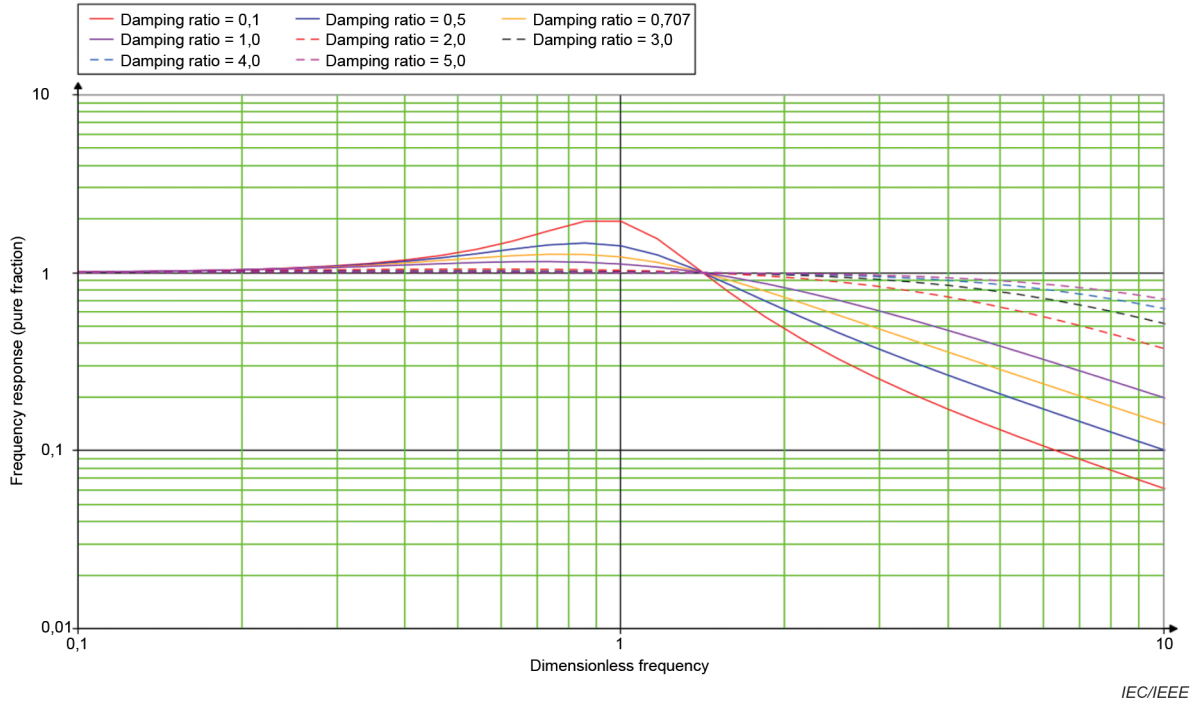


Figure C.3 – Detail of frequency response for the control system of Figure C.1 for dimensionless frequency in the range 0,1 to 10

In addition to undamped natural frequency ω_n and damping ratio ζ , the parameters 3 dB bandwidth and gain peaking are often used when specifying clock performance. The 3 dB bandwidth is defined as the value of frequency for which the frequency response is equal to -3 dB. Since dB is given by 10 multiplied by the logarithm to base 10 of the power ratio, which is 20 multiplied by the logarithm to base 10 of the amplitude ratio, -3 dB corresponds to the value $10^{-3/20}$. The 3 dB bandwidth can be computed by setting Formula (C.9) equal to $10^{-3/20}$ and solving for x in terms of ζ . This is equivalent to setting the quantity in parentheses (i.e. inside the square root) in Formula (C.9) equal to $10^{-3/10}$ and solving for x . Now, $10^{-3/10}$ is approximately equal to 0,501 2, i.e. it is very close to $\frac{1}{2}$. Then the 3dB bandwidth can be obtained by solving the following formula for x in terms of ζ :

$$\frac{4\zeta^2 x^2 + 1}{(1 - x^2)^2 + 4\zeta^2 x^2} = \frac{1}{2} \quad (\text{C.10})$$

or

$$x^4 - 2(2\zeta^2 + 1)x^2 - 1 = 0 \quad (\text{C.11})$$

The result is:

$$x = \left[2\zeta^2 + 1 + \sqrt{(2\zeta^2 + 1)^2 + 1} \right]^{1/2} \quad (\text{C.12})$$

or

$$\omega_{3dB} = \omega_n \left[2\zeta^2 + 1 + \sqrt{(2\zeta^2 + 1)^2 + 1} \right]^{1/2} \quad (C.13)$$

The gain peaking is the maximum value of the frequency response, in dB. It is computed by differentiating Formula (C.9) with respect to x , setting the result to zero, solving for x , and then substituting this value of x into Formula (C.9) to obtain the maximum. The result is:

$$H_p = \left[1 - 2\alpha - 2\alpha^2 + 2\alpha(2\alpha + \alpha^2)^{1/2} \right]^{-1/2} \quad (C.14)$$

where α is related to damping ratio by:

$$\alpha = \frac{1}{4\zeta^2} \quad (C.15)$$

and H_p is the gain peaking expressed as a pure fraction. The gain peaking in dB is equal to $20 \cdot \log_{10} H_p$. In some cases, it is necessary to compute damping ratio from gain peaking. The result for this is:

$$\alpha = \frac{(1-q)(1+\sqrt{1-q})}{2q} \quad (C.16)$$

where

$$q = \frac{1}{H_p^2} \quad (C.17)$$

Damping ratio is obtained from α using Formula (C.15).

If 3 dB bandwidth and gain peaking are given, damping ratio can be obtained using Formulas (C.15), (C.16) and (C.17). Undamped natural frequency can then be obtained using Formula (C.13).

Figure C.4 shows gain peaking, expressed as a pure fraction, as a function of damping ratio. Figure C.5 shows gain peaking in dB as a function of damping ratio.

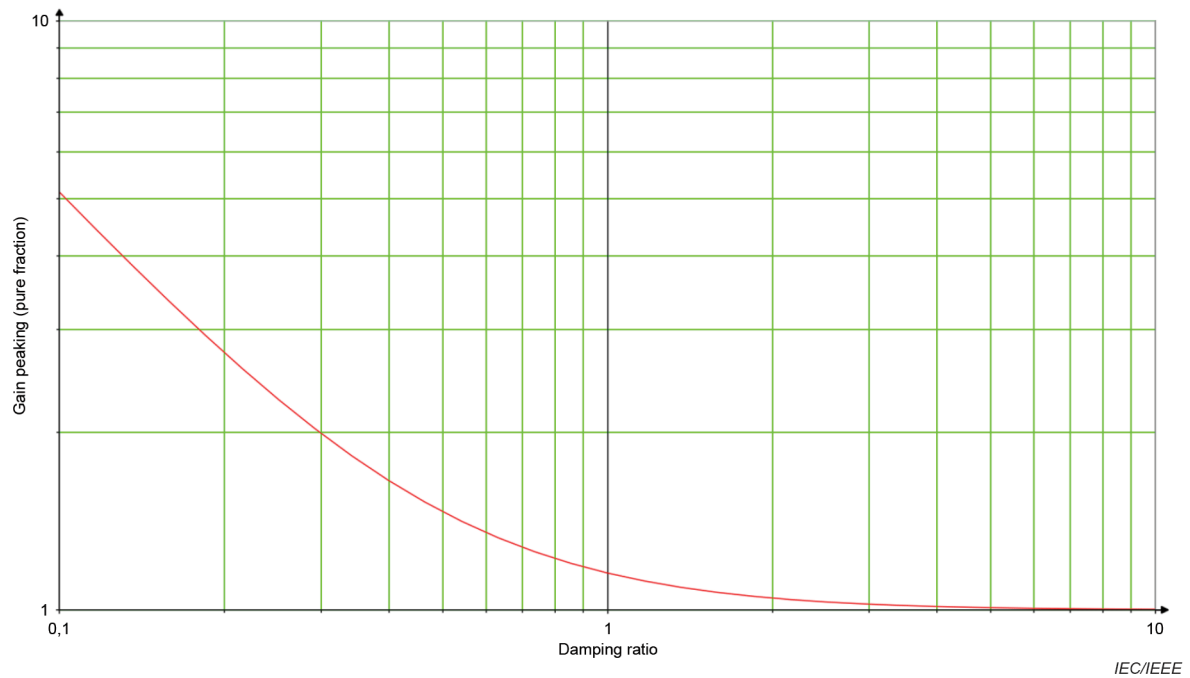


Figure C.4 – Gain peaking (pure fraction) as a function of damping ratio

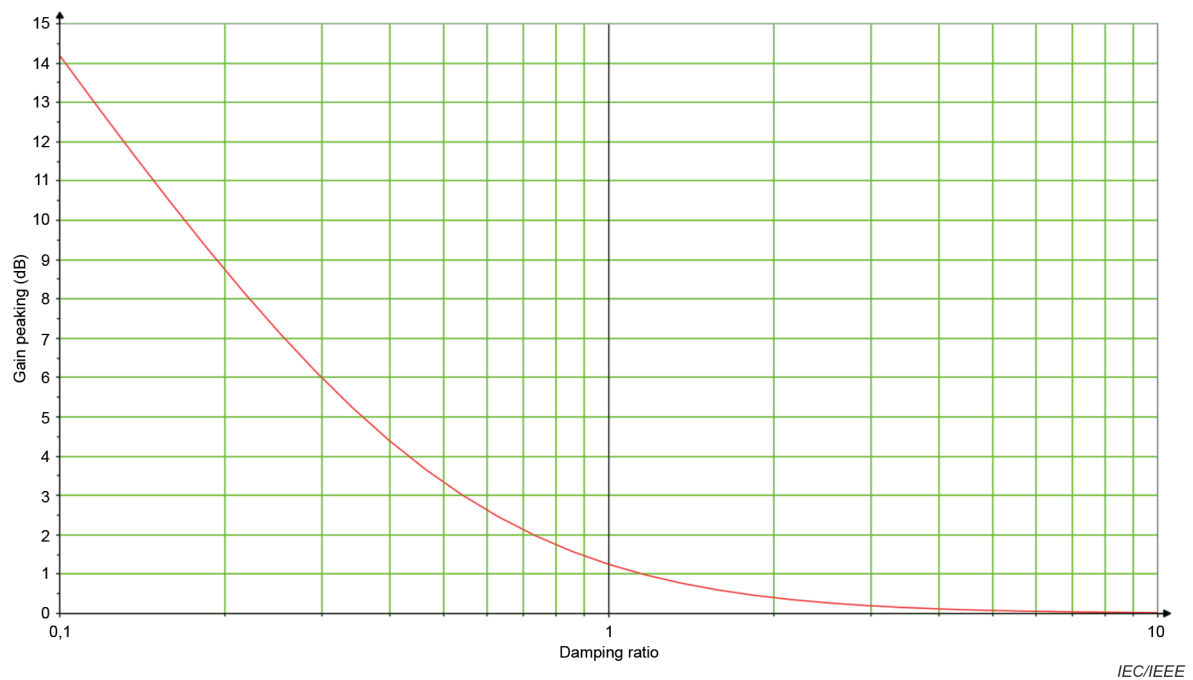


Figure C.5 – Gain peaking in dB as a function of damping ratio

The performance of the clock control system can be described using the frequency response as follows:

- Maximum 3 dB bandwidth in Hz,
- Maximum gain peaking in dB, and
- Frequency response plot (mask) corresponding to (a) and (b) that is not to be exceeded.

C.4 Example

Consider a clock control system with $K_p K_o = 4,23 \text{ rad/s}$ and $K_i K_o = 9,62 (\text{rad/s})^2$. The undamped natural frequency and damping ratio are:

$$\omega_n = \sqrt{K_i K_o} = \sqrt{9,62 (\text{rad/s})^2} = 3,10 \text{ rad/s} \quad (\text{C.18})$$

$$\zeta = \frac{K_p K_o}{2\sqrt{K_i K_o}} = \frac{4,23 \text{ rad/s}}{2\sqrt{9,62 (\text{rad/s})^2}} = 0,682 \quad (\text{C.19})$$

The gain peaking is obtained from:

$$\alpha = \frac{1}{4(0,682)^2} = 0,537 \quad (\text{C.20})$$

$$H_p(\text{purefraction}) = \left[1 - 2(0,537) - 2(0,537)^2 + 2(0,537)\sqrt{2(0,537) + (0,537)^2} \right]^{-\frac{1}{2}} = 1,28803 \quad (\text{C.21})$$

$$H_p(\text{dB}) = 20 \log_{10}(1,29) \text{ dB} = 2,2 \text{ dB} \quad (\text{C.22})$$

The 3dB bandwidth is:

$$f_{3\text{dB}}(\text{Hz}) = \frac{\omega_n}{2\pi} \left[1 + 2\zeta^2 + \sqrt{(1 + 2\zeta^2)^2 + 1} \right]^{1/2} \quad (\text{C.23})$$

$$= \frac{3,10}{2\pi} \left[1 + 2(0,682)^2 + \sqrt{(1 + 2(0,682)^2)^2 + 1} \right]^{1/2} = 1,0 \text{ Hz}$$

The frequency response is shown in Figure C.6.

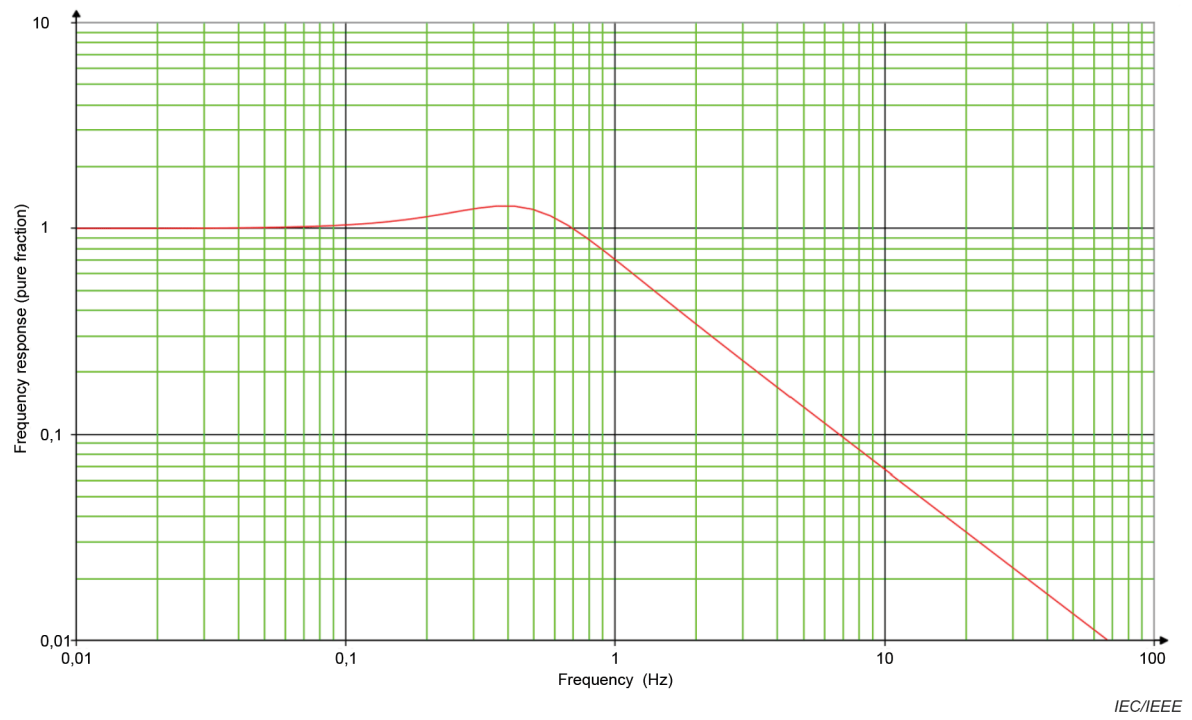


Figure C.6 – Example frequency response

Annex D (informative)

Time synchronization annex

D.1 Overview

Annex D describes how a network of compliant devices can achieve a time synchronization accuracy, at the application level, of $\pm 1 \mu\text{s}$, relative to the Clock Source at the Grandmaster, over 100 network hops. To achieve this, it allocates the overall error budget of 1 000 ns as described in Table D.1.

Table D.1 – Time synchronization error budget

Network Aspect	Error Type	Network-Level Error Budget (ns)
All PTP Instances	Constant time error	200
	Dynamic time error	600
All PTP links	Constant time error	200
	Dynamic time error	

A chain of 1 Grandmaster PTP Instance, 99 PTP Relay Instances and 1 PTP End Instance (100 network hops) that all comply with the normative requirements of 6.2.2, 6.2.3, 6.2.4, and 6.2.5 generates a network-level Time Error at or below the Error Budget for All PTP Instances.

Clause D.2 describes the principles of operation this document assumes.

Clause D.3 provides additional information on specific normative requirements.

The principles of operation include the use of crystal oscillators (XOs) as opposed to more accurate, stable, and costly options such as temperature-compensated crystal oscillators (TCXOs).

Clause D.4 describes a potential approach to testing the normative requirements. It is not a test specification, but rather a high-level overview of one potential approach that can be adopted by a full test specification.

The use of XOs means that some of the normative requirements are difficult or impossible to meet without employing algorithms that track Neighbor Rate Ratio (NRR) drift and Rate Ratio drift and compensate for consequent errors in calculating Rate Ratio and Correction Field.

Clause D.5 provides examples of algorithms that can be used for this purpose, and which have been shown to enable compliance with the normative requirements.

Implementations that employ TCXOs or other more accurate, stable oscillators can still find some of the normative requirements difficult or impossible to meet without employing algorithms to track and compensate for errors due to clock drift. This is because other PTP Instances that use XOs can still cause the implementation to experience Neighbor Rate Ratio drift, Rate Ratio drift or both.

There is no normative requirement to use the algorithms described in Clause D.5; an implementation can employ alternative algorithms provided the normative requirements are met. Clause D.5 describes the potential risks of deploying a network whose instances employ a mix of different algorithms. It is the responsibility of implementers to mitigate the risks and ensure alternative algorithms deliver the desired network-level performance.

This document does not include normative requirements for PTP Links. Annex D.2.4 describes PTP Link characteristics that influence achieving 1 μ s time synchronization accuracy. It includes some examples using common PTP Link characteristics.

This document's normative requirements regarding instance-level error generation are intended to ensure not just an overall level of dTE generation at each node, but also the performance of drift tracking and error compensation algorithms and the amount of dTE generation due to timestamp error versus clock drift. The algorithms are employed to mitigate errors due to clock drift but cannot mitigate timestamp errors.

D.2 Principles of operation

D.2.1 General

Achieving ± 1 μ s time synchronization accuracy across 100 network hops involves managing the accumulation of errors in the preciseOriginTimestamp plus correctionField and the Rate Ratio as they are passed, via Sync or Follow_Up messages, down the chain of PTP Instances and are then used by the PTP End Instance to keep its ClockTarget in line with the ClockSource at the Grandmaster PTP Instance. The majority of significant errors can ultimately be traced back to one of three sources: timestamp error, clock drift, or path delay asymmetry. The selection of PTP protocol parameters often involves trading off one source of error against the other. This document requires specific PTP protocol configurations, and assumes the use of mechanisms (algorithms), that reduce dTE due to timestamp error but would also – without additional measures – increase dTE due to clock drift to the point where the latter exceeds the allocated error budget. However, this document also assumes additional measures to minimize some sources of dTE due to clock drift, and mechanisms and to track and compensate for errors from other sources to a sufficient degree that the error budget is not exceeded.

The specific protocol configurations and other measures, along with their intended effects, are described in Table D.2.

Table D.2 – Protocol configurations & other measures to achieve dTE budget

Configuration or measure	Description and intended effect(s)
Sync Interval 125 ms	Effects: 1) Calibrates the balance between dTE from timestamp error vs error due to clock drift. Larger intervals lead to less timestamp error and more error due to clock drift. 2) Keeps below acceptable limits the impact of errors in Rate Ratio and Rate Ratio Drift estimation when keeping ClockTarget in line with ClockSource between arrival of Sync messages. Larger intervals increase the impact of any errors.
Drift_Tracking TLV – syncEgressTimestamp	Effect: Enables calculation of NRR using Sync message timestamps, which eliminates error due to NRR clock drift that would otherwise occur between calculation of NRR using Pdelay_Resp messages and use during Sync message processing (i.e. calculation of Rate Ratio and output Correction Field values).

Configuration or measure	Description and intended effect(s)
NRR Smoothing	Description: Algorithm to use timestamps from multiple past Sync messages when calculating NRR. Effect: Reduces the amount of error in the estimate of NRR due to timestamp error while increasing the amount of error due to clock drift.
NRR Drift Tracking & Compensation	Description: Algorithm to use timestamps from multiple past Sync messages to estimate NRR drift rate and then apply compensation to correct for consequent errors in NRR calculation. Effect: Mitigates the effect of errors due to clock drift when calculating and using the estimated NRR.
Drift_Tracking TLV – rateRatioDrift	Description: Carries estimate of Rate Ratio drift rate from one node to the next. Effect: Allows each node to estimate its own Rate Ratio drift rate by combining the incoming Rate Ratio drift rate with the local estimate of NRR drift rate.
RR Drift Compensation	Description: Algorithm that uses the estimate of RR drift rate to compensate for that drift, adjusting the estimated RR over time according to the drift rate. Effect: For PTP Relay Instances, minimizes errors in the Correction Field caused by Rate Ratio drift. For PTP End Instances, a similar approach can reduce errors in keeping ClockTarget in line with ClockSource between arrival of Sync messages, but is outside the scope of this document.
Pdelay Interval Consistency	Description: This document requires tighter control of the interval between Pdelay messages generated at the Grandmaster PTP Instance than the defaults in IEEE Std 802.1AS-2020. Effect: This document requires the use of Sync messages to calculate NRR (see above). However, when a sufficient number of Sync messages are not available, for example on startup or after a reconfiguration, Pdelay_Resp messages can be used instead. In such cases, errors due to clock drift at Relay Instances have a tendency to cancel out. A clock drift that generates a positive error in NRR measurement on receipt of a Pdelay_Resp message generates a negative error in NRR measurement at the next node. The degree of cancellation depends on the consistency of the intervals over which NRR is measured at neighboring nodes. Tighter control of the Pdelay Interval increases the consistency of the measurement interval and thus decreases the amount of error.
Mean Residence Time	Description: This document defines a mean Residence Time requirement, which is significantly lower than the default maximum Residence Time in IEEE Std 802.1AS-2020. Effect: The amount of error in the Correction Field at the PTP End Instance due to clock drift is proportional to the cumulative meanLinkDelay and residenceTime experienced by a Sync message during transit from the Grandmaster PTP Instance to the PTP End Instance. Specifying a lower mean residenceTime reduces this source of error.

D.2.2 Grandmaster PTP Instance implementation

Depending on implementation, a Grandmaster PTP Instance can:

- a) contain a single oscillator used for both Local Clock and Clock Source,
- b) contain separate oscillators for Local Clock and Clock Source, or
- c) contain only an oscillator for Local Clock and accept an external input for Clock Source.

In some cases, a Grandmaster PTP Instance can support more than one mode of operation and transition between them depending on changes in network configuration (see Splitting, Joining and Aligning Time Domains).

In the first case, the rateRatio transmitted by the Grandmaster PTP Instance is one (i.e. 0 ppm) the rateRatioDrift is 0 ppm/s, reflecting the fact there is no difference between the Local Clock and Clock Source frequencies.

In the second and third cases there can be differences between the Local Clock and Clock Source frequencies. Any differences are reflected in the rateRatio and rateRatioDrift fields transmitted by the Grandmaster PTP Instance. This means that the Grandmaster PTP Instance tracks rateRatio over time in order to calculate rateRatioDrift, similarly to PTP Relay Instances and PTP End Instances. The exact implementation can vary.

D.2.3 Splitting, joining and aligning time domains

D.2.3.1 General

Modular machines or production cells can allow the splitting and combining of machines if this is required by the production process. When separate, the ClockSources of two machines run separately, each with its own time domain. If both ClockSources are traceable to the same PTP timescale, the difference between the ClockSources can be relatively small. If traceable to different timescales, especially if one or both are ARB timescales, there can be a very large difference between the ClockSources.

When two machines are joined, the first machine's time domain remains unaffected, and it can continue operation without disruption. There are two typical approaches to how the second machine behaves. In the first case, at a time of the end user's choosing, the second machine's time domain ceases to exist, with its PTP Instances becoming part of the first machine's time domain. In the second case, the second machine's time domain is gradually aligned with the first machine's time domain such that control loop cycles are coordinated. In the first case the second machine's time domain is unaffected, and it can continue production even if the machines are accidentally connected, until the end user chooses to join the time domains. In the second case the second machine can continue production while its time domain is being aligned.

D.2.3.2 Joining machines with single time domain

In the first case, where the second machine's time domain ceases to exist, a discontinuity in timing for the second machine's PTP Instances can occur, as they switch to use the first machine's Grandmaster. Some implementations implement measures to limit such timing discontinuities, but these measures are outside the scope of this document. Typically, in this case, the second machine is not operational while it is joined to the first. It resumes operation once its PTP Instances have synchronized with the first machine's Grandmaster.

D.2.3.3 Joining machines with multiple coordinated time domains

In the second case, where the second machine's time domain is gradually aligned with the first machine's time domain, this typically requires both machines to be implementing the same control loop cycle time. The goal is that, once coordinated, each control loop cycle of the first machine is aligned with the start of a control loop cycle of the second machine, even though the two machines maintain separate time domains and there can be a large time difference between their Clock Sources.

In this case, after being joined together, the first machine effectively drives the second machine's Clock Source faster or slower, during an alignment period, until coordination is achieved. During the alignment period, this drive from the first machine can result in the second machine's Clock Source temporarily exceeding the usual normative requirement on range of fractional frequency offset relative to the nominal frequency of ± 50 ppm. The usual normative requirement on range of rate of change of fractional frequency offset of ± 1 ppm/s, applicable when split (i.e. independent) or coordinated (i.e. joined and stable, after the alignment period), may also be temporarily exceeded. However, if the value stays within the range ± 3 ppm/s, the network-level performance of $1\ \mu\text{s}$ time synchronization accuracy can be maintained. For this reason, this document specifies a separate normative requirement for temporary, externally driven, rate of rate of change of fractional frequency offset.

Since the second machine experiences no time discontinuities and the network-level performance is maintained the second machine can, if desired, continue operation during the alignment period.

Once coordinated, the first machine continues to drive the Clock Source of the second machine to maintain coordination. In this stable, coordinated mode of operation the normal range of ± 1 ppm/s is not exceeded.

The mechanism by which the first machine drives the Clock Source of the second machine is not addressed in this document.

D.2.3.4 Splitting machines

In the first case, where the second machine's time domain ceased to exist while joined to the first, splitting machines means that the second machine creates its own time domain again. The second machine's Clock Source typically starts at the PTP Grandmaster Instance's last, best estimate of the first machine's Clock Source. The goal is for no discontinuities in time sync to occur; however, depending on implementation, it can take some time to before the time synchronization accuracy of all the second machine's PTP Instances relative to its Grandmaster can be relied upon. For this reason, it is possible the second machine is not operational during the split. Hot Standby can be employed to mitigate this transition time, but the details of how to do so are out of scope for this document.

In the second case, where the second machine maintains its time domain while joined to the first, splitting machines means that the first machine ceases driving the second machine's Clock Source to maintain coordination of control loop cycle times. Without this drive, the two time domains can drift relative to each other resulting in loss of coordination. Time synchronization performance within the second machine is maintained during the split and the second machine can, if desired, continue operation throughout the process.

D.2.4 PTP link characteristics

A vast majority of time synchronization error due to PTP link characteristics is caused by asymmetrical path delay in one direction versus the other. The mechanism to measure path delay assumes the link is symmetrical and cannot detect asymmetry, thus asymmetry causes an error. The potential maximum asymmetry and thus error typically scales linearly with physical path length.

The error budget due to PTP link characteristics for an entire network is 200 ns. In any specific network this budget can be allocated as required with some links allocated a higher budget (typically longer length) than others.

A typical specified maximum delay skew for Category 6 Ethernet cables is 50 ns per 100 m. Some Ethernet MAU types such as 100BASE-TX use different pairs for transmit and receive. If such cables are used, a maximum total cable length between Clock Source and Clock Target with 99 PTP Relay Instances between them (i.e. 100 network hops) is 400 m. Extending the cable length beyond 400 m without jeopardizing network-level performance would require the use of cables with less delay skew or asymmetry compensation for delay skew.

It is possible for the delay skew in one section of cable to cancel all or part of a delay skew in the opposite direction from prior section but, depending on how cables are manufactured and deployed, it is feasible for the delay skews of every cable segment between a Grandmaster PTP Instance and a PTP End Instance to be additive.

D.3 Notes on normative requirements

D.3.1 Oscillator requirements

Clock drift at the Grandmaster PTP Instance causes greater dTE than the same amount of clock drift at a PTP Relay Instance or the PTP End Instance. This document therefore requires tighter limits on maximum fractional frequency offset for an oscillator at the Grandmaster PTP Instance than at other instances.

This document does not place requirements on operational temperature range or other environmental factors. The required oscillator behavior is delivered for the operational conditions across which a device claims it is compliant. These conditions typically include temperature range but can also include rate of change of ambient temperature, supply voltage stability, amount of vibration and others.

D.3.2 Timestamp Granularity Error

Timestamp Granularity Error (TSGE) is the error in timestamping each incoming and outgoing message due to the maximum timestamp resolution of which an implementation is capable. It is typically directly related to an implementation's clock rate.

For example: a clock rate of 125 MHz typically results in a maximum resolution of 8 ns. Depending on implementation the consequent TSGE range can be -8 ns to 0 ns, 0 ns to 8 ns, or anything in between. In some implementations, offsets are applied to ensure the average TSGE is 0 ns with, assuming uniform error distribution, a range of -4 ns to $+4$ ns

Similarly, a clock rate of 500 MHz results in a maximum resolution of 2 ns; a consequent TSGE range between -2 ns to 0 ns and 0 ns to 2 ns; and, if a suitable offset is applied to ensure a TSGE average of 0 ns, a range of -1 ns to $+1$ ns.

A minimum resolution of 8 ns, i.e. minimum clock rate of 125 MHz is assumed. It is further assumed that TSGE for the sum of the `preciseOriginTimestamp` and `followUpCorrectionField` at the Grandmaster PTP Instance (see IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdm-2024, 10.2.9.2.1) has an average of 0 ns and that the TSGE averages for other timestamps are stable and consistent across all a PTP Instance's ports. No assumption regarding the value of the TSGE average for these other timestamps is necessary as they are always used to measure intervals such that any stable, consistent offset cancels out.

D.3.3 Dynamic Timestamp Error

Dynamic Timestamp Error (DTSE) is the, effectively random, error in timestamping each incoming and outgoing event message due to an implementation's inherent inaccuracies, excluding TSGE. It is assumed to vary between a minimum of -6 ns and a maximum of $+6$ ns with an average of 0 ns. Lower levels of DTSE are better.

If an implementation timestamps an incoming or outgoing message at a point other than as specified in IEEE Std 802.3-2022 as amended by IEEE Std 802.3cx-2023 and IEEE Std 802.3de-2022, any variability in delay between that point and the delay point specified in 802.3 translates to DTSE. Some common implementations were not designed to limit this variability. If care is not taken to avoid implementations with high variability, the assumed DTSE range is easily exceeded. Such implementations can find some of the normative requirements difficult or impossible to meet.

D.3.4 Grandmaster PTP Instance error generation

Table 12 sets normative requirements for error generation at a Grandmaster PTP Instance that ensure the relevant fields in the Sync and Follow_Up messages it transmits are sufficiently accurate to deliver the network-level performance. Table D.3 describes how the normative requirements align with major sources of error.

Table D.3 – Main sources of error affecting normative requirements for Grandmaster PTP Instance

Item	Normative requirement	Main sources of error
1	preciseOriginTimestamp + correctionField vs Direct measurement of Working Clock at Grandmaster (acting as a Clock Source)	Timestamp error relative to Clock Source plus accuracy measuring any internal delay between generation of the preciseOriginTimestamp and Sync message transmission.
2	rateRatio vs Direct measurement of Rate Ratio of Clock Source vs Local Clock	Accuracy of internal mechanism to measure Rate Ratio of Clock Source vs. Local Clock, potentially including algorithms that track RateRatioDrift and modify Rate Ratio accordingly ^a
3	syncEgressTimestamp vs Direct measurement of Local Clock	Timestamp error relative to Local Clock
4	(responseOriginTimestamp – requestReceiptTimestamp) vs Direct measurement of Local Clock interval between reception of Pdelay_Req and transmission of Pdelay_Resp	Timestamp error relative to Local Clock
^a Only applicable if Clock Source and Local Clock are not locked to the same frequency by the implementation. If they are locked, then rateRatio is 0 ppm and rateRatioDrift is 0 ppm/s.		

D.3.5 PTP Relay Instance error generation

Table 13 sets normative requirements for error generation at a PTP Relay Instance that ensure the relevant fields in the Sync and Follow_Up messages it transmits as part of Sync processing are sufficiently accurate to deliver the network-level time sync performance. The requirements include the ability to mitigate errors in rateRatio and rateRatio drift that would otherwise occur due to clock drift at the current PTP Relay Instance, an adjacent PTP Relay Instance, or the Grandmaster PTP Instance. Table D.4 describes how the normative requirements align with major sources of error.

Table D.4 – Main sources of error affecting normative requirements for PTP Relay Instance

Item	Normative requirement	Clock drifts	Main sources of error
1	preciseOriginTimestamp + correctionField vs Direct measurement of Clock Source at Grandmaster PTP Instance	None	Timestamp errors relative to Local Clock when measuring Residence Time, i.e. Sync message ingress and egress. Accuracy of meanLinkDelay measurement. Errors in Rate Ratio used when translating Residence Time measured in terms of Local Clock to Residence time in terms of Clock Source, although these are typically orders of magnitude smaller than those from Timestamp Errors.
2	rateRatio vs Direct measurement of Rate Ratio of Clock Source vs Local Clock	None	Timestamp error affecting measurement of NRR when there is no NRR Drift. The effect should be low. This normative requirement is a baseline for the next two requirements.
3		Clock Source (RR Drift)	Accuracy of measurement of NRR when there is no NRR Drift (as above). Accuracy of calculation of rateRatio, including algorithms for RR Drift tracking & error compensation.
4		Clock Source and Local Clock at previous PTP Instance (RR Drift & NRR drift)	Accuracy of measurement of NRR when there is NRR Drift, including algorithms for NRR Drift tracking & error compensation Accuracy of calculation of rateRatio, including algorithms for RR Drift tracking & error compensation. Combined with test 3 this effectively requires a level of performance regarding NRR Drift tracking & error compensation, whether the source of the NRR drift is the Local Clock of the current PTP Instance or the previous PTP Instance.
5	rateRatioDrift vs Direct measurement of Rate Ratio Drift of Clock Source vs Local Clock	None	Timestamp error affecting measurement of NRR Drift when there is no NRR Drift. The effect should be low. This normative requirement is a baseline for the next two requirements.
6		Clock Source (RR Drift)	Accuracy of measurement of NRR Drift when there is no NRR Drift (as above). Accuracy of calculation of rateRatioDrift, including algorithms for RR Drift tracking & error compensation.
7		Clock Source and Local Clock at previous PTP Instance (RR Drift & NRR drift)	Accuracy of measurement of NRR Drift when there is NRR Drift, including algorithms for NRR Drift tracking & error compensation. Accuracy of calculation of rateRatioDrift, including algorithms for RR Drift tracking & error compensation. Combined with test 6 this effectively requires a level of performance regarding NRR Drift tracking & error compensation, whether the source of the NRR drift is the Local Clock of the current PTP Instance or the previous PTP Instance.

Item	Normative requirement	Clock drifts	Main sources of error
8	syncEgressTimestamp vs Direct measurement of Local Clock	None	Timestamp error relative to Local Clock
9	(responseOriginTimestamp – requestReceiptTimestamp) vs Direct measurement of Local Clock interval between reception of Pdelay_Req and transmission of Pdelay_Resp	None	Timestamp error relative to Local Clock
10	Mean Link Delay	None	Timestamp error affecting measurement of meanLinkDelay. Each Pdelay measurement can be subject to relatively large errors from both the local and neighboring node. This normative requirement ensures a suitable algorithm mitigates the impact of any such single Pdelay measurement.

D.3.6 PTP End Instance error generation

Table 14 sets normative requirements for error generation at a PTP End Instance that ensure the ClockTarget it generates from incoming Sync and Follow_Up messages is sufficiently accurate to deliver the network-level time sync performance. Table D.5 describes how the normative requirements align with major sources of error.

Table D.5 – Main sources of error affecting normative requirements for PTP End Instance

Item	Normative requirement	Clock drifts	Main sources of error
1	ClockTarget vs ClockSource	None	Timestamp error affecting measurement of NRR Drift when there is no NRR Drift. The effect should be low. This normative requirement is a baseline for the next two tests.
2		Clock Source (RR Drift)	Accuracy of measurement of NRR Drift when there is no NRR Drift (as above). Accuracy of calculation of rateRatioDrift, including algorithms for RR Drift tracking & error compensation.
3		Clock Source and Local Clock at previous PTP Instance (RR Drift & NRR drift)	Accuracy of measurement of NRR Drift when there is NRR Drift, including algorithms for NRR Drift tracking & error compensation. Accuracy of calculation of rateRatioDrift, including algorithms for RR Drift tracking & error compensation. Combined with test 2 this effectively requires a level of performance regarding NRR Drift tracking & error compensation, whether the source of the NRR drift is the Local Clock of the current PTP Instance or the previous PTP Instance.
4	Mean Link Delay	None	Timestamp error affecting measurement of meanLinkDelay. Each Pdelay measurement can be subject to relatively large errors from both the local and neighboring node. This normative requirement ensures a suitable algorithm mitigates the impact of any such single Pdelay measurement.

D.3.7 Explanation for the asymmetric normative requirements for the allowable range of dTE in Table 14, rows 2 and 3

The PTP End Instance requirements for dTE in Table 14, rows 2 and 3, are asymmetric, i.e. the allowable range in both cases is -145 ns to +15 ns. These tests are performed with the fractional frequency offset of the Working Clock (acting as the ClockSource) at the Grandmaster PTP Instance increasing at the rate of 1 ppm/s. dTE is measured at the output of the PTP End Instance clock control system after steady-state is reached, i.e. after any transients due to initiating the 1 ppm/s frequency drift rate have decayed. The clock control system is a second-order filter, with bandwidth and gain peaking requirements given in Table 11. Table 11 also indicates that the filter has a minimum roll-off of 20 dB/decade. The worst-case dTE response can be analyzed using the second-order filter model described in Annex C. The transfer function for this filter is given by Formula (C.5). The example in C.4 shows that the damping ratio, ζ , for a filter whose gain peaking is 2,2 dB, i.e. the maximum allowed in Table 11, is 0,682 (see Formula (C.19)). The allowable range for filter 3dB bandwidth in Table 11 is 0,9 Hz – 1,0 Hz. The example in C.4 indicates that the undamped natural frequency, ω_n corresponding to the maximum bandwidth of 1,0 Hz is 3,10 rad/s (see Formula (C.18)).

It is well-known that if a frequency drift is applied to a linear, second-order filter with 20 dB/decade roll-off, the steady-state output has a time offset. This means that if the input to the PTP End Instance from the ClockSource (see Figure D.5) has a frequency drift of 1 ppm/s, the measured dTE output of the ClockTarget has a steady-state offset. Any errors due to other sources (e.g. timestamp granularity, dynamic timestamp error, clock noise) are in addition to this. If these other errors are symmetric about zero, the addition of the steady-state offset results in an asymmetric range of dTE.

The requirement in Table 14 is on the Working Clock (acting as ClockTarget) at the PTP End Instance minus the Working Clock (acting as ClockSource) at the Grandmaster. This difference, which is the error between the input and output and is denoted by $E(s)$ in the s-domain, is given by:

$$\begin{aligned}
 E(s) &= Y(s) - U(s) \\
 &= H(s)U(s) - U(s) \\
 &= (H(s) - 1)U(s) \\
 &= H_e(s)U(s)
 \end{aligned}
 \tag{D.1}$$

where $Y(s)$ and $U(s)$ are the Laplace transforms of the output and input, respectively, $H(s)$ is the transfer function between the input and output and is given by Formula (C.5), and $H_e(s)$ is the transfer function between the input and error. Using Formula (C.5), the error transfer function is given by:

$$H_e(s) = -\frac{s^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}
 \tag{D.2}$$

The phase drift corresponding to an $A = 1$ ppm/s = 1 000 ns/s² frequency drift is $0,5At^2$, where t is the time in seconds and the phase drift is in ns if A is in ns/s². The Laplace transform of this waveform is $U(s) = A/s^3$. The steady-state value of the filter output, $y_{\text{steady-state}}$, due to this drift is obtained from the final value theorem:

$$y_{\text{steady-state}} = \lim_{s \rightarrow 0} sH(s)U(s) = \lim_{s \rightarrow 0} s \cdot \left(-\frac{s^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \right) \cdot \frac{A}{s^3} = -\frac{A}{\omega_n^2} \quad (\text{D.3})$$

Since the 3dB bandwidth is proportional to the undamped natural frequency (see Formula (C.23)), Formula (D.32) indicates that the worst case, i.e. largest steady-state error occurs for the minimum 3dB bandwidth, i.e. 0,9 Hz in Table 11. In the example of C.4, the undamped natural frequency of 3,10 rad/s and damping ratio 0,682 correspond to a 3dB bandwidth of 1,0 Hz. Then, due to the direct proportionality between 3dB bandwidth and undamped natural frequency (for the same damping ratio), a 3dB bandwidth of 0,9 Hz corresponds to an undamped natural frequency of:

$$\omega_n = \left(\frac{0,9\text{Hz}}{1,0\text{Hz}} \right) (3,10\text{rad/s}) = 2,79\text{rad/s} \quad (\text{D.4})$$

Then, using Formula (D.32), the steady-state error corresponding to a frequency drift of 1 ppm/s is:

$$y_{\text{steady-state}} = -\frac{1000\text{ns/s}}{(2,79\text{rad/s})^2} = -128\text{ns} \quad (\text{D.5})$$

Then, assuming the error without the filter is 15 ns as given in Table 14, row 1, the lower end of the range of dTE when the filter is present is:

$$\text{dTE} = -128\text{ns} - 15\text{ns} = -143\text{ns} \approx -145\text{ns} \quad (\text{D.6})$$

D.4 Approach to testing normative requirements

D.4.1 General

This document does not specify tests to ensure conformance with the normative requirements. However, it is important that the requirements are, in principle, testable. Clause D.4 describes, at a high level, approaches a test specification can take to testing conformance with some of the requirements related to time synchronization.

It is assumed that test equipment can precisely measure the output of the ClockSource (at a Grandmaster PTP Instance), ClockTarget (at a PTP End Instance) and Local Clock (at any PTP Instance) to ensure conformance with frequency offset and frequency offset drift requirements. This might be via a Pulse per Second (PPS) plus Time-of-Day information or another mechanism.

It is also assumed that test equipment can generate sequences of PTP messages with precise timing and content (for testing PTP Relay Instances and PTP End Instances) and receive, log, and process sequences of PTP messages with precise timing measurement (e.g. of message arrival).

D.4.2 Testing Grandmaster PTP Instance

Figure D.1 illustrates an approach to testing the three normative requirements discussed in D.3.4.

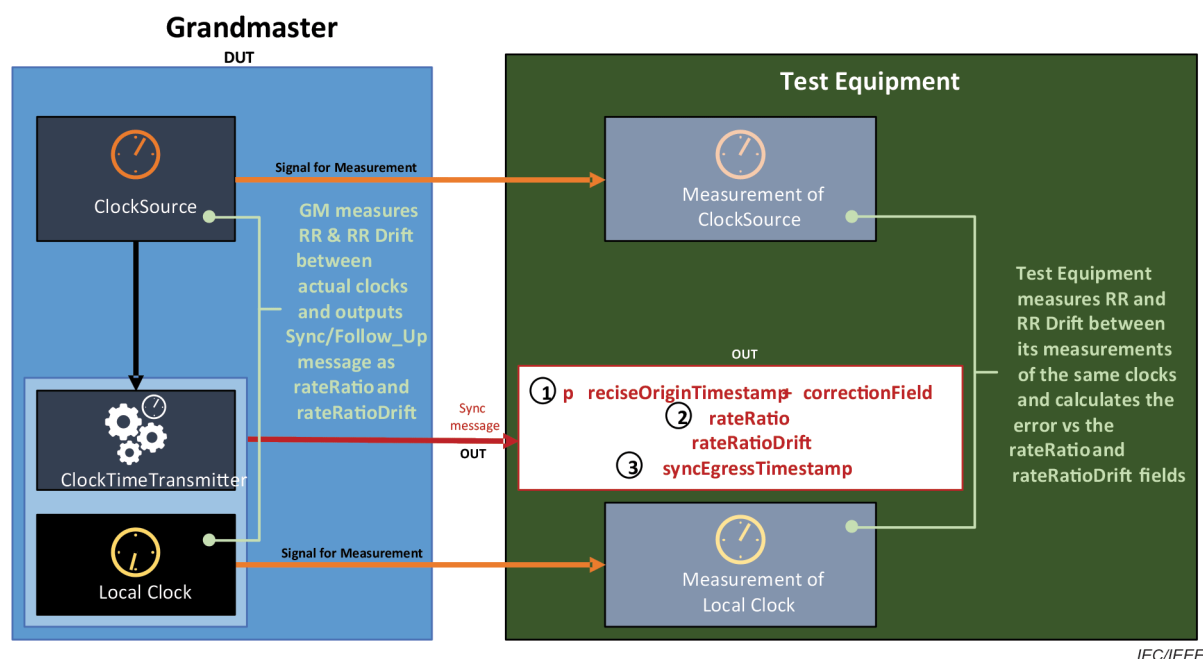


Figure D.1 – Approach to testing normative requirements for Grandmaster PTP Instance

The test equipment can calculate the time the Sync message is output at the DUT by subtracting the link delay from the measured arrival time at the test equipment.

For test 1, the test equipment compares the value of the preciseOriginTimestamp + correctionField against its measurement of the ClockSource.

For test 2, the test equipment compares the value in the rateRatio field with its calculation of the equivalent value based on its measurement of the ClockSource.

For test 3, the test equipment compares the value of the syncEgressTimestamp against its measurement of the Local Clock.

D.4.3 Testing PTP Relay Instance

Figure D.2 illustrates an approach to testing normative requirements 1, 2, 5 and 8 discussed in D.3.5.

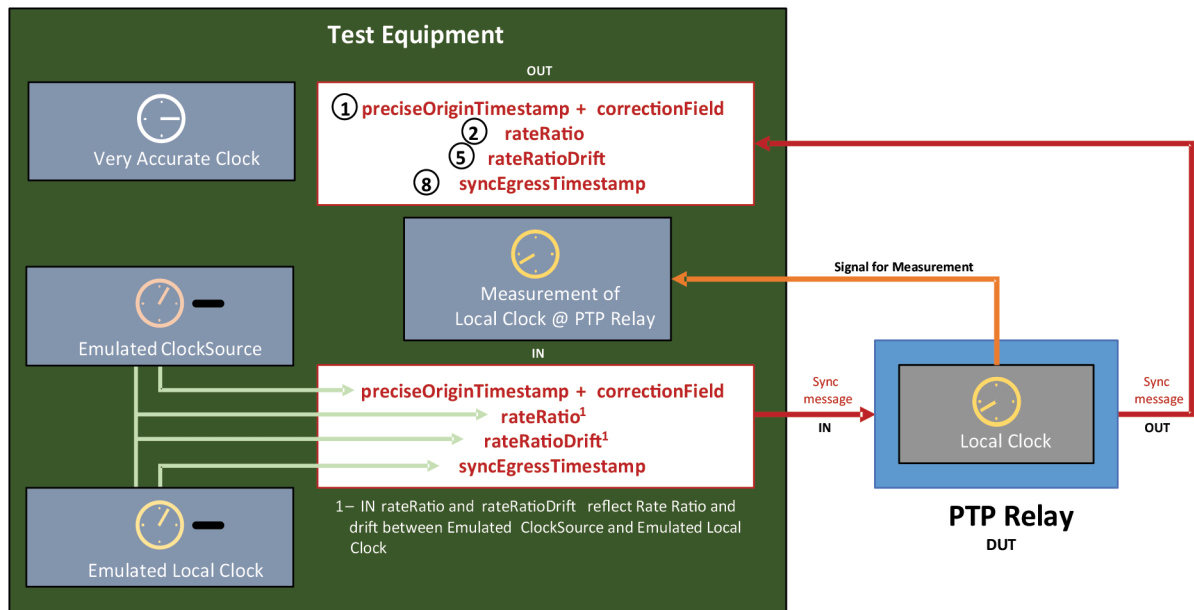


Figure D.2 – Approach to testing normative requirements for PTP Relay Instance – 1

The test equipment can compare the DUT's output Sync message to the expected result given the measurement of the Local Clock and the timing of the input PTP timing message transmission and output PTP timing message reception.

For these four tests, the Emulated ClockSource and Emulated Local Clock are stable and in sync. In practice, both can be equal to the test equipment's Very Accurate Clock. In the input Follow_Up information TLV, $rateRatio$ is 0 ppm, and in the input Drift_Tracking TLV $rateRatioDrift$ is 0 ppm/s. If the Local Clock of the PTP Relay Instance is also stable, it measures NRR of 0 ppm and NRR Drift of 0 ppm/s.

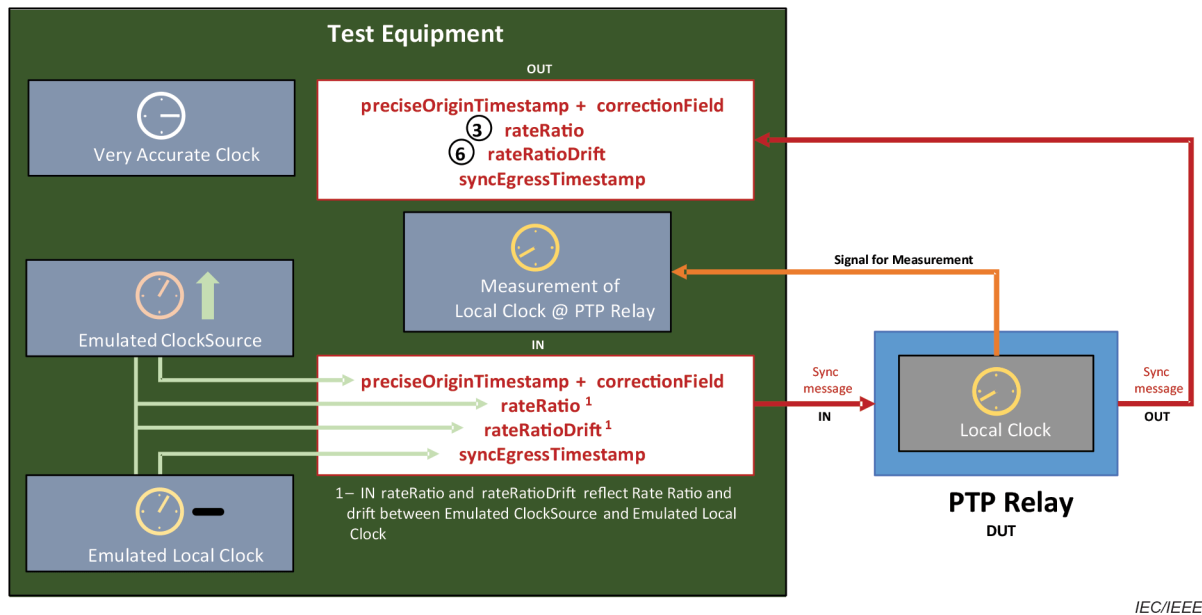
The test equipment can calculate the time the output Sync message is output at the DUT by subtracting the link delay from the measured arrival time at the test equipment.

For test 1, the test equipment can compare the increase in the value of the $correctionField$ to the measured $meanLinkDelay$ (from the test equipment to the DUT) plus $residenceTime$. The test equipment accounts for the additional delay between the PTP Relay Instance's transmission of the input PTP timing message and its reception by the test equipment.

For tests 2 and 5, the test equipment can compare the $rateRatio$ and $rateRatioDrift$ fields in the output PTP timing message with the equivalent calculated values between the measured Local Clock and the Emulated ClockSource.

For test 8, the test equipment can compare $syncEgressTimestamp$ value in the output PTP timing message with its measurement of the Local Clock.

Figure D.3 illustrates an approach to testing normative requirements 3 and 6 discussed in D.3.5.



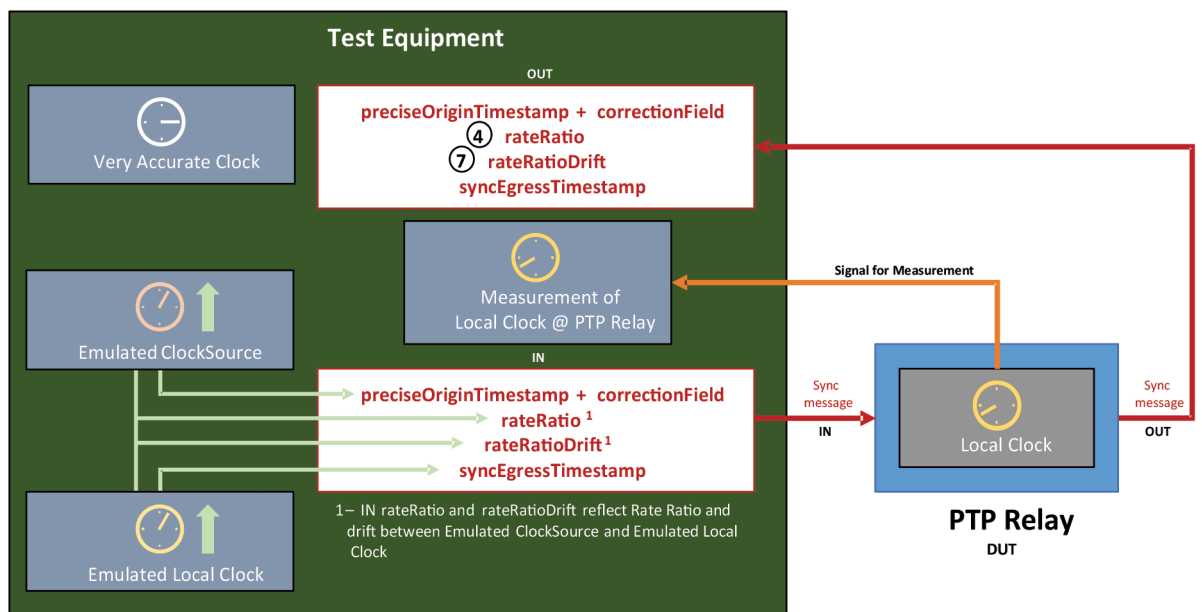
IEC/IEEE

Figure D.3 – Approach to testing normative requirements for PTP Relay Instance – 2

For these two tests, the fractional frequency offset of the Emulated ClockSource is increasing at a defined ppm/s rate relative to the Very Accurate Clock. The Emulated Local Clock is stable; in practice, it can be equal to the test equipment's Very Accurate Clock. In the output Follow_Up information TLV, the rateRatio field increases over time, and in the output Drift_Tracking TLV, the rateRatioDrift field maintains a matching positive value. If the Local Clock of the PTP Relay Instance is also stable, it measures a NRR of 0 ppm and a NRR Drift of 0 ppm/s.

For tests 3 and 6, the test equipment can compare the rateRatio and rateRatioDrift fields in the output Follow_Up information TLV and Drift_Tracking TLV respectively with the equivalent calculated values between the measured Local Clock and the Emulated ClockSource.

Figure D.4 illustrates an approach to testing normative requirements 4 and 7 discussed in D.3.5.



IEC/IEEE

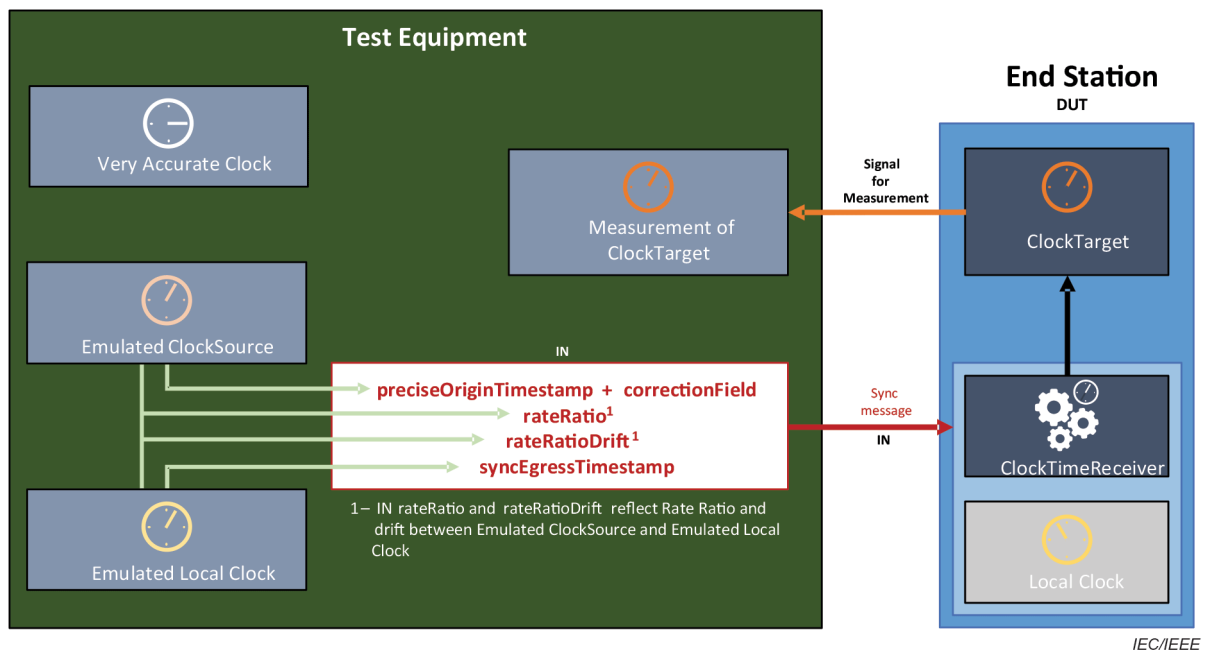
Figure D.4 – Approach to testing normative requirements for PTP Relay Instance – 3

For these two tests, the fractional frequency offsets of the Emulated ClockSource and the Emulated Local Clock are equal and increasing at a defined ppm/s rate relative to the Very Accurate Clock. In the output Follow_Up information TLV, the rateRatio field is 0 ppm, and in the output Drift_Tracking TLV the rateRatioDrift field is 0 ppm/s. If the Local Clock of the PTP End Instance is stable, the NRR it measures increases over time and the NRR Drift it measures maintains a matching positive value.

For tests 4 and 7, the test equipment can compare the rateRatio and rateRatioDrift fields in the output Follow_Up information TLV and Drift_Tracking TLV respectively with the equivalent calculated values between the measured Local Clock and the Emulated ClockSource.

D.4.4 Testing PTP End Instance

Figure D.5 illustrates an approach to testing the three normative requirements discussed in D.3.6.



IEC/IEEE

Figure D.5 – Approach to testing normative requirements for PTP End Instance

The test equipment can compare its measurement of the DUT's ClockTarget to the Emulated ClockSource. It accounts for the additional delay between its transmission of the input Sync message and the reception of the message by the DUT.

For test 1, the Emulated ClockSource and Emulated Local Clock are stable and in sync. In practice, both can be equal to the test equipment's Very Accurate Clock. In the input Follow_Up information TLV, rateRatio is 0 ppm, and in the input Drift_Tracking TLV, rateRatioDrift is 0 ppm/s. If the Local Clock of the PTP End Instance is also stable, it measures NRR of 0 ppm and NRR Drift of 0 ppm/s.

For test 2, the fractional frequency offset of the Emulated ClockSource is increasing at a defined ppm/s rate relative to the Very Accurate Clock. The Emulated Local Clock is stable; in practice, it can be equal to the test equipment's Very Accurate Clock. In the output Follow_Up information TLV, the rateRatio field increases over time, and the output Drift_Tracking TLV, the rateRatioDrift field maintains a matching positive value. If the Local Clock of the PTP Relay Instance is also stable, it measures NRR of 0 ppm and NRR Drift of 0 ppm/s.

For test 3, the fractional frequency offsets of the Emulated ClockSource and the Emulated Local Clock are equal and increasing at a defined ppm/s rate relative to the Very Accurate Clock. In the output Follow_Up information TLV, the rateRatio field is 0 ppm, and in the output output Drift_Tracking TLV, the rateRatioDrift field is 0 ppm/s. If the Local Clock of the PTP Relay Instance is stable, the NRR it measures increases over time and the NRR Drift it measures maintains a matching positive value.

D.4.5 Testing meanLinkDelay Accuracy

D.3.5 and D.3.6 both discuss a normative requirement to test the accuracy of meanLinkDelay measurements for PTP Relay Instances and PTP End Instances respectively. This can be achieved by using a cable of known length and asymmetry to connect the test equipment to the DUT. The test equipment then responds to the DUT's Pdelay_Req messages with Pdelay_Resp messages, allowing the DUT to estimate the path delay.

The DUT's estimate of the path delay can be accessed via the meanLinkDelay global variable (see IEEE 802.1AS-2020, Clause 14.8.8).

The error in a single path delay measurement via a Pdelay_Req/Pdelay_Resp exchange can be large relative to the normative requirement, but an estimate with significantly less error can be generated by considering more than one measurement, hence the "mean" in "meanLinkDelay". Two consequences of this are noteworthy.

Firstly, since errors can be introduced by Timestamp Errors at both the DUT and a neighboring node, and the DUT is responsible for mitigating both, the test equipment simulates typical errors on its timestamps for receiving Pdelay_Req and transmitting Pdelay_Resp messages to provide a complete test.

Secondly, since the meanLinkDelay measurement can take some time to stabilize based on multiple Pdelay_Req/Pdelay_Resp exchanges, leaving sufficient time for these multiple exchanges before assessing meanLinkDelay accuracy results in a more accurate test.

D.5 Example algorithms

D.5.1 General

This document does not place normative requirements on the use of specific algorithms. However, the normative requirements assume the use of algorithms to reduce the effect of errors in meanLinkDelay and to track clock drift and compensate for consequent errors. PTP Instances that do not implement algorithms can find it difficult or impossible to meet the normative requirements.

D.5 provides examples of algorithms that can be used for:

- tracking NRR drift;
- correcting for errors in measured NRR (mNRR) due to NRR drift;
- calculating RR drift;
- correcting for errors in measured RR (mRR) due to RR drift;
- reducing the effect of errors in meanLinkDelay.

D.5.2 Algorithm for tracking NRR drift

For measured NRR, measured RR, and meanLinkDelay, an example for how startup behavior can be handled is provided.

NRR Drift Tracking and Error Correction is carried out for each network hop, i.e. at every node other than the Grandmaster. It is based on pairs of timestamps with each pair associated with a Sync message transmitted from the previous node ($n-1$) to the current node (n).

- t_{s1outP} – Timestamp of the Sync message egress from the previous node ($n-1$), timestamped by that node's Local Clock. Unit: ns.
- t_{s2in} – Timestamp of the Sync message ingress to the current node (n), timestamped by that node's Local Clock. Unit: ns.

All timestamps are affected by Timestamp Errors.

The algorithm uses information from the 32 most recent Sync messages. However, a node only keeps track of the 9 most recent pairs of timestamps from the most recent (x) to the 9th most recent ($x-8$) Sync message. The algorithm generates one measurement of NRR using the prior 2 s of Sync message data (on average, based on a nominal Sync Interval of 125 ms), and a second measure based on the 2 s of Sync message data prior to that. It then uses the difference in the two measurements over the interval between the effective measurement points to calculate the NRR drift rate.

On arrival of a new Sync message (x), or Follow_Up in the case of two-step time transport, a node executes a NRR calculation:

$$NRRcalc(x) = \left(\frac{t_{s1outP}(x) - t_{s1outP}(x-8)}{t_{s2in}(x) - t_{s2in}(x-8)} - 1 \right) \times 10^6 \quad (D.7)$$

where

- $NRRcalc$ is the calculated Neighbor Rate Ratio, expressed in ppm;
- x is the most recent Sync message;
- t_{s1outP} is the timestamp of the Sync message egress from the previous node ($n-1$), timestamped by that node's Local Clock, expressed in ns;
- t_{s2in} is the Timestamp of the Sync message ingress to the current node (n), timestamped by that node's Local Clock, expressed in ns.

with an associated effective measurement point:

$$NRRcalcT(x) = \frac{t_{s2in}(x) + t_{s2in}(x-8)}{2} \quad (D.8)$$

where

- $NRRcalcT$ is the effective measurement point, expressed in ns;
- t_{s1outP} is the timestamp of the Sync message egress from the previous node ($n-1$), timestamped by that node's Local Clock, expressed in ns;
- t_{s2in} is the Timestamp of the Sync message ingress to the current node (n), timestamped by that node's Local Clock, expressed in ns.

A node keeps track of the 24 most recent NRR calculations and effective measurement points, from the most recent (x) to the 24th most recent ($x-23$).

After of a new most-recent NRR calculation, a node calculates an NRR drift rate:

$$NRR_{averageA} = \sum_{i=x-7}^x \frac{mNRR_{calc}(i)}{8} \quad (D.9)$$

where

$NRR_{averageA}$ is the average of the 8 most recent Neighbor Rate Ratio calculations, expressed in ppm.

$$NRR_{averageB} = \sum_{i=x-23}^{x-16} \frac{mNRR_{calc}(i)}{8} \quad (D.10)$$

where

$NRR_{averageB}$ is the average of the 8 least recent Neighbor Rate Ratio calculations, expressed in ppm.

$$NRR_{driftInterval} = \sum_{i=x-7}^x \frac{mNRR_{calcT}(i)}{8} - \sum_{i=x-23}^{x-16} \frac{mNRR_{calcT}(i)}{8} \quad (D.11)$$

where

$NRR_{driftInterval}$ is the period across which Neighbor Rate Ratio drift is measured, expressed in ns.

$$NRR_{driftRate}(n) = \left(\frac{NRR_{averageA} - NRR_{averageB}}{NRR_{driftInterval}} \right) \times 10^9 \quad (D.12)$$

where

$NRR_{driftRate}(n)$ is the the NRR drift rate for the current Node n , expressed in ppm/s.

D.5.3 Algorithm to compensate for errors in measured NRR due to clock drift

D.5.3.1 General

The algorithm to measure NRR uses data from the previous 1 s of Sync message data, combined with the NRR drift estimate from the previous step. This smaller amount of data (vs that used for either of the NRR measurements in the previous step) is employed as it improves responsiveness to sudden changes in NRR drift with minimal loss of accuracy.

On arrival of a Sync message (x), or Follow_Up in the case of two-step time transport, a node executes a NRR calculation:

$$mNRR_{calc}(x) = \left(\frac{t_{s1outP}(x) - t_{s1outP}(x-4)}{t_{s2in}(x) - t_{s2in}(x-4)} - 1 \right) \times 10^6 \quad (D.13)$$

with an associated effective measurement point:

$$mNRRcalcT(x) = \frac{t_{s2in}(x) + t_{s2in}(x-4)}{2} \quad (D.14)$$

A node keeps track of the 4 most recent mNRR calculations and effective measurement points, from the most recent (x) to the 4th most recent ($x-3$). The mNRR calculations use information from the 5 most recent Sync messages, but the node is already keeping track of information from the 9 most recent Sync messages for the NRR drift tracking algorithm.

The node then calculates an error corrected measured NRR value.

For $i = x$ to $(x-3)$

$$mNRRcorrected(i) = mNRRcalc(i) + \left(NRRdriftRate(n) \times \frac{(t_{s2in}(x) - mNRRcalcT(i))}{10^9} \right) \quad (D.15)$$

where

$mNRRcorrected(i)$ is the error-corrected measured NRR value at message i , expressed in ppm.

$$mNRR(n) = \sum_{i=x-3}^x \frac{mNRRcorrected(i)}{4} \quad (D.16)$$

where

$mNRR$ is the error-corrected measured NRR value, expressed in ppm.

The result is a measured NRR value, error-corrected to the time when the most recent Sync message was received.

D.5.3.2 Measured NRR algorithm – Startup behavior

NRR is used when calculating meanLinkDelay and output Sync/Follow_Up message fields. The first NRR drift calculation is only be available after receipt of 32 Sync/Follow_Up messages, i.e. after approximately 4 seconds of operation given the 125 ms Sync Interval. During this time meanLinkDelay and output Sync/Follow_Up messages fields have not been calculated, so an alternative is used, even if it cannot deliver the same assurances regarding network-level performance.

If measured NRR from Sync/Follow_Up message information is unavailable but equivalent information from Pdelay_Resp messages is available, it may be substituted for Sync/Follow_Up message information. However, measuring NRR using Pdelay_Resp messages is vulnerable to additional error due to clock drift between the time NRR is measured, on receipt of the latest Pdelay_Resp message, and use of the measurement during Sync message processing. This is the reason using Sync/Follow_Up message information is preferable. It also means that a switch to using Sync/Follow_Up message information as soon as possible is desirable. It is technically possible to calculate a NRR using a combination of Pdelay_Resp and Sync messages but this can be risky due to the potential for very short intervals between messages and resulting high error due to timestamp errors, so it not recommended.

It is the responsibility of implementers to decide whether and when to use Pdelay_Resp message information and when to switch to using Sync/Follow_Up message information. The normative requirements in this document can be achieved after 32 Sync/Follow_Up messages have been received and assume the use of the algorithms in D.5.2 and D.5.3, or more effective algorithms. Implementations that continue to use Pdelay_Resp message information to calculate NRR after 32 messages have been received can find some of the normative requirements difficult or impossible to meet.

The following describes potential startup behavior applicable to either Sync/Follow_Up or Pdelay_Resp message information.

- a) At least two messages must be received before calculating a NRR value.
- b) Prior to two messages being received, NRR = 1 (i.e. 0 ppm) should be used.
- c) Once two messages have been received, NRR should be calculated using the formula:

$$\text{2nd message: } mNRR = \left(\left(\frac{t_3(x) - t_3(x-1)}{t_4(x) - t_4(x-1)} \right) - 1 \right) \times 10^6 \quad (\text{D.17})$$

where

- $t_3(i)$ is the timestamp of the i^{th} Pdelay_Resp message on egress from the previous node ($n-1$), timestamped by that node's Local Clock, expressed in ns;
- $t_4(i)$ is the timestamp of the Pdelay_Resp message on ingress to the current node (n), timestamped by that node's Local Clock, expressed in ns.

- d) When three to four messages have been received, NRR should be calculated using the following formula:

$$\text{3rd message: } mNRR = \left(\left(\frac{t_{1\text{outP}}(x) - t_{1\text{outP}}(x-2)}{t_{2\text{in}}(x) - t_{2\text{in}}(x-2)} \right) - 1 \right) \times 10^6 \quad (\text{D.18})$$

$$\text{4th message: } mNRR = \left(\left(\frac{t_{1\text{outP}}(x) - t_{1\text{outP}}(x-3)}{t_{2\text{in}}(x) - t_{2\text{in}}(x-3)} \right) - 1 \right) \times 10^6 \quad (\text{D.19})$$

- e) On arrival of the 5th Sync/Follow_Up message the first $mNRRcalc$ and $mNRRcalcT$ calculations can take place and should be used for NRR:

$$mNRRcalc(x) = \left(\frac{t_{s1outP}(x) - t_{s1outP}(x-4)}{t_{s2in}(x) - t_{s2in}(x-4)} - 1 \right) \times 10^6 \quad (D.20)$$

$$mNRRcalcT(x) = \frac{t_{s2in}(x) + t_{s2in}(x-4)}{2} \quad (D.21)$$

$$5^{th} \text{ Sync message: } mNRR = mNRRcalc(x) \quad (D.22)$$

- f) As the 6th, 7th and 8th messages arrive an average can be taken and used for NRR, so:

$$6^{th} \text{ message: } mNRR = \sum_{i=x-1}^x \frac{mNRRcalc(i)}{2} \quad (D.23)$$

$$7^{th} \text{ message: } mNRR = \sum_{i=x-2}^x \frac{mNRRcalc(i)}{3} \quad (D.24)$$

$$8^{th} \text{ message: } mNRR = \sum_{i=x-3}^x \frac{mNRRcalc(i)}{4} \quad (D.25)$$

- g) For the 9th to the 31st message, the Formula (D.19) can be used.
h) Once the 32nd message arrives, the regular formulas with NRR drift tracking and error correction can be used.

D.5.4 Algorithm for tracking RR Drift

A Sync or Follow_Up message carries the rateRatio field, which informs each node of the previous node's estimate of its (the previous node's) Rate Ratio. This document also requires support for the Drift_Tracking TLV, which carries the rateRatioDrift field and informs each node of the previous node's estimate of its (the previous node's) Rate Ratio Drift.

If the implementation of the Grandmaster PTP Instance means the ClockSource and Local Clock (at the Grandmaster PTP Instance) are linked such that the two are always operating at the same frequency, the rateRatio field received by the first node (Node 1) is always 0 ppm and the rateRatioDrift field is always 0 ppm/s. Thus, at Node 1, RR equals NRR, RR Drift equals NRR Drift, and therefore D.5.3 and D.5.2 describe how to calculate RR and RR Drift at Node 1.

If the implementation of the Grandmaster PTP Instance means the ClockSource and Local Clock (at the Grandmaster PTP Instance) can operate at different frequencies, the implementation populates the rateRatio and rateRatioDrift field with values reflecting those differences.

In either case all PTP Instances, other than the Grandmaster PTP Instance, calculate an estimate of the local Rate Ratio Drift when the latest Sync/Follow_Up Message is received, based on the received rateRatioDrift field and the local measure of NRR Drift. The Rate Ratio Drift Rate from the previous node is in ppm/s relative to the timebase of its Local Clock (i.e. the "s" in "ppm/s"). For highest precision, this can be converted to the timebase of the current node's Local Clock.

$$rateRatioDrift(n) = \frac{rateRatioDrift(n-1)}{\left(1 + \frac{mNRR(n)}{10^6}\right)} + NRRdriftRate(n) \quad (D.26)$$

where

rateRatioDrift is the Rate Ratio drift rate, expressed in ppm/s.

However, given that adding ppm/s already lacks the precision of multiplying actual ratios, this simplification delivers similarly accurate results.

$$rateRatioDrift(n) = rateRatioDrift(n-1) + NRRdriftRate(n) \quad (D.27)$$

D.5.5 Algorithm to compensate for errors in measured RR due to clock drift

On receipt of a Sync or Follow_Up message, all PTP Relay Instances estimate a measured RR ($mRR(n)$) based on the received rateRatio field ($mRR(n-1)$) and the local measure of NRR ($mNRR(n)$). An $mRR(n)$ value is used to translate the sum of meanLinkDelay and residenceTime from Local Clock timebase into Grandmaster timebase. An $mRR(n)$ value is also passed in the transmitted Sync or Follow_Up message's rateRatio field to the next node. Errors in these estimates due to clock drift can be reduced by taking account of RR Drift. Since the optimal point in time for each estimate is different, the amount of applicable RR Drift is different, and hence the estimates are different.

(For discussion of how different Grandmaster PTP Implementations affect the behavior of a PTP Relay Instance at Node 1 – or not – see D.5.4.)

A PTP End Instance is similar in that it estimates $mRR(n)$ on receipt of a Sync message, subsequently uses an $mRR(n)$ value, and errors in the latter due to clock drift can be mitigated by taking account of RR Drift. However, unlike a PTP Relay Instance, the $mRR(n)$ value is used to keep the ClockTarget in line with the ClockSource and transmitting a rateRatio field to a subsequent node is unnecessary.

For a PTP Relay Instance there are three points in time of interest:

- Point a: receipt of the Sync Message by the current node (Node n),
- Point b: mid-point between transmission of the Sync message by the previous node (Node $n-1$) and transmission of the consequent Sync message by the current node (Node n), and
- Point c: transmission of the Sync Message by the current node (Node n).

Figure D.6 illustrates these points and the associated calculations.

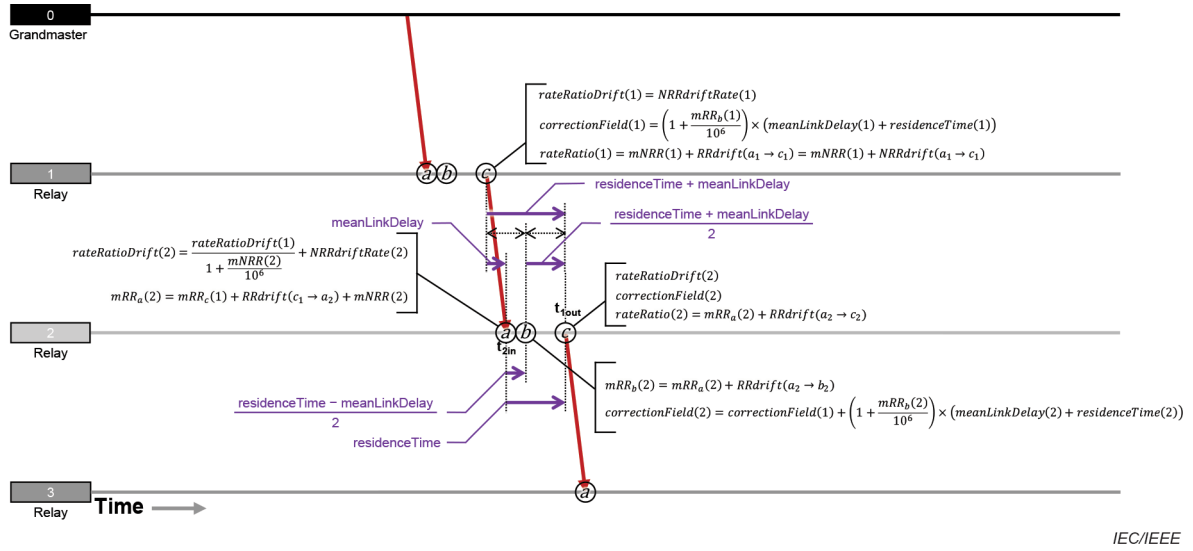


Figure D.6 – RR Drift tracking and error compensation calculations – PTP Relay Instance

The estimate of RR when the Sync message arrives can be calculated as follows:

$$mRR_a(n) = rateRatio(n-1) + RRdrift(c_{n-1} \rightarrow a_n) + mNRR(n). \quad (D.28)$$

$$= rateRatio(n-1) + \left(rateRatioDrift(n-1) \times \left(1 + \frac{mNRR(n)}{10^6} \right) \times meanLinkDelay(n) \right) + mNRR(n)$$

where

mRR_a is the estimate of RR when the Sync message arrives at node n , expressed in ppm;

$RRdrift(c_{n-1} \rightarrow a_n)$ is the amount $rateRatio(n-1)$ drifts between transmission of the Sync message at Node $n-1$ and reception at Node n , expressed in ppm/s.

$RRdrift(c_{n-1} \rightarrow a_n)$ is equivalent to $rateRatioDrift(n-1)$ multiplied by meanLinkDelay but, since meanLinkDelay is measured in terms of Node n 's Local Clock and rateRatioDrift is in terms of Node $n-1$'s Local Clock the former should be multiplied by the NRR at Node n for the highest accuracy.

However, given that adding ppm/s already lacks the precision of multiplying actual ratios, the following simplification delivers similarly accurate results:

$$mRR_a(n) = rateRatio(n-1) + (rateRatioDrift(n-1) \times meanLinkDelay(n)) + mNRR(n) \quad (D.29)$$

Once the time when Node n transmits the consequent Sync message is known, the correctionField value can be calculated.

$$mRR_b(n) = mRR_a(n) + RRdrift_n(a \rightarrow b) \quad (D.30)$$

$$= mRR_a(n) + \left(rateRatioDrift(n) \times \frac{residenceTime(n) - meanLinkDelay(n)}{2} \right)$$

where

$mRR_b(n)$ is the estimate of RR when Node n transmits the consequent Sync message, expressed in ppm.

The correctionField is calculated as follows:

$$\begin{aligned} correctionField(n) = & correctionField(n-1) + \left(1 + \frac{mRR_b(n)}{10^6} \right) \\ & \times (meanLinkDelay(n) + residenceTime(n)) \end{aligned} \quad (D.31)$$

where

$correctionField(n)$ is the value of the correction field transmitted by node n , expressed in ns.

The rateRatio field is calculated as follows:

$$\begin{aligned} rateRatio(n) = & mRR_a(n) + RRdrift_n(a \rightarrow c) \\ = & mRR_a(n) + (RRdriftRate(n) \times residenceTime(n)) \end{aligned} \quad (D.32)$$

where

$rateRatio(n)$ is the value of the rateRatio field transmitted by node n , expressed in ppm.

D.5.6 Algorithm to compensate for errors in measured RR due to clock drift at PTP End Instance

Figure D.7 illustrates a possible approach to applying similar RR drift tracking and error compensation at a PTP End Instance.

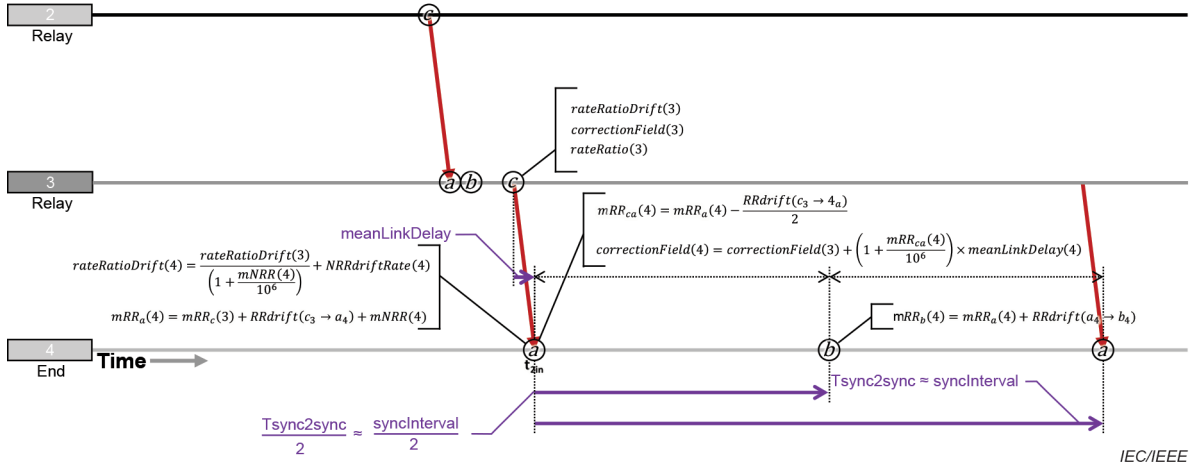


Figure D.7 – RR Drift tracking and error compensation calculations – PTP End Instance

The initial calculations for $rateRatioDrift$ and mRR_a are exactly the same as for a PTP Relay Instance. Instead of using RR to translate $meanLinkDelay + residenceTime$ from the Local Clock timebase to the Grandmaster timebase – as is done at a PTP Relay Instance – a PTP End Instance uses mRR to translate $meanLinkDelay$ to the Grandmaster timebase (there is no $residenceTime$ at an End Instance), adding the result to the incoming $correctionField$ to obtain an estimate of the ClockSource at the time the Sync message arrives, then uses mRR to keep its ClockTarget in line with the ClockSource until arrival of the next Sync message. The optimal mRR value for translating $meanLinkDelay$ is halfway between $meanLinkDelay$'s transmission (at c_{n-1} , i.e. point C at the previous node) and reception (at a_n , i.e. point A at the current node); in the formulas below, this value is referred to as mRR_{ca} .

$$\begin{aligned}
 mRR_{ca}(n) &= mRR_a(n) - \frac{RRdrift(c_{n-1} \rightarrow a_n)}{2} \\
 &= mRR_a(n) - \left(rateRatioDrift(n) \times \frac{meanLinkDelay(n)}{2} \right) \quad (D.33)
 \end{aligned}$$

where

$mRR_{ca}(n)$ is the estimate of RR calculated based upon one-half of the $meanLinkDelay$, expressed in ppm.

The correction field at the PTP End Instance is given by:

$$correctionField(n) = correctionField(n-1) + mRR_{ca}(n) \times meanlinkDelay(n) \quad (D.34)$$

The optimal value of mRR for keeping the ClockTarget in line with the Clock Source is mRR_b , where Point B is halfway between the most recently received Sync message and the next Sync message.

Of course, the exact interval until the next Sync message's arrival ($T_{\text{sync2sync}}$ in Figure D.7) can't be known before it happens, but the Rate Ratio value is required as soon as possible after arrival of the most recent Sync message. The solution is to use the nominal value of the interval, i.e. syncInterval , which is 125 ms.

$$\begin{aligned}
 mRR_b(n) &= mRR_a(n) + RRdrift_n(a \rightarrow b) \\
 &= mRR_a(n) + \left(rateRatioDrift(n) \times \frac{\text{syncInterval}}{2} \right) \\
 &= mRR_a(n) + (rateRatioDrift(n) \times 0,0625)
 \end{aligned} \tag{D.35}$$

where

syncInterval is the nominal value of the interval between sync messages, expressed in ms.

It is also possible to use more complex algorithms that repeatedly or continuously adjust the mRR value between Sync messages, but such an approach is not addressed in this document.

D.5.7 Mean Link Delay averaging

The actual Path Delay from one node to the next – for a wired connection – is very stable and errors measuring it due to Timestamp Error average to zero. Thus, taking a long average or applying a low-pass filter with a low bandwidth is an effective way to reduce error in meanLinkDelay . Care is taken during system startup or after any other initialization of the algorithm, to quickly converge on a stable value.

The basic Pdelay calculation, used by the Common Mean Link Delay service, remains the same. Figure D.8 illustrates it.

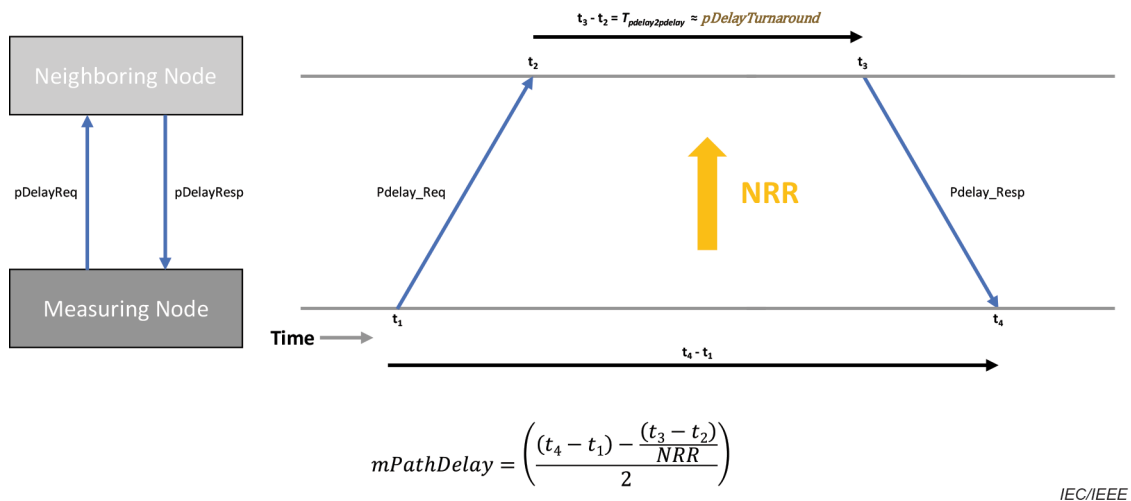


Figure D.8 – Signals and timestamps to measure path delay

Following each Pdelay_Req – Pdelay_Resp exchange, the measured path delay ($mPathDelay$) is calculated.

For the x^{th} message after initialization

$$mPathDelay(x) = \frac{(t_4 - t_1) - \frac{(t_3 - t_2)}{NRR}}{2} \quad (D.36)$$

where

- mPathDelay* is the measured path delay between the measuring node and the neighboring node expressed in ns;
- t_1 is a measurement point as defined in Figure D.8, expressed in ns;
- t_2 is a measurement point as defined in Figure D.8, expressed in ns;
- t_3 is a measurement point as defined in Figure D.8, expressed in ns;
- t_4 is a measurement point as defined in Figure D.8, expressed in ns.

The meanLinkDelay is then updated via an IIR (Infinite Impulse Response) filter. For the first measurement, the filter is initialized:

$$meanLinkDelay(x) = mPathDelay(x) \quad (D.37)$$

For the next few minutes after initialization (when $x < 1\,000$) the filter is in startup mode. It then transitions to steady-state mode.

If $x < 1\,000$ then $f = x$ else $f = 1\,000$

$$meanLinkDelay(x) = \frac{(meanLinkDelay(x-1) \times (f-1)) + mPathDelay(x)}{f} \quad (D.38)$$

For example:

$$meanLinkDelay(100) = \frac{(meanLinkDelay(99) \times 99) + pDelay(x)}{100} \quad (D.39)$$

$$meanLinkDelay(5\,836) = \frac{(meanLinkDelay(5\,835) \times 999) + pDelay(x)}{1\,000} \quad (D.40)$$

It is possible to automatically reinitialize the algorithm if an *mPathDelay* value, or series of values, deviates too much from the *meanLinkDelay*, but the details are outside the scope of this document.

The behavior of timestamp error means that, for shorter actual link delays, `mPathDelay` can be a negative value. It can seem tempting to reject negative values, since a negative delay is impossible. However, at a device level, including negative values of `mPathDelay` in the input to the IRR filter results in a more accurate filter output, i.e. `meanLinkDelay` value; values lower than the actual delay, even when negative, are balanced by values higher than the actual delay.

Similarly, at a network level, using negative values of `meanLinkDelay`, i.e. the output of the IIR filter, results in a more accurate `correctionField` calculation at the PTP End Instance when there are many networking hops between it and the Grandmaster PTP Instance.

Annex E (normative)

60802 device info TLV definition

E.1 60802 device info TLV

E.1.1 General

In the Type, Length, Value (TLV) definition that follows, any fields that are labelled as “Reserved” are transmitted as zero and ignored on receipt.

The 60802 device info TLV is a mandatory TLV that indicates a device’s configuration capabilities.

Figure E.1 shows the 60802 device info TLV.

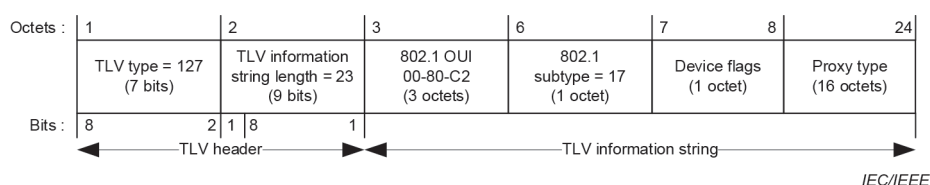


Figure E.1 – 60802 device info TLV

E.1.2 Device flags

The device flags field shall contain a bit map indicating the device configuration capabilities as defined in Table E.1.

Table E.1 – Device flags

Bit	Value/meaning	Function
0	1 = Proxy required 0 = no Proxy required	Proxy required
1	1 = supported 0 = not supported	NETCONF configuration supported
2-7	(set to zero)	Reserved for future standardization

E.1.3 Proxy type

The proxy type field is conditionally mandatory if the device flags indicate that the device requires a proxy. It indicates the proxy type that is required to configure the device. The format is a human readable string, that is specific for a specific proxy and its associated devices.

Annex F (informative)

MAC and PHY delay

F.1 Accumulated delays in industrial automation

F.1.1 General

MAC and PHY delays affect the performance of industrial communication. These delays impact forwarding delay, the accuracy of time-triggered transmission, and synchronization accuracy. Factors contributing to these delays include:

- the clock system between the MAC application, the MAC, the Media Independent Interface (MII), and the PHY,
- the processing of the frame, and
- the transmission of bits from the MAC application over MAC, MII and PHY to the MDI.

These delays consist of two components:

- the propagation delay, and
- the propagation delay variance (often called jitter).

F.1.2 Forwarding delay

Linear topologies, as commonly used in industrial communication, result in cumulative MAC and PHY delays. With 64 hops, there are 130 MAC and PHY delays (64 hops x 2 + 2x end stations) in end-to-end transmission.

For example, a frame sent from the end station at the beginning of a linear topology (64 Hops) to an end station at the end of the line is delayed by $26\ \mu\text{s}$ / $65\ \mu\text{s}$ / $130\ \mu\text{s}$ solely due to a PHY delay of 200 ns/500 ns/1 μs . This means that the lower the PHY delay, the shorter the achievable update interval and thus, the roundtrip delay.

F.1.3 Time-triggered transmission

The reference point for time-triggered transmission is at the MDI. However, the frames are handed over from the transmission control to the MAC in a time-controlled manner. To transmit a frame at the correct time at the reference point, the MAC and PHY delay are compensated by the transmission control.

For example, the magnitude of the propagation delay reduces the computation time available for the application to calculate new process values, as the frames are handed over from the transmission control to the MAC earlier so they can be sent at the correct time at the reference point. Additionally, the frames are also received later by the application, which further reduces the computation time (see 4.1).

The magnitude of the propagation delay variance affects the safety margin of the windows in TAS and is a consideration when configuring PSFP and scheduled traffic. Additionally, it increases the required TAS window size in linear topologies to ensure that all planned frames can pass through their assigned TAS windows.

F.1.4 Synchronization

The reference point for synchronization is also at the MDI. The actual point at which the transmitted or received synchronization frames are timestamped depends on the implementation and can be in the PHY, on the MII, or behind the MAC. Depending on the implementation, part or all the MAC and PHY delay is compensated to determine the time at which the synchronization frame was at the reference point.

For example, during synchronization, the mean link delay and residence time are determined. The propagation delay variance may be different for the transmission of pdelay frames and synchronization frames. This results in an uncompensated delay error for the sync frames, which accumulates in linear topologies.

F.1.5 Recommended delays

Ethernet hardware used for industrial communication typically considers the requirements for MAC and MII propagation delay and propagation delay variance, minimizing these values during development. PHYs, if not integrated, are appropriately developed or selected.

Table F.1 shows the recommended values for PHYs in industrial communication.

Table F.1 – Recommended PHY delay

MAU Type	Recommended propagation delay	Recommended propagation delay variance
10BASE-T1L	maximum < 5 μ s minimum < 500 ns	maximum < 16 ns minimum < 4 ns
100BASE-TX, 100BASE-FX, 1000BASE-SX, and 1000BASE-T	maximum < 1 μ s minimum < 500 ns	maximum < 16 ns minimum < 4 ns

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